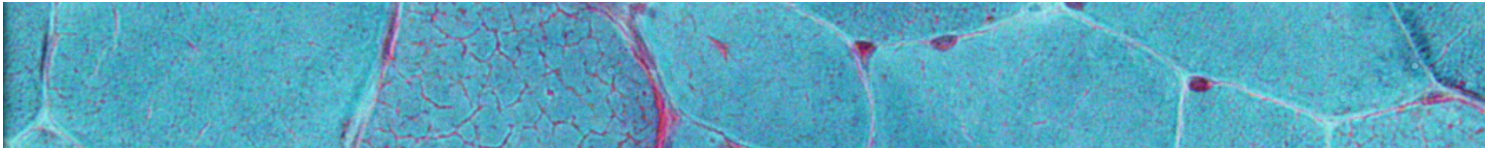




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PHARMACOLOGY LEARNING MODULES

This page contains pharmacology learning modules. Most have a mini-quiz at the end to check your understanding of the content.

MODULES:

Biochemistry pharm – Cystic fibrosis drugs
Biochemistry pharm – Cytoprotective agents

Cardiology pharm – Anti-arrhythmics
Cardiology pharm – Lipid-lowering agents
Cardiology pharm – Mixed HY drugs

Endocrine pharm – Diabetes drugs
Endocrine pharm – HY mixed drugs

General pharm – Alpha (α) adrenergic drugs
General pharm – Beta (β) adrenergic drugs
General pharm – Cholinergic agents (anti-)
General pharm – Cholinergic agents (pro-)
General pharm – Drug antidotes
General pharm – Drug metabolism/elimination
General pharm – Enzyme kinetics
General pharm – Pharmacokinetics
General pharm – Types of drug interactions

GI pharm – Antacids (highest yield points)
GI pharm – HY mixed agents

Heme/onc pharm – Anti-platelet agents, NSAIDs, aspirin
Heme/onc pharm – Heparin, anticoagulants, thrombolytics
Heme/onc pharm – HY mixed anti-cancer agents I
Heme/onc pharm – HY mixed anti-cancer agents II
Heme/onc pharm – Purine + pyrimidine synthesis inhibitors
Heme/onc pharm – Vitamin K and Warfarin

Immunology pharm – Monoclonal antibodies (anti-cancer)
Immunology pharm – Monoclonal antibodies (autoimmune disease therapy)
Immunology pharm – Monoclonal antibodies (miscellaneous application)
Immunology pharm – Non-antibody immunosuppressants

Microbiology pharm – Antifungals
Microbiology pharm – Cephalosporins (highest yield points)

Microbiology pharm – Hepatitis B drugs + HY serology

Microbiology pharm – Hepatitis C drugs

Microbiology pharm – HIV drugs

Microbiology pharm – HY mixed antimicrobials I

Microbiology pharm – HY mixed antimicrobials II

Microbiology pharm – HY mixed antimicrobials III

Microbiology pharm – HY mixed antivirals

Microbiology pharm – Protein synthesis (30/50s) inhibitors

Microbiology pharm – TB drugs

MSK pharm – Gout drugs

MSK pharm – Osteoporosis agents

MSK pharm – Rheumatoid arthritis drugs

Neuro pharm – Epilepsy drugs and hypnotics

Neuro pharm – Parkinson drugs

Psych pharm – Antidepressants + miscellaneous agents

Psych pharm – Antipsychotics

Pulmonary pharm – Asthma drugs

Pulmonary pharm – HY mixed drugs

Renal pharm – Diuretics HY points

Repro pharm – HY mixed agents

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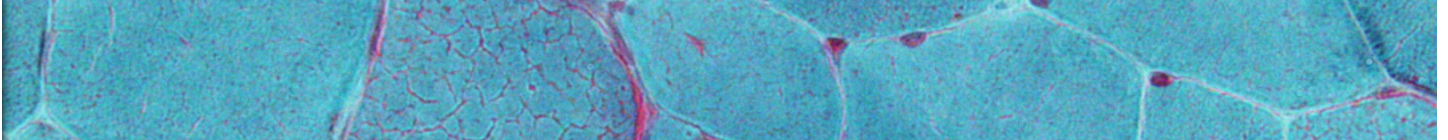
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I am a: *

Which exam(s) are you currently studying for? (select all that apply) *



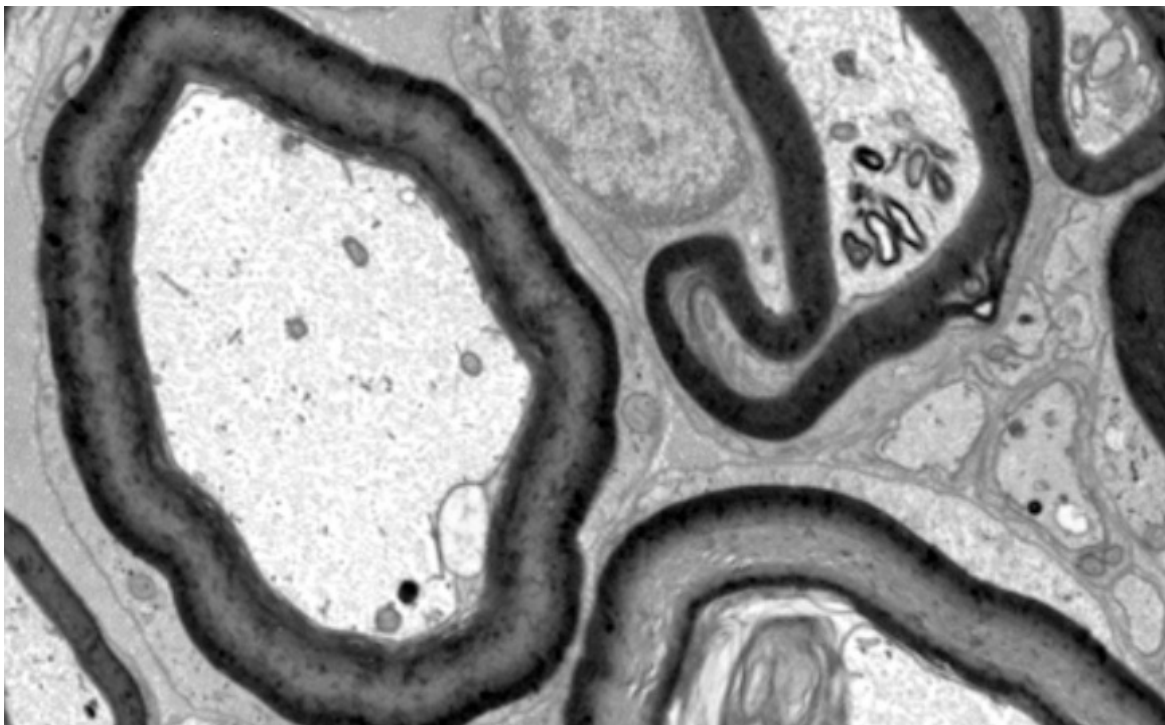
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BIOCHEMISTRY / PHARMACOLOGY

Biochemistry pharm – Cystic fibrosis drugs

by MEHLMANMEDICAL



Note to students: this CF pharm module is no longer required for the pass/fail Step 1. I originally created it during the numerical Step 1 era. So my options are: take down the module, or just keep it up (as I've clearly done) for those of you who like the extra pharm, but I'm letting you know this specific module isn't mandatory anymore.

A 6-year-old Caucasian boy with a history of recurrent respiratory infections, failure to thrive, and failure to pass meconium for 72 hours after birth has genotyping performed in order to be considered for a novel treatment for his condition. Gene sequencing shows a G551D mutation in the *CFTR* gene, in which glycine (G) in position 551 is replaced with aspartic acid (D). He does not carry a Phe508del ($\Delta F508$; deletion of phenylalanine at position 508) mutation. Which of the following agent(s) may be used in this patient as a result of these findings?

☐ Dornase-alfa

- ☐ Guaifenesin
- ☐ Ivacaftor
- ☐ Lumacaftor + Ivacaftor
- ☐ Elexacaftor + Ivacaftor + Tezacaftor

MEHLMANMEDICAL.COM	
Drug names	Mechanism of action
Ivacaftor	CFTR potentiator (↑ opening/activity of channel at cell surface)
Lumacaftor	CFTR corrector (↑ shuttling of defective channel to cell surface)
Tezacaftor	CFTR corrector
Elexacaftor	CFTR corrector
Formulations used in clinical practice	When to use
Ivacaftor	Patients >6 yrs of age with at least one copy of the G551D-CFTR mutation
Lumacaftor + Ivacaftor	Patients who are homozygous for the Phe508del (ΔF508) mutation
Elexacaftor+ Ivacaftor + Tezacaftor	Patients age ≥ 12 years who have ≥ 1 F508del mutation

This patient has cystic fibrosis (CF), an autosomal recessive disorder on chromosome 7 caused by a defective chloride channel.¹

CF is characterized by chronic bacterial respiratory infections, fat malabsorption secondary to exocrine pancreas insufficiency, infertility in males due to congenital bilateral absence of the vas deferens (CBAVD), and elevated concentrations of chloride in sweat.^{2, 3}

Ivacaftor^{4, 5, 6, 7}

- One type of mutation responsible for CF, the G551D missense mutation, leads to a glycine that is replaced by an aspartic acid at position 551.
- This enables the cystic fibrosis transmembrane regulator (CFTR) channel to make it to the cell surface, in contrast to sequestration in the RER/cytosol. The CFTR channel is still defective; it's just merely at the correct location on the cell surface.
- Ivacaftor is a CFTR modulator/potentiator that binds to this misfolded channel and increases the probability it will open.
- Ivacaftor is approved for use in patients >6 yrs of age with at least one copy of the G551D-CFTR mutation.**

Lumacaftor/ivacaftor^{8, 9}

- Approved for the treatment of CF in patients who are homozygous for the Phe508del (ΔF508) mutation.
- Lumacaftor works by increasing the trafficking of CFTR proteins to the cell surface. Ivacaftor works by enabling the opening of what would otherwise be a dysfunctional chloride channel.⁸

Elexacaftor/ivacaftor/tezacaftor^{10, 11, 12, 13, 14}

- Approved by the US FDA for the treatment of cystic fibrosis in patients age ≥ 12 years who have ≥ 1 F508del mutation in the CFTR gene.^{10, 12}
- Tezacaftor and elexacaftor are similar to lumacaftor. They are known as "correctors" and help traffic the defective CFTR channel to the cell surface.^{11, 13, 14}

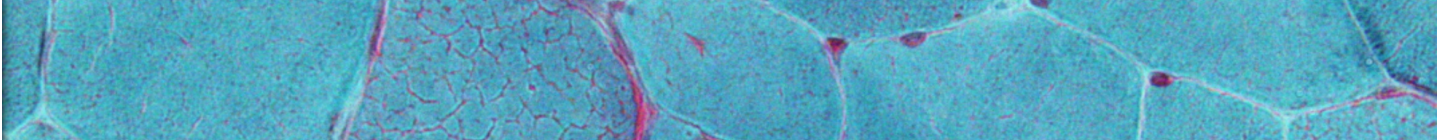
Dornase-alfa is a recombinant deoxyribonuclease used in the treatment of CF. It functions as a mucolytic.^{15, 16}

Guaifenesin is an over-the-counter mucous expectorant that may be used in CF, although there is no evidence that it is effective for any form of lung disease.¹⁷

Bottom line: Ivacaftor is approved as a monotherapy CFTR potentiator in patients with at least one G551D mutation. This mutation leads to the misfolded CFTR channel making it to the cell surface instead of sequestration in the RER. Lumacaftor/ivacaftor and elexacaftor/ivacaftor/tezacaftor are combination therapies used in patients with the $\Delta F508$ mutation. Elexacaftor is a potentiator similar to ivacaftor. Lumacaftor and tezacaftor are CFTR correctors that facilitate the trafficking of the channel to the cell surface.

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Drug names	Mechanism of action
Ivacaftor	CFTR potentiator ($\uparrow$ opening/activity of channel at cell surface)
Lumacaftor	CFTR corrector ($\uparrow$ shuttling of defective channel to cell surface)
Tezacaftor	CFTR corrector
Elexacaftor	CFTR corrector
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Ivacaftor	Patients >6 yrs of age with at least one copy of the G551D-CFTR mutation
Lumacaftor + Ivacaftor	Patients who are homozygous for the Phe508del ($\Delta F508$) mutation
Elexacaftor + Ivacaftor + Tezacaftor	Patients age ≥ 12 years who have ≥ 1 F508del mutation

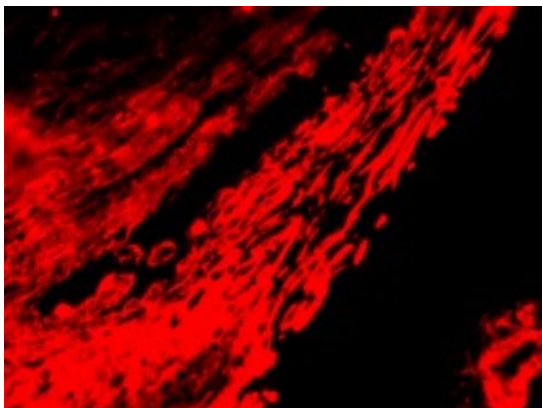
- 1) <https://ghr.nlm.nih.gov/gene/CFTR>
- 2) <https://www.ncbi.nlm.nih.gov/books/NBK546620/>
- 3) <https://www.nejm.org/doi/10.1056/NEJMe020070>
- 4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3626070/>
- 5) <https://err.ersjournals.com/content/22/127/66>
- 6) <https://www.ncbi.nlm.nih.gov/pubmed/24656117>
- 7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4272825/>
- 8) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6650604/>
- 9) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5495103/>
- 10) <https://www.ncbi.nlm.nih.gov/m/pubmed/31784874/>
- 11) [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(19\)32597-8](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(19)32597-8)
- 12) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5768901/>
- 13) <https://pubchem.ncbi.nlm.nih.gov/compound/CFTR-corrector-1>
- 14) <https://www.ncbi.nlm.nih.gov/pubmed/30073878>
- 15) <https://www.ncbi.nlm.nih.gov/pubmed/27043279>



BIOCHEMISTRY / PHARMACOLOGY

Biochemistry pharm – Cytoprotective agents

by MEHLMANMEDICAL



A 19-year-old man being treated with chemotherapy for testicular cancer presents to hospital with tinnitus and high-frequency hearing loss. His creatinine is 1.8 mg/dL. Diuresis with 0.9% NaCl is initiated. Administration of which of the following agents may have prevented these findings?

- ☐ Mesna
- ☐ N-acetylcysteine
- ☐ Leucovorin
- ☐ Dexrazoxane
- ☐ Amifostine

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Chemotherapeutic agent	Toxicity (USMLE-important)	Mitigating / cytoprotective agent
Cisplatin	Ototoxicity; nephrotoxicity	Amifostine
Cyclophosphamide	Hemorrhagic cystitis	Mesna
Acetaminophen	Hepatotoxicity	N-acetylcysteine
Methotrexate	Bone marrow, pulmonary fibrosis, hepato- / nephrotoxicity	Leucovorin (folinic acid)
Doxorubicin (adriamycin)	Dilated cardiomyopathy	Dexrazoxane

This man is being treated with **cisplatin**, a platinum-based compound that is one of the first-line chemotherapeutic agents

used for testicular cancer.¹

It is well-known to damage the cochlear stereocilia, thereby causing **ototoxicity**.² It is also associated with **nephrotoxicity**.³

In an aqueous environment, the chloride ligands of cisplatin are replaced by water molecules. This generates a positively charged electrophilic moiety that damages DNA.^{4, 5, 6}

Hydration is essential for all patients to prevent cisplatin-induced nephrotoxicity: ⁷

- “Specifically, short-duration, low-volume, outpatient hydration with magnesium supplementation and mannitol forced diuresis (in select patients) represent best practice principles for the safe use of cisplatin.”⁸

Amifostine is a cytoprotective adjuvant that decreases incidence of cisplatin-induced toxicity if administered prior to and during cisplatin infusion.^{9, 10}

Mesna decreases the risk of hemorrhagic cystitis in patients being treated with cyclophosphamide.¹¹

N-acetylcysteine decreases the incidence of severe hepatotoxicity caused by acetaminophen overdose. It regenerates reduced glutathione.¹²

Leucovorin “rescue” (folinic acid) may be used to mitigate myelosuppression, gastrointestinal toxicity, neurotoxicity, and acute kidney injury in patients receiving methotrexate.^{13, 14, 15}

Dexrazoxane is a free-radical chelator that decreases the risk of dilated cardiomyopathy in patients receiving doxorubicin.¹⁶

Bottom line:

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Chemotherapeutic agent	Toxicity (USMLE-important)	Mitigating / cytoprotective agent
Cisplatin	Ototoxicity; nephrotoxicity	Amifostine
Cyclophosphamide	Hemorrhagic cystitis	Mesna
Acetaminophen	Hepatotoxicity	N-acetylcysteine
Methotrexate	Bone marrow, pulmonary fibrosis, hepato- / nephrotoxicity	Leucovorin (folinic acid)
Doxorubicin (adriamycin)	Dilated cardiomyopathy	Dexrazoxane

1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5835297/>

2) <https://www.ncbi.nlm.nih.gov/pubmed/15721567/>

3) <https://journals.sagepub.com/doi/abs/10.1177/019262338601400215>

4) <https://www.pnas.org/content/113/41/11507>

5) <https://www.ncbi.nlm.nih.gov/pubmed/4831344/>

6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3153174/>

7) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5423518/>

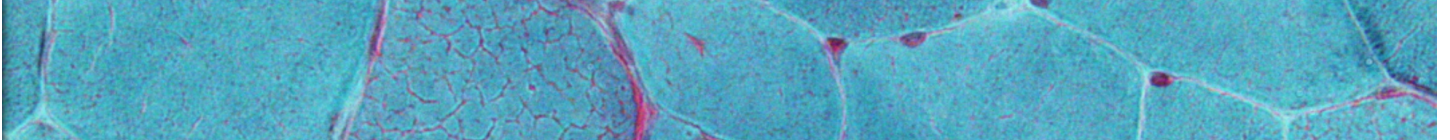


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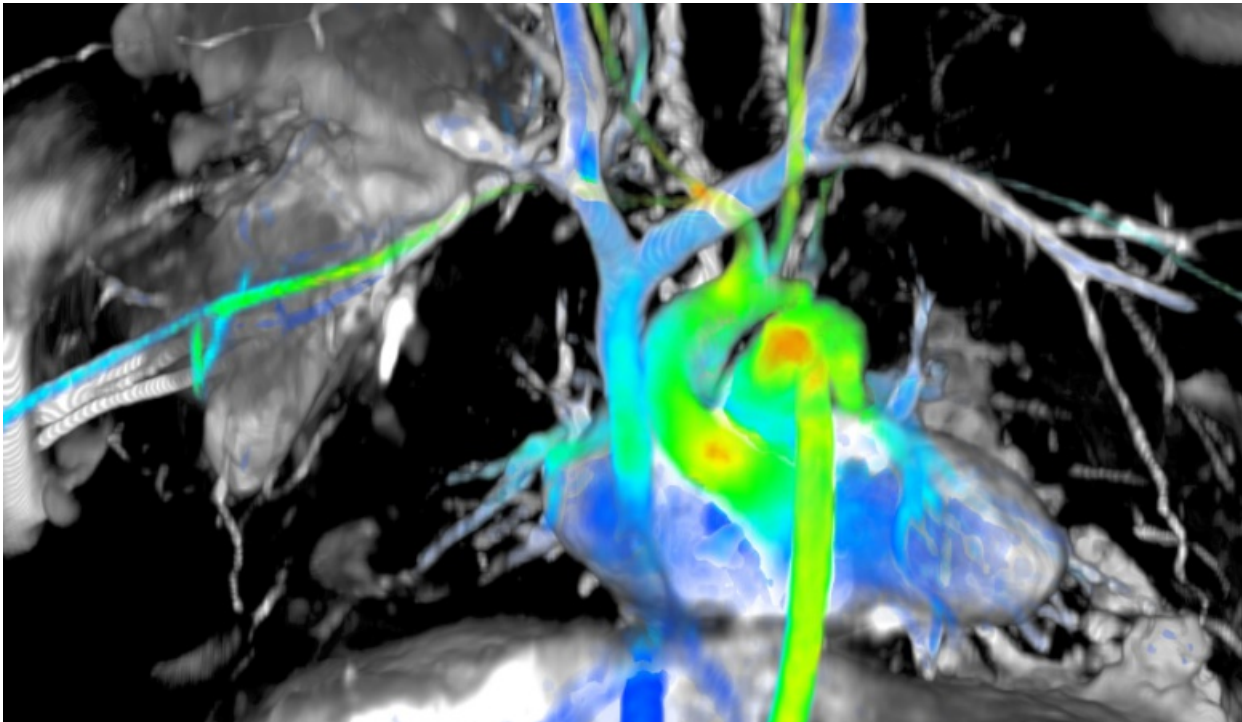
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CARDIO / PHARMACOLOGY

Cardiology pharm – Anti-arrhythmics

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

—

The USMLE doesn't care about much of the nitpicky physiology around anti-arrhythmics.

The USMLE focuses almost exclusively on MOAs and side-effects.

In other words, if you're like, "God I hate the cardio drugs. I don't even know where to start." My response is: "Recognize that if you simply know the MOAs (i.e., sodium vs potassium channel blocker, etc.) + the HY side-effects, that's sufficient for the overwhelming majority of NBME/USMLE questions.

Sodium channel blockers (Class Ia)

The **Queen** **Pro**claims **Diso**'s **pyramid** → **Quinidine, Procainamide, Disopyramide**

- Lengthen action potential duration.
- Increase risk of torsades de pointes (dangerous sinusoidal arrhythmia with high chance of progression to a fatal ventricular fibrillation).

Quinidine

- Causes cinchonism ("quinchonism") → headache + tinnitus.
- Displaces digoxin from bilirubin → can cause digoxin toxicity.
 - 69M + taking digoxin + started on new anti-arrhythmic + now has yellow, wavy vision; which anti-arrhythmic did he commence? → answer = quinidine. Digoxin toxicity classically presents as "Vincent van Gogh" yellow, wavy vision.

Procainamide

- Can cause drug-induced lupus (anti-histone antibodies).
 - 60F + started on anti-arrhythmic a couple months ago + now has arthritis in both hands + mediastinitis + thrombocytopenia; which agent was she started on? → answer = procainamide.
- Can be used for Wolf-Parkinson-White syndrome.

Disopyramide

- Just know this drug blocks sodium channels.

Sodium channel blockers (Class Ib)

I'd buy **Liddy**'s **Mexican** **Tacos** → **Ib** – **Lidocaine, Mexiletine, Tocainide**

- Shorten the action potential duration.
- Preferentially affect ischemic myocardium.
- **Can cause CNS dysfunction.**
 - 72M + started on new anti-arrhythmic + now has dizziness + tremor; which agent was he started on? → answer = lidocaine → that is: choose the Class Ib agent if a patient starts an anti-arrhythmic and gets miscellaneous neuro findings.

Lidocaine

- Can be used as a local anesthetic.
- If injected into a blood vessel, can cause metallic taste in the mouth (on 2CK Obgyn form).
- Associated with dizziness and tremor (as per above example).

Mexiletine, Tocainide

- Just know they're Class Ib agents.
- Less chance of dizziness and tremor compared with lidocaine.

Sodium channel blockers (Class Ic)

Fred Entertains Properly → **Flecainide, Encainide, Propafenone**

- No change to action potential duration.
- Increased risk of torsades de pointes.
- Pro-arrhythmic. Do not use if patient has ischemic heart disease.

Flecainide

- First rhythm-control agent used for atrial fibrillation (AF) if patient has no structural or coronary artery disease.

Encainide, Propafenone

- Just know these are Ic.

Beta-blockers (Class II)

- See [here](#) for a detailed post on beta-blockers and beta-agonists.
- Classified as anti-arrhythmic but considered "rate-control" rather than "rhythm-control" when discussing AF. For AF treatment, we try rate-control before rhythm-control.
- Used as first-line rate-control in atrial fibrillation (i.e., give metoprolol first to manage AF).
 - For AF: use beta-blocker first (rate-control); if contraindicated, use another rate-control agent like verapamil (non-dihydropyridine calcium channel blocker; cardiac-selective).
 - If rate-control fails, go to rhythm control.
 - Flecainide (Ic sodium channel blocker) is the first rhythm-control agent used if patient has no structural or coronary artery disease.
 - If patient has structural (e.g., LVH, DCM, etc.) or coronary artery disease, use a different rhythm-control agent like the Class III potassium channel blockers (discussed below).
- Can cause depression and sexual dysfunction.
- May mask the signs of hypoglycemia in diabetes.
- Contraindicated in depression, diabetes, 2nd/3rd-degree heart block, unstable heart failure, COPD, asthma (latter two: use β_1 -selective only).

Potassium channel blockers (Class III)

- Can cause torsades de pointes (really HY for these drugs).
- Can be used for rhythm control for atrial fibrillation if patient cannot take flecainide (i.e., if patient has structural or coronary artery disease).
- Can be attempted in resuscitation from ventricular fibrillation when defibrillation and epinephrine fail.
- Delay repolarization of cardiac action potential.

Amiodarone, Dronedronerone

- The following side-effects are HY for for USMLE. Memorize all of these.
- Pulmonary fibrosis
- Torsades
- Blue/grey skin discolorations (amiodarone only)
- Drug-induced thyroiditis (both hyper- and hypothyroidism)

- Hepatotoxicity

Dofetilide, Ibutilide, Sotalol

- Merely be aware that these are potassium channel blockers.

Calcium channel blockers (Class IV)

- Called non-dihydropyridine calcium channel blockers (verapamil, diltiazem; cardiac-selective).
- Block L-type calcium channels → results in prolongation of plateau phase of myocardial action potential.
- Don't confuse with dihydropyridine CCBs (i.e., amlodipine, nifedipine; vascular-selective).

Verapamil

- Classic use is rate-control in AF if patient has contraindication to beta-blocker.
- One of the highest yield side-effects for USMLE is knowing that verapamil causes constipation (don't confuse with dihydropyridine CCBs, which cause fluid retention / edema).

Diltiazem

- Technically a mixed cardiac-/vascular-selective agent (i.e., an intermediate between verapamil and nifedipine/amlodipine).
- Verapamil is pure cardiac-selective (i.e., pure non-dihydropyridine).
- Nifedipine/amlodipine are pure vascular-selective (i.e., pure dihydropyridine).
- Diltiazem is in the middle.

—

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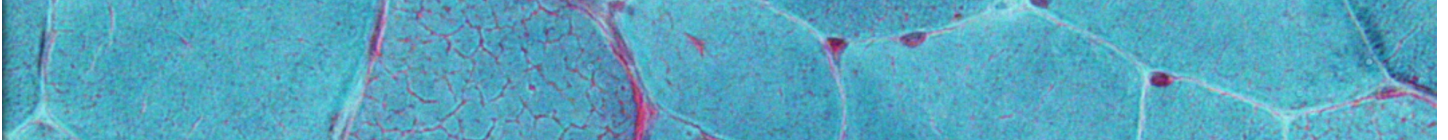


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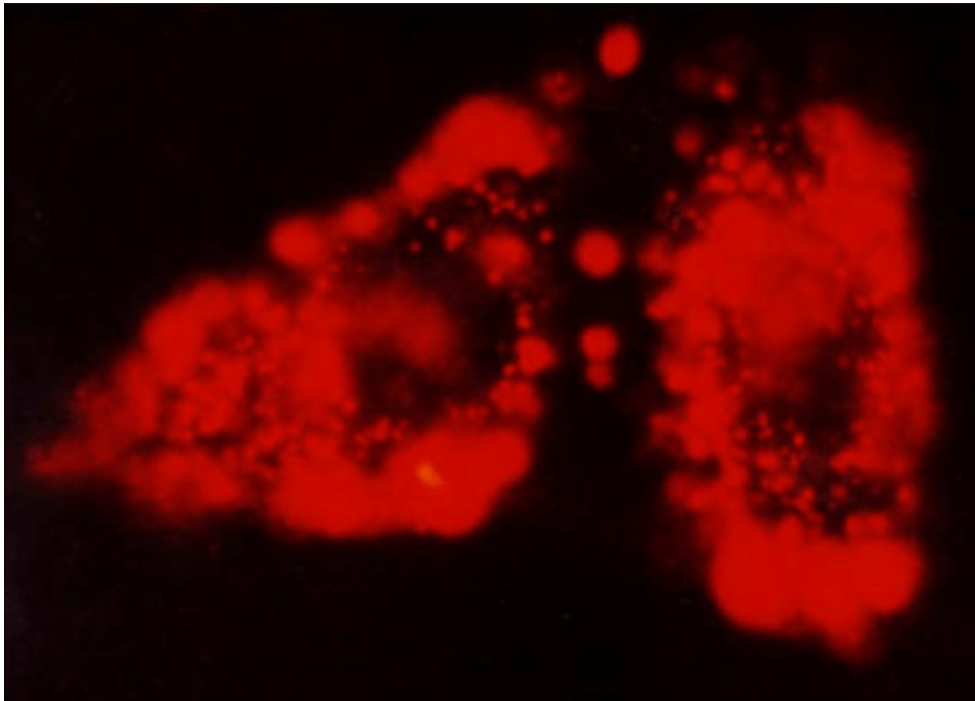
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CARDIO / PHARMACOLOGY

Cardiology pharm – Lipid-lowering agents

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Statins (e.g., atorvastatin)

MOA of statins?

- Competitive, reversible inhibitors of HMG-CoA reductase (rate-limiting enzyme for cholesterol synthesis).
- This causes a compensatory increase in LDL receptor expression on hepatocytes (i.e., if hepatic cholesterol synthesis is inhibited, then it will try to pull more out of the blood to compensate).
- Students will memorize that statins are HMG-CoA reductase inhibitors, but sometimes the USMLE will simply have “increased LDL receptor expression” as the answer for the mechanism. So know both, despite the latter being the indirect effect of the enzyme inhibition.
- USMLE wants you to know that HMG-CoA reductase **mRNA** will increase as a result of giving a statin (i.e., if we inhibit the enzyme, then the cell will attempt to synthesize more enzyme to compensate).

When do we use statins?

- Non-diabetics 20-39: if LDL >190 mg/dL.
- Diabetics 20-39: if LDL >100 mg/dL.
- Anyone 40-75: if LDL >70 mg/dL.
- Age <20: only consider if familial dyslipidemia with severely elevated LDL.
- Age >75: discuss with patient lowering the dose or stopping it altogether (↑ risk of side-effects in elderly + benefit not proven age >75).

Side-effects of statins?

- Hepatotoxicity (↑ LFTs); most common side-effect.
 - Mild transaminitis is normal and expected when commencing hepatotoxic agents. Do not lower dose or discontinue if patient is asymptomatic and has mild elevation of ALT and/or AST.
- Myositis (↑ creatine kinase); more common when combined with fibrates (e.g., fenofibrate).
 - If the USMLE asks for the most common side-effect of statins, choose myopathy over toxic hepatitis.
 - A Step 1 NBME Q asks why myositis is more common when statins are combined with fibrates; answer = “inhibition of P-450 enzymes.”

Out of the lipid-lowering agents, why are statins associated with reduced mortality but others (e.g., ezetimibe, bile acid sequestrants, etc.) are not?

- Statins have an anti-oxidant effect on vascular endothelial cells that transcends the lipid-lowering effect.

Fibrates (e.g., fenofibrate, gemfibrozil)

MOA of fibrates?

- Upregulate lipoprotein lipase (enables the movement of TGAs out of the blood and into the adipocytes).
 - TGAs cannot transcytose vascular endothelium. Lipoprotein lipase breaks down TGAs into monoacylglycerol and three fatty acids. The latter can then move into the adipocytes.
 - The upregulation of lipoprotein lipase occurs following stimulation of PPAR-α.

When do we use fibrates?

- When TGAs are greater than 500 mg/dL.
- **Decrease TGAs more than any other agents.**
- A step 1 NBME Q has both LDL and TGAs elevated in a patient where TGAs are 550. Correct answer is the fibrate; statin is the wrong answer.

Side-effects of fibrates?

- Similar to statins except carry the added side-effect of causing cholesterol gallstones (due to inhibition of 7α hydroxylase, which converts cholesterol to bile acids).
- Hepatotoxicity ($\uparrow$ LFTs); most common side-effect.
 - Mild transaminitis is normal and expected when commencing hepatotoxic agents. Do not lower dose or discontinue if patient is asymptomatic and has mild elevation of ALT and/or AST.
- Myositis ($\uparrow$ creatine kinase); more common when combined with statins.
 - Students tend to remember myositis but don't realize hepatotoxicity is more common. If the USMLE asks for the most common side-effect of fibrates, choose hepatotoxicity, **not** myopathy.
 - A Step 1 NBME Q asks why myositis is more common when statins are combined with fibrates; answer = "inhibition of P-450 enzymes."

Bile acid sequestrants (cholestyramine, colestevlam, colestipol)

MOA of bile acid sequestrants?

- Prevent enterohepatic circulation of bile acids (i.e., bind to bile acids in the small bowel and prevent their reabsorption at the terminal ileum and return to the liver).
- If the liver has an insufficiency of bile acids because it's not receiving them via enterohepatic circulation, then it needs to synthesize more. The way it does this is by converting cholesterol into bile acids via 7α hydroxylase. As a result, more cholesterol is consumed intrahepatically, and in turn the liver will pull more cholesterol out of the blood to compensate.

Ezetimibe

MOA of ezetimibe?

- Selectively inhibits the absorption of cholesterol through the brush border of the small bowel wall.

Niacin (vitamin B3)

MOA of niacin?

- **Increases HDL more than any other agent.**
- Decreases hepatic synthesis and export of VLDL (therefore $\downarrow$ TGAs); also inhibits hormone-sensitive lipase.

Side-effects of niacin?

- Facial flushing (redness); caused by prostaglandin, **not** histamine (makes sense, since aspirin treats the flushing, and aspirin inhibits prostaglandin synthesis).
- Hyperglycemia (insulin resistance).
- Gout (hyperuricemia).
 - Vignette will give a guy with a big cardiovascular history and also a history of podagra (big toe gout), then ask which drug is contraindicated in this patient; answer = niacin.
- Rarely used clinically because of its negative effects on blood glucose (i.e., worsens diabetes, and pushes people who are pre-diabetic into diabetes).

Evolocumab, alirocumab

MOA of evolocumab, alirocumab?

- PCSK9 inhibitors.
- Proprotein convertase subtilisin/kexin type 9 is an enzyme that breaks down LDL receptor.
- LDL receptor is necessary for pulling LDL out of the blood.

- So if we inhibit the enzyme that breaks down LDL receptor, then more cholesterol can be pulled out of the blood.
- Some PCSK9 inhibitors have been shown to so drastically lower LDL (i.e., to 10-30 mg/dL), that longer term studies are needed to determine safety.

Fish oil (DHA/EPA) and Omega-3 fatty acids

- Not prescribed as medication, but scientifically proven to lower triglycerides (merely be aware they can have this effect on blood lipids).

Orlistat

MOA of orlistat?

- Pancreatic lipase inhibitor.
- Can be used for obesity.
- Can lower TGAs.
- Can cause diarrhea.

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HY USMLE Q #965 – Cardiopulm

January 26, 2024

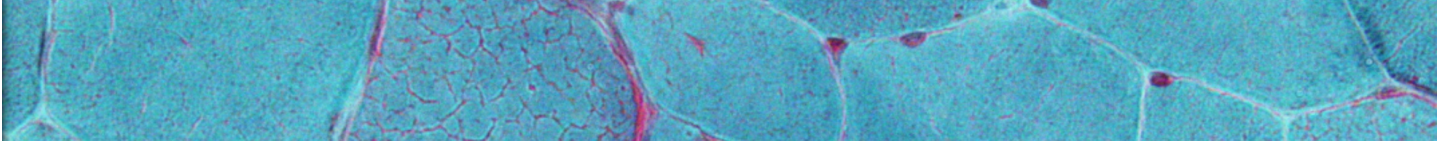


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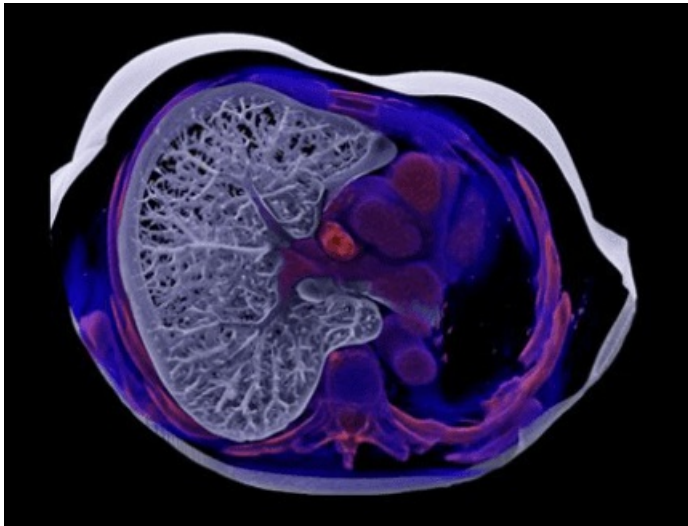
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CARDIO / PHARMACOLOGY

Cardiology pharm – Mixed HY drugs

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

The drugs mentioned in this post fall outside the categorization of [anti-arrhythmics](#) and [lipid-lowering agents](#), but are nevertheless HY for the USMLE.

Dihydropyridine vs non-dihydropyridine calcium channel blockers (CCBs)

MOA of dihydropyridine CCBs (i.e., amlodipine, nifedipine, etc.)?

- **Vascular-selective** (i.e., block calcium channels on arteries/arterioles only).
- Cause arterial vasodilation → ↓ BP + afterload on the left heart → HR increases to compensate (baroreceptor reflex).

When do we use dihydropyridine CCBs?

- To treat hypertension in patients without pre-diabetes, diabetes, proteinuria, cardiovascular/cerebrovascular disease.
 - If patient has any of the above, use an ACEi or ARB first to manage the HTN.
 - If patient doesn't have any of the above, dihydropyridine CCB or a thiazide diuretic is chosen.
 - Sounds weirdly specific, but this is HY notably for the 2CK FM component.
 - 45M + increased fasting sugars with 1+ protein on dipstick + BP of 142/90; which agent should we commence? → answer = ACEi or ARB.
 - 45M + repeated HTN measurements despite lifestyle modifications + no other issues; which agent should we commence? → answer = nifedipine or HCTZ.
 - That's the over-simplified version of it, but that's the gist you need to take away: if the patient has pre-diabetes, diabetes, proteinuria, cardiovascular/cerebrovascular disease → use an ACEi or ARB; if the patient doesn't, use a dCCB or a thiazide.
- IV nicardipine is a hard-hitting dCCB often used as part of hypertensive emergencies with sodium nitroprusside and fenoldopam (D1 agonist).
- Nimodipine is a dCCB used post-subarachnoid hemorrhage to prevent vasospasm and ischemic stroke.

Major side-effect of dihydropyridine CCBs?

- **Peripheral edema** (fluid retention).
- This is one of the highest yield side-effects for pharmacology on all USMLE Steps.
- Do not confuse with verapamil, which causes constipation.
- 35F + recently started on amlodipine + has puffy limbs + massive paragraph of a vignette with other extraneous info; Q asks the cause of her edema; answer = "side-effect of medication" (amlodipine).

MOA of non-dihydropyridine calcium channel blockers (i.e., verapamil, diltiazem)?

- **Cardiac-selective** (blocks calcium channels on myocardial and nodal tissue only).
- Verapamil is pure non-dihydropyridine.
- Diltiazem is a mixed agent, with effects on both cardiac and arterial calcium channels.

When do we use non-dihydropyridine CCBs?

- Verapamil is used for rate-control for atrial fibrillation when **beta-blockers** are contraindicated or not effective, prior to attempting rhythm-control.
- Diltiazem is not assessed on USMLE beyond just needing to know it's a calcium channel blocker. It can be used for HTN and arrhythmias.

Major side-effect of verapamil?

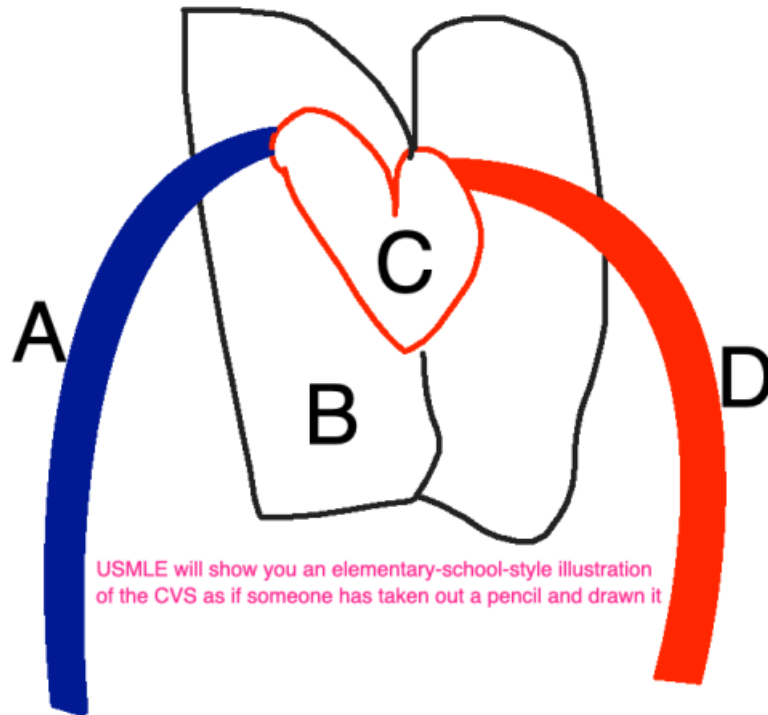
- Constipation.
- This is one of the highest yield side-effects for pharmacology on all USMLE Steps.
- Do not confuse with the nifedipine/amlodipine, which cause peripheral edema (fluid retention).

Other drugs

Hydralazine

MOA of hydralazine?

- Causes disruption of arterial smooth muscle calcium currents, resulting in vasodilation and decreased BP.
- Used for hypertensive emergency in pregnancy.
- Arterial vasodilation → ↓ BP + afterload on the left heart → heart rate increases to compensate (baroreceptor reflex).
- In other words, USMLE will ask you where hydralazine acts (arteries/arterioles) or what will happen to HR when you give it (↑).



- USMLE might show you a drawing similar to the above and then ask you where hydralazine acts (choice D).
- Choice A refers to nitrates (dilate veins).
 - Patient has angina + takes a drug under his tongue to alleviate the pain. Where on the diagram does the drug act? → Answer = A.
- Choice B refers:
 - Bosentan (endothelin-1 receptor antagonist used for pulmonary hypertension).
 - Endothelin is a vasoconstrictor of pulmonary vessels. Antagonizing these receptors will dilate pulmonary vessels.
 - ACE inhibitors if the Q asks where the inhibited enzyme is located (i.e., ACE converts AT I into AT II in the lungs).
- Choice C refers to **anti-arrhythmics**, including non-dihydropyridine calcium channel blockers (i.e., verapamil, diltiazem).
- Choice D refers to:
 - ACE inhibitors if the Q asks about *the effect on* the cardiovascular system (result in less arterial vasoconstriction and decreased afterload).
 - Angiotensin II receptor blockers (e.g., valsartan).
 - Dihydropyridine calcium channel blockers (i.e., nifedipine, amlodipine, etc.)
 - **Hydralazine** (calcium current disruptor used for hypertensive emergency in pregnancy).
 - Dipyridamole (phosphodiesterase inhibitor used for pharmacologic stress testing [i.e., dipyridamole + thallium]).
 - Dipyridamole acts on arteries → vasodilation → drop in BP → heart rate increases as part of baroreceptor reflex → increased cardiac workload → can reveal areas of ischemia using an uptake-radioisotope such as thallium.
 - Cilostazol (phosphodiesterase inhibitor used for intermittent claudication). It works on arterial smooth

muscle.

- Cilostazol is almost always the wrong answer when listed on the USMLE for the management of arterial disease on 2CK Surg shelves. Choose “commence a walking program,” or “commence an exercise program” as the first management for intermittent claudication. If the exercise program fails, cilostazol can be commenced.
- Sodium nitroprusside, unlike other nitrates, acts on arteries > veins.

Adenosine

MOA of adenosine?

- Increases potassium release from cells, resulting in hyperpolarization.

When is adenosine used?

- To treat supraventricular tachycardia (SVT) if carotid massage (vagal maneuvers) or ice pack to the face (in young kids) fails to re-establish sinus rhythm.

Side-effect of adenosine?

- Causes severe chest pain for a few seconds, then the patient is fine.

Miscellaneous important info about adenosine:

- Produced endogenously by muscles as a result of cellular metabolism.
- **Adenosine is the molecule responsible for cardiac pain with angina and MI.** USMLE likes this. Essentially: myocardium becomes ischemic, where ATP is metabolized maximally and adenosine builds up → increased pain in the setting of ischemia.

Ivabradine

MOA of ivabradine?

- Inhibits mixed Na/K “funny currents” at the sinoatrial node, resulting in decreased heart rate.
- Reduces chronotropy (HR) without reducing inotropy (contractility).
 - This is in contrast to beta-blockers and verapamil, which reduce both HR and contractility.

When is ivabradine used?

- Can be used to manage angina in patients who cannot take beta-blockers (BB contraindicated in: diabetes, depression, sexual dysfunction, asthma, 2nd/3rd degree heart block, unstable heart failure).

Side-effect of ivabradine?

- “Luminous phenomena” → sensations described by patients as enhanced brightness despite a fully maintained visual field.

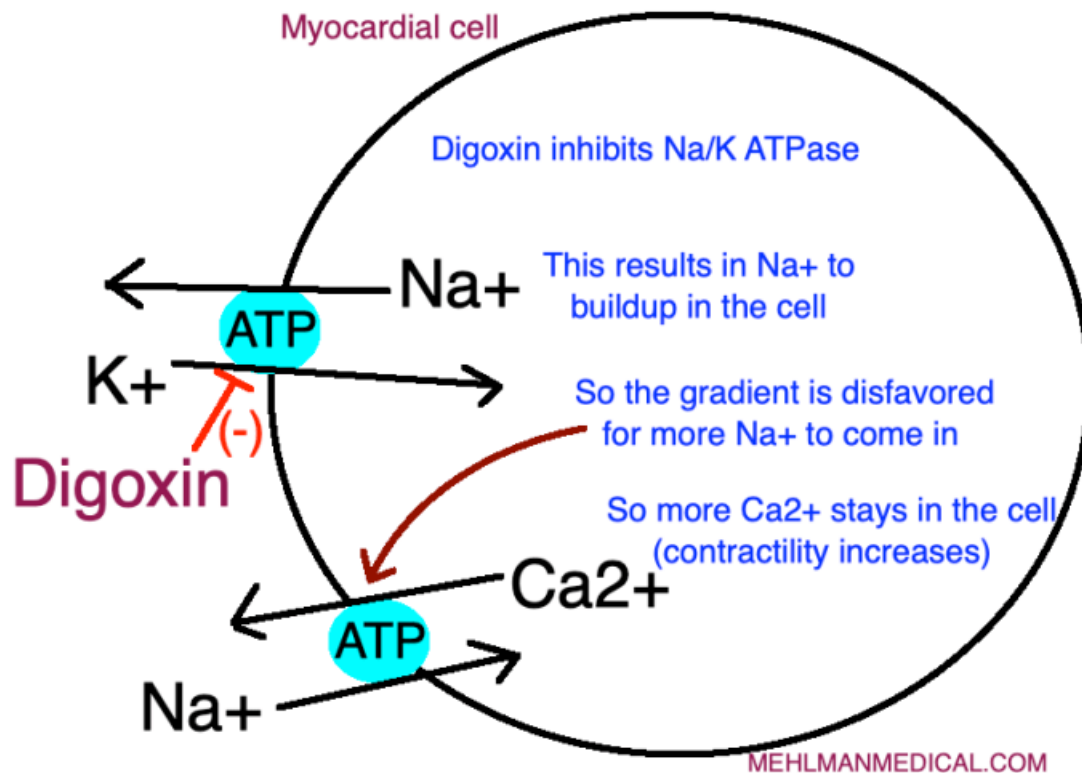
Digoxin

MOA of digoxin?

- Inhibits myocardial Na/K ATPase pumps **by binding to the extracellular K⁺ binding site** → causes buildup of sodium in the cell → creates a less favorable gradient for sodium to move into the cell via the Ca/Na ATPase pump

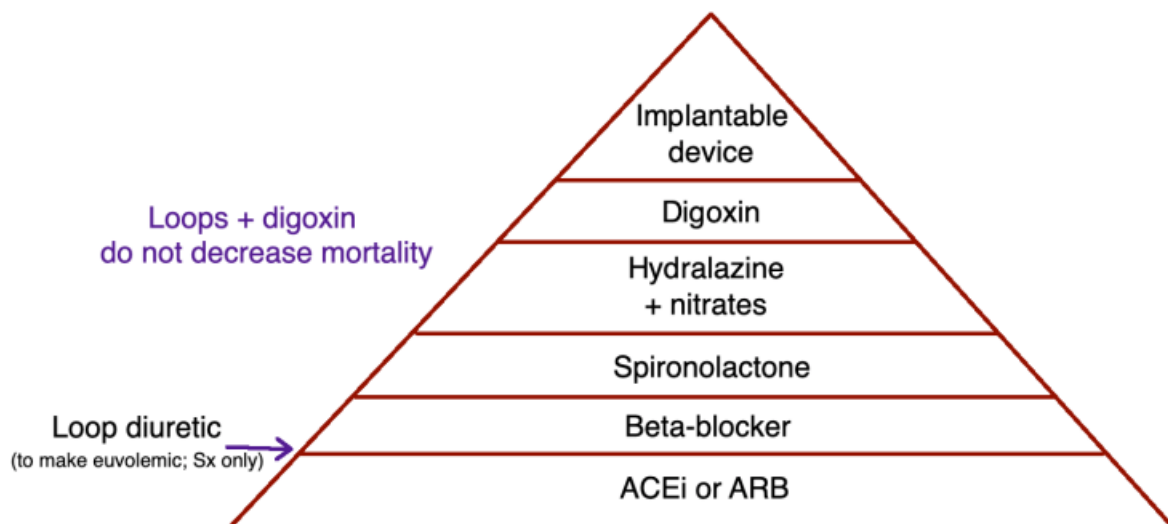
→ leads to indirect inhibition of the Ca²⁺/Na⁺ ATPase → calcium stays in the cell → **increased myocardial contractility**.

- **Slows heart rate** due to increasing parasympathetic potentiation at nodal tissue.
- In other words: digoxin ↑ inotropy but ↓ chronotropy.



When do we use digoxin?

- For heart failure in patients failing other management.



Hierarchy for drugs in heart failure (simplified but HY for USMLE):

- Start with ACE inhibitor or an angiotensin II receptor blocker (ARB) to improve ejection fraction (normal is 55-70%).
- If the patient is fluid overloaded (i.e., peripheral edema or pulmonary edema), attempt to achieve euvolemia by *adding* furosemide (loop diuretic).
- If insufficient ejection fraction with the ACEi or ARB, *add* a beta-blocker (metoprolol XR, bisoprolol, carvedilol, and nebivolol decrease mortality in heart failure; propranolol does not).
- If insufficient ejection fraction with ACEi/ARB + beta-blocker, *add* spironolactone (aldosterone receptor antagonist).

- If insufficient ejection fraction, *add* the **combination** of hydralazine + nitrates. Only the combo decreases mortality in HF. It is especially efficacious in African Americans (tend to have stiffer vessels).
- If insufficient ejection fraction when already on ACEi/ARB + beta-blocker + spironolactone + hydralazine + nitrates, *add* digoxin.
- If pharmacologic therapy insufficient, use implantable device.

Side-effects of digoxin?

- Yellow, wavy “Vincent van Gogh” vision.
- Hyperkalemia (prevents K⁺ from entering the cell).

What causes digoxin toxicity?

- Hypokalemia (since K⁺ and digoxin compete for the extracellular K⁺ binding site on the Na/K ATPase pump, if less K⁺ is floating around, then more digoxin binds per unit time).
- Quinidine (displaces digoxin from bilirubin → less protein-bound digoxin → more digoxin available to bind to Na/K ATPases).

Treatment for digoxin toxicity?

- Normalize serum potassium.
- Give anti-digoxin immunoglobulin Fab fragment.
- Give Mg²⁺.

Ranolazine

MOA of ranolazine?

- Blocks sodium channels.

When is ranolazine used?

- Angina refractory to other therapies (i.e., nitrates, beta-blockers).

Sacubitril

MOA of sacubitril?

- Neprilysin inhibitor.
- Neprilysin is an enzyme that breaks down atrial and brain natriuretic peptides (i.e., ANP and BNP).
- ANP and BNP act as natural diuretics that are secreted by atria and ventricles, respectively, in response to increased chamber stretch. They function to decrease sodium retention at the distal kidney.

When is sacubitril used?

- Sacubitril is combined with valsartan (an angiotensin II receptor blocker) as sacubitril/valsartan for treatment of heart failure.

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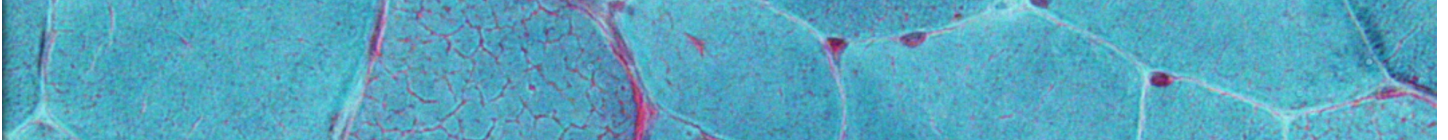


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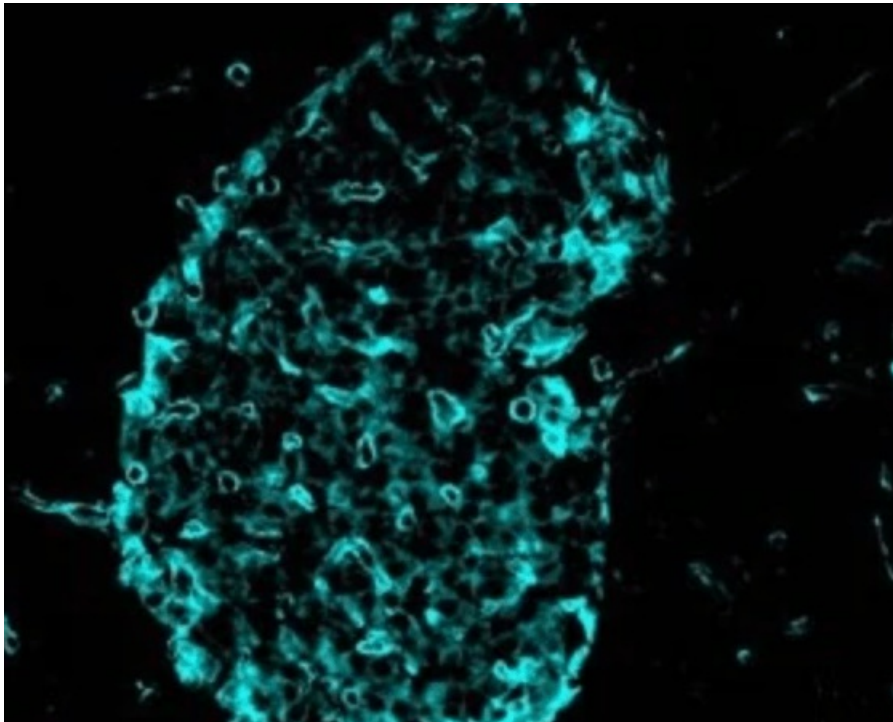
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ENDOCRINE / PHARMACOLOGY

Endocrine pharm – Diabetes drugs

by MEHLMANMEDICAL



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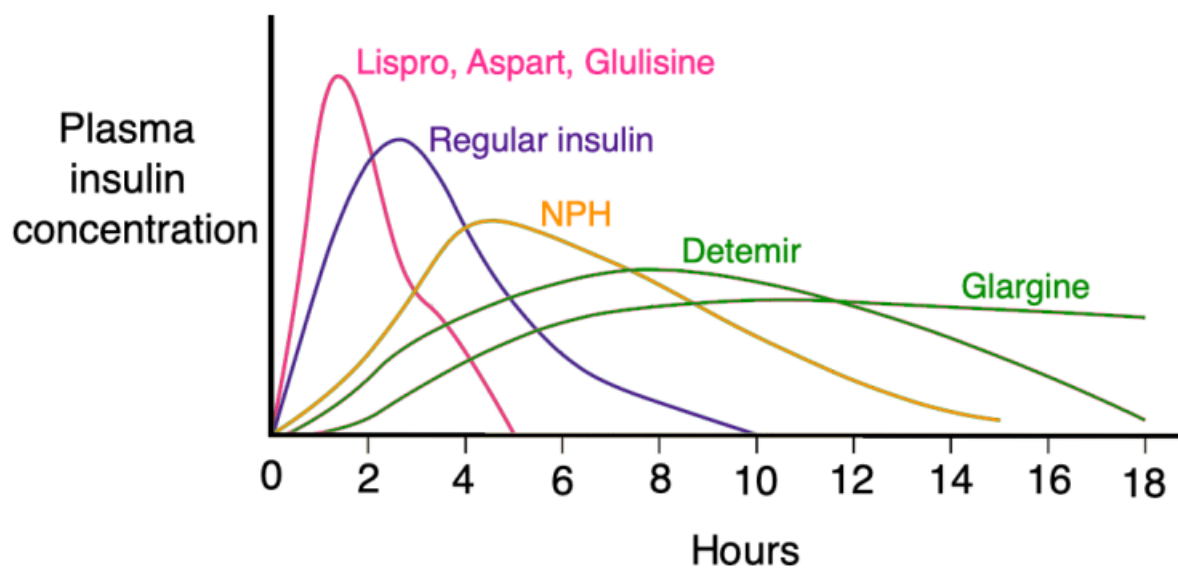
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HY points about each drug followed by a quiz at the end

—
[Insulin preparations](#)

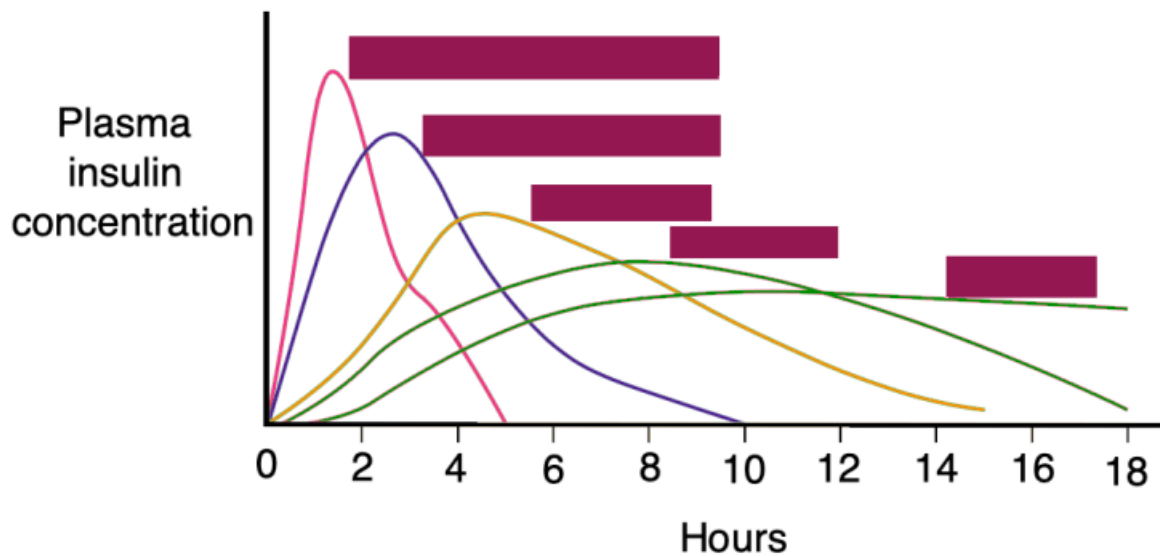
- Insulin is the only treatment for type I diabetes (insulin deficiency).
- Insulin is used last resort for type II diabetes (insulin resistance).

Insulin preparations		
Type	When after administration does peak effect occur?	Name
Rapid-acting	1 hour	LAG : Lispro, Aspart, Glulisine
Short-acting	2-3 hours	Regular insulin
Intermediate-acting	4-10 hours	NPH
Long-acting	Detemir (6-12 hours); Glargine (no real peak)	Detemir, Glargine



Quiz yourself:

Insulin preparations		
Type	When after administration does peak effect occur?	Name
Rapid-acting	_____ hours	_____
Short-acting	_____ hours	_____
Intermediate-acting	_____ hours	_____
Long-acting	_____ (6-12 hours); _____ (no real peak)	_____



Type II diabetes meds (oral hypoglycemics)

Metformin

MOA of metformin?

- Inhibits mGPD (a mitochondrial enzyme called glycerophosphate dehydrogenase).
- “Increases glycolysis and decreases gluconeogenesis” is the MOA answer on the USMLE.

Side-effects of metformin?

- Lactic acidosis (HY)
 - Increased risk in patients with renal insufficiency.
 - 45M + T2DM + bicarb of 20 + creatinine of 1.6; next best step? → answer = stop metformin.

Thiazolidinediones (pioglitazone, rosiglitazone)

MOA of thiazolidinediones?

- Activate PPAR- α , leading to increased insulin sensitivity.

Side-effects of thiazolidinediones?

- Heart failure
- Weight gain
- Edema

Sulfonylureas

- **First generation:** Tolbutamide, Chlorpropamide
- **Second generation:** Glyburide, Glipizide, Glibenclamide
- **Third generation:** Glimepiride

MOA of sulfonylureas?

- Insulin secretagogues (i.e., promote insulin release); require functioning beta-islet cells. If patient is on a sulfonylurea and his/her HbA1c is very high (e.g., 9+), it's because he/she has most likely already lost the beta-

islet cells.

- Close K^+ channel on β -islet cells $\rightarrow$ more K^+ trapped in the cell $\rightarrow$ cell depolarizes $\rightarrow$ Ca^{2+} then enters cell $\rightarrow$ insulin-containing vesicles are released from the cell.

Side-effects of sulfonylureas?

- First generation: disulfiram-like reaction with EtOH.
- Second + third generation: hypoglycemia (especially 2nd-gen glibenclamide).

Meglitinides (Repaglinide, Nateglinide)

MOA of meglitinides?

- Same as sulfonylureas but structurally different.
- Just know the MOA.

GLP-1 analogues (Liraglutide, Exenatide)

MOA of GLP-1 analogues?

- $\downarrow$ glucagon release; $\uparrow$ glucose-dependent insulin release.
- GLP-1 (glucagon-like peptide-1) is an endogenous GI hormone that suppresses glucagon release and increases insulin release in response to oral glucose.

Side-effects of GLP-1 analogues?

- Pancreatitis

DPP-4 inhibitors (Linagliptin, Saxagliptin, Sitagliptin)

MOA of DPP-4 inhibitors?

- DPP-4 is an enzyme that breaks down GLP-1.
- Therefore the -gliptins help maintain levels of GLP-1.
- Since GLP-1 $\downarrow$ glucagon release and $\uparrow$ glucose-dependent insulin release, DPP-4 inhibitors produce the same effect as the GLP-1 analogues liraglutide and exenatide.

SGLT-2 inhibitors (Dapagliflozin, Canagliflozin)

MOA of SGLT-2 inhibitors?

- Inhibit sodium-glucose transporter 2 (SGLT-2), which normally reabsorbs glucose in the PCT of the kidney.
- These agents decrease serum glucose by promoting glycosuria.

Side-effects of SGLT-2 inhibitors?

- Increased risk of vulvovaginal candidiasis and UTIs (disruption of normal flora and pH balance).

α -glucosidase inhibitors (Acarbose, Miglitol)

MOA of α -glucosidase inhibitors?

- Decrease small bowel absorption of glucose at the brush border, resulting in $\downarrow$ postprandial hyperglycemia.

Side-effects of α -glucosidase inhibitors?

- Diarrhea

Amylin analogues (Pramlintide)

MOA of pramlintide?

- Amylin is a peptide normally co-secreted with insulin. It inhibits glucagon secretion, delays gastric emptying, and promotes satiety.
- It is deficient in type I diabetics (and late type II after the pancreas burns out).

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January 24, 2024

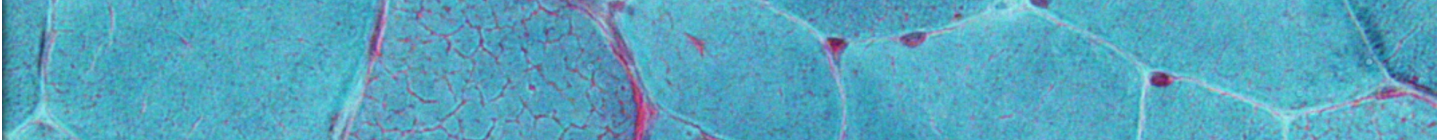


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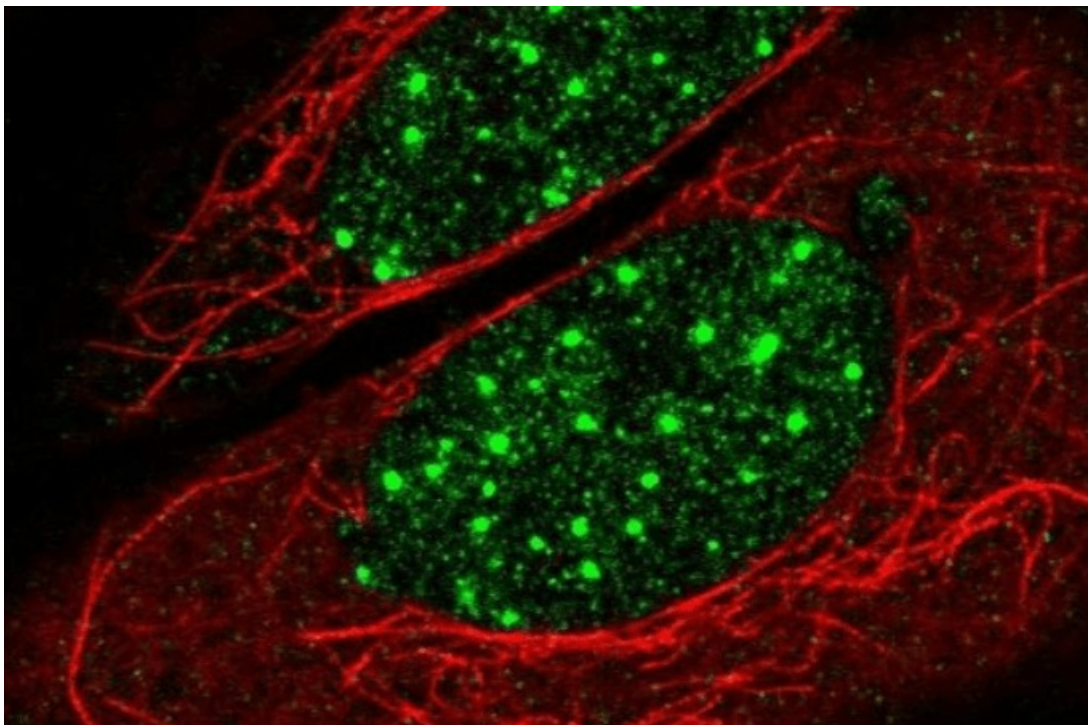
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ENDOCRINE / PHARMACOLOGY

Endocrine pharm – HY mixed drugs

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Thyroid drugs

Thionamides

- Propylthiouracil (PTU)
- Methimazole

MOA of PTU methimazole?

- Both inhibit thyroperoxidase.
- PTU has a second MOA: inhibits 5-deiodinase.
- Thyroperoxidase converts iodide to iodine within the thyroid gland so that it can be added to thyroglobulin for thyroid hormone synthesis.
- 5-deiodinase converts T4 to T3 peripherally.

When are PTU and methimazole used?

- Graves, toxic adenoma, and toxic multinodular goiter (all ↓ TSH, ↑ T3, ↑ T4).
- TSH-secreting tumor (rare) prior to any type of surgery (↑ TSH, ↑ T3, ↑ T4).
- Do **not** treat subacute granulomatous thyroiditis (deQuervain) presenting with hyperthyroidism with thionamides; use aspirin or steroids.

Major side-effects of PTU and methimazole?

- **Agranulocytosis (neutropenia)** → presents as mouth ulcers, sore throat, fever.
 - 38F + treated four days ago for hyperthyroidism in hospital + now has a fever and a mouth ulcer; why? → answer = “drug-induced neutropenia” (PTU or methimazole). This is on 2CK IM form.
- Hepatotoxicity (PTU > methimazole).
- Methimazole is teratogenic in first trimester (causes aplasia cutis congenita).
- In pregnant woman with Graves, PTU is used first trimester; methimazole is then used 2nd and 3rd trimester to spare the patient from PTU's greater risk of hepatotoxicity with continued use.

Propranolol

- Beta-blockade decreases peripheral conversion of T4 to T3.
- Propranolol the classic beta-blocker used to treat tachycardia in hyperthyroidism.
- See [here](#) for a detailed post on beta-blockers and -agonists.

Levothyroxine (T4), Triiodothyronine (T3)

- Levothyroxine is a T4 analogue (thyroxine is endogenous T4).
- Triiodothyronine is same as endogenous T3.

When are levothyroxine and triiodothyronine used?

- Hashimoto thyroiditis (↑ TSH, ↓ T3, ↓ T4).
 - Increase dose by 50% in pregnancy (USMLE asks this detail on Step 3).
- Subclinical hypothyroidism (↑ TSH, normal T3 + T4) *if* 1) anti-Hashimoto antibodies, 2) pregnant, or 3) TSH >10 mIU/L (NR 0.5-5.0).
- Drug-hypothyroidism (e.g., due to lithium or amiodarone).
 - 25F + started on medication for bipolar disorder + cold intolerance + weight gain + TSH is high + serum lithium levels within normal limits; what do we do? → answer = keep dose of lithium the same + commence levothyroxine.

USMLE wants you to know important physiology with use of these agents:

- 40F + Hashimoto + started on levothyroxine; how will her TSH, T3, and T4 change as a result of this treatment? → answer = ↓TSH, ↑ T3, ↑ T4; once again, these arrows are the *mere changes* in the patient's serum values as a result of the T4 administration. Untreated (or insufficiently treated) Hashimoto will have (↑ TSH, ↓ T3, ↓ T4).
- 40F + Hashimoto + started on triiodothyronine; how will her TSH, T3, and T4 change as a result of this treatment? – > answer = ↓TSH, ↑ T3, ↓ T4. Student says, "Wait, what? Why is T4 down." Because T4 is converted to T3 peripherally, but not the other way around. If we give thyroxine (T4), of course T4 is high, but so is T3 because it's converted T4 → T3. However if we give T3, it's not converted to T4, so T4 (which is already low) will go down further because the T3 induces negative feedback on TSH secretion, and the lower TSH will mean the thyroid gland is suppressed.

Vasopressin (anti-diuretic hormone; ADH) drugs

Desmopressin (DDAVP)

- Vasopressin analogue.

When is it used?

- Central diabetes insipidus (replaces vasopressin).
- von Willebrand disease (stimulates vWD production + release from vascular endothelium).
- Hemophilia A (stimulates factor VIII production + release from the liver and vascular endothelium).

Conivaptan, Tolvaptan

MOA of conivaptan and tolvaptan?

- Vasopressin receptor antagonists.

When are they used?

- SIADH (usually caused by small cell bronchogenic carcinoma or head trauma).

Demeclocycline

MOA of demeclocycline?

- Technically a tetracycline antibiotic, but not used for antimicrobial purposes because it causes nephrogenic diabetes insipidus (i.e., antagonizes vasopressin receptors). However this side-effect is advantageous and capitalized upon in order to treat SIADH.
- For USMLE, demeclocycline and the -vaptan agents are both used for SIADH.

Aldosterone drugs

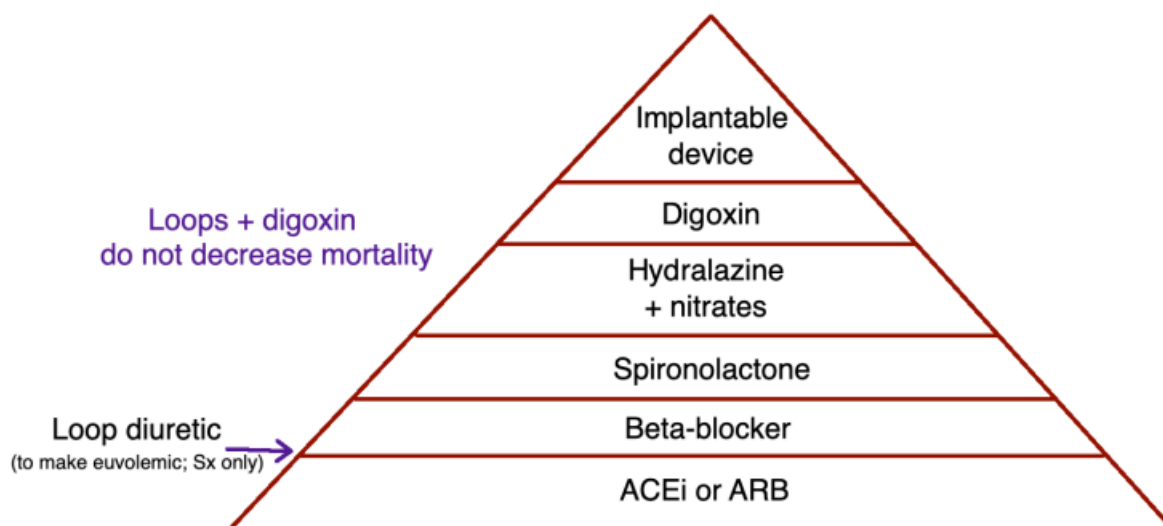
Spirolactone, Eplerenone

MOA of spironolactone and eplerenone?

- Aldosterone receptor antagonists.

When are they the answer for USMLE?

- Spironolactone or eplerenone used as potassium-sparing diuretic in patients *already on* furosemide (the latter causes potassium wasting).
 - 44M + liver failure + peripheral edema + already taking furosemide; what drug could be added to help with this patient's edema? → answer = spironolactone.
 - (It should be noted that triamterine and amiloride, which are ENaC inhibitors, are correct answers for this Q as well; I've seen this Q with numerous answers).
- Hyperaldosteronism due to adrenal hyperplasia or Conn syndrome (aldosterone-secreting tumor).
- Spironolactone is used in heart failure when patient has low ejection fraction despite already being on an ACEi/ARB and a beta-blocker.



Hierarchy for drugs in heart failure (simplified but HY for USMLE):

- Start with ACE inhibitor or an angiotensin II receptor blocker (ARB) to improve ejection fraction (normal is 55-70%).
- If the patient is fluid overloaded (i.e., peripheral edema or pulmonary edema), attempt to achieve euvoolemia by *adding* furosemide (loop diuretic).
- If insufficient ejection fraction with the ACEi or ARB, *add* a beta-blocker (metoprolol XR, bisoprolol, carvedilol, and nebivolol decrease mortality in heart failure; propranolol does not).
- If insufficient ejection fraction with ACEi/ARB + beta-blocker, *add* spironolactone (aldosterone receptor antagonist).
- If insufficient ejection fraction, *add* the **combination** of hydralazine + nitrates. Only the combo decreases mortality in HF. It is especially efficacious in African Americans (tend to have stiffer vessels).
- If insufficient ejection fraction when already on ACEi/ARB + beta-blocker + spironolactone + hydralazine + nitrates, *add* digoxin.
- If pharmacologic therapy insufficient, use implantable device.

Side-effects of spironolactone?

- Gynecomastia (HY for USMLE); eplerenone has much lesser risk.
- Hyperkalemia (potassium-sparing diuretic, but can inadvertently cause hyperkalemia in some patients).

Calcium/Phosphate drugs

Cinacalcet

MOA of cinacalcet?

- Calcimimetic (i.e., sensitizes calcium-sensing receptor on the parathyroid glands).
- Causes a decrease in serum PTH, thereby reducing serum calcium levels.

When is cinacalcet used?

- Secondary hyperthyroidism due to renal failure in patients on dialysis (i.e., once electrolytes are stabilized in patients with renal failure, cinacalcet can help lower the PTH levels).
- Primary hyperparathyroidism refractory to surgery.

What do I need to know for USMLE?

- USMLE is way more likely to ask *what effect* giving cinacalcet would have on a patient versus why it's used → answer = ↓ PTH; ↓ serum Ca^{2+} .

Sevelamer

MOA of sevelamer?

- Phosphate-binder that binds phosphate in the small bowel and prevents its absorption.
- Decreases phosphate levels in patients with chronic renal disease.

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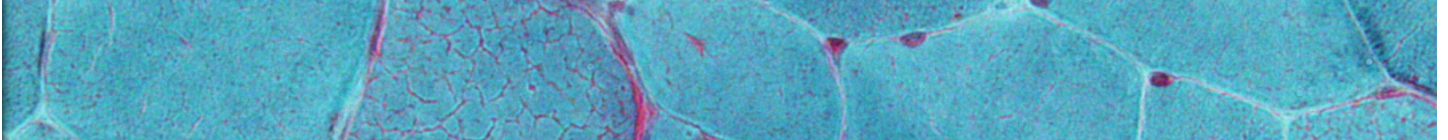


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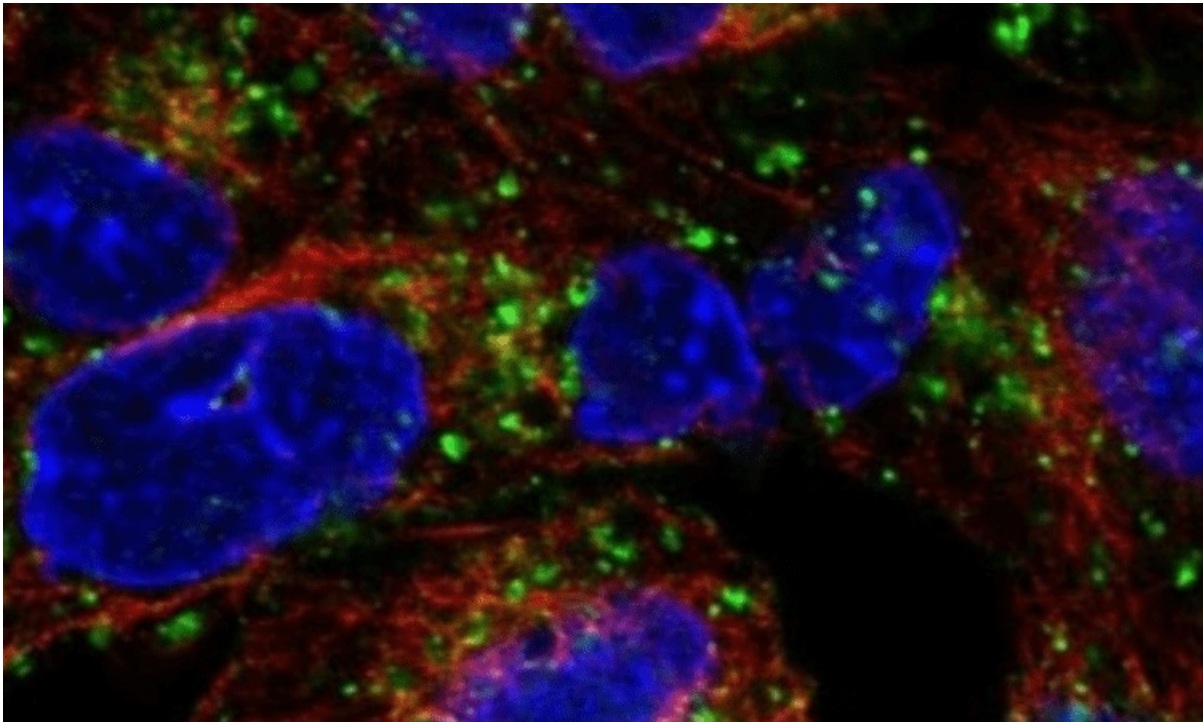
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PHARMACOLOGY

General pharm – Alpha (α) adrenergic drugs

by MEHLMANMEDICAL



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HY points followed by a quiz at the end

—

What are α and β receptors?

- Catecholamines are hormones produced by adrenal medulla – i.e., epinephrine, norepinephrine, and dopamine.
- α and β receptors are bound by catecholamines.
- They are known as **adrenergic** receptors because the **adrenal** medulla hormones bind them.

USMLE wants you to know which receptors the catecholamines bind to:

- Epinephrine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$.
- Norepinephrine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, **but not $\beta 2$** .
- Dopamine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$, D1.
 - At low-dose dopamine, D1 is bound most strongly (makes sense); as dose increases, β and α receptors are also bound.
 - If the USMLE Q tells you dopamine is used to increase someone's BP, choose $\alpha 1$ as the answer (discussed below).

$\alpha 1$ -receptors

Where are $\alpha 1$ receptors predominantly found?

- Arterioles.
- Internal sphincters (bladder, anal)

What HY $\alpha 1$ agonists do I need to know for USMLE?

- Phenylephrine, Oxymetazoline, Midodrine (**POM**) are HY $\alpha 1$ agonists.

What are the highest yield points about $\alpha 1$ agonism? ("Cut to the chase. I don't want superfluous detail.")

Causes arteriolar vasoconstriction → increases BP → reflex bradycardia.

- In other words, epinephrine, norepinephrine, and dopamine binding to $\alpha 1$ receptors on arterioles is *how* basal blood pressure is maintained.
- Should be noted that cortisol is *permissive* of the effects of catecholamines. That is, cortisol upregulates $\alpha 1$ receptors and allows catecholamines to do their job.
- 62F + receives intravenous midodrine intraoperatively. What might be expected in this patient? → answer = ↑ BP; ↓ HR.
- 34F + receives IV phenylephrine; what happens to blood flow through capillary beds? → answer = decreases.
- 22M jumps into cold water; what happens to central blood volume? → answer = increases; we expect $\alpha 1$ agonism to constrict peripheral vessels in order to decrease surface area for heat exchange; this will shift blood toward the core.

Used for nasal decongestion → $\alpha 1$ agonism causes small vessel constriction in the nasal mucosa → decreased hydrostatic pressure within vessels → decreased extravasation of transudate into nasal interstitium → **nasal decongestion**.

- 13F + nasal congestion + uses intranasal spray; what's the most likely agent she used? → answer = oxymetazoline, phenylephrine, midodrine.
- 16M + viral infection + has been using nasal spray for 5 days straight to treat nasal congestion + stops using the spray and gets rebound nasal congestion; what agent was he using? → answer = oxymetazoline, phenylephrine, midodrine.
 - Diagnosis is rhinitis medicamentosa → rebound nasal congestion after continual intranasal use of $\alpha 1$ agonists.
 - Mechanism is tachyphylaxis (tolerance) → down-regulation of intranasal $\alpha 1$ receptors.

- Prevention is by not using intranasal α_1 agonists for more than 3 days + use as infrequently as possible.

α_1 agonists can be used to treat epistaxis → similar mechanism as for nasal decongestion → decreased extravasation of blood to mitigate nose bleeding.

- 24M + nose is bleeding in the emergency department; next best step? → answer = nasal packing (not drugs).
- 24M + nose is bleeding in the emergency department; which of the following drugs may help treat this patient? → answer = choose the α_1 agonist listed → Phenylephrine, Oxymetazoline, Midodrine (**POM**).

Should be avoided in patients with Raynaud phenomenon (sounds weirdly specific, but asked on NBME) → if patient has abnormal peripheral vasoconstriction resulting in Raynaud, then the patient should avoid meds that vasoconstrict.

- 50F + history of dysphagia and pulmonary fibrosis + fingers turn color in the cold; what drug should be avoided in this patient? → answer = choose the α_1 agonist listed → Phenylephrine, Oxymetazoline, Midodrine (**POM**).

Constrict **internal** sphincters (i.e., bladder and anal) → USMLE wants you to know **internal** sphincters are under sympathetic control (i.e., catecholamines bind to α_1 receptors in order to constrict them; these are **not** under voluntary control). In contrast, **external** sphincters are under voluntary (somatic) control (nicotinic receptors binding acetylcholine).

- 34F + pregnant + performing Kegel exercises to strengthen pelvic floor muscles; which muscle is **not** strengthened by Kegel exercises? → student says, "Huh wtf? I'm supposed to know Kegel exercises at that level of detail?" → No. But of the random muscles they list, you'll notice that "internal anal sphincter" is one of them → then you say, "Wait, well, internal sphincters aren't under voluntary control, **so it's impossible that they could be strengthened by a voluntary exercise.**" This is a forest for the trees scenario: the USMLE doesn't expect you to be an obstetrician; they're just seeing if you know that internal sphincters are sympathetic-controlled.

What HY α_1 antagonists do I need to know for USMLE?

Phentolamine, phenoxybenzamine, -osins (prazosin, tamsulosin, terazosin).

What are the highest yield points about α_1 antagonism?

Causes decreased arteriolar vasoconstriction (not the same as vasodilation) → decreases BP → reflex tachycardia. Antagonizing the α_1 receptors doesn't cause an active vasodilation of the arterioles; it merely attenuates the degree of vasoconstriction. The catecholamines only function to increase the tone/constriction of the arterioles.

Phenoxybenzamine is **irreversible and non-competitive**. It is used for pheochromocytoma (catecholamine-secreting tumor of adrenal medulla). The other agents are **reversible and competitive**.

- 34M + periodic pounding headaches and palpitations + has history of **NF1** + BP of 120/80; what medication might help this patient? → answer = phenoxybenzamine. Student says, "Wait, how the fuck does the BP of 120/80 make sense?" → pheochromocytoma causes *paroxysmal* elevations in BP – i.e., sometimes it will be 220/120; other times it will be 120/80. USMLE Qs like to slam students by giving normal BP in pheochromocytoma vignettes.
- 28F + periodic palpitations and headaches + BP of 180/100 + treated for kidney stone last year; what's the mechanism of the agent that could treat the BP? → answer = irreversible α_1 antagonism (phenoxybenzamine). Patient has MEN2A (pheochromocytoma, hyperparathyroidism, medullary thyroid cancer). The kidney stone would be from hypercalcemia from the hyperPTH.

Phentolamine is used as the classic USMLE example of an α_1 antagonist that's reversible and competitive. Why they single this one out? → no fucking idea, but the do. So basically think: "Ok, phenoxybenzamine = irreversible + non-competitive; phentolamine = reversible + competitive."

Tamsulosin and terazosin = the answers for α_1 antagonists used for benign prostatic hyperplasia (BPH).

- 74M + interruption of urinary stream and dribbling; mechanism of agent that is most appropriate for this patient? → answer = competitive α_1 antagonist (i.e., terazosin or tamsulosin). The 5 α -reductase inhibitor finasteride can also be used. Irreversible α_1 antagonism (phenoxybenzamine) is not correct.

TCA's, anti-psychotics (usually 2nd gen), and first-gen H1 blockers all can cause three types of side-effects: α_1 antagonism (orthostatic hypotension + fainting), anti-cholinergic (hot, red, dry patient with a full bladder +/- dry mouth and dry eyes), and anti-H1 (sedation).

- 68F + being treated for depression with new agent + now has full bladder + is hot, red, dry + fainting spells; which agent was she given? → answer = TCA → fainting spells = α_1 antagonism from the TCA; the urinary retention and anhydrosis are anti-cholinergic side-effects.
- 24F + taking a drug for allergies + has a fainting spell + full bladder; next best step? → answer = stop anti-cholinergic medications → diphenhydramine (first-gen H1 blocker) has strong anti-cholinergic effects. Fainting spell due to anti- α_1 effects.
- 45M + being treated for schizophrenia with risperidone + fainting spell + suprapubic mass; why is the patient fainting? → answer = adverse effect of medication.

Any other weird factoids about α_1 receptors I should know?

- Lower yield, but rarely they will ask about G-proteins (if you want >260 on Step 1).
- α_1 receptors are G- α_q G-proteins, which means that agonism causes $\uparrow$ inositol triphosphate (IP3) and antagonism $\downarrow$ IP3.
- 72M + started on tamsulosin for BPH; what type of molecular effect does this have? → answer = $\downarrow$ IP3.
- 31F + receives midodrine during surgery; what molecular effect does this have? → answer = $\uparrow$ IP3.

α_2 -receptors

- Sort of the "odd one out" in that they serve a negative-feedback function on presynaptic terminals.
- In other words, if a catecholamine binds to an α_2 receptor, then the result is decreased release of additional catecholamine from that presynaptic terminal. That is:
 - $\uparrow$ α_2 agonism = $\downarrow$ catecholamine release = $\downarrow$ BP.
 - $\uparrow$ α_2 antagonism = $\uparrow$ catecholamine release = $\uparrow$ BP.

What HY α_2 agonists do I need to know for USMLE?

Methyldopa, clonidine.

- Methyldopa is used to treat hypertension during pregnancy.
 - It's frequent that I'll ask students, "What's methyldopa?" And they say it's for Parkinson disease. Completely unrelated. What you're thinking of is carbidopa/levodopa.
- Clonidine can be used to treat HTN, but is notably used as a psych drug where it has utility in treating Tourette syndrome.

What HY α_2 antagonists do I need to know for USMLE?

Mirtazapine.

- Stimulates appetite. Used to treat patients with anorexia who also have depression.

Any other weird factoids about α_2 receptors I should know?

- α_2 receptors are G- α_i G-proteins, which means that agonism causes $\downarrow$ cAMP and antagonism $\uparrow$ cAMP.
- 34F + receives methyldopa during pregnancy; what molecular effect does this have? $\rightarrow$ answer = $\downarrow$ cAMP.
- 17F + BMI of 14 + depression + receives mirtazapine; what molecular effect does this have? $\rightarrow$ answer = $\uparrow$ cAMP.

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January 26, 2024

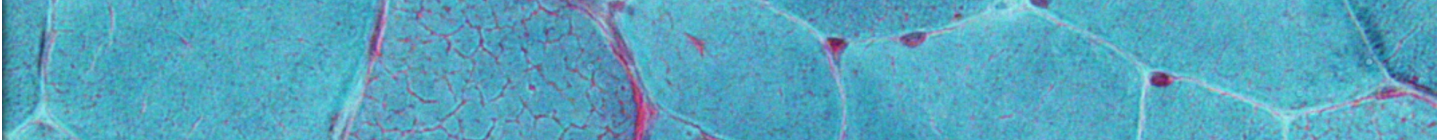


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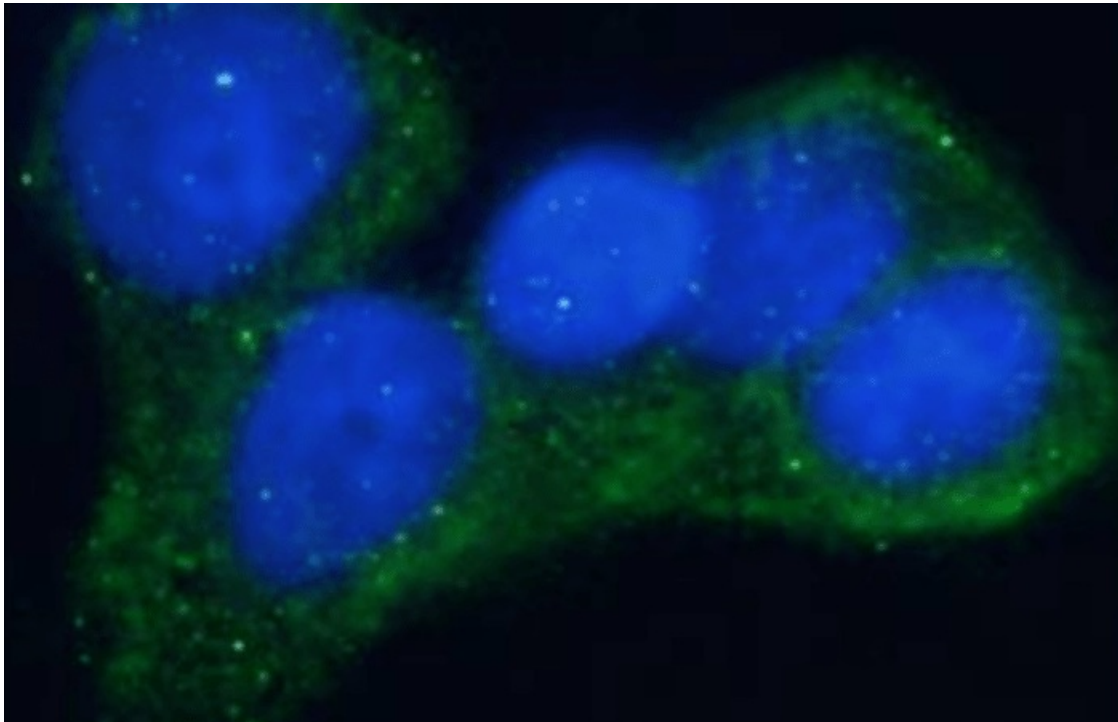
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PHARMACOLOGY

General pharm – Beta (β) adrenergic drugs

by MEHLMANMEDICAL



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HY points followed by a quiz at the end

—

What are α and β receptors?

- Catecholamines are hormones produced by adrenal medulla – i.e., epinephrine, norepinephrine, and dopamine.
- α and β receptors are bound by catecholamines.
- They are known as **adrenergic** receptors because the **adrenal** medulla hormones bind them.

USMLE wants you to know which receptors the catecholamines bind to:

- Epinephrine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$.
- Norepinephrine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, **but not $\beta 2$** .
- Dopamine agonizes: $\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$, D1.
 - At low-dose dopamine, D1 is bound most strongly (makes sense); as dose increases, β and α receptors are also bound.
 - If the USMLE Q tells you dopamine is used to increase someone's BP, choose **$\alpha 1$ as the answer**.

$\beta 1$ receptors

Where are $\beta 1$ receptors predominantly found?

- Cardiac muscle.
- Juxtaglomerular cells of the kidney.

What HY $\beta 1$ agonists do I need to know for USMLE?

Dobutamine.

What's the main effect of $\beta 1$ agonism?

$\uparrow$ chronotropy (heart rate), $\uparrow$ inotropy (contractility), $\uparrow$ oxygen demand on the heart.

- Dobutamine is classically used as “dobutamine-echo” for stress tests in patients who cannot do an exercise stress test. Dobutamine will increase the heart rate, therefore increasing oxygen demand, and then the cardiac function can be evaluated using a transthoracic echo.

Causes increased renin release from the kidney.

- 22M + runs 10 miles and is dehydrated; which of the following would be expected in this patient? $\rightarrow$ answer = “increased sympathetic nerve stimulation at the kidney.” This is because we'd have upregulation of the renin-angiotensin system to help restore blood volume, so $\uparrow$ $\beta 1$ agonism via sympathetic nerve stimulation at JG cells $\rightarrow$ increased renin release.

What HY $\beta 1$ blockers do I need to know for USMLE?

Beta-blocker discussion is a bit lengthy but very HY.

Beta 1-selective blockers: **A BEAM** of beta-blockers $\rightarrow$ **A**tenolol **B**isoprolol, **E**smolol, **A**cebutolol, **M**etoprolol.

- If it starts with a letter before M in the alphabet and ends in -olol, it's beta 1-selective.

Beta 1/2 non-selective blockers: everything else $\rightarrow$ i.e., **Propranolol**, Pindolol, Timolol, Nebivolol.

Beta-blockers with mixed alpha-blocking effect (i.e., mostly block beta, but also block alpha to a lesser degree): labetalol, carvedilol.

- Beta-blockers end with -olol. If the ending is not -olol (as with labetalol + carvedilol), that's how you can remember

it's mixed.

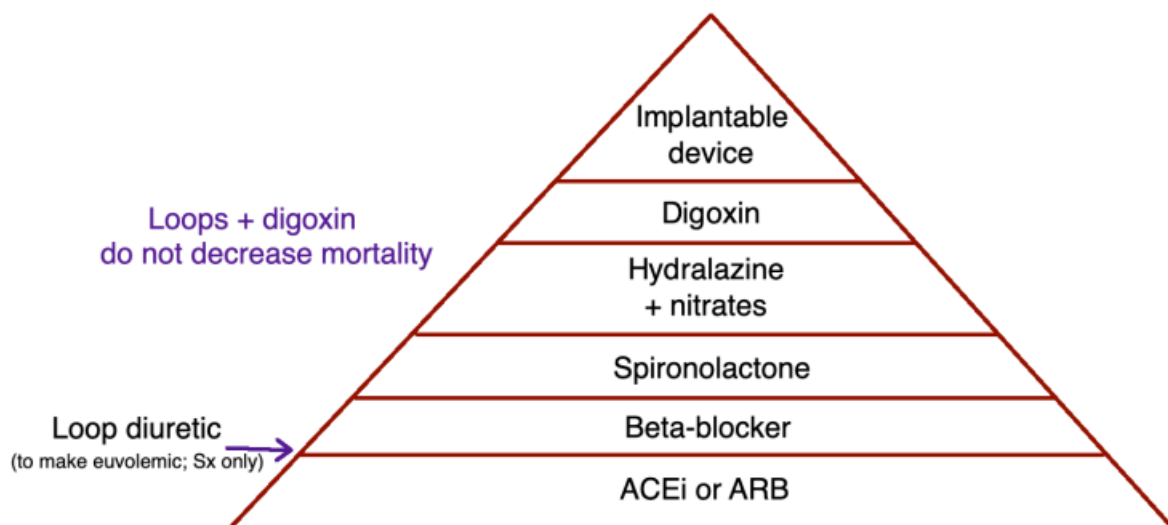
What HY things do I need to know about β 1 blockers?

The four beta-blockers that decrease mortality in heart failure: metoprolol XR, bisoprolol, carvedilol, nebivolol.

- USMLE will only ever write “metoprolol.” But some cardiologists on the ward will freak the fuck out if you don't specify “extended release” as the metoprolol formulation when discussing mortality reduction in heart failure. “Regular” metoprolol is used for atrial fibrillation.

Hierarchy for drugs in heart failure (simplified but HY for USMLE):

- Start with ACE inhibitor or an angiotensin II receptor blocker (ARB) to improve ejection fraction (normal is 55-70%).
- If the patient is fluid overloaded (i.e., peripheral edema or pulmonary edema), attempt to achieve euvolemia by *adding* furosemide (loop diuretic).
- If insufficient ejection fraction with the ACEi or ARB, *add* a beta-blocker (metoprolol XR, bisoprolol, carvedilol, or nebivolol).
- If insufficient ejection fraction with ACEi/ARB + beta-blocker, *add* spironolactone (aldosterone receptor antagonist).
- If insufficient ejection fraction, *add* the **combination** of hydralazine + nitrates. Only the combo decreases mortality in HF. It is especially efficacious in African Americans (tend to have stiffer vessels).
- If insufficient ejection fraction when already on ACEi/ARB + beta-blocker + spironolactone + hydralazine + nitrates, *add* digoxin.
- If pharmacologic therapy insufficient, use implantable device.



- 68M + shortness of breath on exertion + peripheral edema + JVD + echo shows EF of 35%; what drug should be initiated to improve EF? → answer = ACEi or ARB; beta-blocker is the **wrong** answer on USMLE.

Metoprolol is used for rate control in atrial fibrillation.

- When treating AF, we need to address both blood viscosity and the actual AF arrhythmia.
- For blood thinning: if low 0-1 **CHADS2** score, give aspirin; if 2+ CHADS2 score, give warfarin or NOAC. Variations of the score exist, but the hyper-simple CHADS2 always works for USMLE Qs. **CHF**; **HTN**; **Age 75+**; **Diabetes**; Hx of **S**troke/TIA. The latter is worth 2; the others are all worth 1. It's to my observation students tend to forget the age (they always say 70 or 65).
- For the actual AF, do rate control before rhythm control. For rate control, use “regular” metoprolol (in contrast to metoprolol XR for HF).
- If contraindication to beta-blocker (discussed below), verapamil (non-dihydropyridine CCB) is frequently used.
- If rate control insufficient, go to rhythm control.

- Flecainide (type Ic Na channel blocker) is used if the patient has AF with no LVH, structural heart disease, or coronary artery disease.
- If patient has LVH, structural heart disease, or coronary artery disease, other anti-arrhythmics, e.g., amiodarone or dofetilide (type III K channel blockers) can be used.

HY contraindications to beta-blockers:

- Hx of depression (beta-blockers can cause depression).
- Diabetes → epinephrine is responsible for symptoms of hypoglycemia; with beta-blockade, symptoms can be masked. Diabetics on beta-blockers require diligent monitoring.
- COPD.
- Asthma (can use beta 1-selective, but cannot use non-selective) → should be noted that a Q on one of the 2CK psych forms has benzo as the answer for social phobia, over propranolol, in a patient with simple asthma, so for USMLE purposes, asthma period, equals no beta-blocker).
- 2nd or 3rd-degree heart block.
- Unstable heart failure resulting in hemodynamic instability.
- Peripheral vascular disease *no longer* a contraindication.

Propranolol has many HY uses for USMLE:

- Can be used to treat essential or idiopathic tremors.
 - 22F + bilateral resting tremor + mom also has tremor + self-medicates with alcohol to decrease tremor; Tx? → answer = propranolol.
 - Essential tremor is a bilateral resting tremor; it's autosomal dominant; patients frequently self-medicate with alcohol, which decreases tremor.
- Tachycardia in hyperthyroidism.
 - Beta-blockade decreases the conversion of T4 to T3.
- Prophylaxis for esophageal varices bleeding.
 - 59M + alcoholic + vomits copious bright red blood that requires banding; what could be given to help prevent future episodes? → answer = "beta blockade" (propranolol) → decreases portal blood flow and, therefore, pressure.
 - Treatment is banding and sometimes octreotide. But *prophylaxis* for a bleed is propranolol.
- Hypertrophic obstructive cardiomyopathy (HOCM).
 - 20F + 2/6 mid-systolic murmur in upper chest + murmur worsens with Valsalva; what could be given to this patient to reduce risk of death? → answer = propranolol → slows heart rate → increases time for diastolic filling → lessens chance of low preload state where the murmur (LV outflow tract obstruction) worsens.
- Akathisia secondary to anti-psychotic use.
 - 39M + schizophrenia + started taking olanzapine a week ago + feels constantly restless; Tx? → answer = propranolol.
- Social phobia.
 - 29M + needs to speak in front of large audience in upcoming presentation for work; Tx? → propranolol.
 - 2CK psych form gives two near-identical Qs – one mentions Hx of asthma; the other doesn't – if patient has asthma, use benzo instead of beta-blocker for social phobia.
 - Atenolol can be used for stage fright instead of propranolol (both are correct).
- Migraine *prophylaxis*
 - 40F + high BP after lifestyle modification + Hx of migraines; which new medication do we start? → answer = "beta blockade" (propranolol). This is on one of the 2CK FM forms.
 - *Treatment* for migraines is NSAIDs followed by -triptans; *prophylaxis* is beta-blocker.
 - Cluster headache prophylaxis is verapamil; beta-blocker is the **wrong** answer for cluster (low-30s%-correct Q in Qbank IIRC).

Topical timolol can be used for glaucoma.

- Beta-blockade decreases aqueous humor production.
- USMLE just wants you to know that beta-blockade, *in general*, can be used for glaucoma and reduces aqueous humor production.
- Propranolol is not used for glaucoma.

Any other weird factoids about β_1 receptors I should know?

- Lower yield, but rarely they will ask about G-proteins (if you want >260 on Step 1).
- β_1 receptors are G- α_s G-proteins, which means that agonism causes $\uparrow$ cAMP and antagonism $\downarrow$ cAMP.
- 34M + long Hx of migraines + started on new drug for HTN; what's the molecular effect of this agent? $\rightarrow$ answer = $\downarrow$ cAMP (propranolol).

β_2 receptors

Where are β_2 receptors predominantly found?

- Bronchiole smooth muscle (cause dilation).
- Peripheral arterioles (cause dilation).
- Uterus (cause relaxation).

What HY β_2 agonists do I need to know for USMLE?

Beta 2-selective agonists: Albuterol, Salmeterol, Ritodrine.

Beta 2 non-selective agonists: Terbutaline, Isoproterenol.

What's the main effect of β_2 agonism?

Dilates bronchioles.

- Albuterol is a short-acting beta 2 agonist (SABA) used as the initial Tx in asthma.
- Whereas with asthma, albuterol is always the first med, in COPD it is *sometimes* chosen as the first med. Ipratropium, a muscarinic receptor antagonist, can also be used first-line in COPD (answer on one of the 2CK IM forms).
- Salmeterol is a long-acting beta 2 agonist (LABA) used after inhaled fluticasone in outpatient asthma management.
- For a more detailed post on asthma management, see [here](#).
- Epinephrine is used as the Tx for anaphylactic shock because the β_2 agonism functions to open the lungs.

Dilates peripheral arterioles.

- Selective β_2 agonists can decrease BP in theory, but are not used for this purpose.
- If epinephrine is administered alone, BP will go up. However BP will decrease with epinephrine *if alpha 1 is blocked first*.
- "What do you mean? That's confusing." I know. Chill the fuck out. I'll explain:
- If we give epinephrine (which agonizes α_1 , α_2 , β_1 , and β_2), **the strong vasoconstrictive effect of α_1** on arterioles overpowers the peripheral vasodilatory effect of β_2 , so BP goes up with epinephrine administration alone.
- That is: effect of α_1 outpowers β_2 , where effect of constriction > dilation, so BP goes up.
- But if we first block alpha 1 with phentolamine, *then* give epinephrine, the effect on β_2 is no longer countered by α_1 , so we get dilation and BP decreases.
- "But Michael, how does the USMLE actually assess that?"
 - Scientist is running an animal model experiment. Drug X is given and BP goes up. He then performs the experiment a second time. In this case, Drug Y is administered before drug X. BP goes down slightly with

Drug Y. It then decreases further with Drug X. What are Drug X and Y? → Answers: Drug X = epinephrine.
Drug Y = phentolamine.

Can be used for tocolysis (cause relaxation of the uterus).

- Tocolytic agents (i.e., ritodrine) delay labor and buy the expectant mother time in certain delivery situations.
 - Woman at 32 weeks gestation is in labor + steroids have not yet been administered for fetal lung maturity + a β_2 agonist is given as a tocolytic; which of the following agents is best described here? → answer = ritodrine.

What HY things do I need to know about β_2 blockers?

- There's no such thing as a "beta 2 blocker." It's more that: of the standard β_1 blockers we use for a variety of reasons, some are non-selective and also antagonize β_2 (e.g., propranolol). But we don't use selective β_2 antagonists in medicine.
- According to Wikipedia, butaxamine is a select β_2 blocker, but its use is limited to experimental research rather than actual medicine.
- The effect on G-proteins is the same as β_1 – i.e., β_2 activates G- α_s G-proteins, which means that agonism causes $\uparrow$ cAMP and antagonism $\downarrow$ cAMP.

β_3 receptors

- Yeah. Weird and esoteric. But you need to know there's a drug called mirabegron, which is a β_3 agonist used for urge incontinence.
- Classic Tx for urge incontinence (overactive detrusor muscle) is oxybutynin, a muscarinic receptor antagonist. But mirabegron has been showing up on students' USMLEs and shelf exams as well.
- So I want you to say: "Treatment for urge incontinence is almost always oxybutynin, a muscarinic receptor antagonist. But mirabegron is a beta 3 agonist that is newer and now shows up on the USMLE."

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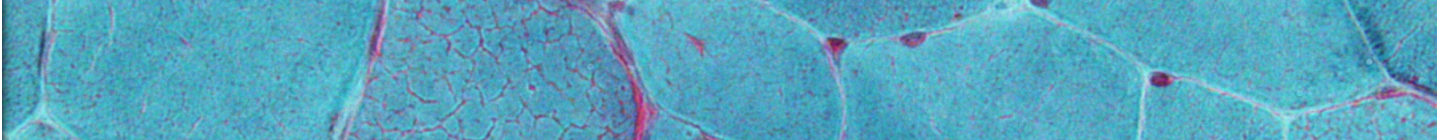


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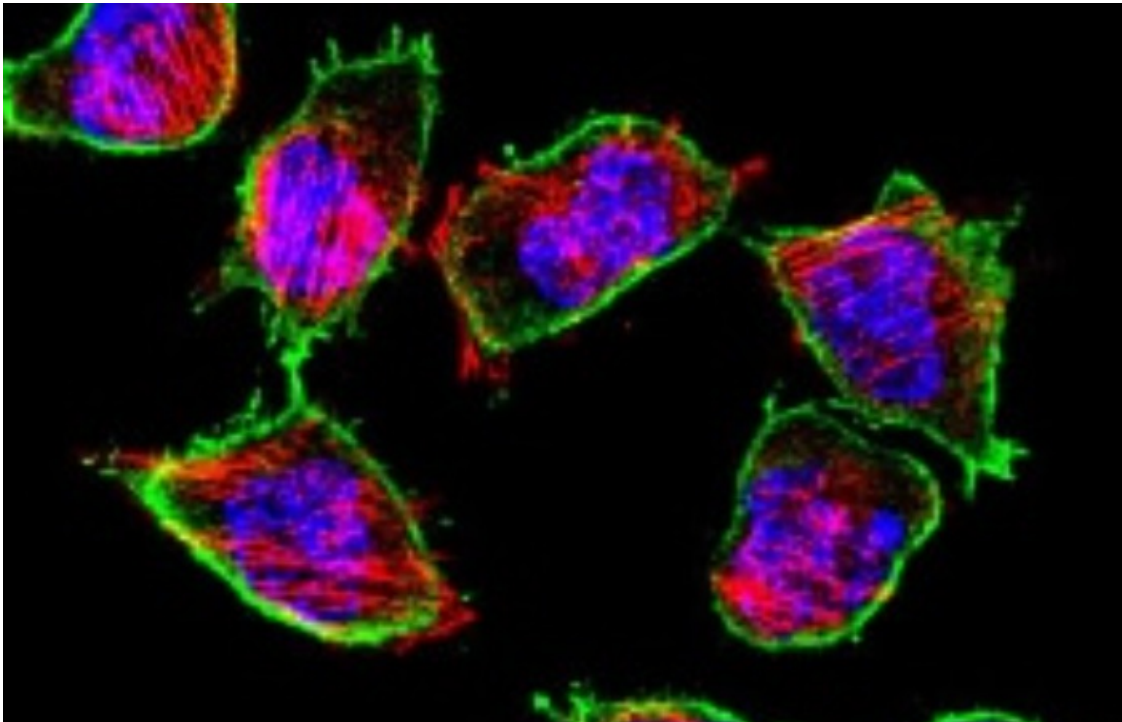
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PHARMACOLOGY

General pharm – Cholinergic agents (anti-)

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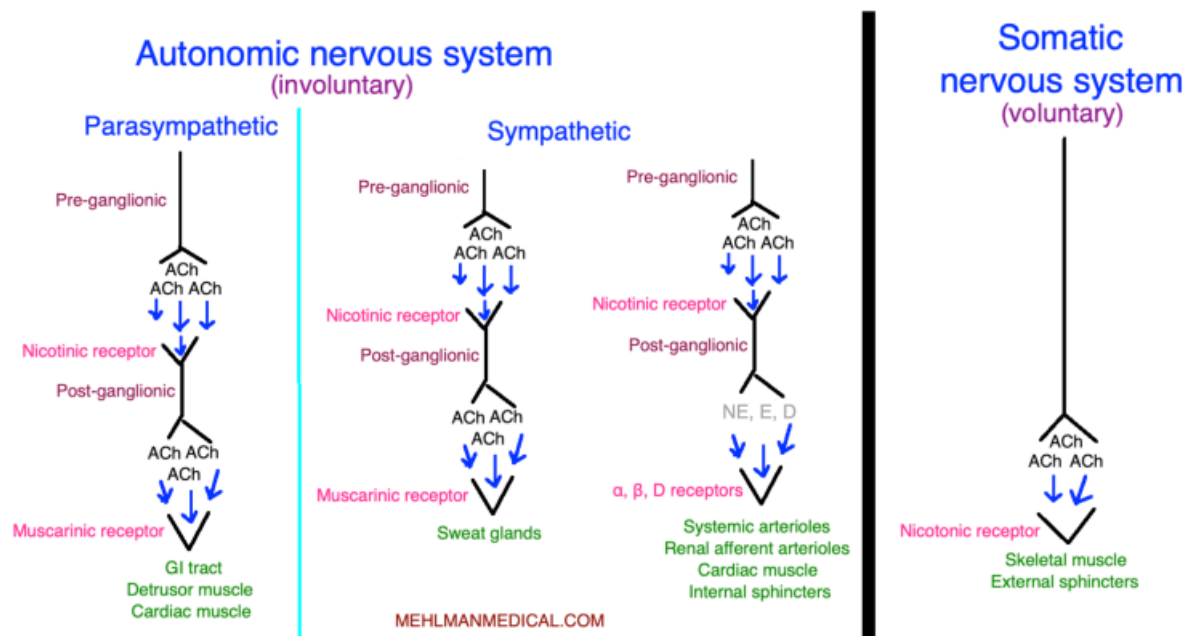
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HY points followed by a quiz at the end

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What are cholinergic receptors?

- Called cholinergic because they bind acetylcholine.
- Muscarinic and nicotinic types.
- Nicotinic are located at most preganglionic nerve terminals and **at skeletal muscle**.
- USMLE focuses probably 49 times out of 50 Qs on muscarinic, which stimulate the **GI tract, urinary bladder, and sweat glands**.
- USMLE cares most that you know the parasympathetic nervous system agonizes muscarinic receptors.
- USMLE also wants you to be aware of pro- vs anti-cholinergic drugs and what effects they'll have.



What does cholinergic vs anti-cholinergic mean?

- Cholinergic agents either: 1) agonize cholinergic receptors directly, or 2) increase the potentiation of ACh binding to its receptors by inhibiting acetylcholinesterase (the enzyme that breaks down ACh).
- If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms → **D**iarrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating.
- Anti-cholinergic agents block cholinergic receptors directly. **They will cause the opposite of DUMBBELSS** → Constipation, Urinary retention, Mydriasis, Tachycardia, *Not bronchodilation* (β_2 agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis.

Muscarinic receptor antagonists

- Block muscarinic receptors directly.
- Cause “opposite of DUMBBELSS” effects.
- Many drugs in this category, but the aim is to get you actual points on the USMLE, not to flood you with superfluous drug names that will never show up.

Atropine

- Highest yield anti-muscarinic for USMLE.
- 50F + intraoperatively atropine is administered; what effect will this have on the patient’s vital signs? → answer = $\uparrow$ heart rate.
- Parasympathetic activity (i.e., increased endogenous ACh binding to M2 receptors on atria) will normally slow the heart rate, so blocking this activity will increase heart rate.
- Used as first-line Tx for organophosphate poisoning; *then* use pralidoxime.

Benztropine

- Used to treat cogwheel rigidity in Parkinson disease and acute dystonia due to antipsychotic use.
- High-yield drug for USMLE.
- 72M + shuffling gait + unilateral resting tremor + stiff arm muscles; a drug with which type of MOA will improve his muscle stiffness? → answer = anti-cholinergic (benztropine).
- 40M + Hx of schizophrenia + started on fluphenazine + a few hours later has abnormal eye movements; Tx? → answer = benztropine. Patient has oculogyric crisis, which is a type of acute dystonia. Torticollis is another type of acute dystonia, which is an abnormally tight and crooked neck.
- **Diphenhydramine** is a first-generation H1 blocker that has strong anti-cholinergic side-effects and is sometimes the answer if benztropine not listed. USMLE will not list both at the same time.

Ipratropium, Tiotropium

- First-line inhaled muscarinic antagonists used for COPD.
- Don't bronchodilate as much as they merely decrease any degree of bronchoconstriction.
- Albuterol, a β_2 agonist, can also be used first-line for COPD.
- 45M + long Hx of smoking + hyper-inflated lungs; next best step in pharmacologic Tx? → answer = ipratropium, tiotropium, or albuterol. On one of the 2CK IM forms, ipratropium is correct.

Oxybutynin

- Super-HY for USMLE and shows up all over the NBME exams.
- Used for urge incontinence (hyperactive detrusor).
- Don't confuse with bethanechol, which is a muscarinic agonist used to treat neurogenic (flaccid) bladder.
- 32F + blurry vision + clonus in right foot + leakage of urine; Tx for urinary leakage? → answer = oxybutynin. (IV methylprednisolone is a steroid used to treat acute flares of multiple sclerosis, but oxybutynin will specifically address the urge incontinence classically seen MS).

Scopolamine

- Classic muscarinic antagonist used for motion sickness.
- 42F + going on a whale watch + Hx of motion sickness; an agent with which type of receptor binding will help prevent motion sickness in this patient? → answer = muscarinic receptor antagonism (scopolamine).

Hyoscyamine

- Anti-muscarinic used sometimes for irritable bowel syndrome.

Super-HY points about agents that cause anti-cholinergic side-effects

- USMLE wants you to know three classes of drugs that cause three different types of side-effects.
- Can seem annoying to learn at first, but super-HY and loaded with points on the Steps.
- Tricyclic antidepressants (TCAs; e.g., amitriptyline), first-generation H1 blockers (e.g., diphenhydramine, chlorpheniramine), and antipsychotics **all cause anti-cholinergic, anti- α_1 , and anti-H1 side-effects.**

"Huh? I'm confused." Chill. I'll explain:

- TCAs (mixed serotonin/NE reuptake inhibitors).
- 1st-gen H1 blockers (histamine 1 receptor blockers).
- Anti-psychotics (D2 receptor antagonists).

Memorize the above three classes as being associated with three different types of side-effects. They are as follows:

- Anti-muscarinic → “opposite of DUMBELSS.”
- Anti- α_1 → orthostatic hypotension / fainting.
- Anti-H1 → sedation.

“Still a bit confused. Can you give examples?” No problem:

Amitriptyline

- TCA anti-depressant with potent anti-cholinergic side-effects. Also first-line agent for diabetic neuropathic pain.
- 22F + just started on new drug for depression + now is hot, red, dry; which drug was she started on? → answer = amitriptyline → “hot, red, dry” is classic combo for anti-muscarinic effects → anhidrosis (not sweating) means the patient doesn’t dissipate heat well.
- 78M + history of BPH and T2DM + on amitriptyline for neuropathic pain + has suprapubic mass + elevated creatinine; next best step? → answer = “stop all anti-cholinergic medications” → amitriptyline is a TCA used first-line for diabetic neuropathic pain → anti-cholinergic side-effects can cause urinary retention (suprapubic mass = full bladder). After the TCA is stopped, a catheter will need to be inserted (Surg shelf).

Diphenhydramine, Chlorpheniramine

- First-generation H1 blockers with such strong anticholinergic side-effects that they frequently show up in USMLE Qs as just being anticholinergics straight-up (i.e., no mention of technically being anti-histamines).
- Used to treat acute dystonia with anti-psychotic use *if benztropine not listed*.
- 44M + schizophrenia + started on aripiprazole + now has diffuse muscle stiffness/rigidity + temperature is 98.6F; treatment? → answer = benztropine **or diphenhydramine**. The latter is a first-gen H1 blocker with potent anticholinergic side-effects. This patient has acute dystonia, **not** neuroleptic malignant syndrome. NMS will have fever. USMLE slams students on this. Rigidity + fever = NMS; but rigidity + no fever = acute dystonia.
- 38F + schizophrenia + started on olanzapine + now neck is crooked and stiff; Tx? → answer = diphenhydramine.

Prochlorperazine

- Antipsychotic med used as an anti-nausea medication.
- The anti-nausea effect comes from the anti-cholinergic side-effects. As discussed above, scopolamine is an anti-cholinergic used for motion sickness (i.e., it prevents nausea). Even though prochlorperazine is an anti-psychotic (D2 antagonist), because its anti-cholinergic effects are so significant, that’s actually a *good thing* when we want to treat nausea (i.e., it acts like scopolamine).
- Anti-cholinergic, anti- α_1 , and anti-H1 effects.
- 50F + just had surgery + has nausea + started on prochlorperazine + faints when she stands up; why? → answer = “antagonism at α_1 receptors” (orthostatic hypotension).

Pralidoxime

- Regenerates active acetylcholinesterase in patients with organophosphate poisoning.
- Organophosphates act as acetylcholinesterase inhibitors → cause DUMBELSS in patient working on fruit farm or who “drank fluid” attempting to commit suicide.
- Pralidoxime will kick out the phosphate group from the organophosphate that’s bound to and inhibiting the acetylcholinesterase → regenerates active acetylcholinesterase.
- 20F + working on fruit farm + has constricted pupils + bradycardia + drooling; what’s the Tx for this patient’s condition? → answer = atropine, followed by pralidoxime.
- Atropine blocks muscarinic receptors directly. This is the answer *before* pralidoxime. If the USMLE Q lists both answers, **choose atropine first**.

Jimson weed

- 16M + tactile and/or visual hallucinations + hot, red, dry + chewing on weeds; Dx? → answer = Jimson weed → illicit drug used by some teenagers → weird, but tested on USMLE → contains atropine compounds.

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HY USMLE Q #965 – Cardiopulm

January 26, 2024

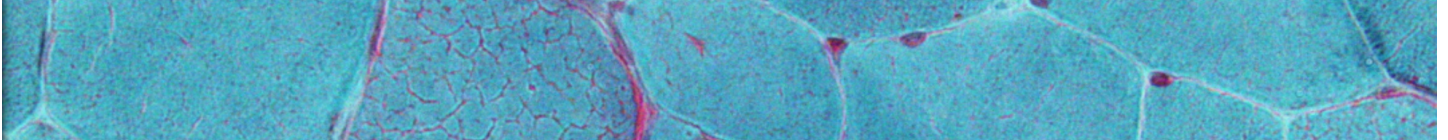


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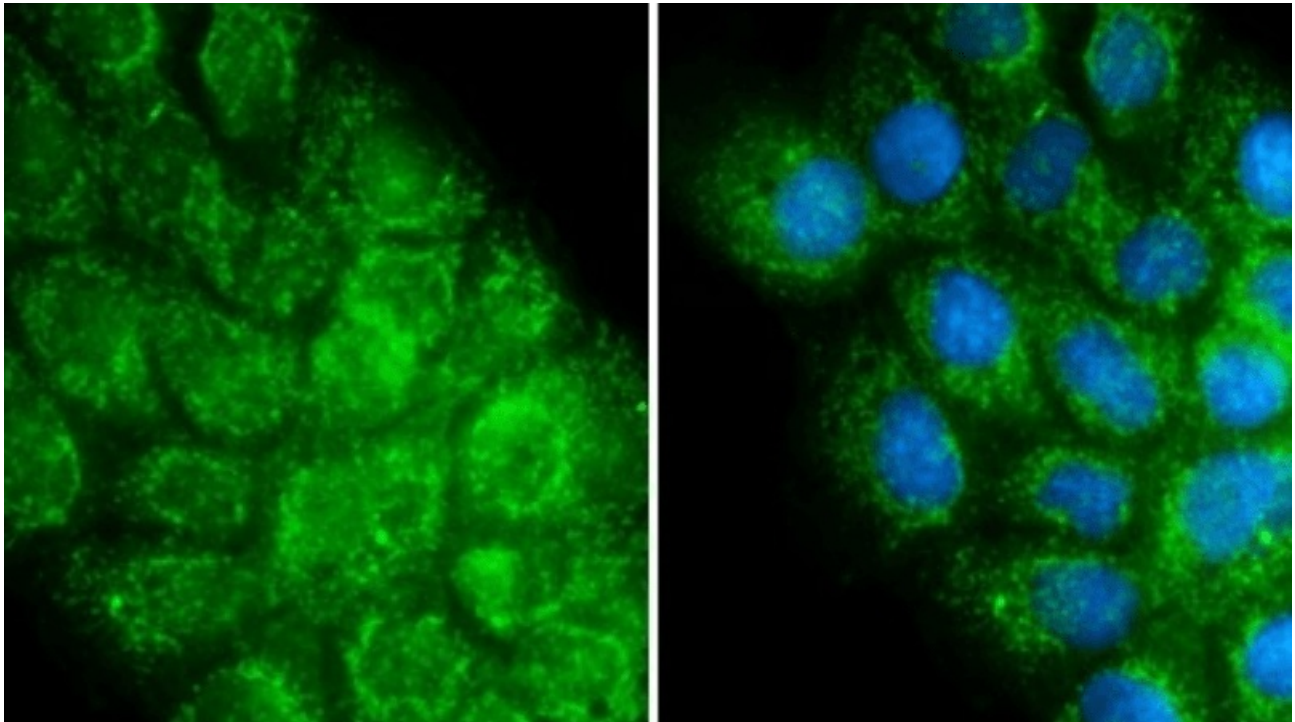
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PHARMACOLOGY

General pharm – Cholinergic agents (pro-)

by MEHLMANMEDICAL



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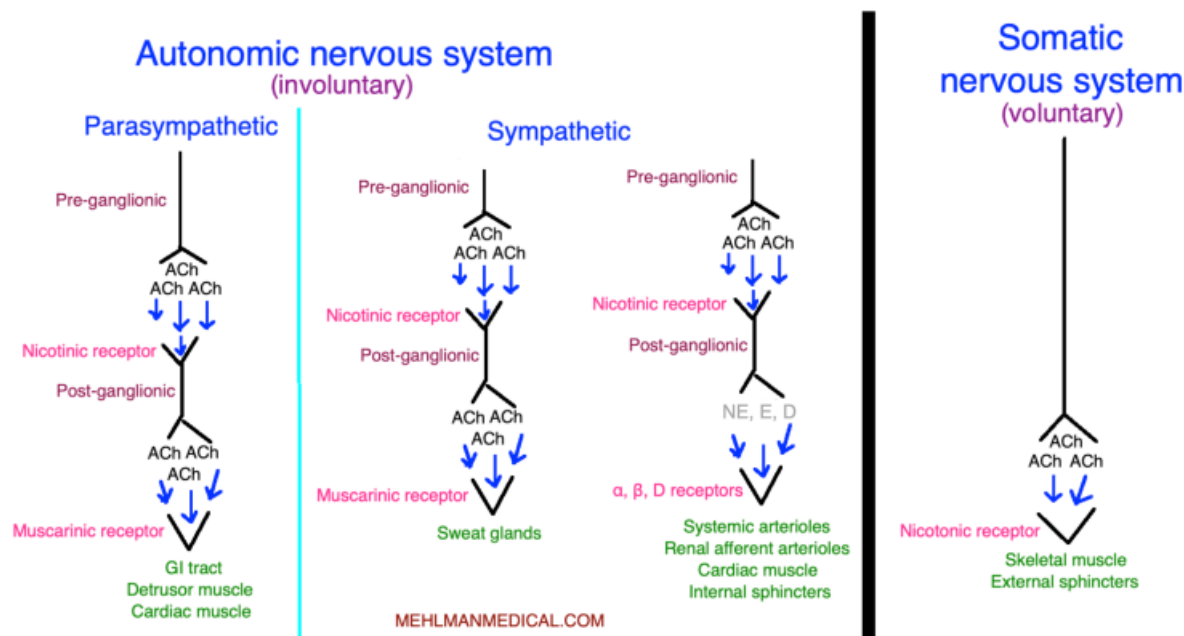
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HY points followed by a quiz at the end

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What are cholinergic receptors?

- Called cholinergic because they bind acetylcholine.
- Muscarinic and nicotinic types.
- Nicotinic are located at most preganglionic nerve terminals and **at skeletal muscle**.
- USMLE focuses probably 49 times out of 50 Qs on muscarinic, which stimulate the **GI tract, urinary bladder, and sweat glands**.
- USMLE cares most that you know the parasympathetic nervous system agonizes muscarinic receptors.
- USMLE also wants you to be aware of pro- vs anti-cholinergic drugs and what effects they'll have.



What does cholinergic vs anti-cholinergic mean?

- Cholinergic agents either: 1) agonize cholinergic receptors directly, or 2) increase the potentiation of ACh binding to its receptors by inhibiting acetylcholinesterase (the enzyme that breaks down ACh).
- If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms → **D**iarrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating.
- Anti-cholinergic agents block cholinergic receptors directly. **They will cause the opposite of DUMBBELSS** → Constipation, Urinary retention, Mydriasis, Tachycardia, *Not bronchodilation* (β₂ agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis.

Muscarinic receptor agonists

Bethanechol

- Answer for neurogenic bladder (hypocontractile bladder), classically in diabetes.
- 49F + T2DM + can't feel her feet or ankles + HbA1c of 11 + urinary studies show post-void volume of 300 mL; drug to Tx her condition? → answer = bethanechol → stimulates detrusor muscle.

Carbachol

- Carbachol is used to constrict the pupil and lower intraocular pressure in open-angle glaucoma.
- Don't confuse with bethanechol, which is used for neurogenic bladder.

Pilocarpine

- Answer for closed angle glaucoma when needing to lower intraocular pressure acutely and fast.
- Constricts the pupil.

- Very powerful agent.
- 38M + left eye is painful and red + pupil is fixed and slightly dilated; which potent mitotic agent is most appropriate for this patient? → answer = pilocarpine.

Methacholine

- Used to diagnose asthma.
- Muscarinic agonist that can induce bronchoconstriction and reproduce asthma symptoms in those with asthma.
- Never use during acute episodes (clearly).
- USMLE wants you to know the MOA is muscarinic receptor agonist.

Acetylcholinesterase inhibitors

- Acetylcholinesterase is the enzyme in the synaptic cleft that breaks down acetylcholine. So if you block this enzyme, you increase the amount of ACh available in the synaptic cleft → increased binding to cholinergic receptors.

Donepezil

- Used to treat Alzheimer.
- Alzheimer is characterized by ↓ ACh-producing neurons in the basal forebrain (nucleus basalis of Meynert).
- 69M + one-year Hx of worsening cognitive decline + MMSE is 22/30 + neurologic exam shows no abnormalities; which drug is most appropriate for this patient? → answer = donepezil.

Rivastigmine, Galantamine

- Slightly lower yield than donepezil, but also used to treat Alzheimer.

Edrophonium (aka Tensilon)

- Short-acting.
- Tensilon test can help diagnose myasthenia gravis.
- A patient with MG gets improvement of muscle weakness with edrophonium administration.
- However, if the USMLE asks how to diagnose myasthenia gravis, **choose antibodies against post-synaptic ACh receptors over Tensilon test.**

Pyridostigmine

- Treatment for myasthenia gravis.
- 40F + worsening diplopia and ptosis by the time she finishes work late-afternoon; Tx? → answer = pyridostigmine.

Physostigmine

- Known to have good blood-brain barrier penetration.
- Treatment for atropine overdose (atropine is a muscarinic receptor antagonist).

Super-HY points about agents that cause cholinergic side-effects

Organophosphates

- Organophosphates act as acetylcholinesterase inhibitors → cause DUMBELSS in patient working on fruit farm or who “drank fluid” attempting to commit suicide.
- Pralidoxime will kick out the phosphate group from the organophosphate that's bound to and inhibiting the

acetylcholinesterase → regenerates active acetylcholinesterase.

- 20F + working on fruit farm + has constricted pupils + bradycardia + drooling; what's the Tx for this patient's condition? → answer = atropine, followed by pralidoxime.
- Atropine blocks muscarinic receptors directly. This is the answer *before* pralidoxime. If the USMLE Q lists both answers, **choose atropine first**.

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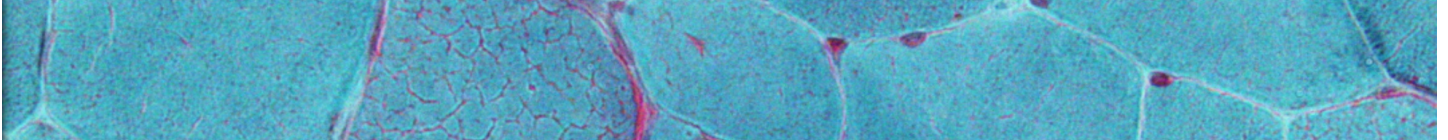


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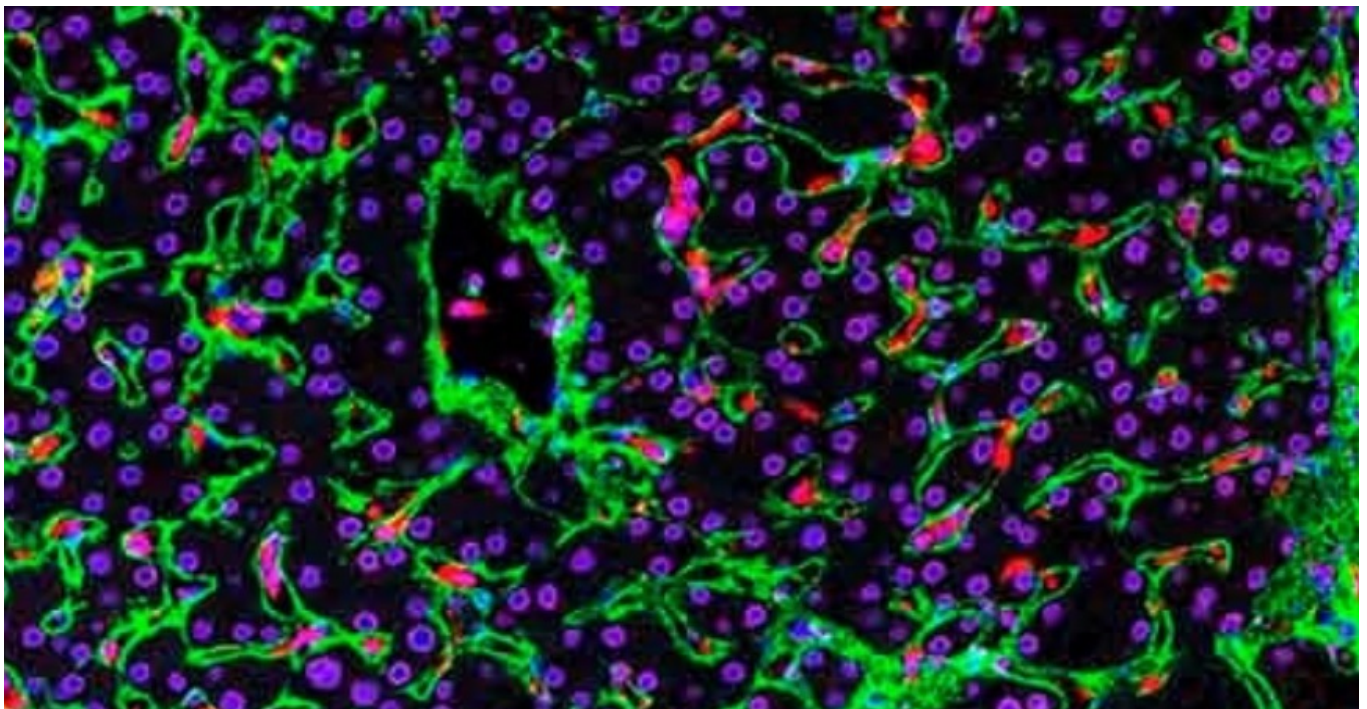
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INTERNAL MEDICINE / PHARMACOLOGY

General pharm – Drug antidotes

by MEHLMANMEDICAL



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HY points followed by a quiz at the end

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Acetaminophen

N-acetylcysteine

Acetylcholinesterase inhibitors (e.g., neostigmine)	Atropine
Anthracyclines (i.e., doxorubicin, hydroxydaunorubicin)	Dexrazoxane
Atropine	Physostigmine
Arsenic	Dimercaprol
Benzodiazepines (e.g., diazepam)	Flumazenil
β -blockers (e.g., propranolol)	Glucagon
Carbon monoxide	Hyperbaric oxygen
Cisplatin	Chloride diuresis (normal saline); amifostine
Copper	Penicillamine
Cyanide	Amyl nitrite; hydroxocobalamin; thiosulfate
Cyclophosphamide	Mesna; <i>N</i> -acetylcysteine
Digoxin	Normalize K^+ ; Anti-Digoxin antibody (immunoglobulin Fab fragment); Mg^{2+}
Ethylene glycol	Fomepizole
Heparin	Protamine sulfate
Iron	Deferoxamine, deferasirox
Lead	Dimercaprol; EDTA; succimer
Mercury	Dimercaprol; EDTA; succimer
Methanol	Fomepizole > ethanol
Methemoglobinemia	Methylene blue; vitamin C
Methotrexate	Folinic acid (leucovorin rescue)
Organophosphates	Atropine first, <i>then</i> pralidoxime
Opioids / heroin	Naloxone
Salicylates (i.e., aspirin)	Sodium bicarbonate
TCA antidepressants	Sodium bicarbonate
Warfarin	Vitamin K; fresh frozen plasma
5-fluorouracil (5-FU)	Thymidine; uridine

Some concise, HY points regarding the above:

Acetaminophen

- Causes hepatic necrosis due to NAPQI production (a highly toxic metabolite). Hepatic glutathione neutralizes NAPQI.
- USMLE wants you to specifically know the reason *N*-acetylcysteine works is because **it regenerates reduced glutathione in the liver.**
- Species with thiol (-SH) groups are in the reduced form and can act as reducing agents (i.e., can act as anti-oxidants).
- Species with disulfide bonds (-S-S-) are in the oxidized form.
- Hepatic glutathione has a thiol (-SH) group. So does *N*-acetylcysteine. After glutathione neutralizes a toxin, it becomes oxidized and inactive.
- The thiol group on *N*-acetylcysteine then binds to and frees up the sulfurs on glutathione so that they can once again neutralize toxins.
- This mechanism may sound cumbersome, but it's HY for USMLE. Probably because *N*-acetylcysteine is such an important antidote used clinically.
- Activated charcoal can be used in select acute circumstances. There's a Qbank Q floating around where activated charcoal is the answer over *N*-acetylcysteine. **On the USMLE, however, always choose *N*-acetylcysteine for acetaminophen.**

Acetylcholinesterase inhibitors (e.g., neostigmine)

- Acetylcholinesterase breaks down acetylcholine. Therefore AChE inhibitors increase the amount of acetylcholine in the synaptic cleft able to bind to post-synaptic ACh receptors.
- Atropine, a muscarinic receptor antagonist, can therefore act as an antidote by competitively inhibiting ACh receptors.

Anthracyclines (i.e., doxorubicin, hydroxydaunorubicin)

- Doxorubicin (Adriamycin) used in the treatment of Hodgkin lymphoma; hydroxydaunorubicin used in non-Hodgkin lymphoma).
- Cause free radical formation that can lead to dilated cardiomyopathy.
- Dexrazoxane chelates free radicals.

Atropine

- Would cause "opposite of DUMBBELSS" effects.
- Blocks muscarinic receptors directly.
- AChE inhibitors are used as antidotes because they increase ACh available so that it can outcompete atropine for its own receptor.

Benzodiazepines (e.g., diazepam)

- Cause respiratory depression, leading to **normal** A-a gradient respiratory acidosis (i.e., arterial oxygen is low *because* the alveolar oxygen is low; this makes sense, as the patient is hypoventilating).
- Flumazenil is a benzodiazepine receptor antagonist.
- Some resources say it should not be used in those with benzo addiction as it can precipitate seizure, but a 2CK NBME Q has flumazenil as the answer used in the emergency setting in a patient with Hx of benzo prescription.
- In other words, on the USMLE, don't over-think this question: just choose flumazenil if they ask for the pharmacologic Tx for the respiratory depression in someone taking benzos.

β-blockers (e.g., propranolol)

- An easy way to remember that glucagon is the antidote (if that seems weird) is because β-blockers ↓ cAMP by inhibiting G-α-s G-proteins, whereas glucagon ↑ cAMP.

- “Michael, that G-protein point actually makes it more confusing.” → “Well that G-protein point is sometimes asked for Step 1 as well.”

Carbon monoxide

- Has a much greater affinity for hemoglobin than oxygen, so the only way to attempt to dislodge it from the hemoglobin is to give 100% hyperbaric oxygen → maximally saturates any binding sites on hemoglobin not already occupied by CO.

Cisplatin

- Chemotherapeutic agent, often for testicular cancer.
- Can cause “toxic neuropathy” (i.e., chemo-induced neurotoxicity); this is on the Neuro shelf forms for 2CK.
- Chloride diuresis (normal saline) is first-line as the Tx for cisplatin toxicity. Amifostine is used second.

Copper

- Penicillamine used as the copper chelator in Wilson disease.
- Penicillamine can cause drug-induced lupus (anti-histone antibodies).

Cyanide

- One of the Step 1 NBMEs has amyl nitrite as the answer for cyanide toxicity.
- Amyl nitrite and thiosulfate can be used together to free the cyanide that’s bound to hemoglobin → causes chromaturia (pink/magenta urine).
- Hydroxocobalamin + cyanide → cyanocobalamin (vitamin B12).

Cyclophosphamide

- Guanine-N7 alkylating agent that is highly toxic. Used classically in non-Hodgkin lymphoma as part of CHOP regimen (Cyclophosphamide, Hydroxydaunorubicin, Oncovin [vincristine], Prednisone).
- Causes hemorrhagic cystitis (hematuria → red urine) due to bladder toxic metabolite called acrolein.
- Mesna, like glutathione and *N*-acetylcysteine, has a thiol group. Rarely, if Mesna is not listed as an answer for cyclophosphamide antidote, *N*-acetylcysteine will be correct.

Digoxin

- Hypokalemia can cause digoxin toxicity (both compete for potassium binding site on Na/K-ATPase pump at the heart, so ↓ K means ↑ digoxin binding). If the patient has a low potassium and has yellow/wavy “Vincent van Gogh” vision, normalizing the potassium is the first answer.
- USMLE will ask which of the following agents’ toxicity is treated with an immunoglobulin, and the answer is digoxin (anti-Digoxin immunoglobulin Fab fragment).
- Quinidine is a type Ia anti-arrhythmic (sodium channel blocker) that can displace digoxin from bilirubin and cause digoxin toxicity (i.e., yellow/wavy “Vincent van Gogh” vision in someone also taking an anti-arrhythmic).
- Magnesium is given after potassium is normalized and anti-Digoxin Fab fragment is administered.

Ethylene glycol (anti-freeze)

- Can cause calcium oxalate urolithiasis and high-anion gap metabolic acidosis (the E in MUDPILES → Methanol, Uremia, DKA, Phenformin, Iron/INH, Lactic acidosis, Ethylene glycol, Salicylates).
- Fomepizole inhibits alcohol dehydrogenase.

Heparin

- Protamine (sulfate) is positively charged. It binds to the large, acidic, negatively-charged heparin molecule.

Iron

- Deferoxamine and deferasirox are used as iron chelators when the patient has **secondary** hemochromatosis (i.e., transfusional siderosis from chronic blood transfusions).
- It is the **wrong** answer for hereditary hemochromatosis (the AR familial disorder). For the latter, the answer is serial phlebotomy. Students fuck this up all the time.
- 34M + hereditary hemochromatosis. Tx? → answer = serial phlebotomy.
- 12M + beta-thalassemia major + transfusional siderosis. Tx? → deferoxamine or deferasirox.

Lead

- If the Q gives you a kid eating paint chips after moving into a new house with his family, that's frankly too easy.
- Lead can cause **microcytic anemia**. → 44M + hunter + cognitive decline over many months + microcytic anemia.
- Weird detail, but chelation is not correct unless serum lead levels are >44 ug/dL.
- Succimer is classically used for kids, whereas dimercaprol and EDTA are for adults, but the USMLE won't make this distinction. The correct answer will be the only lead chelator listed.

Mercury

- Toxicity seen in hatters (mercury used sometimes to create a sheen on furs).
- Heavy metal like lead, so the same chelators are correct.

Methanol

- Toxicity can present as stomach pain and blindness.
- USMLE wants you to know methanol can be in **paint thinner**.
- Fomepizole is used for both methanol and ethylene glycol toxicity.
- Ethanol has greater affinity for alcohol dehydrogenase, so Jack Daniels can technically be used as a Tx if in a rural area where fomepizole not available, but if the USMLE lists both, choose fomepizole only (i.e., not even the combo of the two; literally just "fomepizole alone").

Methemoglobinemia

- Oxygen can only bind to Fe^{2+} oxygen, not Fe^{3+} (methemoglobin).
- Methemoglobinemia will present frequently as "brown blood," or "chocolate blood" on the USMLE.
- Congenital methemoglobinemia is due to deficiency of cytochrome B5 reductase (sounds absurd, but it's on the USMLE).
- Questions on the antidote are easy → will give you a patient with chocolate blood and then the answer is simply "IV methylene blue."

Methotrexate

- Can cause pulmonary fibrosis, hepatotoxicity, and neutropenia (mouth ulcers, sore throat, fever).
- Inhibits dihydrofolate reductase.
- Folinic acid (**not** folic acid) can be used for toxicity. It is also known as "leucovorin rescue."

Organophosphates

- Organophosphates act as acetylcholinesterase inhibitors → cause **DUMBELSS** in patient working on fruit farm

or who “drank fluid” attempting to commit suicide.

- Atropine blocks muscarinic receptors directly. This is the answer *before* pralidoxime. If the USMLE Q lists both answers, **choose atropine first**.
- Pralidoxime regenerates active acetylcholinesterase in patients with organophosphate poisoning.
- Pralidoxime will kick out the phosphate group from the organophosphate that's bound to and inhibiting the acetylcholinesterase → regenerates active acetylcholinesterase.
- 20F + working on fruit farm + has constricted pupils + bradycardia + drooling; what's the Tx for this patient's condition? → answer = atropine first, *then* pralidoxime.

Opioids / heroin

- Cause pinpoint pupils and respiratory depression, leading to **normal** A-a gradient respiratory acidosis (i.e., arterial oxygen is low *because* the alveolar oxygen is low; this makes sense, as the patient is hypoventilating).
- Naloxone antagonizes opioid receptors.

Salicylates (aspirin) vs TCAs

- Salicylates (aspirin) are acidic, meaning they have hydroxyl (-OH) groups converted to oxylate ($-O^-$) groups at alkaline pH.
- The kidney does not reabsorb charged species as well as neutral ones, so sodium bicarbonate can increase salicylate excretion by forcing it into the oxalate form.
- USMLE Q will ask why bicarb is Tx for aspirin toxicity; **answer = “increased excretion through urinary alkalinization.”**
- However this is not the same mechanism for TCAs, despite sodium bicarb also being the antidote.
- TCAs are basic, meaning they have $-NH_3^+$ that are converted to $-NH_2$ at alkaline pH, so the kidney actually *excretes less* when sodium bicarb is given. So why the fuck do we give it then?
- Because TCAs are cardiotoxic and can cause arrhythmia by binding to myocardial sodium channels. So sodium bicarb causes 1) the sodium to outcompete the TCA for its own channel, and 2) the bicarb causes dissociation of the drug from the myocardial sodium channel by making it less charged.
- USMLE Q will ask why bicarb is the Tx for TCA toxicity; **answer = “increased dissociation from myocardial sodium channels.”**
- “Increased excretion through urinary alkalinization” is the **wrong** answer for TCAs.
- I group salicylates (aspirin) and TCAs together here because this distinction is so HY.

Warfarin

- Vitamin K is the answer if the patient is *not* bleeding or is *not* going to have surgery right away. That is, vitamin K causes a *slow* warfarin reversal.
- If the patient is actively bleeding or needs to be rushed to surgery, fresh frozen plasma is the answer.

5-fluorouracil (5-FU)

- Inhibits thymidylate synthase, which catalyzes a methyl-group transfer from dUMP to dTMP.
- Tetrahydrofolate (synthesis inhibited by methotrexate) is required as a cofactor for the thymidylate synthase reaction.
- Folinic acid (leucovorin) is not effective to overcome 5-FU toxicity because 5-FU inhibits the enzyme necessary for the methyl transfer, so even if THF is present in sufficient amounts as a cofactor, the enzyme is still inhibited.
- Uridine can be given as a substrate to help overcome the enzyme deficiency (i.e., just drown out the enzyme with sufficient substrate), or thymidine can be given as the product, where the enzyme is bypassed altogether.

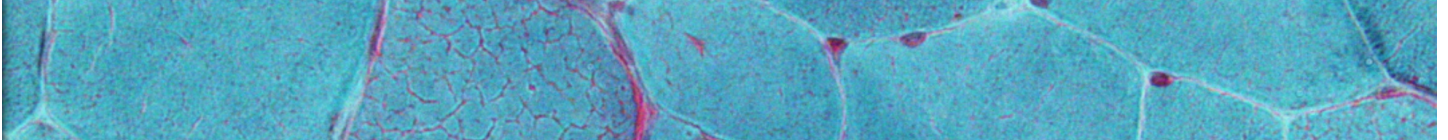


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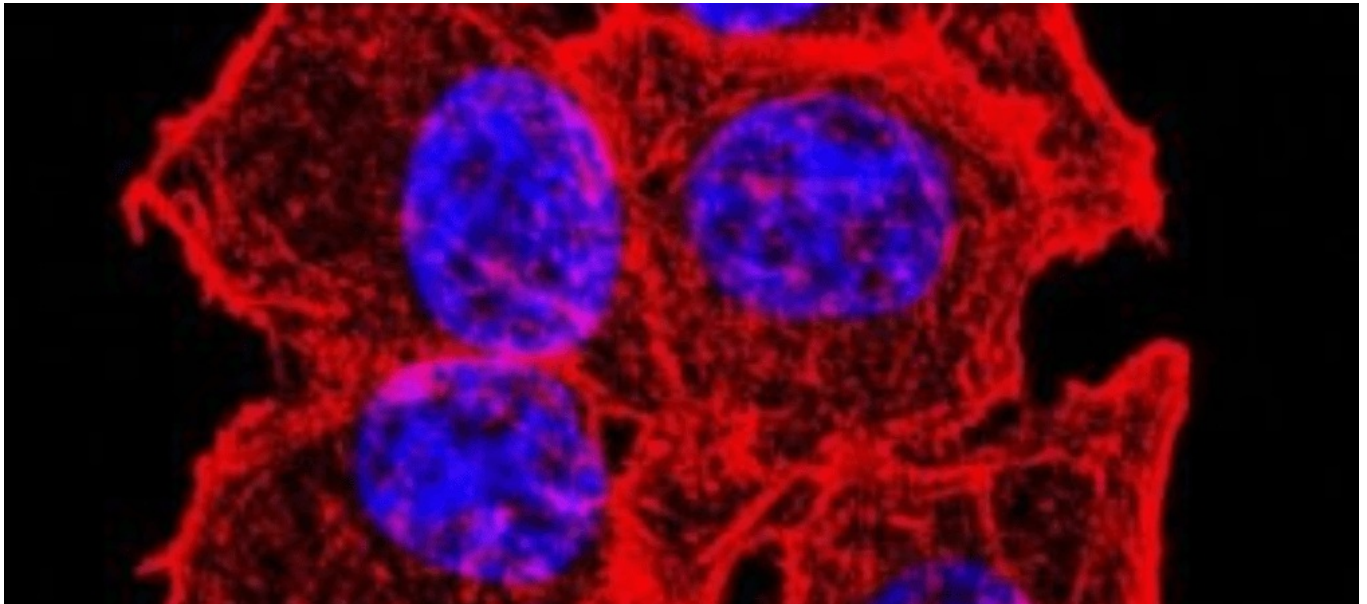
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PHARMACOLOGY

General pharm – Drug metabolism/elimination

by MEHLMANMEDICAL



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[HY points about each drug followed by a quiz at the end](#)

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Drug metabolism

- Phase I & Phase II types of metabolism
- **1 Old HOR, 2 MAGS** ("One old whore, two mags.") ← Weird-as-fuck mnemonic I created, but it literally does help.

Phase I

- Cytochrome P-450-dependent.
- Geriatrics (**Old** people) lose phase I ability.
- **H**ydroxylation, **O**xidation, **R**eduction reactions.

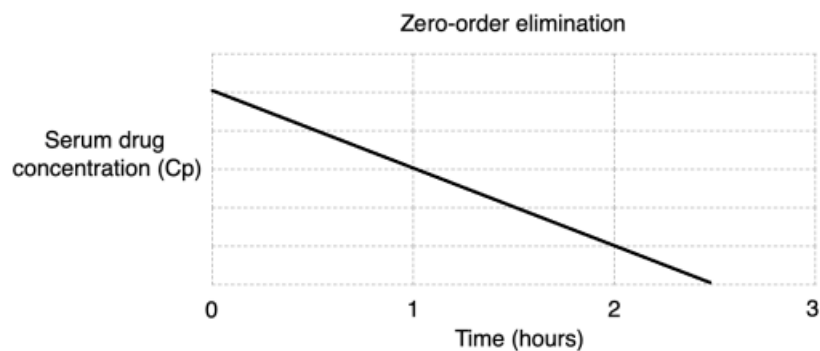
Phase II

- Not P-450-dependent
- **M**ethylation, **A**cetylation, **G**lucuronidation, **S**ulfation
- People who are slow-acetylators (slower Phase II acetylation rate compared to most individuals) can have ↑ side-effects from drugs such as **isoniazid** because of ↓ rate of metabolism.

Drug elimination

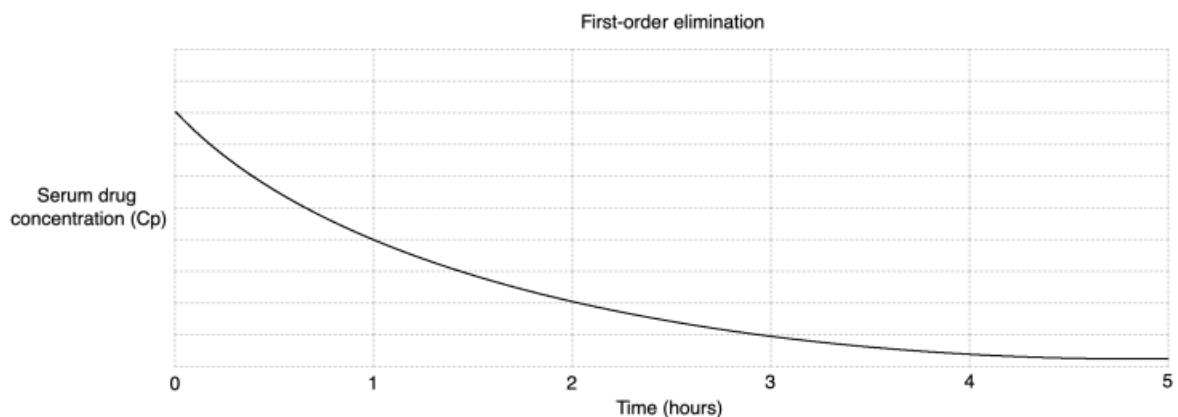
Zero-order elimination

- Same **amount** of drug eliminated per unit time, regardless of dose (e.g., 2mg/hr eliminated always).
- Graph of serum drug concentration (Cp) vs time is a straight line.
- Capacity-limited elimination (i.e., the liver/kidney are saturated/maxed out on how much they can eliminate).
- **PEA** → **P**henytoin, **E**thanol, **A**spirin.



First-order elimination

- Same **fraction** of drug eliminated per unit time (e.g., 20% of drug eliminated per hour always).
- Graph of serum drug concentration (Cp) vs time is a curved line.
- Flow-limited elimination (i.e., the liver/kidney aren't saturated/maxed out on how much they can eliminate; the drug is the rate-limiting factor).



Drug elimination in relation to urine pH

- Key concept is: ionic species (i.e., $\text{-NH}_3^+/\text{-COO}^-$) are poorly reabsorbed; neutral species (i.e., $\text{-NH}_2/\text{-COOH}$) are readily reabsorbed.
- If a species is a weak acid (i.e., has a -COOH group, where $\text{-COOH} \leftrightarrow \text{-COO}^-$):
 - ↓pH favors -COOH
 - ↑pH favors -COO^-
 - **Salicylates (aspirin) are HY example**, where Tx for overdose is sodium bicarbonate → increases pH of urine → traps aspirin as -COO^- form in the urine → decreased reabsorption. USMLE will ask why sodium bicarb is the Tx for aspirin toxicity; answer = **“increased excretion through urinary alkalinization.”**
- If a species is a weak base (i.e., has an -NH_2 group, where $\text{-NH}_3^+ \leftrightarrow \text{-NH}_2$):
 - ↓pH favors -NH_3^+
 - ↑pH favors -NH_2
 - **Tricyclic antidepressants (TCAs) and barbiturates are HY examples.**
 - Tx for TCA toxicity is sodium bicarbonate, **which paradoxically decreases urinary excretion.** Student says, “Wait, well then why the fuck does that make any sense then?” → It’s because the sodium bicarb functions to induce dissociation of the TCA from myocardial sodium channels to decrease risk of arrhythmia; the purpose isn’t to increase urinary excretion. In other words:

42M + consumes bottle of aspirin in suicide attempt; sodium bicarb is administered; what’s the mechanism via which this helps? → answer = increased excretion through urinary alkalinization.

In contrast,

42M + consumes his TCA medication in suicide attempt; sodium bicarb is administered; what’s the mechanism via which this helps? → answer = “increased dissociation from myocardial sodium channels.” The wrong answer is “increased excretion through urinary alkalinization,” which refers instead to salicylates.

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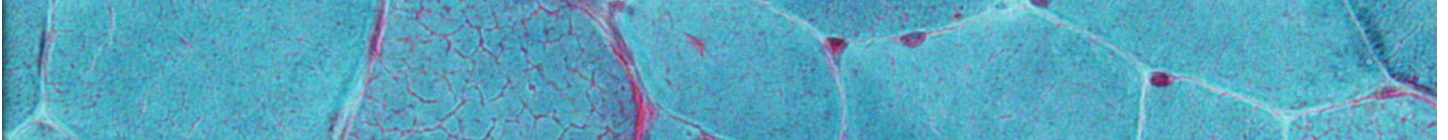


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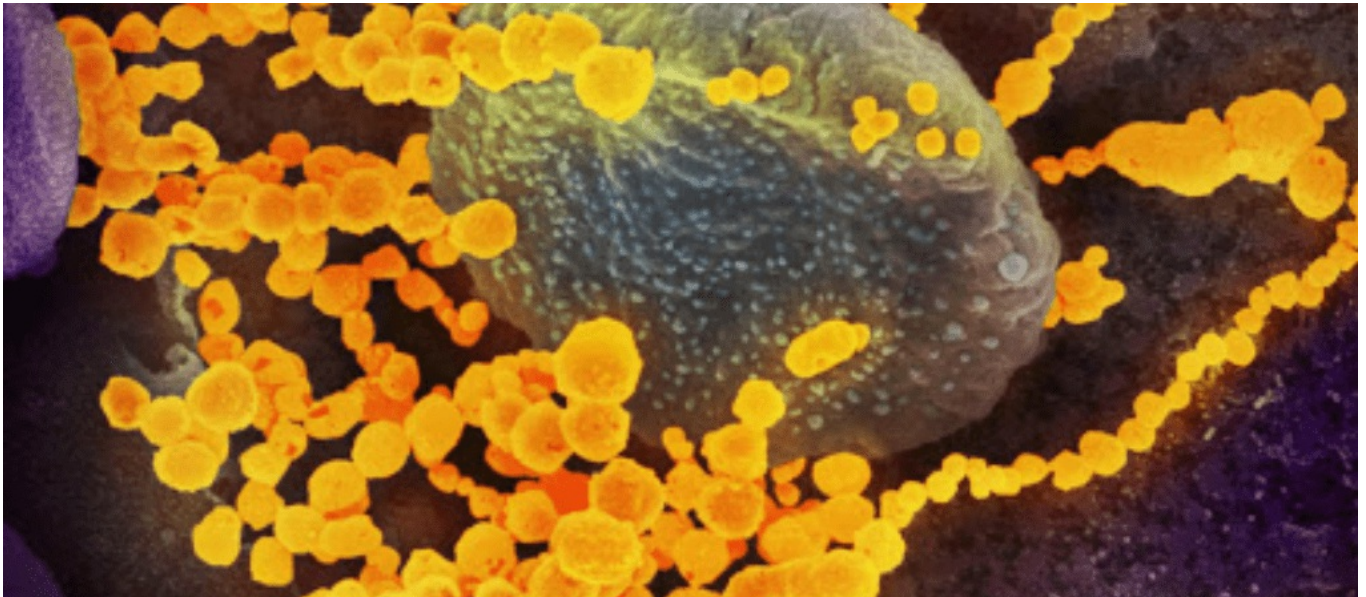
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PHARMACOLOGY

General pharm – Enzyme kinetics

by MEHLMANMEDICAL



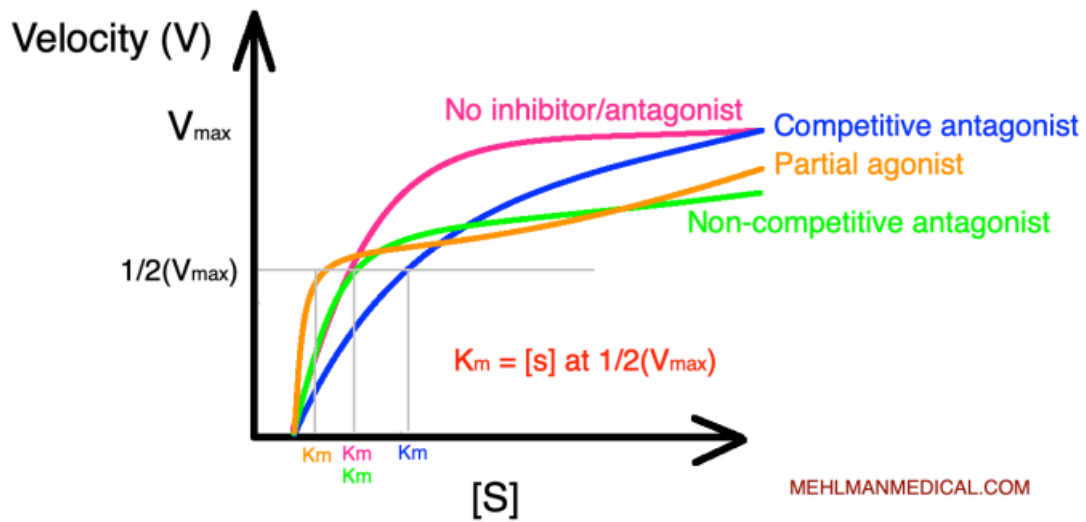
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[HY discussion of concepts followed by a quiz at the end](#)

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V_{max}

- Maximal rate of reaction for a given substrate-enzyme interaction.
- Reflects enzyme **efficacy** (i.e., maximum rate of reaction, or “how high the mountain goes”).
- **That is:** ↑ V_{max} = ↑ **efficacy**; ↓ V_{max} = ↓ **efficacy**.

K_m

- The substrate concentration at 1/2 V_{max}.
- Reflects enzyme affinity (i.e., the stronger the enzyme binds to the substrate, the less substrate is needed to attain half V_{max}).
- **That is:** ↑ K_m = ↓ **affinity**; ↓ K_m = ↑ **affinity**.

Competitive inhibitor/antagonist

- Binds to same site on the enzyme as the substrate itself (i.e., the enzymatic site).
- Reversible (almost always).
- Shifts curve to the right.
- **Increases K_m** → more substrate is needed to achieve half of max reaction rate because competitive inhibitor is occupying some enzyme binding sites.
- **V_{max} unchanged** → as long as enough substrate is present, max reaction rate can occur.
- **HY USMLE examples:**
 - The alpha-1 antagonists **phentolamine**, terazosin, tamsulosin, prazosin.
 - Methotrexate (competitive, reversible inhibitor of dihydrofolate reductase).
 - Statins (competitive, reversible inhibitors of HMG-CoA reductase).

Non-competitive inhibitor/antagonist

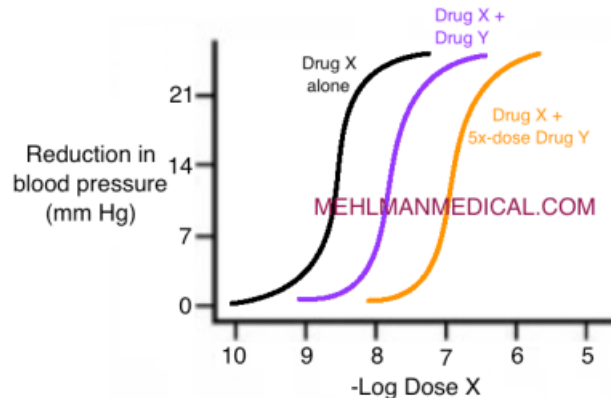
- Binds to a site on the enzyme distinct from the substrate (i.e., an allosteric site).
- Irreversible (almost always).
- Shifts curve down.
- **Decreases V_{max}** → because it binds to a site on the enzyme distinct from the substrate, no matter how much substrate is added into the system, maximal reaction rate cannot occur.
- **No change in K_m** → because it doesn't bind to the enzymatic site (i.e., it's not competing with the substrate), the amount of substrate needed to achieve half of the maximal reaction rate (i.e., K_m) won't be affected.
- **HY USMLE example:**
 - The alpha-1 antagonist **phenoxybenzamine** (used for pheochromocytoma).

Partial agonist

- Binds to the same site as the substrate (therefore competitive).
- **Decreases V_{max}** → produces a positive response, but simply does not carry the same efficacy as the substrate itself. Even if more substrate is added to the system to outcompete the partial agonist, because the latter is capable of producing a partial response, the net response is still attenuated. In the case of a competitive antagonist, it produces no response when bound, but once it dislodges from the enzymatic site, the original substrate can reoccupy that site and induce the max response. This is why V_{max} is normal with competitive antagonists but decreased with partial agonists.
- **Can shift the curve left and down.**
- **K_m can be decreased**, normal, or increased → in other words, the partial agonist, despite reducing the maximal response (V_{max}), **actually can have increased affinity** for the enzyme compared to the original substrate. The USMLE, if they give you a partial agonist question, will give you **decreased K_m** (i.e., left-shifted curve) in order to emphasize that the agent is indeed a partial agonist.
- **The net result of giving a partial agonist is therefore similar to giving an antagonist.**
- **HY USMLE examples:**
 - Clomiphene citrate (estrogen receptor partial agonist at hypothalamus → hypothalamus thinks there's decreased estrogen because the binding isn't as strong as estrogen itself, albeit positive, so GnRH increases).
 - Danazol (androgen-receptor partial agonist at the liver → leads to increased C1-esterase inhibitor production in the treatment of hereditary angioedema; can cause hirsutism).

Now let's look at two examples:

—



Regarding the above findings, which of the following is true?

- A) Drug Y is a non-competitive antagonist of Drug X
- B) Drug X is a non-competitive antagonist of Drug Y
- C) Drug Y is a competitive antagonist of Drug X
- D) Drug X is a competitive antagonist of Drug Y
- E) Drug Y is a partial agonist of Drug X
- F) Drug X is a partial agonist of Drug Y

(The answer is down below in blue)

The main point to observe is that the presence of Drug Y does not decrease the ability of Drug X to achieve its maximum potential effect, even if more Drug X is required to achieve just such.

This means Drug X and Drug Y **are competing** for the same receptor(s) – i.e., as long as you keep increasing the dose of

Drug X, you can eventually overcome and drown out the impact of Drug Y.

Important examples of competitive inhibitors for the USMLE are **methotrexate** (on dihydrofolate reductase)¹, **statins** (on HMG-CoA reductase)², and **almost all α1-blockers** (i.e., tamsulosin³, terazosin⁴, phentolamine⁵, etc.).

If Drug Y were to cause a ↓ in the ability of Drug X to achieve its maximum effect (i.e., a downward shift of the curve), even as the dose of Drug X is continually increased, Drug Y would therefore **not be competing** for the same receptor as Drug X. This could be due to a negative allosteric effect on Drug X, or it could mean Drug Y binds to Drug X’s receptor at a distinct location and prevents Drug X from binding, while simultaneously not inhibiting or potentiating the receptor itself. In other words, with non-competitive inhibition, no matter how much the dose of Drug X is increased, the impact of Drug Y can never be overcome.

The super-HY non-competitive antagonist you need to know for the USMLE is **phenoxybenzamine** at α1 receptors.⁶

The answer is C.

Bottom line: Competitive antagonism = right-shifted curve. Non-competitive antagonism = down-shifted curve. Methotrexate, statins, and almost all α1-blockers are competitive inhibitors. Phenoxybenzamine is non-competitive.

- 1) <https://www.ncbi.nlm.nih.gov/pubmed/3122764>
- 2) <https://www.ncbi.nlm.nih.gov/pubmed/27313057>
- 3) <https://www.ncbi.nlm.nih.gov/pubmed/9117115>
- 4) <https://www.ncbi.nlm.nih.gov/pubmed/2457715>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1282628/>
- 6) <https://www.sciencedirect.com/science/article/pii/B9780323481106000144>

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L-asparaginase is an integral component of treatment of acute lymphoblastic leukemia (ALL) in children. It hydrolyzes L-asparagine to L-aspartate in leukemic cells. A researcher who is investigating this enzyme isolates an actinomycetes strain, *Streptomyces* sp. NEAE-82, which produces L-asparaginase. He then carries out kinetic studies of the purified enzyme and presents his findings in a report:

Researcher’s report	Km (M)	Vmax (Uml ⁻¹ min ⁻¹)
Estimations prior to investigation:	0.02000	95.00
Findings as per kinetic studies:	0.01007	95.08

Which of the following most accurately describes the researcher’s findings of L-asparaginase in comparison to his prior estimations?

- A) ↓ enzyme affinity for substrate; ↑ efficacy
- B) ↓ enzyme affinity for substrate; ↓ efficacy
- C) ↑ enzyme affinity for substrate; ↑ efficacy

D) $\uparrow$ enzyme affinity for substrate; $\downarrow$ efficacy

(The answer is down below in blue)

Once again:

V_{max}

- Reaction velocity (rate of reaction) when the enzyme is fully saturated by substrate, indicating that all the binding sites are being constantly reoccupied.⁷
- Reflective of enzyme efficacy (i.e., maximum rate of reaction).⁸
- **In other words, $\uparrow V_{max} = \uparrow$ efficacy; $\downarrow V_{max} = \downarrow$ efficacy**

K_m

- The substrate concentration at $1/2 V_{max}$ ⁹
- Inversely reflective of affinity (i.e., if the enzyme binds more strongly to the substrate, then you need less substrate to achieve $1/2 V_{max}$).⁹
- **In other words, $\uparrow K_m = \downarrow$ affinity; $\downarrow K_m = \uparrow$ affinity**

The true finding for K_m was less than the researcher's prior estimate, meaning the enzyme has greater affinity for substrate than predicted.

The true finding for V_{max} was greater than the researcher's prior estimate, meaning the enzyme achieves greater efficacy than predicted.

(By the way, the researcher's true findings are actually the real K_m and V_{max} for L-asparaginase.¹⁰)

The answer is C – $\uparrow$ enzyme affinity for substrate; $\uparrow$ efficacy.

Bottom line: K_m = substrate concentration at $1/2 V_{max}$. $\uparrow K_m = \downarrow$ affinity; $\downarrow K_m = \uparrow$ affinity. $\uparrow V_{max} = \uparrow$ efficacy; $\downarrow V_{max} = \downarrow$ efficacy.

7) <https://www.sciencedirect.com/science/article/pii/B9780123971760000017>

8) <https://books.google.co.jp/books?id=9-xeDwAAQBAJ&pg>

9) <https://books.google.co.jp/books?id=MgqI-pfk45QC&pg>

10) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5015098/>

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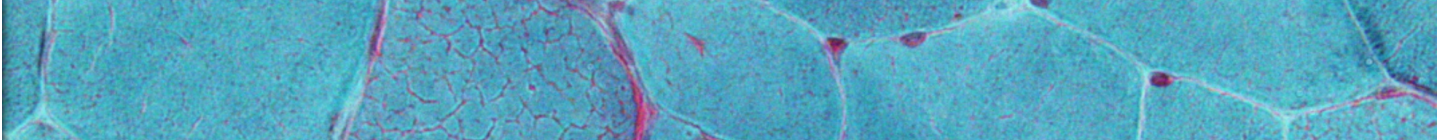


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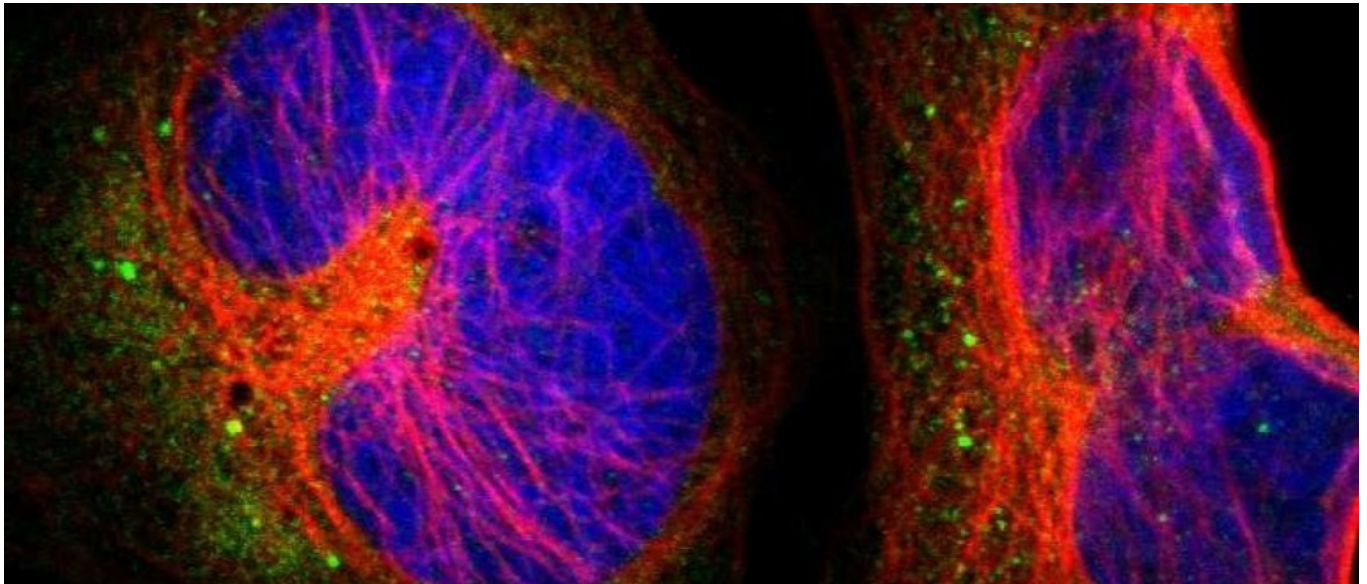
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PHARMACOLOGY

General pharm – Pharmacokinetics

by MEHLMANMEDICAL



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Bioavailability (F)

- The amount of an administered drug that reaches the circulation unaltered.
- 100% for IV administration.
- Oral bioavailability always <100% (due to gut excretion and first-pass metabolism).
- First-pass metabolism → The drug will be absorbed by gut lymphatics, where it travels to the liver and is metabolized before reaching the circulation.

Loading dose (LD)

A 46-year-old patient is receiving an unknown drug intravenously. It is known that the drug is distributed into total body water. What dose of the drug is needed to obtain an initial plasma level of 4 mg/L in a patient weighing 70 kg?

- a) 42 mg
- b) 168 mg
- c) 200 mg
- d) 210 mg
- e) 280 mg

(The answer is below in blue)

In order to answer this question, we need to determine the loading dose of the drug.

The loading dose is the initial higher dose of the drug given to achieve a certain plasma concentration earlier before maintaining it with a lower, maintenance dose.

It is calculated from the following equation: $LD = (C_p \times V_d) / F$, where C_p is the plasma level, V_d is the volume of distribution, and F is the bioavailability of the drug.

The loading dose is proportional to the volume of distribution because the larger the volume of distribution the more drug needs to be given to achieve the same plasma concentration.

The bioavailability of the drug is the opposite. More drug needs to be given if the bioavailability is lower (i.e., drugs with low bioavailability less readily enter systemic circulation so more drug needs to be given).

For drugs administered intravenously, $F = 1$. Since the drug is distributed in total body water, V_d is about 60% of total body weight, or 42 kg (0.6×70 kg). 42 kg of water is 42 L of water.

Therefore, $V_d = 42$ L.

$LD = 4 \text{ mg/L} \times 42 \text{ L} / 1 = 168 \text{ mg}$.

The correct answer is b.

Plasma drug concentration

A 20-year-old male hospitalized for a severe infection is receiving an IV antibiotic that distributes in total body water. He was initially given 1,800 mg of the drug. His body weight is 75 kg. Which of the following represents the plasma concentration of the antibiotic in this patient:

- a) 14 mg/L
- b) 24 mg/L
- c) 40 mg/L
- d) 45 mg/L
- e) 75 mg/L

(The answer is below in blue)

The answer to this question can be calculated from the equation for the loading dose of the drug.

The loading dose is the initial higher dose of the drug given to achieve a certain plasma concentration earlier before reducing the dosage to a maintenance dose.

The loading dose is calculated as: $LD = (C_p \times V_d) / F$, where C_p is the plasma level, V_d is the volume of distribution, and F is the bioavailability of the drug.

The greater the volume of distribution of the drug, the more drug needs to be given to achieve the same plasma concentration, whereas more drug needs to be given if the bioavailability is lower.

For drugs administered intravenously, $F = 1$ as all the drug enters systemic circulation.

Since the drug is distributed in total body water, V_d is about 60% of total body weight, or $0.6 \times 75 \text{ kg} = 45 \text{ kg}$. 45 kg of water is 45 L of water. Loading dose is 1,800 mg.

$$1,800 \text{ mg} = C_p \times 45 \text{ L} / 1$$

$$C_p \times 45 \text{ L} = 1,800 \text{ mg}$$

$$C_p = 1,800 \text{ mg} / 45 \text{ L} = 40 \text{ mg/L}.$$

The correct answer is c.

Maintenance dose

A drug is administered to a 75-year-old patient whose renal function has decreased by 50%. Which of the following best describes the change that would be expected in this patient:

- a) $t_{1/2}$ would be shorter
- b) V_d would decrease
- c) Loading dose would increase
- d) Clearance of the drug would increase
- e) Maintenance dose would be lower

(The answer is below in blue)

The patient in question has decreased renal function, meaning his renal clearance is lower.

The maintenance dose of the drug is the dose of drug given to the patient to maintain a certain plasma concentration.

The equation for the maintenance dose is: $MD = Cl \times C_p \times t_d / F$, where Cl is the clearance, C_p is the steady state concentration, t_d is the dosing interval and F is the bioavailability.

It can be concluded from this equation that the maintenance dose would be lower if the renal clearance was lower.

It also makes sense intuitively as a decrease in renal clearance of the drug indicates that more drug remains in the body for extended periods of time so lower maintenance doses would be needed to maintain a certain plasma concentration. V_d would not change. $t_{1/2}$ would increase due to decreased clearance.

Note that the loading dose equation, in comparison, does not take into account the renal clearance.

The correct answer is e.

Steady state concentration

A 50-year-old patient is started on an IV infusion of a drug at the rate of 300 mg/h. The clearance of the drug is 50 L/h. Which of the following represents the steady state concentration of the drug in question:

- a) 250 mg/L
- b) 75.6 mg/L
- c) 60 mg/L
- d) 6 mg/L
- e) 3.6 mg/L

(The answer is below in blue)

The maintenance dose of the drug is the dose of drug given to the patient to maintain a certain plasma concentration.

It can be calculated as: $MD = Cl \times C_p \times t_d / F$, where Cl is the clearance, C_p is the steady state concentration, t_d is the dosing interval and F is the bioavailability.

Since the drug is given as an IV infusion, there is no need to consider the dosing interval or bioavailability so it can be simplified as:

$$MD = Cl \times C_p$$

$$C_p = MD / Cl = 300 \text{ mg/h} / 50 \text{ L/h} = 6 \text{ mg/L}$$

The correct answer is d.

Volume of distribution / Clearance

A clinical study is done to determine the characteristics of a novel drug that is distributed in total body water. Which of the following represents the total amount of the drug in a body of a 35-year-old participant whose body mass is 80 kg if the plasma concentration of the drug is 10 mg/L:

- a) 480 mL
- b) 480 mg
- c) 640 mL
- d) 640 mg
- e) 800 mg

(The answer is below in blue)

Once a drug enters the body, it distributes within body fluid compartments. Not all drugs distribute equally (i.e., some may remain in plasma only, others may distribute in interstitial fluid, while some others may distribute in total body water).

The volume of distribution is the ratio of the total amount of drug in the body and the concentration of the drug (remember, for any solution $c = \text{amount} / V$; where $V = \text{amount} / c$).

Therefore, $V_d = \text{amount of drug in the body} / C_p \rightarrow \text{amount of drug in the body} = V_d \times C_p$.

Since water represents approximately 60% of body weight, it can be calculated that this patient has 48 kg of water in his body ($0.6 \times 80 \text{ kg}$), or 48 L of water.

Therefore, V_d is 48 L.

$$C_p = 10 \text{ mg/L}$$

$$\text{Amount of drug in the body} = 48 \text{ L} \times 10 \text{ mg/L} = 480 \text{ mg}.$$

The correct answer is b.

A 25-year-old patient is being treated with a drug that distributes in the total body water. Currently, the amount of drug in his body is 600 mg and the plasma concentration of drug is 10 mg/L. Which of the following represents this patient's body weight:

- a) 60 kg
- b) 66 kg
- c) 80 kg
- d) 88 kg
- e) 100 kg

(The answer is below in blue)

Individual drugs distribute differently between the body's fluid compartments. The volume of distribution represents to what degree the drug distributes within the body and can be calculated as the ratio of the total amount of drug in the body and the concentration of the drug (remember, for any solution $c = \text{amount} / V \Rightarrow V = \text{amount} / c$).

Therefore, the volume of distribution of a drug can be calculated from the following formula:

$$V_d = \text{amount of drug in the body} / C_p$$

In this patient, $V_d = 600 \text{ mg} / 10 \text{ mg/L} = 60 \text{ L}$.

Since the drug is distributed in the total body water, the total volume of water in this patient is 60 L. The weight of 60 L of water is 60 kg. Since water represents approximately 60% of body weight, it can be calculated that his body weight is $60 \text{ kg} / 0.6 = 100 \text{ kg}$.

The correct answer is e.

A novel drug is being studied in a clinical trial. It is determined that the half-life of a drug is 4 h and that the V_d of a drug in one of the participants is 30 L. Which of the following represents clearance of the drug in this patient:

- a) 0.18 L/h
- b) 3.15 L/h
- c) 5.25 L/h
- d) 7.50 L/h
- e) 10.71 L/h

(The answer is below in blue)

First-order elimination of the drug means that a constant fraction of the drug is eliminated per unit of time. Therefore, we can determine the half-life of the drug, which is the time frame in which the concentration of the drug in the body halves. The equation used to calculate the half-life of the drug is:

$$t_{1/2} = 0.7 \times V_d / Cl$$

It makes sense that clearance is inversely proportional to the half-life (i.e., greater clearance means more drug is removed from the body, making the half-life shorter).

The clearance of the drug can be calculated from the equation for the half-life:

$$t_{1/2} = 0.7 \times V_d / Cl$$

$$Cl = 0.7 \times V_d / t_{1/2}$$

$$Cl = 0.7 \times 30 \text{ L} / 4 \text{ h} = 5.25 \text{ L/h}$$

The correct answer is c.

Half-life

A 40-year-old patient with a body mass of 70 kg is treated with the drug that is distributed in total body water. The drug has clearance of 10 L/h. Which of the following best represents the half-life of the drug in question:

- a) 1 h
- b) 2 h
- c) 3 h
- d) 4 h
- e) 5 h

(The answer is below in blue)

The half-life is calculated for the drugs that follow first-order kinetics, meaning that a constant fraction of the drug is eliminated per unit of time. It represents the time frame in which the concentration of the drug in the body halves.

The equation used to calculate the half-life of the drug is:

$$t_{1/2} = 0.7 \times V_d / Cl$$

Since the drug is distributed in total body water, V_d is the total volume of water in the body. Since water accounts for approximately 60% of body weight, the weight of water in this patient's body is approximately 42 kg ($0.6 \times 70 \text{ kg}$). 42 kg have the volume of 42 L. Cl is 10 L/h.

Therefore, $t_{1/2} = 0.7 \times 42 \text{ L} / 10 \text{ L/h} = 2.94 \text{ h}$, or roughly 3 hours.

The correct answer is c.

Rate of elimination

A 35-year-old patient is participating in a clinical trial involving a novel drug that distributes in total body water. The plasma concentration of the drug, its clearance and half-life are measured. Which of the following represents the correct way of calculating the rate of elimination of the drug from the body:

- a) clearance x plasma concentration
- b) clearance / plasma concentration
- c) clearance x volume of distribution / half-life
- d) clearance x half-life
- e) clearance x half-life / volume of distribution

(The answer is below in blue)

The rate of elimination can be calculated from the following equation:

$$Cl = \text{rate of elimination} / C_p$$

Rate of elimination = $Cl \times C_p$; where Cl is clearance and C_p is plasma concentration.

It makes sense that the rate of elimination of the drug is proportional to the clearance and the concentration of the drug. If there is more drug in the plasma and more plasma gets cleared, more drug will be eliminated.

The correct answer is a.

A 40-year-old patient is treated with a novel drug that distributes in total body water. The body mass of the patient is 80 kg. It is determined that the half-life of the drug is 5 hours. The plasma drug concentration is 10 mg/L. Which of the following represents the rate of elimination of the drug:

- a) 6.72 L/h
- b) 67.2 mg/h
- c) 48 L/h
- d) 160 mg/h
- e) 40 mg/h

(The answer is below in blue)

The rate of elimination can be calculated from the following equation:

$$Cl = \text{rate of elimination} / C_p$$

Rate of elimination = $Cl \times C_p$; where Cl is clearance and C_p is plasma concentration.

The rate of elimination of the drug is proportional to the clearance and the concentration of the drug (the more drug in the plasma and the higher the clearance of such plasma – the more drug gets eliminated).

First-order elimination of the drug means that a constant fraction of the drug is eliminated per unit of time, making it possible to determine the half-life of the drug, which is the time frame in which the concentration of the drug in the body reduces by 50%.

The equation used to calculate the half-life of the drug is:

$$t_{1/2} = 0.7 \times V_d / Cl, \text{ where } t_{1/2} \text{ is the half-life and } V_d \text{ volume of distribution.}$$

Clearance of the drug can be calculated from that equation:

$$t_{1/2} = 0.7 \times V_d / Cl$$

$$Cl = 0.7 \times V_d / t_{1/2}$$

The drug is distributed in total body water, which accounts for 60% of the body weight, or 48 kg (0.6 x 80 kg). 48 kg of water have the volume of 48 L. Therefore, $V_d = 48 \text{ L}$.

$$Cl = 0.7 \times 48 \text{ L} / 5 \text{ h} = 6.72 \text{ L/h}$$

$$\text{Rate of elimination} = 6.72 \text{ L/h} \times 10 \text{ mg/L} = 67.2 \text{ mg/h}$$

The correct answer is b.

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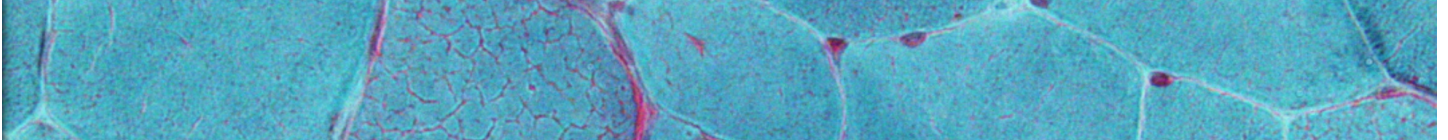


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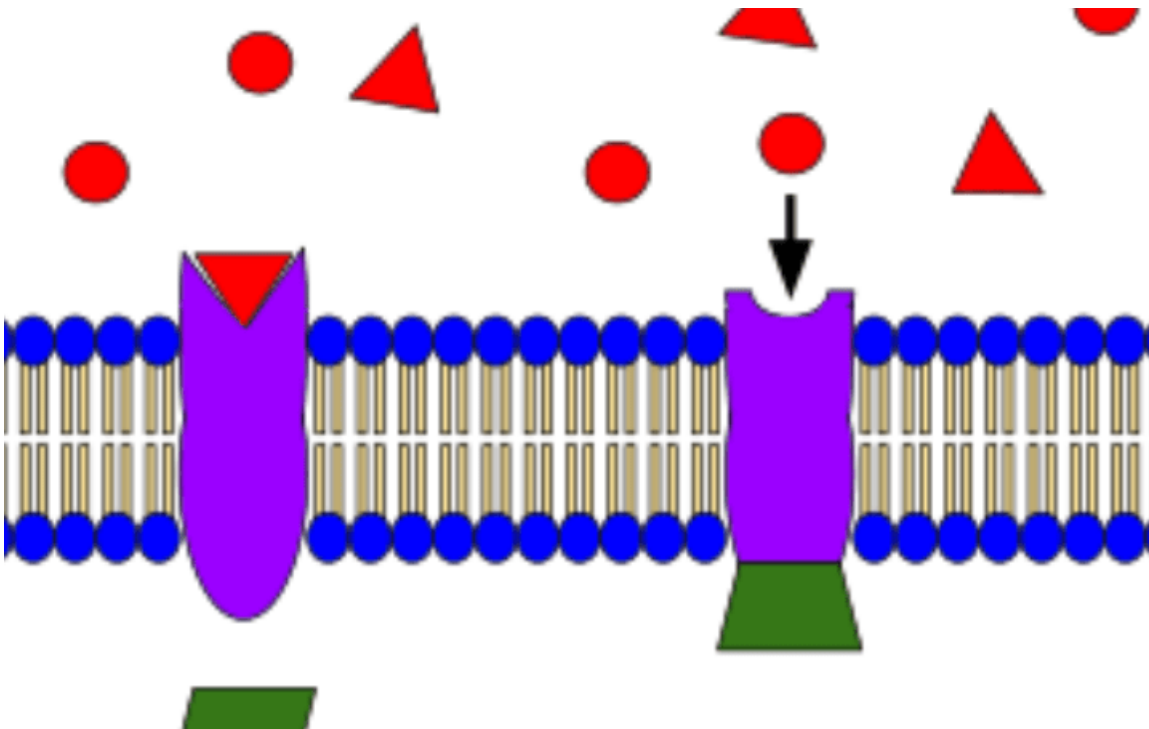
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PHARMACOLOGY

General pharm – Types of drug interactions

by MEHLMANMEDICAL



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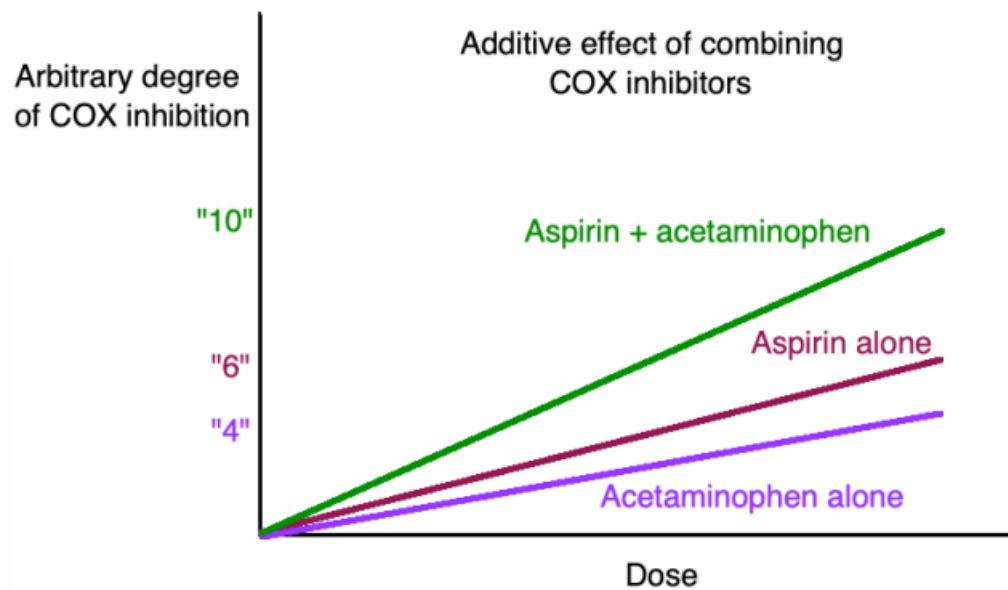
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HY points followed by a quiz at the end

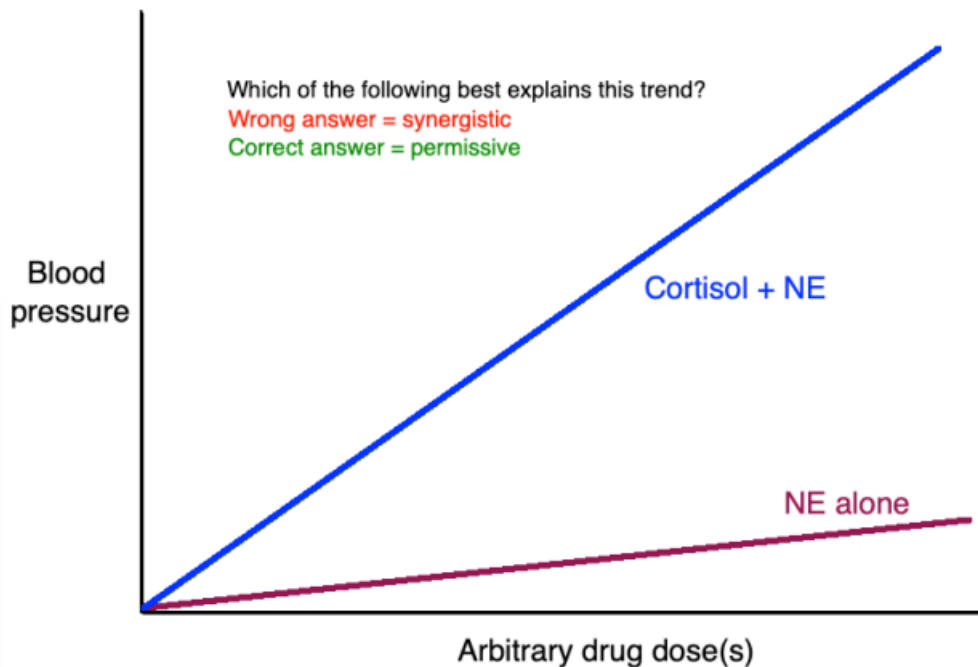
Additive

- Combined effect of Drug A + Drug B is equal to the sum of their individual effects ("2+2 = 4").
- **HY USMLE example: aspirin + acetaminophen.**
 - Degree of COX inhibition is the sum of both drugs individually.



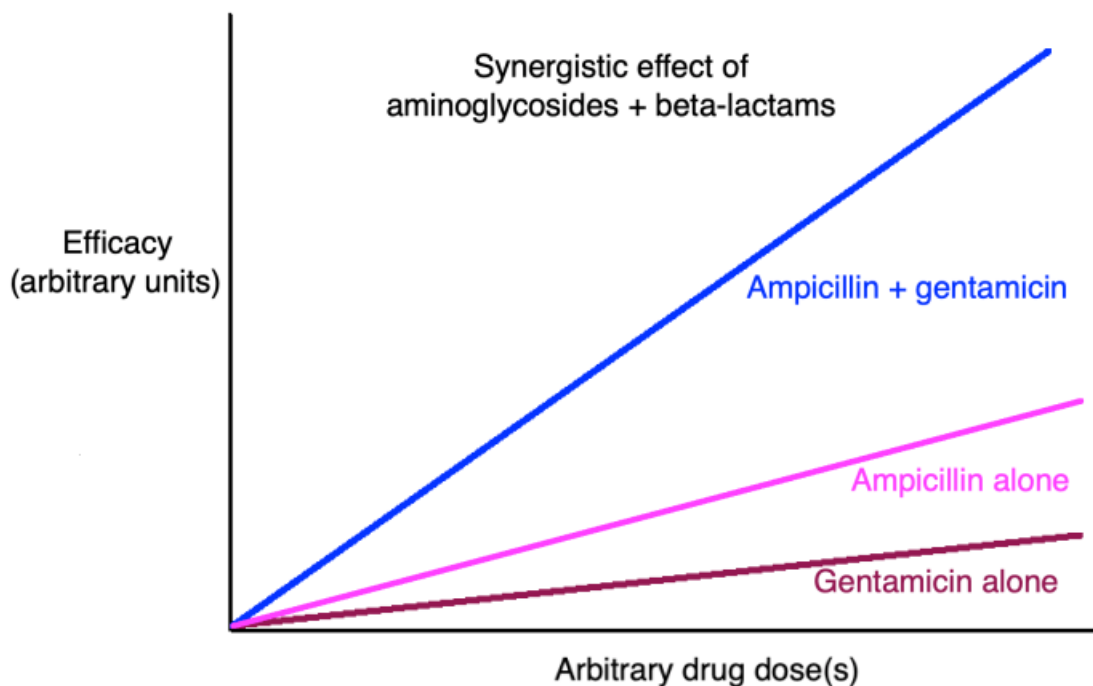
Permissive

- Molecule A must be present so that Molecule B can do its job.
- **HY USMLE example: cortisol is permissive of the effects of catecholamines (i.e., norepinephrine and epinephrine).**
 - Alpha-1 agonism causes arteriolar vasoconstriction and an increase in BP.
 - Norepinephrine and epinephrine bind to alpha-1 receptors on arteriole endothelium.
 - Cortisol helps maintain blood pressure, *not* by binding to cortisol receptor where that directly increases BP; instead, cortisol upregulates alpha-1 receptors on peripheral arterioles.
 - Norepinephrine and epinephrine *can now do their job*.
 - Without cortisol present, there is decreased expression of alpha-1 receptors on vascular endothelium, so even if NE and E are floating around, there is insufficient alpha-1 receptor expression and BP cannot be maintained.
 - USMLE wants you to know the phrase: "cortisol is permissive of the effects of catecholamines," where cortisol *permits* NE and E to do their job.
 - 43F + SLE + has surgery + intraoperatively her BP falls to 80/40; IV fluids are administered; what drug should be given? → answer = hydrocortisone. Students always get this wrong. Patients with autoimmune disease (i.e., SLE, IBD, RA) managed with chronic corticosteroids (prednisone) will have suppression of the zona fasciculata and a decreased ability to mount a cortisol stress response during surgery or trauma – > corticosteroid in the blood is consumed → NE and E can't do their job → BP falls → give hydrocortisone (cortisol analogue) to restore alpha-1 receptor expression.



Synergistic

- Combined effect of Drug A + Drug B is greater than the sum of their individual effects (“2+2 > 4”).
- **HY USMLE examples: inhibition of platelet function by clopidogrel + aspirin.**
 - Clopidogrel inhibits ADP2Y12 receptor on platelets.
 - Aspirin inhibits COX.
- **Ampicillin + gentamicin.**
 - Ampicillin inhibits bacterial cell wall synthesis/cross-linking.
 - Gentamicin inhibits protein synthesis by inhibiting the 30S ribosomal subunit.
 - Ampicillin enables greater penetration of gentamicin into the cell.
- **Gastric acid secretion (not pharmacologic but HY).**
 - Acetylcholine binding to M3 receptors on parietal cells.
 - Histamine binding to H2 receptors on parietal cells.
 - Gastrin binding to gastrin receptors on parietal cells.
 - Combined effect of all three mechanisms is synergistic for gastric acid secretion.



Tachyphylaxis (tolerance)

- Desensitization after repeated/continual administration.
- **HY USMLE examples: alpha-1 agonists causing rhinitis medicamentosa.**
 - Alpha-1 agonism causes vasoconstriction.
 - Intranasal alpha-1 agonists (oxymetazoline, phenylephrine, midodrine) decrease blood flow through nasal mucosa, resulting in decreased inflammation. These agents are used as nasal decongestants.
 - Continual intranasal use for >4-5 days can result in decreased effect + rebound nasal congestion once stopped.
 - USMLE will ask: 16F + stuffy nose + using intranasal drug for 5 days continuously + stops using it + now has rebound nasal congestion; what drug did she use? → answer = oxymetazoline.
- **Leuprolide administered continuously for prostate cancer.**
 - Leuprolide is a GnRH receptor agonist.
 - Initial effect is an increase in LH and FSH.
 - After a couple days, LH and FSH production decrease due to GnRH receptor desensitization (tachyphylaxis).
 - Should be noted that flutamide, an androgen receptor blocker, is administered concurrent to leuprolide during the first couple days in order to nullify the transient surge in LH and FSH.
- **Nitrates for angina.**
 - Nitrates cause upregulation of guanylyl cyclase in vascular endothelium (predominantly veins).
 - This results in increased cGMP → increased protein kinase G activity → dephosphorylation of myosine light-chain kinase → relaxation of venous smooth muscle → pooling of blood in veins → decreased venous return to right atrium → decreased preload → decreased myocardial oxygen demand → alleviation of cardiac pain.
 - Repeated use of nitrates results in diminished venodilatory effect.

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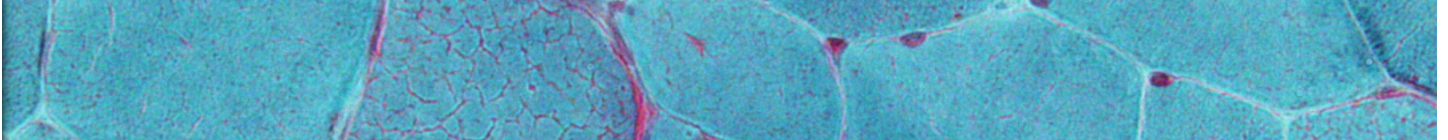


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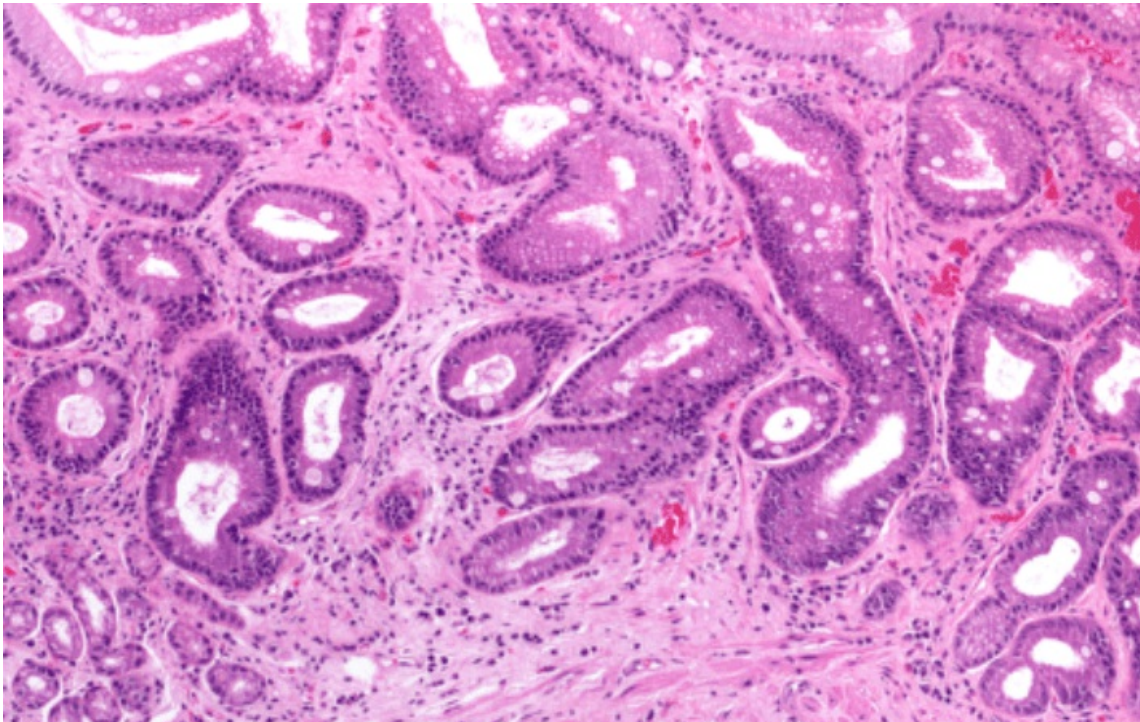
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GASTROINTESTINAL / PHARMACOLOGY

GI pharm – Antacids (highest yield points)

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HY points about each drug followed by a quiz at the end

H2 blockers (Cimetidine, Ranitidine, Famotidine)

- Don't confuse with second-generation H1 blockers (e.g., loratadine), which are used for allergies.
- H2 blockers are "idine"; the second-gen H1 blockers are "adine."

MOA of H2 blockers?

- Reversible, competitive antagonists at Histamine 2 receptor.
- H2 receptor is located on parietal cells.
- Agonism by endogenous histamine stimulates acid secretion.
- Therefore blocking these receptors decreases acid secretion.

Highest yield points about H2 blockers:

- **Cimetidine causes gynecomastia (anti-androgenic).**
- Cimetidine inhibits P-450 >>> more than ranitidine (strange detail asked on a retired Step 1 NBME).
- USMLE wants you to know H2 blockers are competitive and reversible, whereas proton pump inhibitors are non-competitive and irreversible.
- If the USMLE asks which drug is more efficacious, choose the proton pump inhibitor.

Proton pump inhibitors (PPIs) (Omeprazole, Esomeprazole)

- Irreversible, non-competitive antagonists of the proton pump (H/K-ATPase antiporter) on parietal cells.
- More efficacious than H2 blockers.

PPIs in the management of *H. pylori*:

- First-line treatment of *H. pylori* is **CAP** → **C**larithromycin, **A**moxicillin, **P**roton pump inhibitor (e.g., omeprazole).
- After CAP is used for 7-10 days, a urea breath test is performed 4 weeks later to see if the *H. pylori* has been eradicated. If not, then assume resistance to the antibiotics. Next best step in management = substitute out the clarithromycin and amoxicillin for metronidazole, tetracycline, and bismuth.
- In other words, use CAP; if doesn't work, use metronidazole, tetracycline, bismuth, and a PPI as the second-line quad combo.
- H2 blockers can rarely be used in place of the PPI for *H. pylori* treatment, but that is rare and will not be tested.

Other points about PPIs:

- Low pH of the stomach is needed for sucralfate to work (coats ulcers). PPIs can decrease the effectiveness of sucralfate.
- PPIs will increase serum gastrin levels. Normally, low pH of stomach acid negatively feeds back on gastrin secretion. Gastrin functions to stimulate acid production from parietal cells directly. It also binds to G receptors on enterochromaffin-like cells, inducing them to secrete histamine, which in turn binds to H2 receptors on parietal cells.
- PPIs can be used to treat Zollinger-Ellison syndrome (gastrinoma of the pancreas) as an alternative to surgery.

GERD vs diabetic gastroparesis

- Diagnosis of GERD is done with 2-week trial of PPIs **or** H2-blockers. PPIs are classically chosen for the 2-week trial, but one of the 2CK NBME Qs has trial of H2 blockers as an answer (and PPIs not listed). Both are correct answers. If you're forced to choose between the two, however, choose the PPI, as mentioned above.
 - 55M + burning sensation in throat after meals past three months + otherwise healthy and no cardiovascular issues; next best step in pharmacologic therapy? → answer = trial of PPIs or H2 blockers.
- Don't confuse with diabetic gastroparesis, which can present like GERD, but in someone who has bad diabetes. USMLE will slam you on this.
 - 55M + burning sensation in throat after meals past three months + T2DM + HbA1c of 9.6 + lack of pinpoint

discrimination up to the ankles; next best step in pharmacologic therapy? → answer = metoclopramide;
wrong answer = omeprazole.

- If the patient has poor glycemic control and neuropathy, the latter extends to the nerves innervating the GI tract as well → can present as diarrhea/constipation or GERD-like symptoms.
- First step is endoscopy to rule out physical obstruction. If negative, then do gastric-emptying scintigraphy (scintigraphic gastric-emptying assay) to confirm delayed gastric emptying. First step in treatment is smaller meals. If insufficient, start using **metoclopramide** (D2 antagonist), which is a pro-kinetic agent (stimulates peristalsis) and anti-emetic; after metoclopramide, can also use **erythromycin** (macrolide antibiotic that is also a motilin-receptor agonist).
- Bottom line is:
 - GERD-like Sx in someone without diabetes, answer = GERD; do trial of PPIs or H2 blocker.
 - GERD-like Sx in someone with bad diabetes; answer = diabetic gastroparesis; do metoclopramide, followed by erythromycin.

Bismuth, Sucralfate

- Both coat ulcers / bind to base of ulcers, thereby decreasing pain.
- Bismuth used in second-line quad therapy for *H. pylori* (metronidazole, tetracycline, bismuth, and a PPI); first-line is CAP (Clarithromycin, Amoxicillin, PPI).
- Sucralfate requires low pH of stomach acid for activation; PPIs ↓ effectiveness of sucralfate.

Aluminum

- Antacid
- **Causes constipation** → “Aluminum amount of feces.”

Magnesium (Milk of magnesia)

- Antacid
- Causes diarrhea.
- Milk of magnesia can actually be used to treat constipation (i.e., the side-effect of diarrhea is advantageous).

Calcium carbonate

- Antacid
- Can cause milk-alkali syndrome → patient takes too many Tums → causes hypercalcemia + metabolic alkalosis – > increased risk of calcium phosphate kidney stones.
- Can cause rebound acid hyper-secretion.

Misoprostol

- Prostaglandin E1 analogue
- ↑ production of gastric mucous and bicarbonate; ↓ acid production.
- Used for NSAID-induced gastric ulcers (NSAIDs normally inhibit COX, the enzyme that synthesizes prostaglandins).

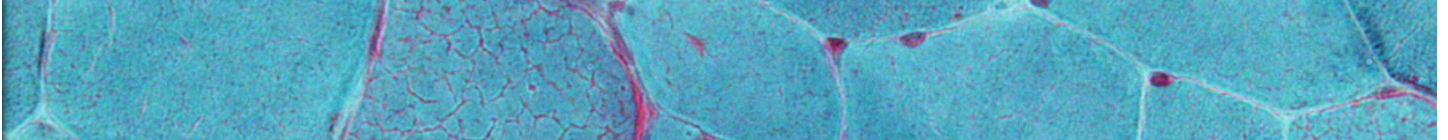


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GASTROINTESTINAL / PHARMACOLOGY

GI pharm – HY mixed agents

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HY points about each drug followed by a quiz at the end

Metoclopramide

MOA of metoclopramide?

- **D2 antagonist** used as a pro-kinetic (↑ peristalsis in diabetic gastroparesis) and anti-emetic (if you're stimulating the GI tract to move things downward, then they're less likely to come back up).
- Students tend to forget this MOA, often saying, "umm.....agonist?"
- Way to remember it's a D2 antagonist is because it carries the same adverse effects as the anti-psychotic agents.

Adverse effects of metoclopramide?

- Hyperprolactinemia
- Tardive dyskinesia (extra-pyramidal side-effects)
- Prolongs QT-interval on ECG (HY, along with macrolides, TCAs, anti-psychotics, and ondansetron).

GERD vs diabetic gastroparesis (this subsection is also in the antacid post because of the yieldness):

- Diagnosis of GERD is done with 2-week trial of PPIs **or** H2-blockers. PPIs are classically chosen for the 2-week trial, but one of the 2CK NBME Qs has trial of H2 blockers as an answer (and PPIs not listed). Both are correct answers. If you're forced to choose between the two, however, choose the PPI, as mentioned above.
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 - 55M + burning sensation in throat after meals past three months + T2DM + HbA1c of 9.6 + lack of pinpoint discrimination up to the ankles; next best step in pharmacologic therapy? → answer = metoclopramide; **wrong answer** = omeprazole.
 - If the patient has poor glycemic control and neuropathy, the latter extends to the nerves innervating the GI tract as well → can present as diarrhea/constipation or GERD-like symptoms.
 - First step is endoscopy to rule out physical obstruction. If negative, then do gastric-emptying scintigraphy (scintigraphic gastric-emptying assay) to confirm delayed gastric emptying. First step in treatment is smaller meals. If insufficient, start using **metoclopramide** (D2 antagonist), which is a pro-kinetic agent (stimulates peristalsis) and anti-emetic; after metoclopramide, can also use **erythromycin** (macrolide antibiotic that is also a motilin-receptor agonist).
 - Bottom line is:
 - GERD-like Sx in someone without diabetes, answer = GERD; do trial of PPIs or H2 blocker.
 - GERD-like Sx in someone with bad diabetes; answer = diabetic gastroparesis; do metoclopramide, followed by erythromycin.

Loperamide

- Mu-opioid agonist used as an anti-diarrheal agent (i.e., promotes constipation).
- **Low addiction potential** → USMLE will, weirdly enough, make this into a "plus sign question," where there are varying numbers of (+), and the answer is "++++" for mu-agonism, and only "+" for addictive potential; in contrast, meperidine is a highly addictive opioid, with "++++" for addictive potential.

Sulfasalazine

MOA of sulfasalazine?

- Combination of sulfapyridine (an antibiotic) and 5-ASA (5-aminosalicylic acid; aka mesalamine).
- Used for inflammatory bowel disease (IBD; i.e., ulcerative colitis and Crohn).
- USMLE just wants you to know that sulfasalazine = 5-ASA = mesalamine = NSAID. **Don't worry about the sulfapyridine.**
- Sulfasalazine and mesalamine are interchangeable on the USMLE.
- Sometimes Qbanks will get pedantic about when to use NSAID vs steroids for IBD; the USMLE won't. You will not

see both an NSAID and steroid as answers to the same question, where you have to pick between the two. And if you did, the vignette would give you, e.g., renal failure or peptic ulcers, where NSAIDs would be contraindicated anyway.

Octreotide

MOA of octreotide?

- Somatostatin analogue.
- Longer half-life than endogenous somatostatin.
- Somatostatin is a hormone that generally shuts off the action of many endocrine hormones, notably growth hormone. Think of it as a general endocrine suppressor. → “Somatostatin...oh right, that’s the one that shuts off other hormones, right?”

When is octreotide the answer?

- Highest yield → pharmacologic treatment for bleeding esophageal varices.
 - Acutely bleeding varices *treatment* = rubber banding + octreotide.
 - Varices bleeding *prophylaxis* = propranolol (beta-blockade).
- Can also be used for acromegaly, VIPoma, and carcinoid syndrome (remember: somatostatin shuts off hormones, so this makes sense right?).

Ondansetron

MOA of ondansetron?

- 5-HT₃ (serotonin) receptor antagonist.
- 5-HT₃ receptors are located at the chemoreceptor trigger zone (vomiting center) in the caudal (lower) medulla. If the USMLE question shows you a sagittal brain MRI and asks about ondansetron MOA, choose the caudal medulla letter as the answer.

When is it used?

- Very strong anti-emetic (stronger than metoclopramide).
- Used mostly in patients undergoing chemotherapy who have nausea and vomiting.

Adverse effects of ondansetron?

- QT prolongation (HY, along with metoclopramide, macrolides, TCAs, and anti-psychotics).

Aprepitant

MOA of aprepitant?

- Blocks NK1 receptors.
- Substance P normally binds to NK1 receptors to induce pain and nausea.
- **Therefore aprepitant antagonizes the effect of Substance P.**
- NK1 = neurokinin-1, aka tachykinin.

When is aprepitant used?

- Similar to ondansetron, can be used for nausea and vomiting in patients undergoing chemotherapy (stronger than metoclopramide).

Orlistat

MOA of orlistat?

- Pancreatic lipase inhibitor (i.e, prevents breakdown of fats in lumen of small bowel → decreased absorption).

When is it used?

- Can be used for obesity.
- Usually a distractor, rather than correct answer, on USMLE questions. But because it shows up in answer choices, students frequently say, "What's that?"

Adverse-effects of orlistat?

- Can cause steatorrhea and fat-soluble vitamin deficiency in theory.

Pancrelipase

What is pancrelipase?

- Pancreatic enzyme replacement composed of amylase, lipase, and proteases.

When is it used?

- Given to patients with chronic pancreatitis (i.e., pancreas burnout from multiple bouts of acute pancreatitis, usually in the setting of alcoholism).

What's a notable point about the presentation of patients with chronic pancreatitis?

- USMLE vignette will give an alcoholic with Hx of multiple acute pancreatitis episodes who has steatorrhea (i.e., unable to absorb fats because he lacks pancreatic lipases) + the Q will say amylase and lipase levels are **low or normal** (chronic pancreatitis is characterized by organ burnout, not acute elevations in enzymes).
- (Tangentially, if you get a vignette of hepatic cirrhosis, ALT and AST will be low, not high, because the liver is burned out).

—

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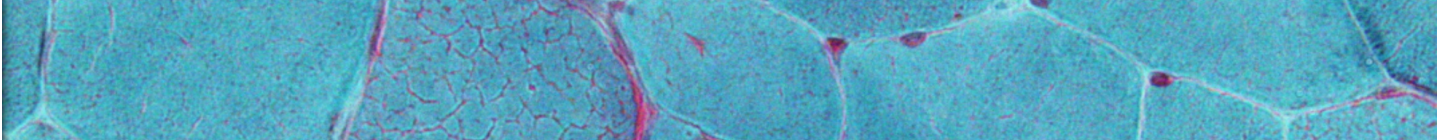


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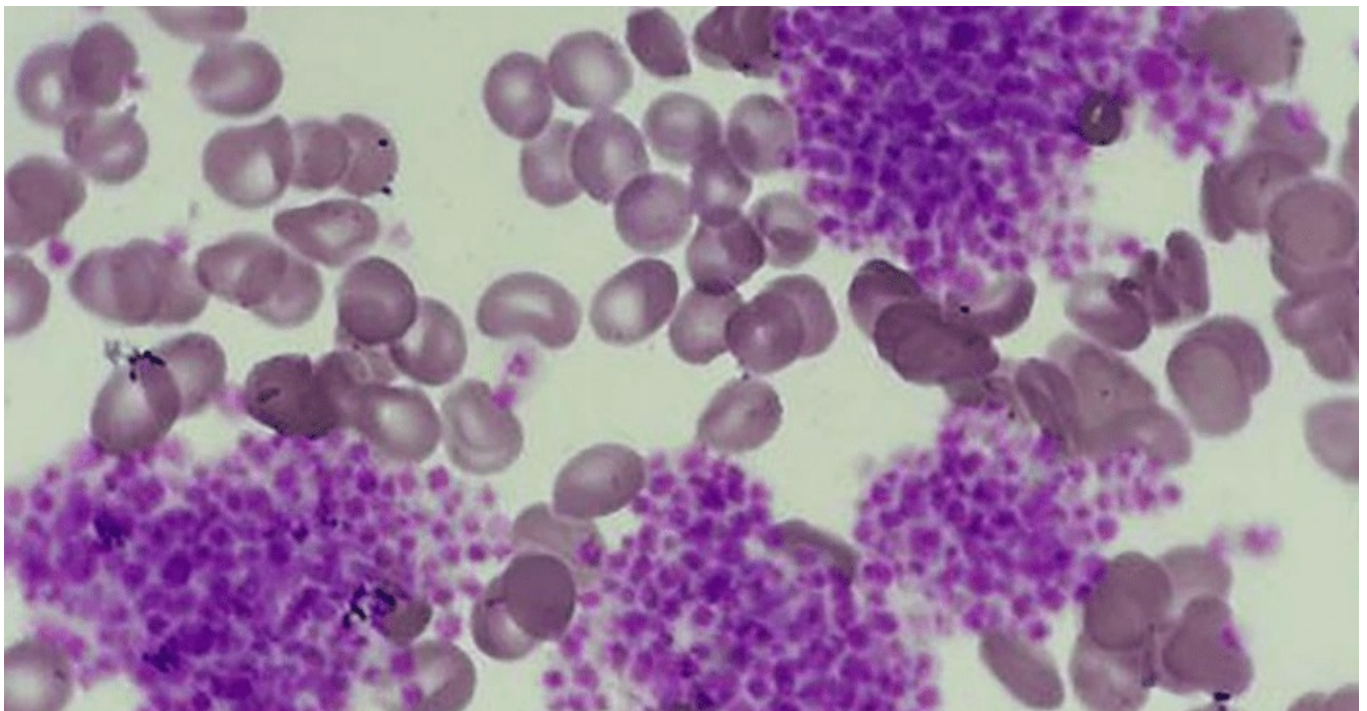
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Heme/onc pharm – Anti-platelet agents, NSAIDs, aspirin

by MEHLMANMEDICAL

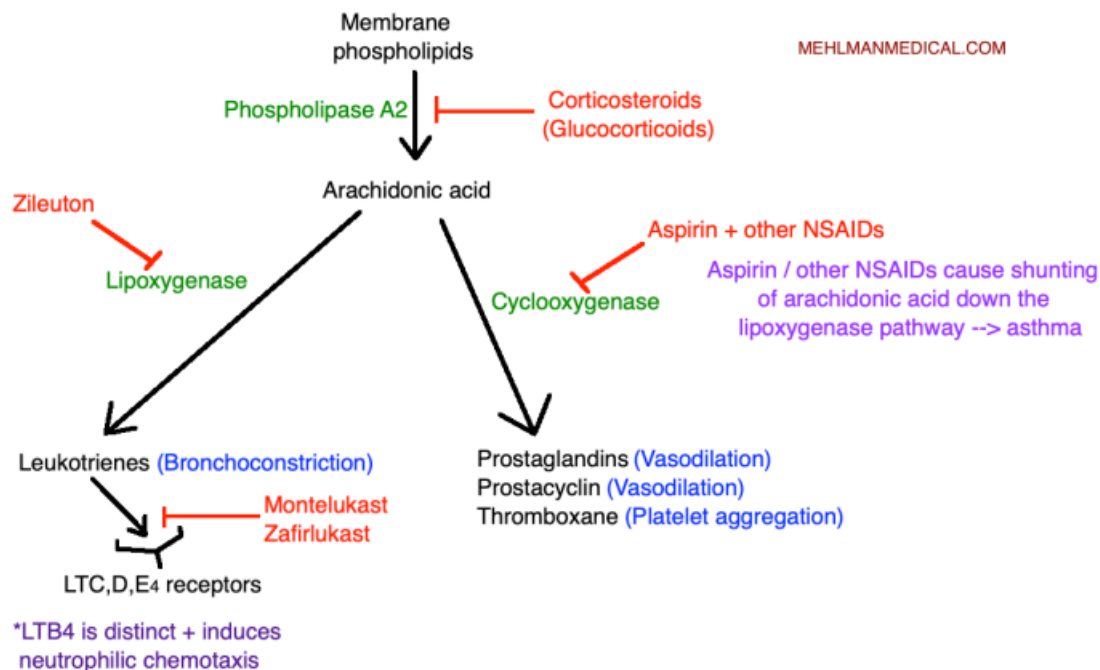


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- The highest yield point to note from this diagram is that aspirin and other NSAIDs (non-steroidal anti-inflammatory drugs) cause shunting of arachidonic acid down the lipoxygenase pathway. The USMLE loves to assess aspirin-/NSAID-induced asthma.
 - 16M + went snowboarding for 8 hours + next day has sore muscles + took aspirin for pain relief + now has wheezing; which agent should be avoid in the future? → answer = ibuprofen (an NSAID); wrong answer = acetaminophen (not an NSAID; acts centrally).

Aspirin (salicylate) + other NSAIDs

- Other HY NSAIDs: diclofenac, ketorolac, ibuprofen, indomethacin, naproxen, mesalamine.
- Acetaminophen is not an NSAID (discussed later).
- I lump aspirin + other NSAIDs together because of their considerable overlap. But important differences are noted.

MOA of aspirin?

- Irreversible** inhibition of cyclooxygenase 1 and 2 (COX1/2).

MOA of other NSAIDs?

- Reversible** inhibition of COX1/2.

Physiologic effects of COX inhibition by aspirin + other NSAIDs:

- Results in ↓ production of prostaglandins and platelet thromboxane.
- ↓ platelet thromboxane means ↑ bleeding time (NR 2-7 minutes, so >7 minutes; no change in platelet count).
- ↓ prostaglandins (vasodilators) means ↓ diameter of arterioles (i.e., less vasodilation of afferent arterioles of the kidney → ↓ renal perfusion).
- ↓ COX1/2 activity means a shunting of arachidonic acid down the lipoxygenase pathway → ↑ leukotriene production → bronchoconstriction → asthma in some patients.

What's the difference between COX1 and 2?

- COX1 is found in gastric mucosa and platelets. Thromboxane A2 is the major product of COX1 in platelets.
- COX2 is found in vascular endothelium and inflammatory cells; inhibition of COX2 is the primary mechanism for ↓

pain.

HY answers for aspirin on USMLE:

- The agent given in acute MI.
 - After aspirin is administered, give an ADP P2Y₁₂ blocker (i.e., clopidogrel) for dual anti-platelet therapy.
- Atrial fibrillation (AF) in patients who have a low CHADS₂ score.
 - CHADS₂ score is used to assess whether patients with AF need anti-platelet (aspirin) or anti-coagulation therapy (usually warfarin).
 - **C**ongestive heart failure, **H**ypertension, **A**ge >75, **D**iabetes, **S**troke/TIA (each worth 1 point; the latter is worth 2 points).
 - If 0 or 1 points, give aspirin to AF patients.
 - If 2+ points, give warfarin (anticoagulation answer on USMLE). Patients who have valvular AF (i.e., AF in the setting of, e.g., mitral regurg) must receive warfarin. Patients with non-valvular AF (i.e., idiopathic or, e.g., in the setting of hyperthyroidism) can receive other anticoagulants such as [apixaban](#).
- Angina (ischemic heart disease) to decrease risk of MI.
 - Stable angina: used *instead of* daily baby aspirin in patients who can't take the latter (e.g., because of renal insufficiency).
 - Unstable angina: *added to* daily baby aspirin. The combo of aspirin + clopidogrel is superior to aspirin alone in unstable angina; this relationship has not yet been proven for stable angina.
- Cerebrovascular disease or carotid stenosis to prevent ischemic stroke (three acceptable regimens are: aspirin alone; aspirin + dipyridamole; or clopidogrel alone).
- Kawasaki disease (+/- IVIG).
 - 4M + five days of fever + cervical lymphadenopathy + edema of the dorsum of the hands + red lips + red eyes; Tx? → answer = aspirin + IVIG.

HY uses of other NSAIDs:

- First-line for pain relief in rheumatoid arthritis; steroids are used for pain if NSAIDs are insufficient. For RA, there are two arms of pharmacologic management:
 - Symptom management only: **NSAIDs first; if insufficient, add steroids**. NSAIDs + steroids do not alter the course of the disease.
 - DMARDs (disease-modifying anti-rheumatic drugs); not used for symptoms; these slow the course of the disease → **methotrexate first almost always**; if insufficient, add other agents like anti-TNF-alpha drugs.
- Toxic synovitis (aka transient synovitis) → hip pain in a kid after a viral infection. Sounds weirdly specific, but it's on the NBME.
 - 6M+ runny nose for 3 days + now has hip pain + low-grade fever; Tx? → answer = ibuprofen (NSAIDs)
 - Septic arthritis, in contrast, will present in peds as a kid who has history of juvenile rheumatoid arthritis, and is almost always the knee, not the hip. Plus, if the USMLE wants toxic synovitis, they'll always mention viral infection preceding the hip pain.
- Subacute granulomatous (de Quervain) thyroiditis → tender/painful thyroid + hypo- or hyperthyroidism, usually after viral infection (but the USMLE will frequently not mention the infection).
 - 28M + tender thyroid + HR 92 + a bit tremulousness; Dx + Tx? → deQuervain; NSAIDs +/- steroids (but choose NSAIDs first on USMLE).
- Pain relief in general (e.g., sore muscles, headache).
- Can be used in osteoarthritis *after* acetaminophen (i.e., choose acetaminophen first on USMLE if NSAID also listed).
 - If patient is overweight, always choose weight loss first as the treatment (obesity = biggest risk factor for OA).
 - But if question asks for first-line pharmacologic management, answer = acetaminophen first, **not** NSAIDs. If acetaminophen alone is insufficient, NSAIDs can be added.
 - Osteoarthritis is degenerative joint disease. The pain will never go away. So if the patient takes chronic

NSAIDs for pain, he or she is prone to renal damage (i.e., peripheral edema in someone taking naproxen for OA).

- Anti-pyretic (fever reduction).
- **Indomethacin** is first-line for acute gout.
 - If patient has renal insufficiency, choose corticosteroids over indomethacin.
 - If indomethacin and steroids are both not listed but patient has acute attack, answer = colchicine.
- Pseudogout → NSAIDs, then steroids, then colchicine (same as acute gout, but NSAID need not be indomethacin).
- Pericarditis → NSAIDs, then steroids, then colchicine (same as acute gout and pseudogout; NSAID need not be indomethacin).

Notable time to **NOT** give aspirin specifically:

- Viral infection in children → aspirin can cause Reye syndrome (cerebral edema, liver damage) due to inhibition of mitochondrial beta-oxidation enzymes.
 - 8M + viral infection + fever + given drug by mom to treat fever + now has confusion and papilledema + LFTs are increased.

Notable time to **NOT** give NSAIDs in general (including aspirin):

- Hx of peptic ulcer disease, gastritis, or upper GI bleeding → prostaglandins normally protect the mucosal barrier of the stomach by increasing production of alkaline mucous in the stomach.
- Renal insufficiency → vasodilating prostaglandins maintain patency of the renal afferent arterioles → NSAIDs decrease renal blood flow by decreasing prostaglandin synthesis.
 - 82F + taking naproxen for 6 weeks for pain relief + now has peripheral edema; why? → answer = “decreased renal excretion of sodium,” or “increased renal retention of sodium”; ↓ renal blood flow means kidney attempts to ↑ fluid retention; it accomplishes this by reabsorbing sodium in the PCT, since water follows sodium → edema + $\text{FeNa} < 1\%$ (fractional excretion of sodium).
 - In pre-renal etiologies of renal insufficiency – i.e., in the setting of decreased renal perfusion – a HY detail is that you know urinary FeNa is $< 1\%$, which would be seen with NSAID use.
- History of aspirin/NSAID-induced asthma → knocking out COX causes shunting of arachidonic acid down lipoxygenase pathway → leukotriene synthesis → bronchoconstriction.

Adverse effects unique to aspirin:

- Reye syndrome (as mentioned above).
- **Tinnitus** (ringing of the ears); first *symptom* of salicylate (aspirin) toxicity (i.e., what the patient reports subjectively).
- Tachypnea (↑ respiratory rate); first *sign* of salicylate toxicity (i.e., what we notice on examination objectively; the patient doesn't report tachypnea).
- Acid-base disturbance (discussed below).

Acid-base disturbance caused by aspirin (exceedingly HY for all Steps):

- Respiratory alkalosis **early** (i.e., first 20 minutes, answer = ↓ pCO_2 ; normal HCO_3^- ; ↑ pH); **bicarb does not change** in acute respiratory acid-base disturbances; it takes minimum 24 hours to really move (e.g., with altitude sickness).
- Mixed respiratory alkalosis-metabolic acidosis **late** (i.e., after 20 minutes, answer = ↓ pCO_2 ; ↓ pH 80% of the time; 20% of the time it's in the normal range; ↓ HCO_3^-). Student says, “Wait, I thought you just said it takes ~24 hours for bicarb to really move, so how is it down already?” Because **aspirin itself causes a metabolic acidosis** (↓ HCO_3^-) that is now superimposed on the respiratory alkalosis (↓ pCO_2). **It's coincidental** that the aspirin-induced metabolic acidosis that occurs with slight delay happens to look like metabolic compensation for the respiratory

alkalosis, **but this is NOT compensation**. Aspirin directly upregulates respiratory centers (tachypnea) + is an acid itself (salicylic acid).

Adverse effects of NSAIDs (including aspirin):

- Gastric ulcers, gastritis, GI bleeding.
- Various renal pathologies:
 - Analgesic nephropathy (any adverse change in the kidney due to NSAIDs); can present as renal papillary necrosis (dark urine) or interstitial nephropathy (eosinophils in the urine).
 - Pre-renal azotemia → starts off as patients getting peripheral edema due to NSAIDs because of decreased renal blood flow (as discussed above); the edema is not a renal pathology in and of itself, but reflects pre-renal changes in the kidney that may eventually cause elevations in creatinine and BUN. Once the renal function tests change, the patient now has pre-renal azotemia.
- Aspirin/NSAID-induced asthma (Samter triad = NSAID hypersensitivity, NSAID-induced asthma, nasal polyps).

Tx for aspirin toxicity?

- Sodium bicarbonate → alkalinity traps the salicylic acid as the oxalate (O^-) form → kidney cannot reabsorb charged species as well as neutral ones → increased urinary excretion.
- If the USMLE asks you the mechanism for how sodium bicarb works for aspirin toxicity, the answer = “increased excretion through urinary alkalization.”
- Do **not** confuse this mechanism with sodium bicarb in the treatment of TCA toxicity. TCAs cause cardiotoxicity by binding to myocardial sodium channels. If the USMLE asks you the answer for the mechanism in this case, the answer = “increased dissociation from myocardial sodium channels”; the wrong answer is “increased excretion through urinary alkalization,” which is aspirin. Students fuck this up and choose the latter because they’ve memorized it as HY for aspirin, but don’t realize it’s wrong for TCAs.

Acetaminophen

MOA of acetaminophen?

- Inhibits COX **in the CNS**; inactivated peripherally.
- Not considered an NSAID → does not act as an anti-inflammatory agent peripherally. The ability to decrease pain is linked to ↓ prostaglandin effects in the CNS, not at peripheral tissues.

When is acetaminophen the answer on USMLE?

- First-line pharmacologic treatment for osteoarthritis (second-line is NSAIDs). Should be noted that USMLE likes simple “weight loss” as the first treatment overall for osteoarthritis, but if a drug is given, choose acetaminophen over NSAIDs.
- Pain relief in patients with renal insufficiency (does not act peripherally so does not kill the kidneys like NSAIDs do).
- Febrile non-hemolytic transfusion reaction (most common type of transfusion reaction). Sounds weirdly specific, but it’s on the NBME.
 - Mechanism is antibodies against donor leukocyte antigens.
 - 30F + fever 2 hours after receiving blood transfusion + Coombs test is negative; Tx? → answer = acetaminophen.
- Can be used safely in children to avoid Reye syndrome caused by aspirin.

Adverse effect of acetaminophen?

- Severe hepatic necrosis. Acetaminophen is metabolized into a toxic metabolite called NAPQI that is immediately neutralized by reduced glutathione in the liver. If the patient overdoses on acetaminophen, the reduced glutathione is depleted and then the NAPQI oxidizes and diffusely kills the hepatocytes.

- In short, whilst NSAIDs and aspirin kill the kidneys, acetaminophen kills the liver.

Treatment for acetaminophen toxicity?

- *N*-acetylcysteine.
- USMLE wants you to specifically know the reason *N*-acetylcysteine works is because **it regenerates reduced glutathione in the liver.**
- Species with thiol (-SH) groups are in the reduced form and can act as reducing agents (i.e., can act as anti-oxidants).
- Species with disulfide bonds (-S-S-) are in the oxidized form.
- Hepatic glutathione has a thiol (-SH) group. So does *N*-acetylcysteine. After glutathione neutralizes a toxin, it becomes oxidized and inactive.
- The thiol group on *N*-acetylcysteine then binds to and frees up the sulfurs on glutathione so that they can once again neutralize toxins.
- This mechanism may sound cumbersome, but it's HY for USMLE. Probably because *N*-acetylcysteine is such an important antidote used clinically.
- Activated charcoal can be used in select acute circumstances. There's a Qbank Q floating around where activated charcoal is the answer over *N*-acetylcysteine. **On the USMLE, however, always choose *N*-acetylcysteine for acetaminophen.**

Celecoxib

MOA of celecoxib?

- Selective COX2 inhibitor.
- Considered to be an NSAID.
- Called a COX2-selective NSAID.

When is it used?

- In patients who have contraindications to NSAIDs (i.e., peptic ulcer disease, gastritis, GI bleeding, renal insufficiency) but who need pain relief (e.g., in rheumatoid arthritis).

Adverse effects of celecoxib?

- Increases risk of MI and thrombosis → selective inhibition of COX2 decreases vasodilatory effects of prostaglandins and prostacyclins without inhibiting thromboxane A2 in platelets (COX1).
- USMLE vignette will mention specifically that a physician is apprehensive about giving celecoxib to a patient because of increased MI risk with this drug; then they'll ask why this is the case → answer on NBME = "inhibition of prostacyclin (PGI₂) formation without inhibition of thromboxane A2 in platelets."
- Similar risk of renal complications as non-selective NSAIDs.

ADP P2Y₁₂ blockers

- Clopidogrel, Prasugrel, Ticagrelor, Ticlopidine
- ADP P2Y₁₂ receptors are on platelets and lead to the expression of platelet glycoproteins IIb/IIIa, which enable platelet aggregation.
- Despite ADP P2Y₁₂ agonism leading to increased expression of GpIIb/IIIa, **do not** confuse these with the actual GpIIb/IIIa inhibitors.
- Clopidogrel will almost always be agent mentioned / asked about on the USMLE in relation to this class of drugs.

When are these the answer on USMLE?

- Second drug given after aspirin for the acute treatment of MI (aspirin given first).
- Technically used for stenting procedures, and this is an important use that's widely propagated in USMLE resources, but I'm yet to have seen this on any NBME assessment as an actual answer.
- Stable angina: used *instead of* daily baby aspirin in patients who can't take the latter (e.g., because of renal insufficiency).
- Unstable angina: *added to* daily baby aspirin. The combo of aspirin + clopidogrel is superior to aspirin alone in unstable angina; this relationship has not yet been proven for stable angina.
- Cerebrovascular disease or carotid stenosis to prevent ischemic stroke (three acceptable regimens are: aspirin alone; aspirin + dipyridamole; or clopidogrel alone).

Adverse effects of ADP P2Y₁₂ blockers?

- Clopidogrel and prasugrel can cause TTP (thrombotic thrombocytopenic purpura) – i.e., can precipitate the inactivation of ADAMTS13, leading to schistocytosis, thrombocytopenia, and renal insufficiency.
- Hemorrhagic stroke (prasugrel > clopidogrel).
- Ticlopidine can cause neutropenia (mouth ulcers, sore throat fever).

GpIIb/IIIa inhibitors

- Abciximab, Tirofiban, Eptifibatide
- GpIIb/IIIa on platelets enable platelet *aggregation* (i.e., platelets sticking to each other). Do not confuse this with platelet *adhesion* (i.e., platelets sticking to vascular endothelium) which is mediated by GpIb.
- Physiologically, fibrinogen is what bridges GpIIb/IIIa on adjacent platelets; therefore the GpIIb/IIIa inhibitors have been described as fibrinogen analogues on USMLE.

When is a GpIIb/IIIa inhibitor the answer on the USMLE?

- Can technically be used for stenting procedures, but I have not seen the USMLE specifically assess this. The exam cares more that you know the *mechanism of action* of these drugs. That is, know the drug names (especially abciximab) and that they simply inhibit GpIIb/IIIa, thereby decreasing platelet *aggregation*.

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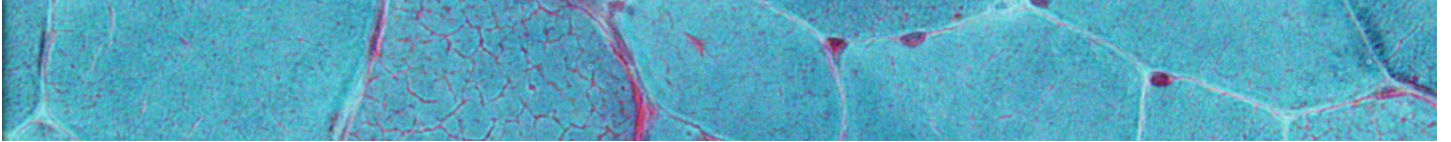


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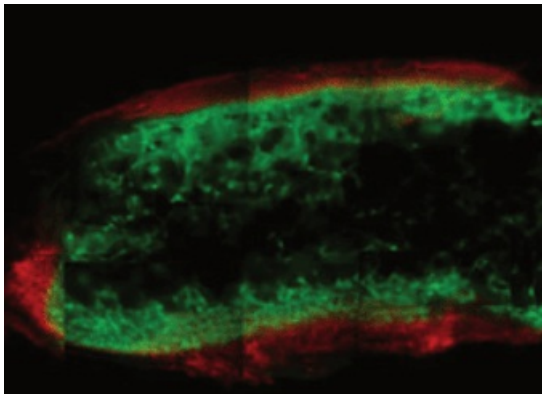
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[HEMATOLOGY](#) / [PHARMACOLOGY](#)

Heme/onc pharm – Heparin, anticoagulants, thrombolytics

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Heparin

- Large, anionic, acidic molecule.
- Heparin upregulates antithrombin, which binds to and inhibits factor IIa (thrombin) and factor Xa.
- **Unfractionated heparin**
 - Greater inhibition of factor IIa relative to factor Xa.
 - Cleared hepatically; used in renal insufficiency.
 - Easily reversible with protamine sulfate.

- Given IV.
- **Low-molecular weight heparin (enoxaparin, dalteparin)**
 - Greater inhibition of factor Xa relative to factor IIa.
 - Cleared renally; contraindicated in renal insufficiency.
 - Not easily reversible compared to unfractionated heparin.
 - Can be given subcutaneously.

MEHLMANMEDICAL.COM	LMWH	Unfractionated heparin
MOA	Inhibits Xa > IIa	Inhibits IIa > Xa
How is it cleared?	Renally (contraindicated in renal failure)	Hepatically
Easily reversible with protamine sulfate?	No	Yes
Administration	SC	IV

HY uses of heparin for USMLE?

- DVT + PE
 - 31F + recent surgery and convalescing in hospital + painful, swollen left leg + sinus tachycardia + shortness of breath; what do we do? → answer = heparin first, *then* spiral CT of the chest (CT angiogram); in other words, don't do the imaging before the heparin.
 - If the patient is already on warfarin and has a PE, do a spiral CT of the chest to confirm *before* inserting an IVC filter. In other words, don't just jump on IVC filter; remember you need to actually know the patient has a PE before you insert metal into their vena cava.
 - If pregnant + PE, give heparin first, *then* do ventilation-perfusion (V/Q) scan, *not* a spiral CT. If a V/Q scan shows segmental defects (suggestive of PE) and the Q asks for the next best step in diagnosis, the answer is spiral CT of the chest. Student says, "Wait wtf? I thought we don't do spiral CT in pregnancy." You're right, we don't. But if they ask for the next best step in diagnosis after V/Q scan, it's still the answer, even if we aren't going to do it.
- Superficial thrombophlebitis (ST)
 - 45F + history of varicose veins + painful 1-cm palpable cord at the ankle; next best step? → answer = "subcutaneous enoxaparin"; compression stockings are the **wrong** answer. The latter are the correct answer for first step in management for venous insufficiency and varicose veins, but if the patient has an active clot (ST or DVT), you need to give heparin.
 - 45F + painful cord tracking from the ankle to the knee; next best step? → answer = subcutaneous enoxaparin.
- Prior to surgery in patients who need anticoagulation
 - On 2CK Surg NBME, they will tell you that a patient soon to be undergoing surgery needs anticoagulation. Ultrasound of the legs shows chronic venous occlusion; the answer is "low dose heparin prophylaxis"; the **wrong** answer is "therapeutic heparin dose." The latter would be used for an active DVT or ST, but for patients at mere increased risk, use lower dose.
- Myocardial infarction
 - Haven't seen any Qs on this from NBME, but it's a known, important use of heparin. Tangentially, USMLE wants you to know that you give aspirin *first* for MI, followed by adding an ADP2Y12 blocker (i.e., clopidogrel) for double anti-platelet therapy. Therapies such as beta-blocker, nitrates, morphine, oxygen are usually used. **Patients also need a statin** (HY).
- Anticoagulation during pregnancy (i.e., antiphospholipid syndrome; APLS)
 - 28F + SLE + Hx of two prior miscarriages + how to decrease risk in current pregnancy? → answer = aspirin +/- heparin (SLE is a cause of APLS, leading to recurrent miscarriages due to clots in the uteroplacental vasculature).

High-yield adverse effect of heparin for USMLE?

- Heparin-induced thrombocytopenia (HIT) → type II hypersensitivity → antibodies against heparin-platelet factor 4 (PF4) complex.
 - 42F + painful, swollen left leg while in hospital + heparin is administered + within two days platelets are <100,000; what do we do? → answer = stop heparin then **give direct-thrombin inhibitor** (i.e., dabigatran, bivalirudin, lepirudin); warfarin is the **wrong** answer.

How to reverse heparin?

- Protamine sulfate → protamine is positively charged; binds to the negatively charged, acidic, anionic heparin.

Heparin vs warfarin

MEHLMANMEDICAL.COM	Heparin	Warfarin
MOA	Upregulates antithrombin ↓ activity factor Xa + IIa	Inhibits activation of clotting factors II, VII, IX, X; and anti-clotting proteins C + S
Location of action	Serum	Liver
Structure	Large, anionic, acidic	Small, hydrophobic/lipophilic
How to reverse	Protamine sulfate	Vitamin K (slow); fresh frozen plasma (fast)
Safe in pregnancy?	Yes	No
Monitoring	aPTT	PT
Administration	SC/IV	PO

Direct-thrombin inhibitors (Dabigatran, Bivalirudin, Lepirudin)

- Directly inhibit factor IIa (thrombin).
- USMLE wants you to know these drug names = directly inhibit thrombin.
- Highest yield point (as mentioned above) is that they are the treatment for HIT. If a patient gets a drop in platelets following heparin administration, the answer is stop the heparin + give a direct-thrombin inhibitor.

Factor Xa inhibitors (Fondaparinux, Apixaban, Rivaroxaban)

- Inhibit solely factor Xa, in contrast to heparin, which has effect on both factor IIa and Xa.
- Can be used for anticoagulation as an alternative to heparin and warfarin.

How to reverse?

- **Andexanet alfa** → new drug approved in 2018 that reverses the Xa inhibitors.

Thrombolytics (tPA, Streptokinase)

MOA of the thrombolytics?

- Help convert plasminogen to plasmin.
- Plasmin breaks down fibrin meshes.

When are they used?

- **Within 3-4.5 hours of an ischemic stroke.**
- Rarely in PE when there is hemodynamic instability.

Contraindications to thrombolytics?

- Many. However HY ones for USMLE:
- Hemorrhagic stroke
- History of intracranial bleeds
- Severe hypertension

How to reverse thrombolytics?

- With anti-fibrinolytics:
 - **Aminocaproic acid**
 - **Tranexamic acid**

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HY USMLE Q #968 – Heme / Gastro

January 28, 2024

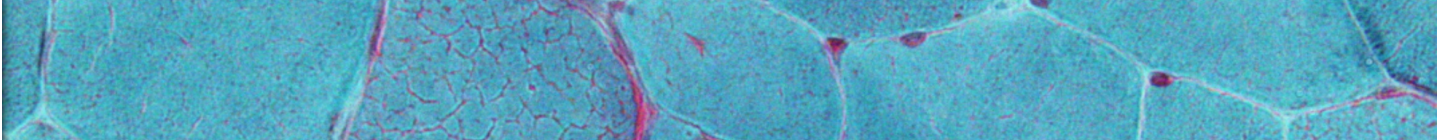


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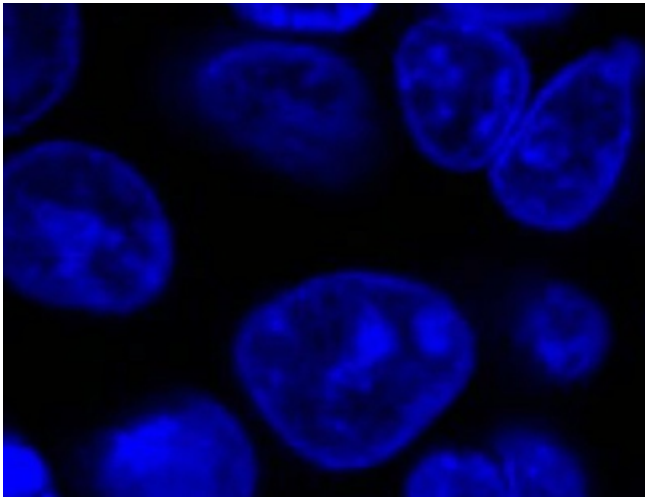
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Heme/onc pharm – HY mixed anti-cancer agents I

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HY points followed by a quiz at the end

Bleomycin

MOA of bleomycin?

- Causes free radical formation → damages DNA.

When is bleomycin used?

- Part of the ABVD regimen for Hodgkin lymphoma (Adriamycin, Bleomycin, Vinblastine, Dacarbazine).

What's the most important point I need to know for USMLE about bleomycin?

- Hands-down that it **causes pulmonary fibrosis**.
- 35F + was treated pharmacologically for Hodgkin disease last year + now has increasing shortness of breath + CXR shows a diffuse reticulonodular pattern; which agent is most likely responsible for this patient's condition? → answer = bleomycin.
- Pulmonary fibrosis on USMLE will be described in a vignette as "reticulonodular" or "reticular" pattern seen on CXR and CT. These are the medical terms for "honeycombing." The latter is often described in resources, but on the USMLE they won't use this buzzword.
- FEV1/FVC will be normal or increased because fibrosis causes restrictive lung disease.
- Other HY agents for USMLE that cause pulmonary fibrosis are: [methotrexate](#), [amiodarone](#), busulfan, and nitrofurantoin.

Doxorubicin (Adriamycin), Daunorubicin, Hydroxydaunorubicin

MOA of doxorubicin?

- Generates free radicals + intercalates into DNA.

When is doxorubicin used?

- Part of the ABVD regimen for Hodgkin lymphoma (Adriamycin, Bleomycin, Vinblastine, Dacarbazine).

What's the most important point I need to know for USMLE about doxorubicin?

- **Causes dilated cardiomyopathy.**
- 44M + started on pharmacologic therapy a couple months ago for Hodgkin disease + now has shortness of breath with exertion + S3 heart sound + dilated cardiac silhouette on CXR; which agent is most likely responsible for this patient's condition? → answer = doxorubicin.

What agent can be used to mitigate the toxicity of doxorubicin?

- [Dexrazoxane](#) is an agent that chelates free radicals and can help prevent dilated cardiomyopathy associated with doxorubicin.

Busulfan

MOA of busulfan?

- Alkylating agent

When is it used?

- Ablates the bone marrow in those undergoing bone marrow transplantation.

What's the most important point I need to know for USMLE about busulfan?

- **Causes pulmonary fibrosis** similar to bleomycin, methotrexate, amiodarone, and nitrofurantoin.

Cyclophosphamide, Ifosfamide

MOA of cyclophosphamide?

- Guanine-N7 alkylating agent

What's the most important point I need to know for USMLE about cyclophosphamide?

- **Causes hemorrhagic cystitis** (red urine).
- Toxic metabolite called acrolein is responsible for the bladder toxicity.
- **Mesna** is a cytoprotective agent that contains thiol (-SH) groups and is used to mitigate the bladder toxicity. If Mesna isn't listed, *N*-acetylcysteine (normally used for acetaminophen toxicity) can rarely be correct (also contains -SH groups).

Nitrosoureas (Carmustine, Lomustine)

MOA of nitrosoureas?

- DNA cross-linking agents

What do I need to know for USMLE?

- That they can treat brain tumors.
- And just know the MOA.

Cisplatin

MOA of cisplatin?

- DNA cross-linking agent.

When is it used?

- Can be used as a treatment for testicular cancer.

What's the toxicity of cisplatin?

- Neurotoxicity (known as "toxic neuropathy"). This is asked on the 2CK Neuro shelf, where a patient receiving cisplatin has miscellaneous neuropathy, and the answer is "toxic neuropathy," which means neuropathy secondary to agents (such as cisplatin or vincristine).
- Ototoxicity (neurosensory hearing loss; tinnitus; vertigo).
- Nephrotoxicity (acute tubular necrosis).

What do I need to know for USMLE that's most HY?

- Treatment of toxicity is first with chloride diuresis (i.e., 0.9% NaCl normal saline).
- After chloride diuresis, amifostine is used.
- **Amifostine** scavenges free radicals.

Microtubule inhibitors

- Not all used for anti-cancer, but all very HY to memorize so I'm grouping them here.
- Vincristine
- Vinblastine
- Taxanes (Paclitaxel, Docetaxel)

- Griseofulvin
- Colchicine
- -Bendazoles (Mebendazole, Albendazole)

Which is the odd one out?

- Taxanes (Paclitaxel, Docetaxel) are the odd ones out. They *hyper-stabilize* microtubules (i.e., prevent disassembly), whereas all of the other agents *prevent formation* of microtubules.
- Because microtubule assembly/disassembly occurs during mitosis, these work during M-phase of the cell cycle.

What do I need to know that's HY for each drug for the USMLE?

- **Vincristine is neurotoxic.** It's used as part of CHOP for non-Hodgkin lymphoma (CHOP → Cyclophosphamide, Hydroxydaunorubicin, Oncovin [Vincristine], Prednisone).
- Vinblastine is used as part of the ABVD regimen for Hodgkin lymphoma (Adriamycin, Bleomycin, Vinblastine, Dacarbazine).
- Taxanes (Paclitaxel, Docetaxel) hyper-stabilize (i.e., prevent disassembly of) microtubules, whereas the others prevent assembly.
- **Griseofulvin** is the treatment for tinea capitis ("cradle cap"; i.e., fungal infection of the scalp). This is asked explicitly on the FM shelf for 2CK.
 - 9M + circular area of alopecia and scaling of the scalp; treatment? → answer = "oral griseofulvin for patient only." (a wrong answer is for patient + close contacts).
- Colchicine is used as one of the treatments for acute gout, pseudogout, and pericarditis.
- -Bendazoles are used for helminth infections. Mebendazole is classically used for nematode (round worm) infections (e.g., ascariasis); albendazole is for neurocysticercosis.

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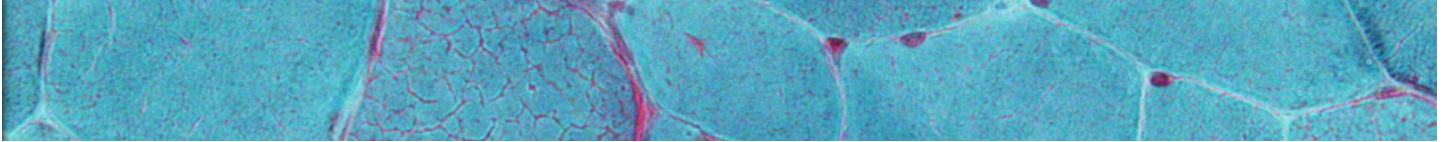


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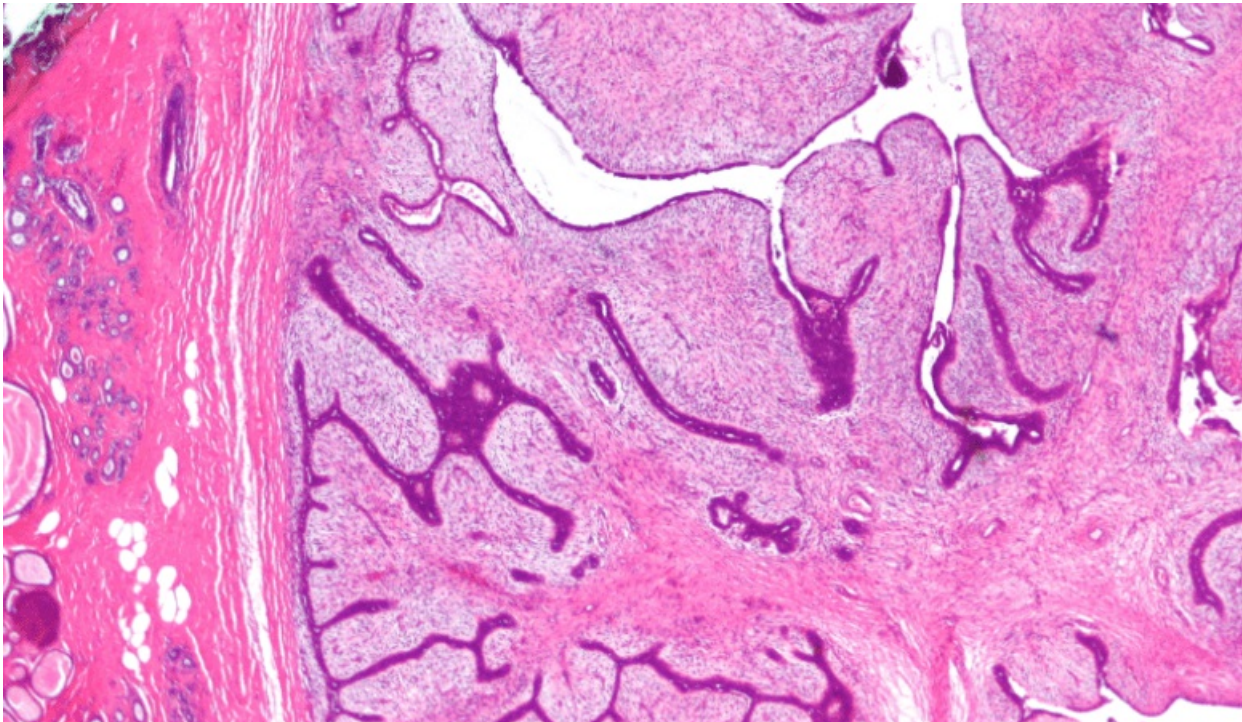
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HY points followed by a quiz at the end

[Irinotecan / Topotecan](#)

MOA of irinotecan / topotecan?

- Inhibit topoisomerase I (**one, not two**)
- Topoisomerase is an enzyme that “nicks” the DNA during replication to prevent supercoiling. Therefore these agents interfere with DNA replication.

How will these be tested?

- Physician is treating a patient for leukemia with an agent called irinotecan. This agent inhibits which of the following enzymes? → answer = topoisomerase I. One of the wrong answers will be “topoisomerase II,” so the USMLE wants you to be able to distinguish.

Etoposide / Teniposide

MOA of etoposide / teniposide?

- Inhibit *eukaryotic* topoisomerase II
- Should be noted that fluoroquinolones (i.e., ciprofloxacin, etc.) are an antibiotic class that also inhibit topoisomerase II, however these are antibiotics, not anti-cancer agents, and inhibit specifically *prokaryotic* topoisomerase II, not eukaryotic. Prokaryotic topoisomerase II is also known as DNA gyrase.
- In other words, fluoroquinolones inhibit DNA gyrase, aka prokaryotic topoisomerase II. In contrast, etoposide / teniposide inhibit simply eukaryotic topoisomerase II (*not* also known as DNA gyrase).

When are these used?

- Can be used for prostate cancer (flutamide + leuprolide is another combo for prostate cancer).
- USMLE is less concerned that you know the specific applications of these anti-cancer agents. **Just make sure you know the MOAs.**

Imatinib

- One of the highest yield anti-cancer agents for Step 1.

MOA of imatinib?

- bcr-abl tyrosine kinase inhibitor used for chronic myelogenous leukemia (CML).

What will USMLE ask me about this drug?

- **CML is caused by the t(9;22) translocation (Philadelphia chromosome).**
- USMLE specifically wants you to know that bcr-abl is an **oncogenic product**.
- That oncogenic product is specifically a **tyrosine kinase**.
- Therefore imatinib inhibits a tyrosine kinase that is a bcr-abl oncogenic product in CML.
- Sounds redundant, but I can't reiterate more that the Step 1 will ask numerous questions around these points.

HY side-effect of imatinib?

- Fluid retention (edema).
- The three highest yield agents causing **fluid retention / edema** for the Step are:
 - Dihydropyridine calcium channel blockers (i.e., amlodipine, nifedipine, etc.)
 - NSAIDs (↓ synthesis of vasodilating prostaglandins → ↓ renal blood flow due to narrowing of renal afferent arterioles → kidney thinks there's low blood volume → PCT attempts to reabsorb more fluid to compensate → accomplishes this by reabsorbing more Na⁺, where water follows sodium → edema).

- Imatinib

Erlotinib

MOA of ertlotinib?

- EGFR tyrosine kinase inhibitor used for non-small cell carcinoma.
- That's all you need to know. The USMLE will ask you straight-up the MOA.
 - 49F + large cell bronchogenic carcinoma + started on drug called erlotinib; which of the following best represents the molecular target of this agent? → answer = "EGFR tyrosine kinase."

Cetuximab

- Monoclonal antibody against EGFR (epidermal growth factor receptor).
- That's all you need to know for cetuximab. Just its MOA.
- "Okay, noted."

Rituximab

MOA of rituximab?

- Monoclonal antibody against CD20 on B cells.
- One of the highest yield anti-cancer drugs for Step 1.

When is it used?

- For lymphomas and leukemias.
- Most lymphomas and leukemias are B cell.
- USMLE wants you to know CD20 is a specific B cell marker.

Bortezomib

MOA of bortezomib?

- Proteasome inhibitor.
- Used rarely in multiple myeloma and mantle cell lymphoma.
- Just know it's a proteasome inhibitor.

Trastuzumab (Herceptin)

MOA of trastuzumab?

- Monoclonal antibody against HER2/neu receptor (produced by *ErbB2* gene).

When is it used?

- Breast cancer that has positivity for HER2/neu receptor.
- Should be noted that HER2/neu (+) breast cancer carries **worse prognosis** than HER2/neu (-) breast cancer, despite Trastuzumab being available as a pharmacologic agent for the former.

Toxicity of trastuzumab?

- **Cardiotoxic.**

- Sounds nitpicky, but is known to show up.
- It's to my observation some students won't even know how diabetes works, but they'll remember trastuzumab is cardiotoxic.

Alemtuzumab

- Monoclonal antibody against CD52 in chronic lymphocytic leukemia (CLL).
- Just memorize the above sentence.

Vemurafenib

MOA of vemurafenib?

- Inhibitor of BRAF serine-threonine kinase.
- Used for metastatic melanoma (+) for BRAF.
- *BRAF* is an **oncogene** encoding BRAF protein.

Tamoxifen / Raloxifene

MOA of tamoxifen / raloxifene?

- Selective-estrogen receptor modulators used for ER (+) breast cancer.
- Both antagonists at breast tissue.
- Both agonists at bone.
- **Tamoxifen only is a partial agonist at endometrium (↑ risk of endometrial cancer).**

Highest yield points for USMLE?

- Tamoxifen increases the risk of endometrial cancer.
- USMLE Q might give you a big paragraph vignette where they casually mention the patient's surgical history (e.g., appendectomy, hysterectomy), and then ask you what element of the patient's history best qualifies her for tamoxifen; answer = "history of hysterectomy" or "the patient's surgical history." → only give tamoxifen to women who do not have a uterus.
- ER (+) breast cancer has **better prognosis** than ER (-) breast cancer.

Adverse effects of tamoxifen / raloxifene?

- Increase the risk of thrombotic events (DVT, MI)
- Vasomotor symptoms (e.g., hot flashes).

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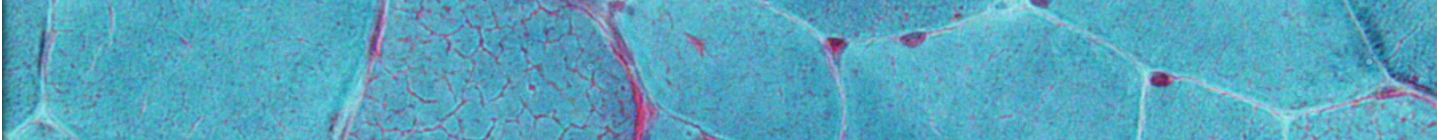


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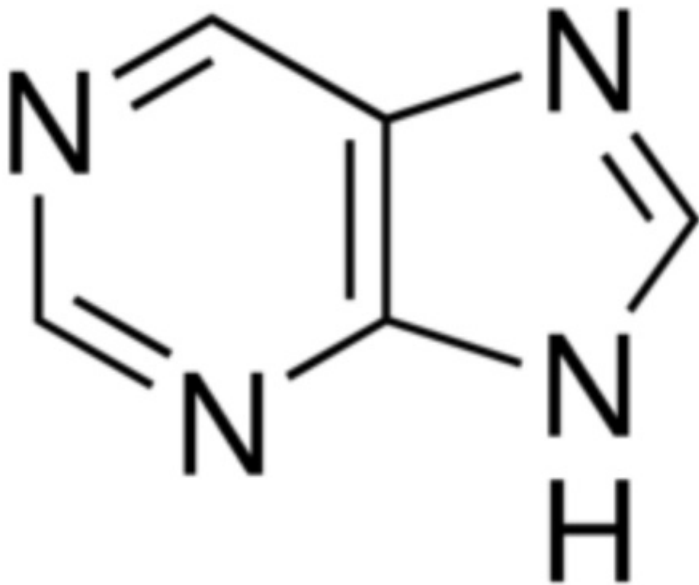
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BIOCHEMISTRY / PHARMACOLOGY

Heme/onc pharm – Purine + pyrimidine synthesis inhibitors

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Methotrexate

Most important drug from “biochemistry pharm” for USMLE is methotrexate.

MOA of methotrexate? → answer = dihydrofolate reductase inhibitor (competitive and reversible) → leads to decreased THF production.

- HY for USMLE: methotrexate, trimethoprim (an antibiotic), and pyrimethamine (used in the treatment of toxoplasmosis) all inhibit dihydrofolate reductase.

Main side-effects of methotrexate? → answer = **pulmonary fibrosis** + neutropenia (presents as mouth ulcers, fever, rash) + hepatotoxicity (increased LFTs).

Main use of methotrexate? → answer = first-line disease-modifying anti-rheumatic drug (DMARD) for rheumatoid arthritis (RA).

- When we treat RA, we need to address the symptoms (i.e., pain relief) separate from the actual disease progression (DMARDs).
- For symptom relief, we use NSAIDs, followed by corticosteroids (prednisone).
- But independent of symptom relief, patient should be started on methotrexate to slow disease progression.

How to overcome toxicity of methotrexate? → answer = folinic acid (**not** folic acid); this is aka leucovorin rescue.

5-fluorouracil (5-FU)

MOA of 5-FU? → answer = thymidylate synthase inhibitor.

How to overcome toxicity of 5-FU? → answer = oral thymidine.

Capecitabine

MOA of capecitabine? → answer = metabolized into 5-FU.

How to overcome toxicity of capecitabine? → answer = oral thymidine (same as 5-FU, since it's metabolized into 5-FU).

6-mercaptopurine (6-MP)

MOA of 6-MP? → answer = PRPP amidotransferase inhibitor (inhibits purine synthesis).

Important USMLE use for 6-MP? → answer = can have utility in the treatment of acute lymphoblastic leukemia (ALL).

Which enzyme activates 6-MP? → answer = HGPRT.

Which enzyme breaks down 6-MP? → answer = xanthine oxidase → drug **cannot** be simultaneously given to a patient on a xanthine oxidase inhibitor (i.e., allopurinol, febuxostat).

- Then you might ask: But why would a patient be on a xanthine oxidase inhibitor and 6-MP at the same time anyway? → treatment of leukemia can cause tumor lysis syndrome, which releases uric acid into the blood, causing gout. Rather than giving a xanthine oxidase inhibitor, patients can be administered a urate oxidase analogue, such as rasburicase or pegloticase, both of which breakdown uric acid directly, rather than prevent its formation.

Azathioprine

MOA of azathioprine? → answer = metabolized into 6-MP, which in turn inhibits PRPP-amidotransferase (inhibits purine

synthesis).

- Same way capecitabine is a pro-drug of 5-FU (i.e., it is metabolized into 5-FU), azathioprine is a pro-drug of 6-MP.
- Therefore the rule about not combining xanthine oxidase inhibitors with 6-MP also applies to azathioprine.

Mycophenolate mofetil

MOA of mycophenolate mofetil? → answer = inhibits IMP dehydrogenase.

Important USMLE use for mycophenolate? → answer = lupus nephritis. Pretty much the only factoid you need to know about this one.

Ribavirin

MOA of ribavirin? → answer = inhibits both RNA polymerase and IMP dehydrogenase.

What are two uses of ribavirin? → answer = hepatitis C and RSV.

- Should be noted that ribavirin is almost always the **wrong** answer when selecting the treatment for RSV bronchiolitis, but the USMLE still wants you to know that it can be used in select cases. Answer “supportive care” for RSV on the USMLE.

Major side-effect of ribavirin? → answer = hemolytic anemia.

Hydroxyurea

MOA of hydroxyurea? → answer = ribonucleotide reductase inhibitor.

Major use of hydroxyurea? → answer = increases HbF in sickle cell.

Leflunomide

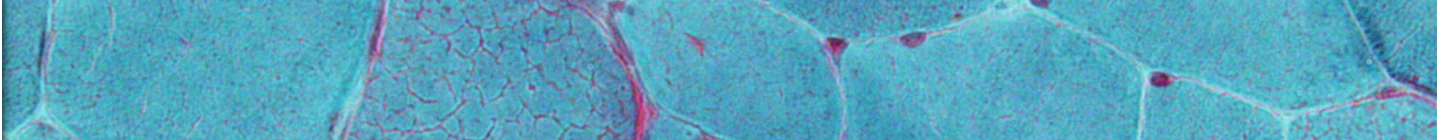
MOA of leflunomide? → answer = dihydroorotate dehydrogenase inhibitor.

Any notable use for leflunomide? → can be used in RA if methotrexate is not effective as first-line DMARD.

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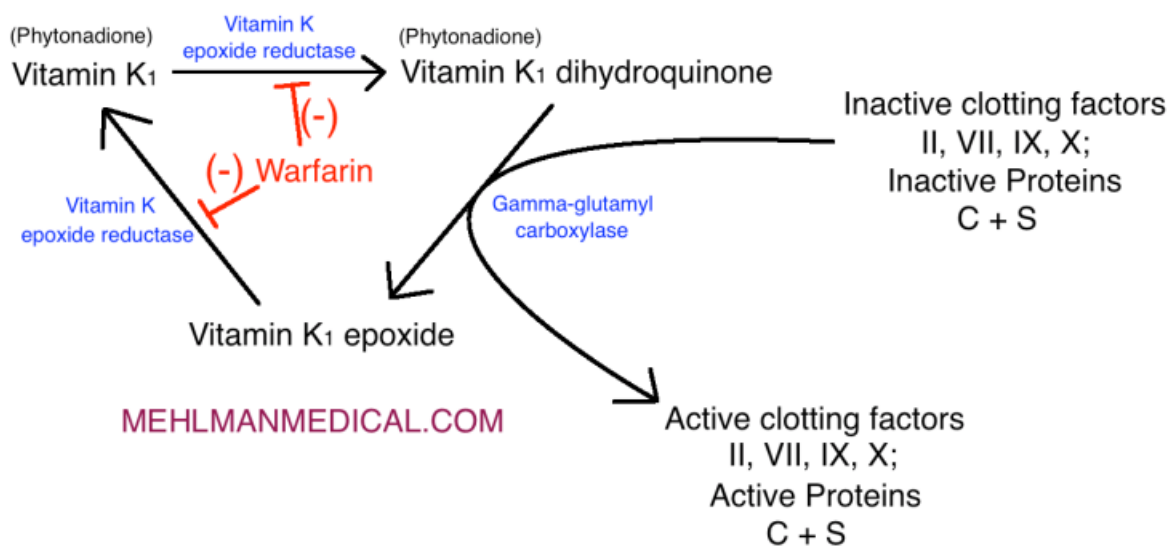
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Vitamin K and Warfarin

by MEHLMANMEDICAL



Vitamin K epoxide reductase

- Enzyme that recycles vitamin K₁ (phytonadione) so that it can be used for the gamma-carboxylation and activation of clotting factors.
- Warfarin inhibits this enzyme.
- In other words, warfarin inhibits the recycling of vitamin K₁ to its active form.
- Warfarin is often erroneously described as a “vitamin K antagonist.” In reality, warfarin doesn’t bind to and inhibit vitamin K directly, so this description is not accurate. Warfarin merely prevents the recycling of vitamin K to its active form, thereby antagonizing its effects indirectly.

Gamma-glutamyl carboxylase

- Vitamin K₁ is a cofactor for γ-glutamyl carboxylase.
- This is the enzyme that γ-carboxylates and activates clotting factors II, VII, IX and X, as well as anti-clotting proteins C & S.

- It γ -carboxylates **glutamic acid** residues on these factors.
- Vitamin K does not enable the hepatic synthesis of these factors; it merely *activates* them.

If you are asked which enzyme has $\downarrow$ activity in vitamin K deficiency, **the answer is γ -glutamyl carboxylase**.

If you are asked which enzyme has $\downarrow$ activity with warfarin use, **the answer is vitamin K epoxide reductase**.

Why prothrombin time (PT) is used to monitor warfarin's effects on vitamin K

- Warfarin is monitored with PT because Factor VII has the shortest half-life (4-7 hours) of the clotting factors, **so PT will prolong before aPTT does**.
- This means warfarin will prolong the extrinsic pathway (monitored with PT) before it prolongs the intrinsic pathway (monitored with aPTT).
- So if we want to accurately ascertain the effect or dose of warfarin in a patient, we need to look at the extrinsic pathway via PT because **it will tell us first** this information.

Vitamin K deficiency AND warfarin = $\uparrow$ PT and $\uparrow$ aPTT

- This is exceedingly HY.
- Because PT is used to monitor warfarin, people often assume warfarin $\uparrow$ PT and that's it. **But warfarin also $\uparrow$ aPTT**.
- Vitamin K deficiency is most often seen in newborns because of their sterile bowels. However it can also be due to **chronic antibiotic use** causing disruption of colonic normal flora.

Factor VII = extrinsic pathway only (PT)

Factor IX = intrinsic pathway only (aPTT)

Factors II, X = **common pathway (PT and aPTT)** $\rightarrow$ This explains why vitamin K deficiency and warfarin prolong both PT and aPTT.

Protein C inactivates Va and VIIIa (active forms) to V and VIII (inactive forms).

Protein S is a cofactor for Protein C.

International normalized ratio (INR)

- INR tells us how long our PT is compared to the expected range in the normal population.
- PT is normally 10-15 seconds. This means if an individual's PT is, e.g., 14 seconds, his or her INR is 1.0.
- We aim for a therapeutic effect of warfarin on the order of INR 2-3.
- So a patient on warfarin would be expected to have a PT of 20-45 seconds in theory.

Warfarin is broken down predominantly by CYP-2C9

- This factoid is so HY it needs its own heading.
- We normally hear about CYP-3A4 in passing. Most drugs are metabolized by this CYP. But warfarin is -2C9.

The USMLE is known to literally ask for which CYP is used for drug metabolism.

Most drugs are CYP-3A4.

Warfarin is CYP-2C9.

The microsomal ethanol oxidizing system (MEOS) is CYP-2E1.

P-450 inhibitors can cause bleeding diathesis

- **Ciprofloxacin (a fluoroquinolone), macrolides, sulfa drugs, and metronidazole** are particularly known to cause bleeding diathesis in patients on warfarin.

Skin necrosis is an adverse effect of warfarin in those who have hereditary Protein C deficiency

- If you get a question on why a patient may have experienced warfarin-induced skin necrosis, the answer is **Protein C deficiency**.
- The anti-clotting Protein C has a short half-life of 8 hours, meaning it will deplete with warfarin use long before II, IX, and X do.
- This means warfarin causes a transient hypercoagulable state (which is why it must be first bridged with heparin).
- Patients who are deficient in Protein C are therefore even more hypercoagulable, leading to greater risk of clotting in cutaneous microvasculature.

Treatment for vitamin K deficiency + the reversal of warfarin

- If the need to reverse warfarin occurs, if the scenario is not emergent (i.e., there is **NO** active bleeding or the patient doesn't require emergency surgery), **vitamin K is merely given**.
- If there **IS** active bleeding or the patient requires emergency surgery, the answer is **fresh frozen plasma (FFP) first, THEN vitamin K**.
- This is because vitamin K-mediated activation of clotting factors takes hours to days, whereas FFP works immediately, so giving vitamin K in emergency situations is futile to quickly reverse warfarin.

USMLE wants you to know vitamin K deficiency can occur in patients who are on chronic broad-spectrum antibiotics. They'll say patient on Abx is requiring a lower dose of warfarin to achieve same INR target range – why? → decrease in colonic normal flora.

Sources of vitamin K

- Exogenously through cruciferous vegetables (i.e., broccoli, Brussels sprouts, cabbage, cauliflower, kale, etc.).
- Endogenously through synthesis by gut flora.
- On the USMLE, if they ever ask you where we get most of our vitamin K from, the answer is always **via synthesis by colonic normal flora**.
- That is, if "green, leafy vegetables" and "colonic normal flora" are both answer choices, **the latter is always correct**.
- "Green, leafy vegetables," in contrast, is always the answer for **folate**.

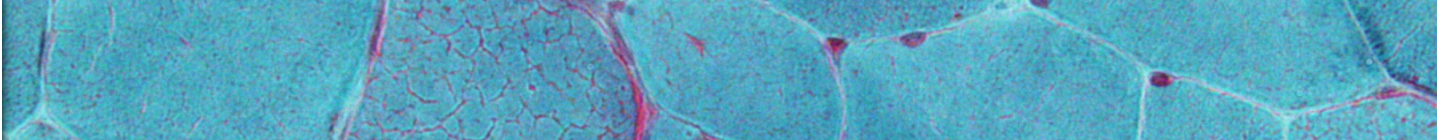


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[IMMUNOLOGY](#) / [PHARMACOLOGY](#)

Immunology pharm – Monoclonal antibodies (anti-cancer)

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HY points about each drug followed by a quiz at the end

Alemtuzumab

MOA of alemtuzumab? → answer = monoclonal antibody against CD52.

Two main uses for alemtuzumab? → answer = chronic lymphocytic leukemia (CLL) + multiple sclerosis (MS).

- “Alymtuzumab” → chronic lymphocytic leukemia

Bevacizumab

MOA of bevacizumab? → answer = monoclonal antibody against vascular endothelial growth factor (VEGF).

Two important uses for bevacizumab? → answer = colorectal cancer + neovascular macular degeneration.

Rituximab

MOA of rituximab? → answer = monoclonal antibody against CD20 on B cells.

Important use for rituximab? → answer = Hodgkin lymphoma.

Rituximab is associated with ↑ risk of which opportunistic infection? → answer = progressive multifocal leukoencephalopathy (PML), caused by JC polyoma virus.

- Natalizumab (monoclonal antibody against alpha-4 integrin in the treatment of MS) also notably increases the risk.

Trastuzumab

MOA of trastuzumab (Herceptin)? → answer = monoclonal antibody against HER2/neu.

Main use of trastuzumab? → breast cancer that has HER2/neu positivity.

Notable toxicity of trastuzumab? → answer = cardiotoxicity.

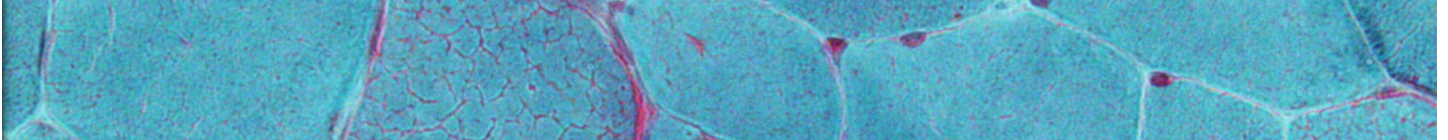
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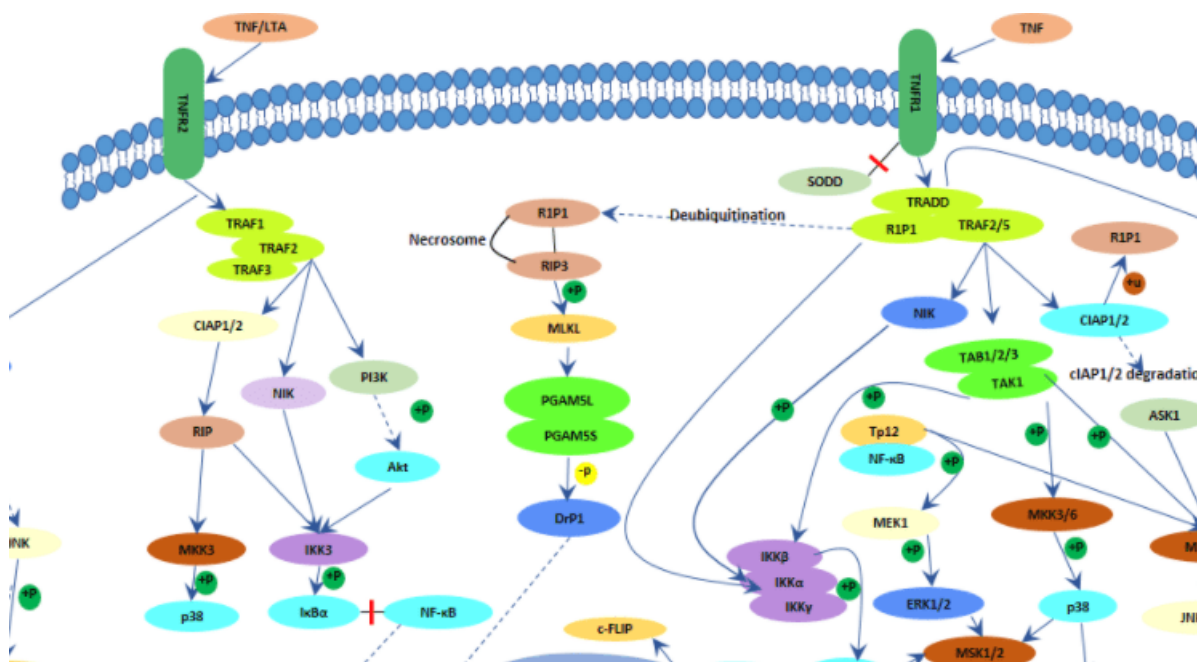
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HY points about each drug followed by a quiz at the end

Anti-TNF-alpha monoclonal antibodies

- Annoyingly, there are five you just need to memorize straight-up:
- **Adal**imumab, **in**fliximab, **cert**olizumab pegol, **goli**imumab.
- “**Adolf** on Net**Flix** is **Cert**ainly a **Gold**en piece of shit.”

MOA of the above antibodies? → answer = inhibit soluble TNF-alpha (not TNF-alpha receptor).

- **Etanercept** is equally as HY as the above antibodies. However it is not technically an antibody; it is a TNF-alpha **recombinant receptor**.
- In other words, etanercept can bind to and mop up soluble TNF-alpha floating around in the plasma.
- Remember that etanercept is the odd one out from the TNF-alpha drugs because it is a **receptor**, not an antibody.

Two notable uses of anti-TNF-alpha drugs?

1. Inflammatory bowel disease (IBD) when NSAIDs and steroids are not sufficient to control disease.
2. Second-line DMARD in rheumatoid arthritis if methotrexate alone is not effective or contraindicated.

When should anti-TNF-alpha drugs notably not be used? → answer = patients with silicosis (silicon ↓ alveolar macrophage function) because they increase the risk of TB.

- Low threshold for performing a PPD test on patients with silicosis who have pulmonary symptomatology.

Eculizumab

MOA of eculizumab? → answer = monoclonal antibody against complement protein C5.

What do we use eculizumab for? → paroxysmal nocturnal hemoglobinuria (PNH).

- PNH is characterized by episodes of hematuria first thing in the morning. It is caused by deficiency of GPI anchor or CD55/59 on RBCs, resulting in increased complement-mediated hemolysis of RBCs. Sounds weird, but it's not. This is HY for USMLE.

Ixekizumab, Secukinumab

MOA of **ixekiz**umab + **secu**kinumab? → answer = both are monoclonal antibodies against IL-17a.

How the fuck to remember these names? → “**Ixe** plays the **kaz**oo in **Seca**ucus.”

Notable use for ixekizumab and secukinumab? → answer = psoriasis.

Natalizumab

MOA of natalizumab? → answer = monoclonal antibody against alpha-4 integrin.

Notable use for **natali**zumab? → multiple sclerosis.

- **Natali**e is a woman's name. Multiple sclerosis usually occurs in women.

Natalizumab notably increases the risk for which opportunistic infection? → answer = progressive multifocal leukoencephalopathy (PML), caused by JC polyoma virus.

- Rituximab (monoclonal antibody against CD20 on B cells in Hodgkin lymphoma) also notably increases the risk.

Ustekinumab

MOA of ustekinumab? → answer = monoclonal antibody against IL-12/-23.

What is ustekinumab used for? → answer = psoriasis.

- Ustekinumab, ixekizumab, and secukinumab are all used for psoriasis.

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HY USMLE Q #957 – Renal

January 18, 2024

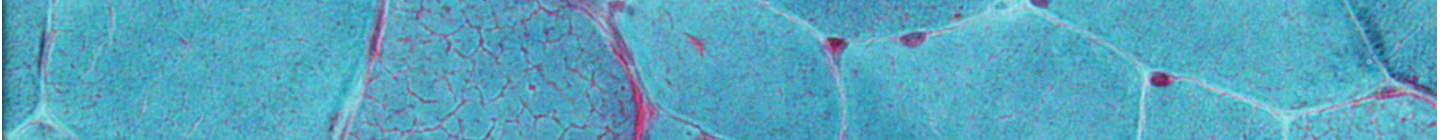


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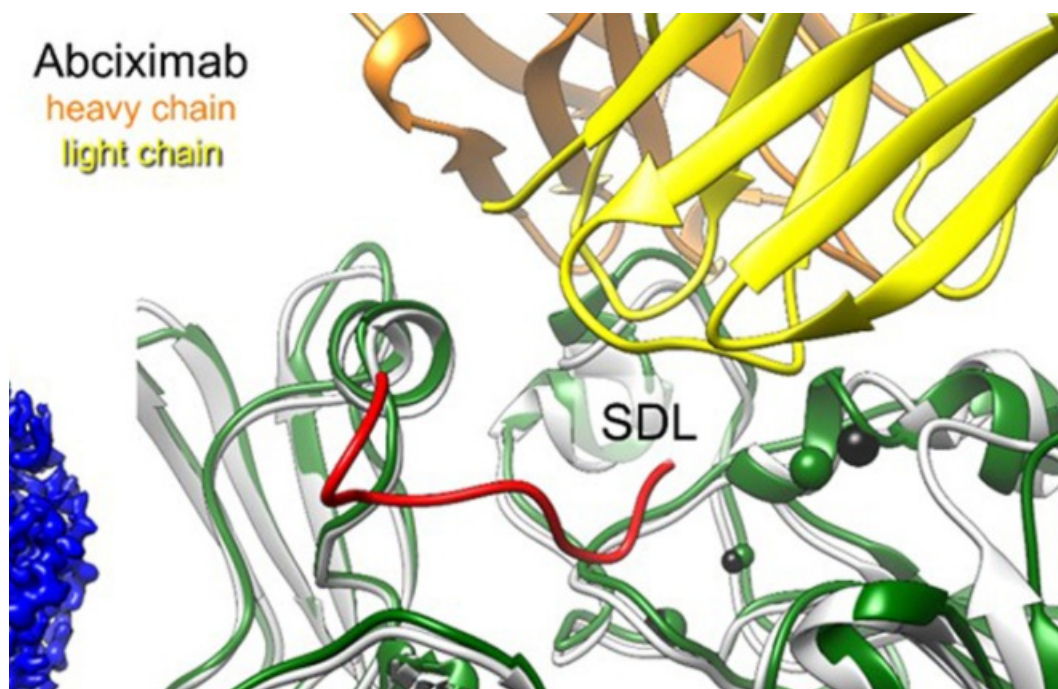
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IMMUNOLOGY / PHARMACOLOGY

Immunology pharm – Monoclonal antibodies (miscellaneous application)

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HY points about each drug followed by a quiz at the end

Abciximab

MOA of abciximab? → answer = monoclonal antibody against GpIIb/IIIa on platelets.

- GpIIb/IIIa on platelets mediate platelet **aggregation** (in contrast, adhesion is GpIb; no testable antibody targets this).
- The normal physiologic entity that binds GpIIb/IIIa on platelets is **fibrinogen**. In other words, fibrinogen is the molecule that bridges GpIIb/IIIa on adjacent platelets and enables their ability to stick together (i.e., aggregation). Therefore, USMLE wants you to know that abciximab is considered a fibrinogen analogue.
- Students often confuse abciximab with adalimumab (also HY), which is instead a monoclonal antibody against soluble TNF-alpha.
- Eptifibatide and tirofiban are two agents that are **not** monoclonal antibodies that also target GpIIb/IIIa.

What is the clinical use for abciximab? → answer = administered to patients undergoing percutaneous coronary intervention (i.e., angioplasty +/- stenting).

Basilixumab, Daclizumab

MOA of basilixumab and daclizumab? → answer = both are monoclonal antibodies against IL-2 receptor.

- Daclizumab was withdrawn from the market in 2018 after reports of encephalitis, but USMLE still wants you to know it's MOA.

What's the main use for basilixumab and daclizumab? → answer = prevention of transplant rejection.

Denosumab

MOA of denosumab? → answer = monoclonal antibody against RANK-L.

Clinical use for denosumab? → answer = advanced osteoporosis with history of pathologic fractures; also used to prevent fractures in those with bone metastases.

Omalizumab

MOA of omalizumab? → answer = monoclonal antibody against the Fc region of IgE (prevents IgE binding to FcεRI).

Clinical use for omalizumab? → answer = severe asthma refractory to all other therapies.

Palivizumab

MOA of palivizumab? → answer = monoclonal antibody against F-protein of RSV (prevents syncytia formation).

- Should be noted that palivizumab will only ever be an answer on the USMLE if the vignette asks specifically about the MOA. Or the vignette might ask which antibody could be used in theory against RSV. However if you are asked blindly how to treat RSV, **“supportive care”** is the correct answer; ribavirin and palivizumab are almost always **wrong** answers.

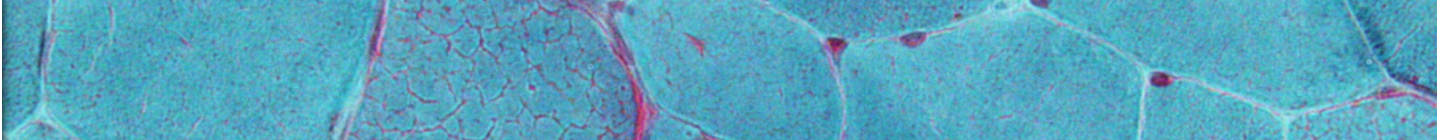
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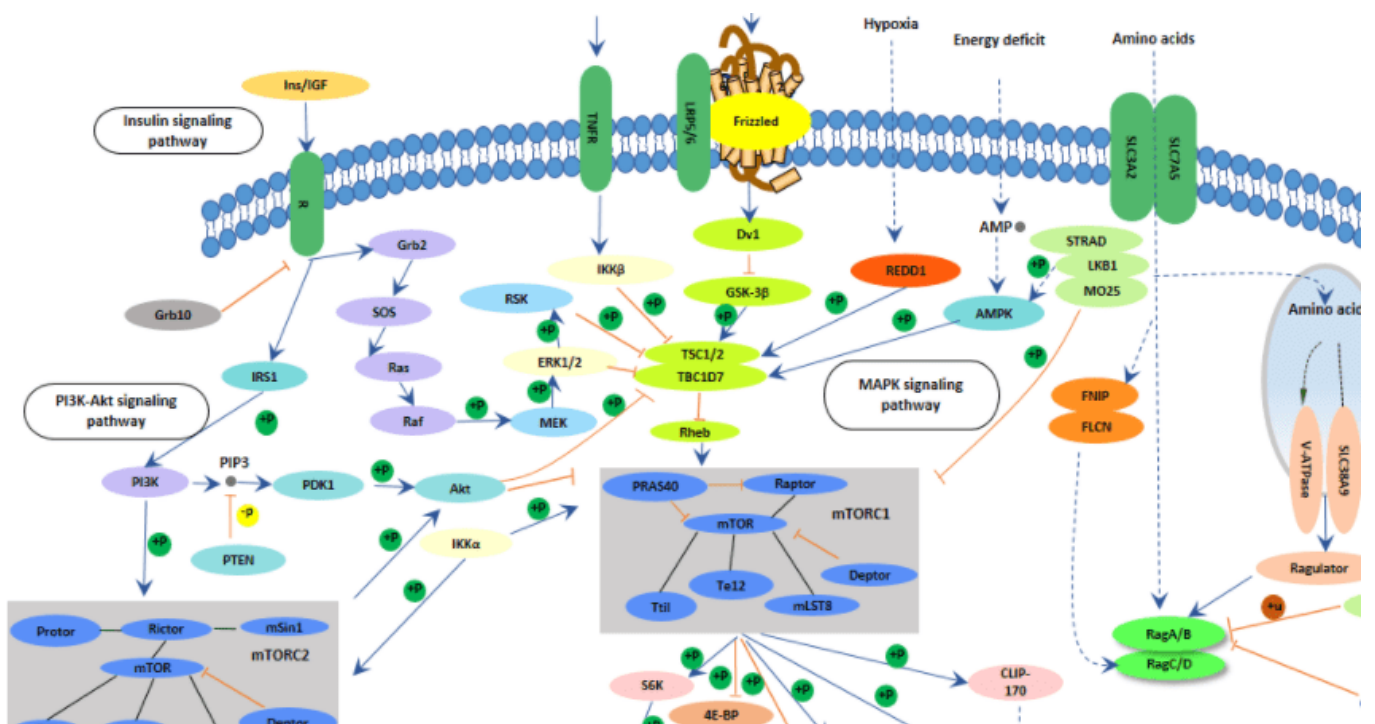
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Immunology pharm – Non-antibody immunosuppressants

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HY points about each drug followed by a quiz at the end

Cyclosporin

MOA of cyclosporine? → inhibits cyclophilin (intracellular protein) → decreases intracellular calcineurin → decreases **IL-2 transcription**.

Important toxicities of cyclosporine? → answer = nephrotoxicity (hyperkalemia), hypertension, hypertrichosis (↑ hair growth), gingival hyperplasia. USMLE wants you to know all of these. This isn't just some random list of arbitrary yieldness.

- USMLE vignette will tell you someone is on an immunosuppressant and that the creatinine has increased → answer = cyclosporine.
- Cyclosporine, verapamil (non-dihydropyridine CCB), and phenytoin (anti-convulsant that blocks Na channels) all are HY for causing gingival hyperplasia.
- Random context: On my peds rotation years ago, I sat in on a consult with a family where the kid was on cyclosporine. I remember the pediatrician looked at the gums, checked the blood pressure, and asked the mom if she noticed any changes in hair growth. I was like, "What do you know? The stuff we study actually does sometimes apply to real life."

Tacrolimus

MOA of tacrolimus? → answer = inhibits FK506 → decreases intracellular calcineurin → decreases **IL-2 transcription**.

Important side-effects of tacrolimus? → answer = **type II diabetes**, nephrotoxicity (hyperkalemia).

- USMLE likes dysglycemia for tacrolimus.
- Random context: met two people in real life who were on tacrolimus. Both had type II diabetes because of it.

Sirolimus

MOA of sirolimus? → answer = blocks mTOR → does **not** decrease intracellular calcineurin → decreases **responsiveness** to IL-2.

Important side-effects of sirolimus? → answer = dyslipidemia; does **not** cause nephrotoxicity.

Summary table

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Cyclosporine	Inhibits cyclophilin	Yes	↓ Transcription	Yes
Tacrolimus	Inhibits FK-506	Yes	↓ Transcription	Yes
Sirolimus	Inhibits mTOR	No	↓ <i>Responsiveness</i>	No

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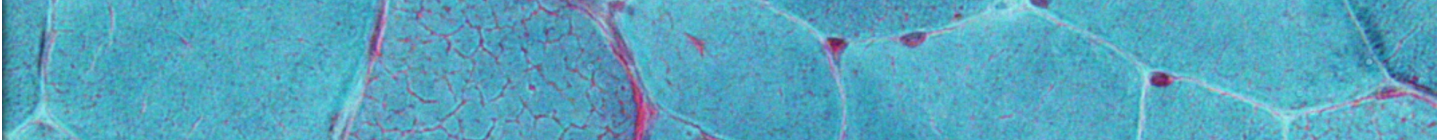


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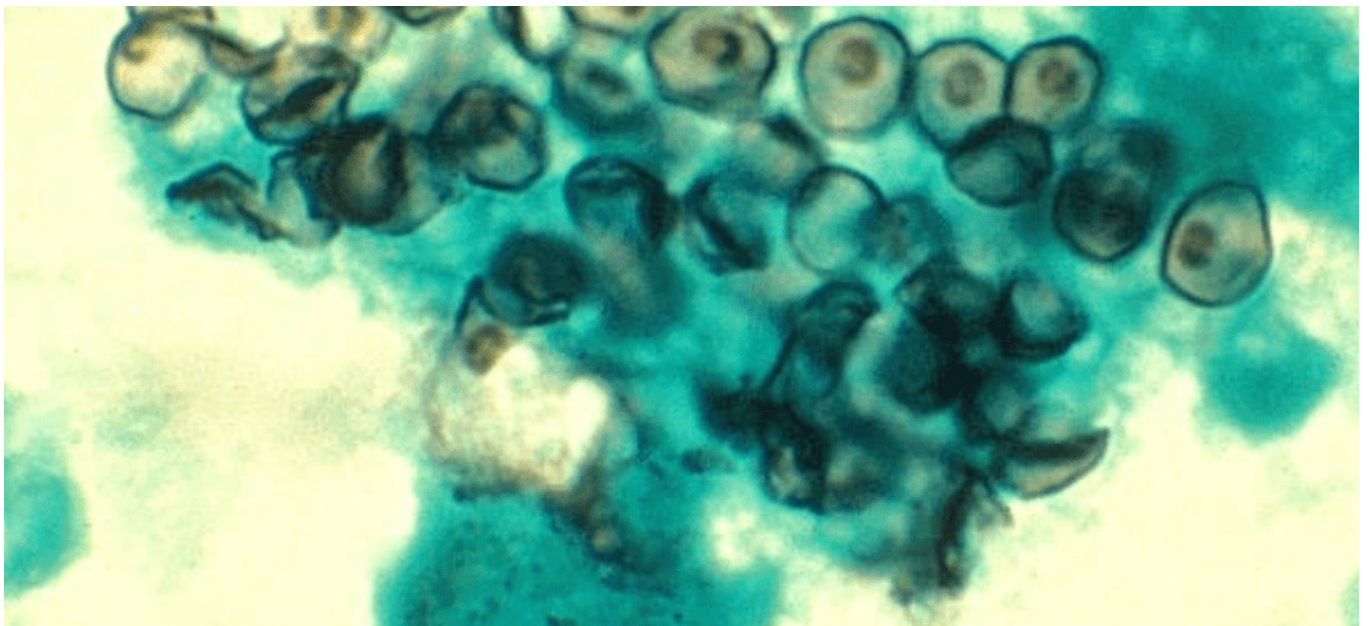
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[MICROBIOLOGY](#) / [PHARMACOLOGY](#)

Microbiology pharm – Antifungals

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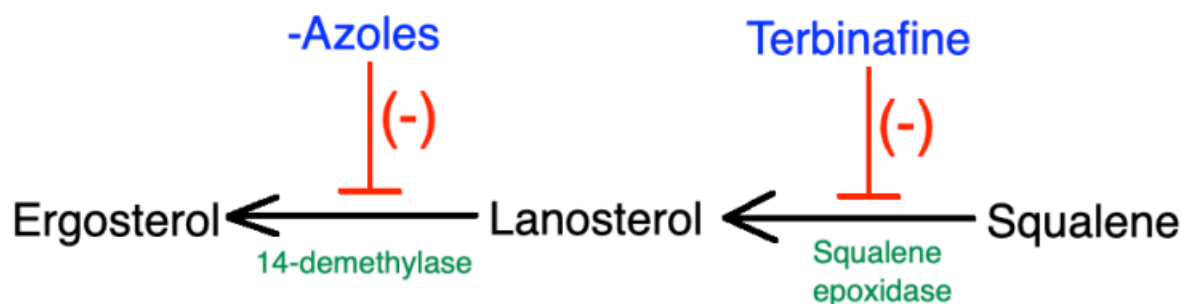
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HY points about each drug followed by a quiz at the end

-Azoles (e.g., fluconazole, itraconazole, ketoconazole)

MOA of -azoles? **(HY for Step 1)**

- Inhibit ergosterol synthesis by inhibiting 14-demethylase.
- Ergosterol is the fungal equivalent of cholesterol.



HY uses for -azoles:

- Tinea corporis (ringworm) → answer = topical -azoles (i.e., clotrimazole or miconazole).
- Candidal skin infections → answer = oral fluconazole.
- Candidal esophagitis → answer = oral fluconazole (in contrast, *oropharyngeal* candidiasis → answer = nystatin mouthwash).
- Vaginal candidiasis → answer = topical nystatin; if not effective, *then* oral fluconazole.
- Uncomplicated *non-Pneumocystis* fungal pneumonias (i.e., coccidioides, histoplasmosis, blastomycosis, mucormycosis, etc.) → answer = oral fluconazole.
 - If the patient has rigors/chills, the answer is instead amphotericin B (rigors imply fungemia + dissemination → need hard-hitting amphotericin B).
 - Some literature suggests that very mild fungal pneumonias can be observed rather than necessitating -azole Tx, but on the USMLE, always treat.
 - Prophylaxis and Tx for *Pneumocystis jirovecii* pneumonia is TMP/SMX.
- *Cryptococcal neoformans* meningitis → answer = amphotericin B + flucytosine, *then* do a fluconazole taper for one-year (yes, one year).
- Tinea pedis (athletes' foot) → topical terbinafine or topical -azoles.
 - I've seen both as answers, but the USMLE won't list both at the same time. In a vacuum, first-line is classically topical terbinafine, but yes, topical -azole is also correct.
- *Sporothrix schenckii* → answer = oral itraconazole.
- Seborrheic dermatitis → answer = ketoconazole or selenium shampoo (both are acceptable).
 - Seborrheic dermatitis presents as weeping papules along the hair lines + scaling of the scalp; in contrast, tinea capitis is classically described as a round-ish area of alopecia + scaling.
- Invasive/disseminated aspergillosis → answer = IV caspofungin or IV voriconazole.

HY side-effects of -azoles:

- Hepatotoxicity.
 - Mild transaminitis (↑ LFTs) is normal and expected; do not discontinue or decrease dose if LFTs go up slightly.
 - Risk for hepatotoxicity is why topical nystatin is used before oral fluconazole for vaginal candidiasis. Because the proper medicine is to technically check the patient's LFTs first if commencing -azoles, whereas topical nystatin can be used right away.
- **Oral** ketoconazole causes gynecomastia.
 - Ketoconazole is an antagonist at androgen receptors + blocks desmolase.
- Oral ketoconazole inhibits P-450 enzymes.
- Voriconazole causes arrhythmia.

Terbinafine

MOA of terbinafine? (HY for Step 1)

- Inhibits lanosterol synthesis by inhibiting squalene epoxidase.

When is terbinafine the answer?

- **Onychomycosis** (fungal infection of nails) → answer = **oral** terbinafine.
 - You do not need to know doses or durations of Tx, but fun fact: for fingernails, give oral terbinafine for 6 weeks; for toenails, 12 weeks.
- Tinea pedis → **topical** terbinafine (i.e., foams, sprays). Topical -azole is also acceptable and sometimes an answer (USMLE will not list both, don't worry), but in a vacuum, **topical** terbinafine is first-line for athletes' foot.

Toxicity of terbinafine?

- Hepatotoxicity.
 - Same as -azoles.
 - Check LFTs before commencing.
 - Mild transaminitis (↑ LFTs) is normal and expected; do not discontinue or decrease dose if LFTs go up slightly.

Griseofulvin

MOA of griseofulvin? (HY for Step 1)

- Inhibits microtubules.
 - USMLE also wants you to know the drugs that relate to microtubules:
 - **Prevent assembly/formation of microtubules:**
 - Vinblastine
 - Vincristine
 - Colchicine
 - -Bendazoles (i.e., mebendazole, albendazole), which are anti-helminthic agents; **do not** confuse with -azoles.
 - Griseofulvin
 - **Hyper-stabilize microtubules / prevent microtubule disassembly:**
 - Taxanes (i.e., paclitaxel, docetaxel) → odd one out, because the others all inhibit *assembly/formation* of microtubules, whereas taxanes *hyper-stabilize / prevent disassembly*.

When is griseofulvin the answer?

- Tinea capitis (cradle cap; scalp infection) → answer = **oral** griseofulvin for **patient only** (i.e., do not give to close contacts; this is on an NBME).

Nystatin

MOA of nystatin?

- Ionophore – i.e., binds to ergosterol in the fungal cell membrane and forms pores/holes.
- Amphotericin B and natamycin have same MOA.

When is nystatin the answer?

- Vaginal candidiasis → topical nystatin is used first; if ineffective, go to oral fluconazole.

- **Oropharyngeal** candidiasis → nystatin mouthwash (not swallowed).
 - Should be noted that **esophageal** candidiasis is treated with oral fluconazole, *not* nystatin mouthwash.

Adverse effects of nystatin?

- Diarrhea, abdominal pain if consumed orally.

Natamycin

MOA of natamycin?

- Ionophore – i.e., binds to ergosterol in the fungal cell membrane and forms pores/holes.
- Amphotericin B and nystatin have same MOA.

When is natamycin the answer?

- **Fungal infections in or around the eye.**
- Used as eyedrops.

Amphotericin B

MOA of amphotericin B?

- Ionophore – i.e., binds to ergosterol in the fungal cell membrane and forms pores/holes.
- Nystatin and natamycin have same MOA.

Most important point to note about amphotericin B?

- Hard-hitting antifungal used for CNS or disseminated infections.
- Often used last resort.
- Wrong answer for uncomplicated fungal infections outside the CNS.

When is amphotericin B the answer?

- CNS infections (*Cryptococcus neoformans* meningitis) → amphotericin B + flucytosine, followed by a one-year fluconazole taper.
- Disseminated fungal infections (i.e., disseminated histoplasmosis, blastomycosis, etc.).
- Fungemia (fungus in the blood) → patient will have rigors/chills + high fever.
 - For example, on USMLE if a patient has a simple fungal pneumonia, answer = oral fluconazole. But if patient has fungal pneumonia + rigors/chills + fever of 103F, answer = amphotericin B, not fluconazole.
- Fungal sepsis → SIRS + fungal infection.
 - SIRS = 2 or more of the following: temp <36C or >38C; HR >90; RR >20; WBCs <4,000 or >12,000.

Side-effects of amphotericin B?

- “Amphoterrible” → hypersensitivity-like reaction with headache, chills, and fever (“shake and bake”).
- **Hypokalemia + hypomagnesemia** → electrolyte disturbances due to renal wasting.

How to decrease risk of amphotericin B toxicity?

- Liposomal formulation → slower intravascular release.

Echinocandins (caspofungin, micafungin)

MOA of echinocandins?

- Inhibit 1-3- β -D-glucan synthase $\rightarrow$ disrupts synthesis of cell **wall**.
 - Cell **membrane**, in contrast, refers to amphotericin B, nystatin, and natamycin (ionophores).

When is caspofungin the answer?

- Notably invasive candidemia or invasive aspergillosis.
- Amphotericin B can also treat these conditions. USMLE will not list both at the same time.

Flucytosine

MOA of flucytosine?

- Fluorinated pyrimidine analogue that disrupts both RNA and DNA synthesis.

When is flucytosine the answer?

- Not an answer on its own.
- Important point is that it is combined with amphotericin B when treating *Cryptococcus neoformans* meningitis.
 - Amphotericin B + flucytosine, followed by a fluconazole taper for one year.

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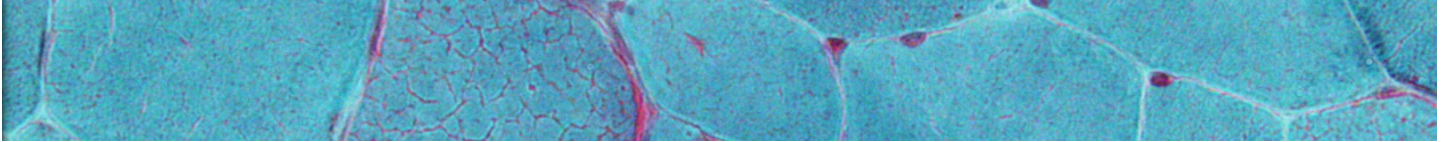


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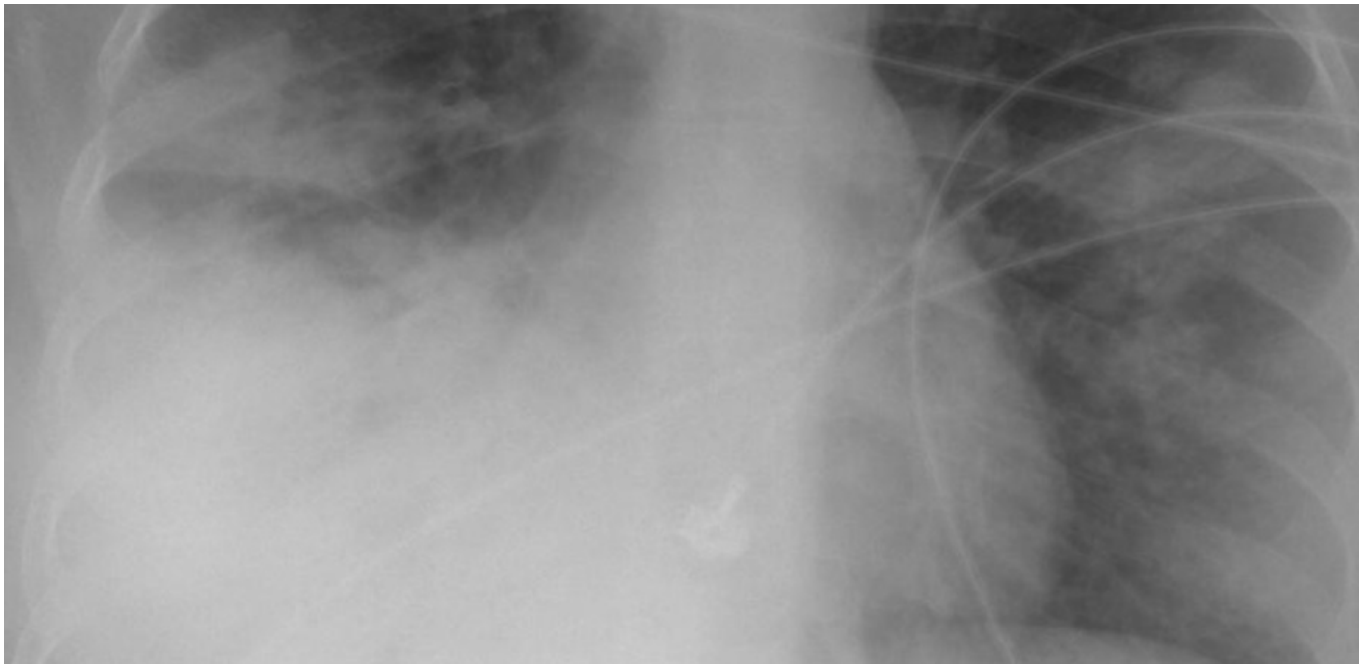
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[MICROBIOLOGY](#) / [PHARMACOLOGY](#)

Microbiology pharm – Cephalosporins (highest yield points)

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

—

You do not need to know every cephalosporin for the USMLE. The following are the highest yield ones and their respective generations.

As the generation # ascends, so does the general breadth of the organism coverage.

1st gen: cephalexin, cefazolin

Cephalexin is oral. Cefazolin is IV. Therefore, cefazolin is the inpatient form of cephalexin.

When do we use these agents?

1st generation cephalosporins **notably cover community Staph** (i.e., methicillin-sensitive *S. aureus*; MSSA) → cellulitis; erysipelas; severe impetigo (mild impetigo we use topical mupirocin).

For cellulitis and impetigo (bullous and non-bullous), *S. aureus* exceeds *Strep pyogenes* (Group A Strep). For erysipelas, Group A strep exceeds *S. aureus*.

Therefore, we need Staph coverage when dealing with skin infections.

Since 90% of MSSA produces beta-lactamase, which means **penicillin, amoxicillin, and ampicillin are not effective treatments against Staph.**

So if we choose to use a beta-lactam for Staph, we need to use beta-lactamase-resistant beta-lactams, such as **dicloxacillin** (oral; outpatient) or **flucloxacillin** (IV; inpatient). **Nafcillin** is classic for confirmed MSSA endocarditis. **Oxacillin** is used for osteomyelitis and septic arthritis.

The first-generation cephalosporins **are essentially the equivalent of the beta-lactamase-resistant beta-lactams.**

In other words, if you want to treat anything that could potentially have MSSA as the culprit (namely skin), you can give oral cephalexin as a satisfactory treatment. Otherwise, just give oral dicloxacillin (do not confuse this with **doxycycline**).

If the patient's skin presentation is severe, you can give IV cefazolin. Otherwise, give IV flucloxacillin.

Bottom line: when you hear cephalexin or cefazolin, the first thing that should go through your mind is: "those are for community Staph coverage; can be used for skin or bone stuff."

"What about if a patient has a beta-lactam allergy? How does that relate to cephalosporins?" What you need to know for USMLE:

- If patient has Hx of rash to beta-lactams (e.g., penicillin), **yes, it's okay** to give a cephalosporin. The crossover for allergy is low (<10%).
- If the patient has Hx of anaphylaxis to beta-lactams, **no, do not** give a cephalosporin. Use another agent instead depending on the condition. For skin, agents such as clindamycin, doxycycline, TMP/SMX, and linezolid are frequently used. These agents are otherwise usually reserved for MRSA of skin (vancomycin as poor skin penetration).

2nd gen: cefoxitin

Only real thing you need to know is that cefoxitin is frequently used as a prophylactic antibiotic prior to surgery, especially in pediatrics.

3rd gen: ceftriaxone, cefotaxime, ceftazidime

These agents are harder hitting and have broader coverage than the previous two generations.

When is ceftriaxone the answer? (Literally. All are HY).

- **Meningitis:**
 - Empiric Tx for bacterial meningitis is ceftriaxone + vancomycin.
 - Confirmed meningococcal meningitis can be managed with ceftriaxone alone (vancomycin is removed because it's only needed for gram-positives).
 - Rifampin, not ceftriaxone, is given to close contacts of patients with meningococcal meningitis.
- **Gonorrhea:**
 - Urethritis and the Dx is uncertain: always co-treat for both chlamydia and gonorrhea with azithromycin (or doxycycline) for the chlamydia + ceftriaxone for the gonorrhea.
 - Urethritis with confirmed gonococcus: co-treat as per above.
 - Urethritis where "nothing grows," but vignette says nothing about gonococcus: co-treat.
 - Urethritis where "nothing grows" and vignette explicitly says gonococcal testing was negative: azithro or doxy only.
 - Gonococcal arthritis presents one of two ways on USMLE:
 - 1) monoarthritis of large joint (such as the knee) in sexually active young-ish patient;
 - 2) triad of polyarthritis, tenosynovitis, and cutaneous papules in young-ish patient;
 - Treatment is ceftriaxone (+ azithro or doxy for potential asymptomatic chlamydia as well).
- **Sickle cell patients with sepsis or bone infection:**
 - Ceftriaxone has gram-negative coverage and can therefore cover *Salmonella* in sickle cell patients.
- **Patients with severe *Strep pneumo* infections:**
 - Ceftriaxone is notably effective against *Strep pneumo*.
 - New 2CK NBME Q adds vancomycin to ceftriaxone for community-acquired *Strep pneumonia* (because of increasing resistance of *Strep pneumo* to ceftriaxone, adding vancomycin, even when the infection isn't nosocomial, is considered acceptable; this is on the NBME so it's not even my opinion).
- **Spontaneous bacterial peritonitis:**
 - Three vignette types here: 1) patient with cirrhosis; 2) patient who recently underwent peritoneal dialysis; 3) patient with nephrotic syndrome.
 - Patient will have fever + diffuse abdominal pain + a fluid wave; antibiotic treatment? → answer = ceftriaxone.
- **Patients with sepsis (2CK level):**
 - Sepsis = SIRS + source of infection.
 - SIRS = systemic inflammatory response syndrome = 2 or more of the following 4 variables:
 - Temperature <36C or >38C;
 - HR >90;
 - RR >20;
 - WBCs <4,000 or >12,000.
 - So for instance, patient has pneumonia + sepsis; Tx? → ceftriaxone common answer (not only covers *S. pneumo* well but patient is also septic).
 - Prostatitis + pyelonephritis are classically treated with ciprofloxacin (a fluoroquinolone), however **if the patient is septic**, then ceftriaxone can be the answer (cipro is still correct, even in sepsis, but the Q won't list both).
 - I.e., 38F + CVA tenderness + fever 38.5C + HR 100; answer = ceftriaxone; student says, "Wait, I thought ciprofloxacin was how we treat pyelo." → Yeah, it is. But the patient's **septic**, which is why ceftriaxone is correct here.
- **Broad-spectrum antibiotic coverage for nosocomial infections:**
 - Patient with hospital-/ventilator-acquired pneumonia or other type of nosocomial infection will receive broad-spectrum antibiotic coverage. Ceftriaxone is a classic agent used in combination with vancomycin.

When is cefotaxime the answer?

- Really HY to know that we **do not** give ceftriaxone to children < age 6 if we can help it (increases displacement of

bilirubin from albumin and can cause jaundice). Age 6 or greater, yes, ceftriaxone is okay.

- Cefotaxime therefore is the ceftriaxone equivalent in peds. USMLE assesses the age:
 - 12-month-old girl with sickle cell + missed dose of penicillin prophylaxis + septic; Tx? → answer = cefotaxime; wrong answer is ceftriaxone (on one of the 2CK IM forms, where both are listed).
 - 6-year-old girl + sepsis; Tx? → answer = ceftriaxone (cefotaxime not listed; on newer 2CK NBME).

When is ceftazidime the answer?

- Covers *Pseudomonas* (HY). low, in your mind, ceftazidime should = *Pseudomonas* coverage.
- Can be combined with vanc for nosocomial infections or general broad-spectrum coverage.

4th gen: cefepime

When is cefepime the answer?

- Hard-hitting agent that covers *Pseudomonas*. That is: both ceftazidime (3rd gen) and cefepime (4th gen) cover *Pseudomonas*.
- Combined with vanc for broad-spectrum nosocomial coverage. Q on one of the 2CK forms gives nosocomial infection, then says, “cefepime is commenced; which other agent should be added?” Answer = vancomycin. Easy because they asked for vanc rather than cefepime, but the point is that this combo is HY for broad-spectrum nosocomial coverage.

5th gen: ceftaroline, ceftobiprole

When would ceftaroline or ceftobiprole be the answers? → newer 5th gen cephalosporins that cover MRSA.

- Have never seen these asked on any NBME material, but they’re important to mention because they cover MRSA and complete the discussion of the cephalosporin generations.
- low, when you hear ceftaroline or ceftobiprole, you just need to think, “Oh, those are 5th gen. They’re the only cephalosporins that cover MRSA.”
- They can be used, same as 3rd and 4th gen, for broad-spectrum nosocomial coverage in combination with vanc.

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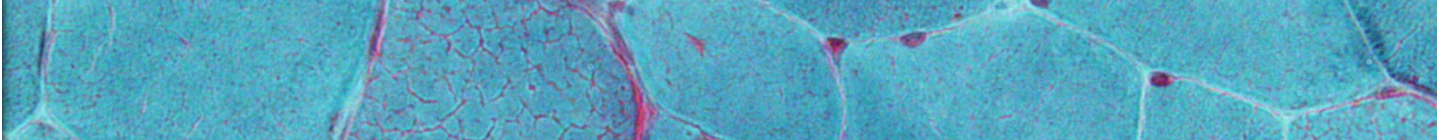


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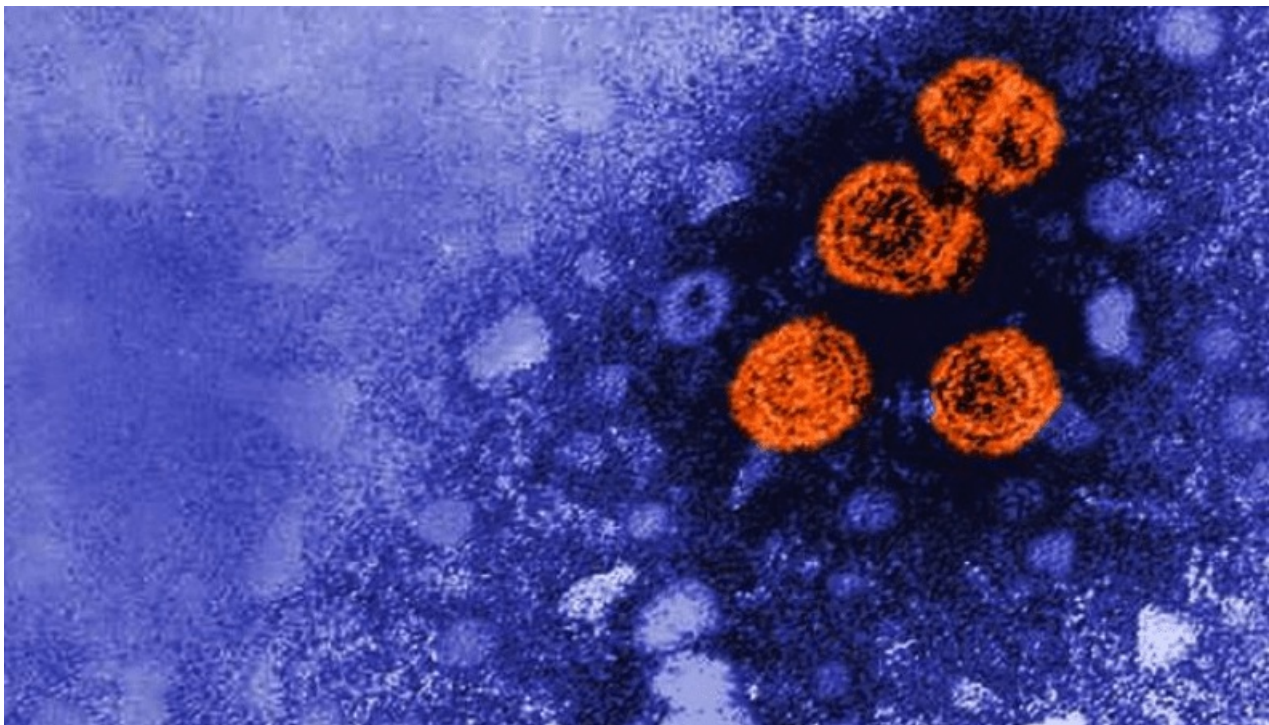
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[MICROBIOLOGY](#) / [PHARMACOLOGY](#)

Microbiology pharm – Hepatitis B drugs + HY serology

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Hepatitis B

- Many drugs used for hepatitis B. USMLE does not expect you to memorize all of them.
- Questions on the USMLE will be more focused on understanding hepatitis B *serology*.

BEATT → Hepatitis **B**, **E**ntecavir, **A**defovir, **T**elbivudine, **T**enofovir

Entecavir, Telbivudine

MOA of entecavir and telbivudine?

- Competitively inhibit HepB DNA polymerase.
- Nucleoside analogues.

Adefovir, Tenofovir

MOA of adefovir and tenofovir?

- Inhibit reverse transcriptase.
- Adenosine nucleotide analogues.
- Tenofovir is also used for HIV as part of [HAART therapy](#).

Interferon-alpha

MOA of interferon-alpha as antiviral therapy?

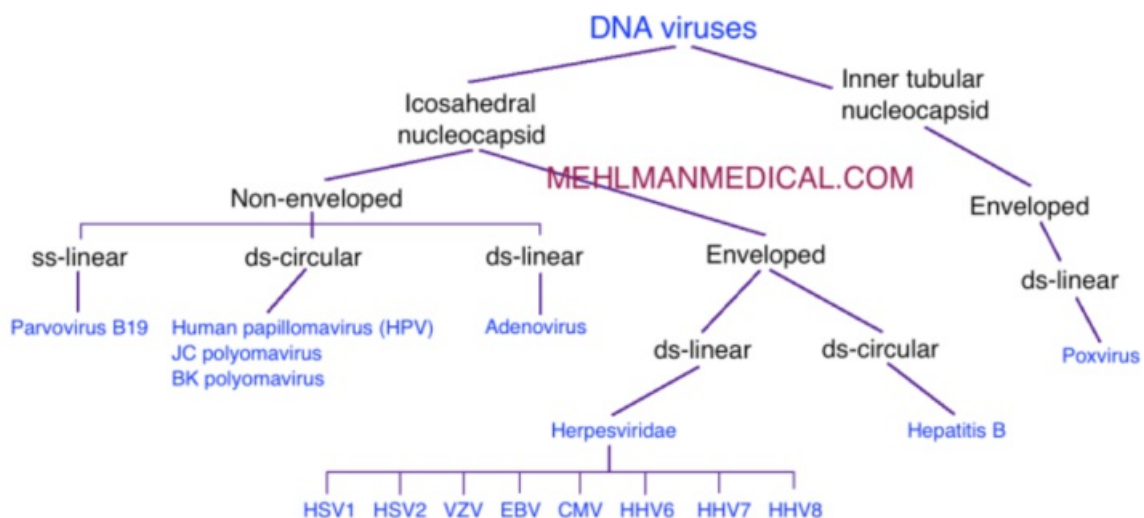
- Causes degradation of viral mRNA, thereby inhibiting protein synthesis.
- Different formulations, including pegylated-interferon-alpha, which can be used for both HepB and C.

Adverse effects of interferon-alpha?

- Neutropenia (mouth ulcers, sore throat, fever).
- Depression.

What do I need to know about the microbiology for hepatitis B?

- USMLE specifically wants you to know that it is DNA, enveloped, and circular (yes, they ask this).



What do I need to know about HepB serology?

Hepatitis B serology interpretation

	HBsAg	HBsAb	HBcAb	HBcAb IgM	HBcAb IgG	HBeAg
Acute infection (low infectivity)	+	-	+	+	-	-
Acute infection (high infectivity)	+	-	+	+	-	+
Chronic infection (low infectivity)	+	-	+	-	+	-
Chronic infection (high infectivity)	+	-	+	-	+	+
Immune (previous infection) (<6 months)	-	+	+	+	-	-
Immune (previous infection) (>6 months)	-	+	+	-	+	-
Immune (vaccinated)	-	+	-	-	-	-
Window period	-	-	+	+	-	-
Susceptible (not immune)	-	-	-	-	-	-

- HBs – surface antibody
 - A marker of immunity.
 - **Positive if immune; negative if not immune.**
 - This should pretty much be the first marker you look at when interpreting HBV lab results. You can immediately just say, “Okay, the patient is immune.” Or, “Ok, he/she is not immune.”
- HBsAg – surface antigen
 - A marker of infectivity.
 - **If positive, indicates current infection, either acute or chronic.**
 - If (+), means the patient cannot be immune because he or she clearly has a current infection.
- HBeAg – “e” antigen
 - Marker of robust infectivity.
 - Correlates with a high level of viral replication.
 - **If positive, means person with HepB can easily transmit virus to a non-immune individual.**
 - Helps guide clinical management.
- HBcAb – core antibody
 - Marker of acute, chronic, or resolved HBV infection.
 - **If positive, means person has had the actual disease before, either now or in the past.**
 - Cannot be positive if the patient has never been infected.
 - **Negative if vaccinated.**
- HBcAb IgM
 - Subclass of HBcAb.
 - Positivity indicates recent infection with HBV (<6 mos).
 - **Positive during the window period.**
- HBcAb IgG

- Subclass of HBcAb.
- Marker of past or current infection.
- If HBsAg is also positive (and HBcAb IgM is negative), indicates chronic HepB infection.

As the patient begins to develop immunity to HepB, he or she will start producing HBsAb. At the same time, as he or she begins to clear the virus, HBsAg will decrease.

The short time period between when HBsAg has been reduced to undetectable levels, while simultaneously, HBsAb has not yet risen to detectable levels, denotes the window period.

If the USMLE asks you which serum marker is most reliable during the window period, the answer is HBcAb IgM because it's the only one that's positive.

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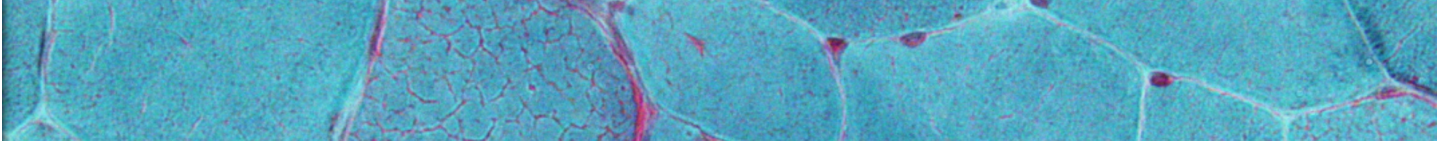


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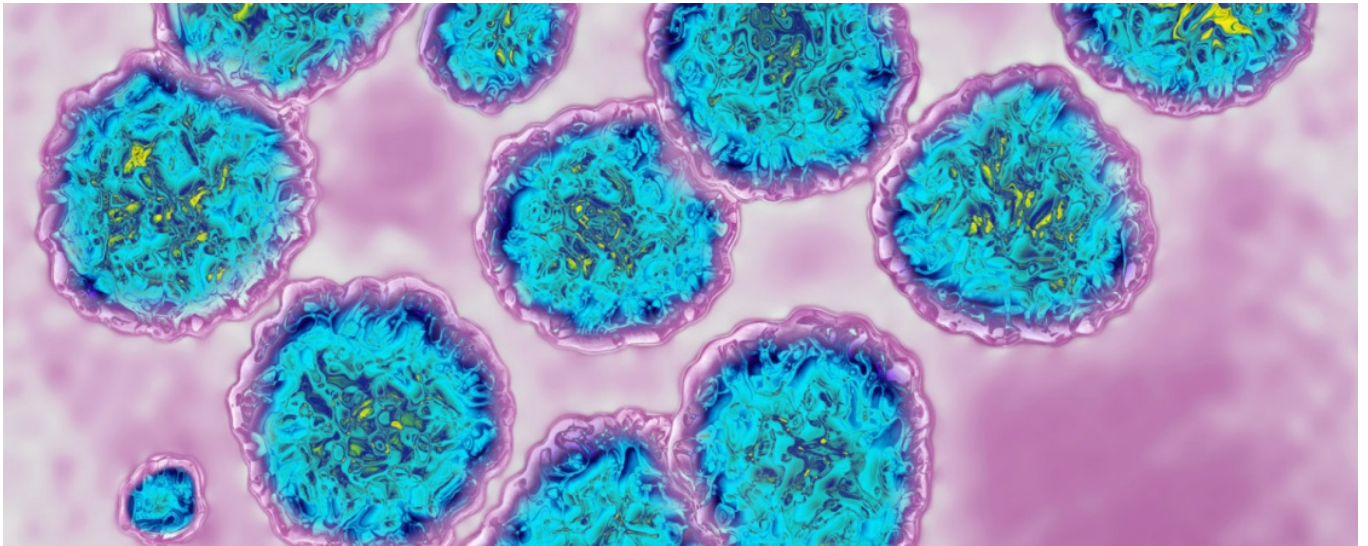
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Hepatitis C

Ledipasvir, Ombitasvir, Velpatasvir

OVAL → Ombitasvir, Velpatasvir, -Asvir, Ledipasvir

MOA of Ledip**asvir**, Ombitas**vir**, Velpatas**vir**?

- Inhibit NS5A, which is a phosphoprotein used in RNA replication.

Sofosbuvir, Dasabuvir

MOA of Sofosb**uvir**, Dasab**uvir**?

- Inhibit NS5**B**, which is an RNA-dependent RNA polymerase.

Grazoprevir, Simeprevir

MOA of grazop**previr**, simep**previr**?

- Inhibit NS3/4A, which is a viral protease.

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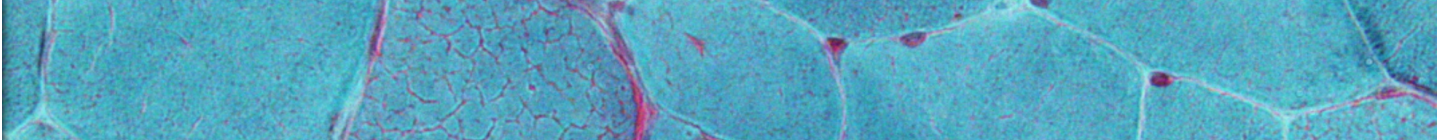


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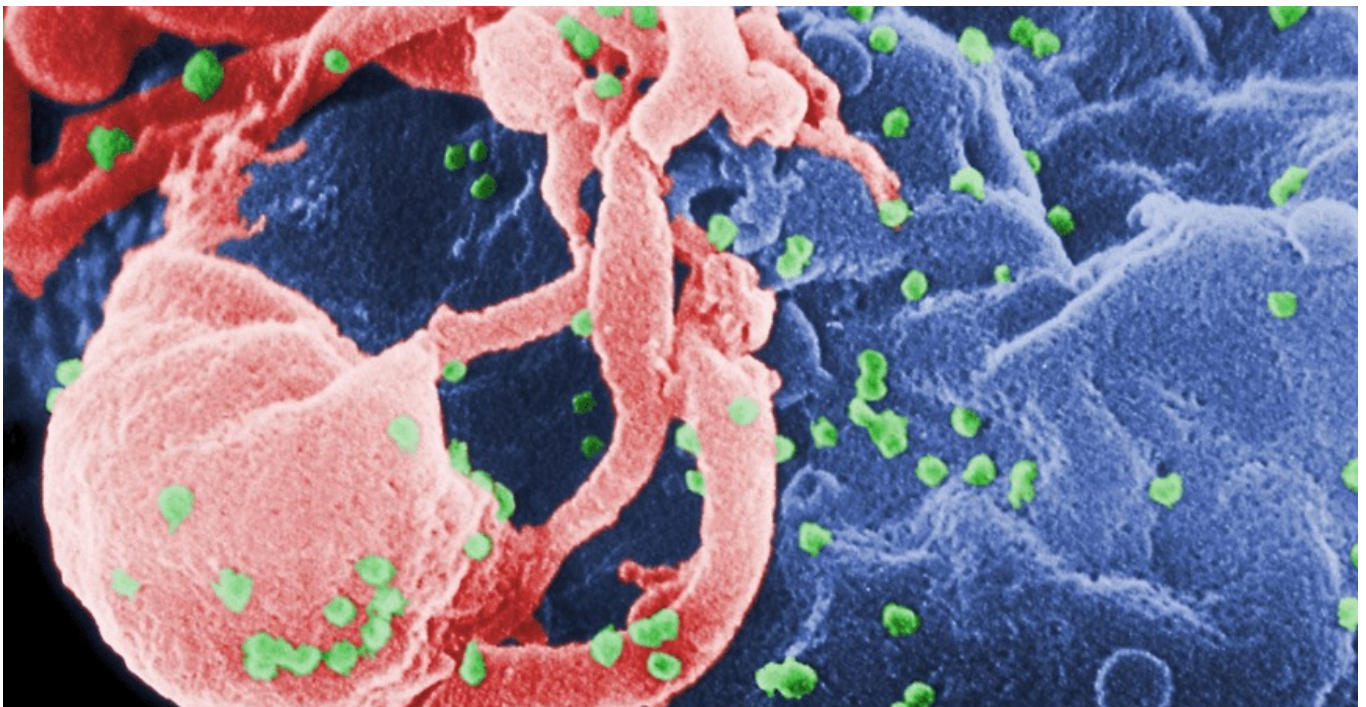
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[Simplified mechanism of HIV entry into CD4 T cell](#)

1. HIV virion produces Gp160 envelope protein via the *env* gene.
2. Gp160 is cleaved into Gp120 and Gp41 (correct, the numbers don't add up).
3. Gp120 is expressed outwardly from the HIV virion cell membrane; it connects to Gp41, which is transmembrane.
4. Gp120 binds to CD4 receptor on CD4+ T cells.
5. This binding enables the cell membranes of the HIV virion and T cell to fuse.
6. After binding to CD4, Gp120 then binds to the T cell co-receptor CCR5 or CXCR4. Without this co-binding, the T cell may sometimes shed the HIV virion particle. Hence this co-binding is a crucial step.
7. Once Gp120 is bound to both CD4 and either CCR5 or CXCR4 on the T cell, Gp41 on the HIV virion is released from its metastable location and inserts itself into the CD4+ T cell membrane.

Entry/fusion inhibitors

Maraviroc

MOA of maraviroc?

- Binds to and inhibits CCR5 on the CD4 + T cell.
- This prevents binding of Gp120 and therefore HIV entry/fusion.

Enfuvirtide

MOA of enfuvirtide?

- Binds to and inhibits Gp41 on the HIV virion.
- This prevents Gp41 from binding to the T cell membrane and therefore HIV entry.

Integrase inhibitors

- Inhibit HIV integrase enzyme.
- Integrase enables the HIV DNA (after it's been produced from RNA via HIV reverse transcriptase) to be incorporated into the DNA of the host cell genome.

Raltegravir

- Can cause ↑ creatine kinase (CK).

Nucleoside reverse-transcriptase inhibitors (NRTIs)

- Nucleoside = nitrogenous base (A, C, G, T) + a sugar (ribose if RNA; deoxy-ribose if DNA).
- Nucleotide = nucleoside + phosphate group(s).
- NRTIs competitively prevent nucleotides from being added to the growing HIV DNA strand during reverse transcription (RNA → DNA).
- The NRTI inserts itself into the growing DNA strand but lacks a 3' OH, so the DNA chain terminates.
- NRTIs require phosphorylation for activation.

Side-effects common to most NRTIs?

- Bone marrow suppression (↓ WBCs, RBCs, platelets; ↑ MCV).
- Lactic acidosis.

Abacavir

- Causes hypersensitivity reaction in patients who are HLA-B5701 positive.
- Sounds really fucking weird, yes. But this is asked for some magical reason.
- 42M + recently diagnosed with HIV + is to be started on abacavir; what to do first? → answer = HLA-B5701 testing.
- 42M + HIV + is positive for HLA-5701; which drug can patient not receive? → answer = abacavir.

Didanosine

- Can cause pancreatitis.

Emtricitabine

- Safe during pregnancy.
- Patients who are pregnant and taking efavirenz (will discuss this below; an NNRTI not safe during pregnancy) can be switched off of efavirenz and onto emtricitabine.

Zidovudine

- Classically the HIV drug given intrapartum to pregnant women.
- Given to neonates for six weeks, starting within 12 hours of birth if mother is HIV + in order to prevent vertical transmission (on 2CK NBME).

Lamivudine, Stavudine

- Just be aware of their names as additional NRTIs, but nothing special about these you need to know for USMLE.

Tenofovir

- The only nucleotide reverse-transcriptase inhibitor you need to know for HIV.
- The other NRTIs are all nucleosides.

Non-nucleoside reverse transcriptase inhibitors (NNRTIs)

- “**N**ever **E**ver **D**eliver Nucleosides” → Nevirapine, Efavirenz, Delavirdine.
- Non-competitively inhibit reverse-transcriptase (i.e., do not cause DNA chain termination via incorporating competitively).
- Do not require phosphorylation for activation (in contrast to NRTIs).

Side-effects common to the NNRTIs?

- Rash.
- Hepatotoxicity.

Efavirenz

- Can cause wild/vivid dreams.
- Not safe during pregnancy. Switch to emtricitabine.
- Varenicline (smoking cessation agent) is the other agent known to cause wild dreams for USMLE.

Protease inhibitors

- All end in -navir.

- HIV-produced proteins must be cleaved into functionally smaller proteins via protease (*pol* gene); protease inhibitors prevent these cleavages.

Side-effects common to the HIV protease inhibitors?

- Hyperglycemia (insulin resistance + type II diabetes).
- Lipodystrophy (fat redistribution with Cushing syndrome-like presentation).

What is a notable point about HIV patients who also need treatment for TB?

- Rifampin used for TB is a P-450 inducer and will decrease serum concentrations of protease inhibitors. Rifapentine is used instead of rifampin in these patients.

Indinavir

- Can cause thrombocytopenia.

Ritonavir

- P-450 inhibitor. Can “boost” serum levels of other HIV drugs taken as part of HAART therapy by decreasing rate of metabolism.

Highly active anti-retroviral therapy (HAART)

- Three drugs: Two NRTIs **plus** either an NNRTI or a protease inhibitor.
- For some reason, a HY question USMLE asks is if someone with HIV should be on a two- or three-drug regimen, and the answer is always three-drug.
- HAART should be commenced **as soon as** someone is diagnosed with HIV, regardless of CD4 count. In other words, the wrong answer is commencing HAART only after CD4 count drops to a certain level. The correct answer is, yes, commence HAART right away.
- Pregnant women should be on HAART, yes. This is the most important way to prevent vertical transmission to the fetus during delivery.
- 32F + pregnant; Q asks how to best prevent vertical transmission → correct answer = HAART therapy.

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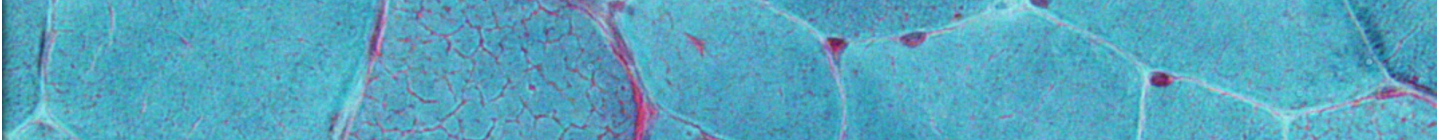


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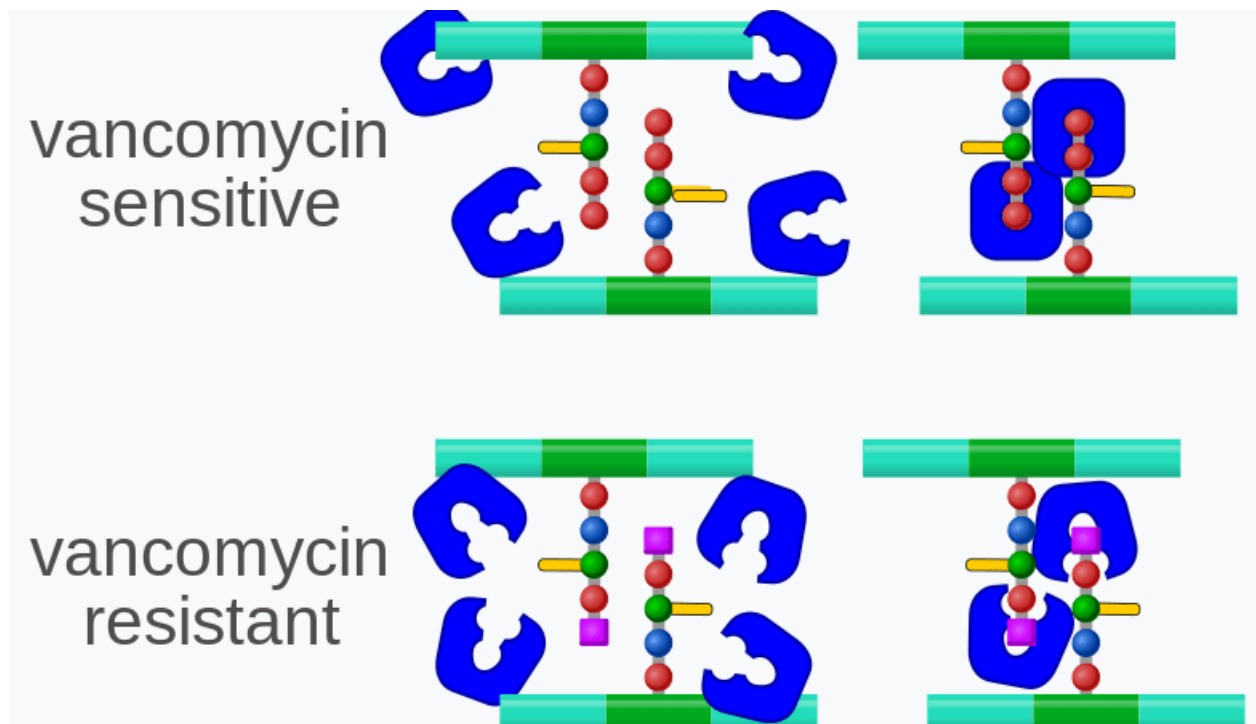
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HY points about each drug followed by a quiz at the end

Beta-lactam HY points

MOA of beta-lactams (e.g., penicillin)

- d-Ala-d-Ala analogues that bind to penicillin-binding proteins (i.e., transpeptidase).
- Inhibit cell wall synthesis by blocking transpeptidase-mediated crosslinking of peptidoglycan.

Mechanism of organism resistance

- Beta-lactamase production.
- MSSA is resistant to methicillin-class beta-lactams because it produces beta-lactamase.
- “How does MSSA → MRSA?” → answer = altered penicillin-binding protein (PBP2); wrong answer = beta-lactamase production. Since MSSA is already resistant to beta-lactamase, that means becoming MSSA → MRSA must be a different mechanism.

Relevance of the methicillin-class

As I talk about in the [cephalosporin post](#), 90% of community Staph (i.e., MSSA, not MRSA) produce beta-lactamase, so penicillin, amoxicillin, and ampicillin will almost always **not** be effective.

- The only time these drugs can be considered for MSSA is if a beta-lactamase inhibitor is added (i.e., amoxicillin/clavulanate [Augmentin]; or ampicillin/sulbactam).
- But the point is: do not straight-up give penicillin or amoxicillin for an infection (usually skin) that could potentially be community Staph.

The methicillin-class agents (i.e., dicloxacillin, flucloxacillin, nafcillin, oxacillin, cloxacillin) are highly steric, so their beta-lactam ring is resistant to beta-lactamase. Therefore these drugs can be used against MSSA.

Skin infections like cellulitis, erysipelas, and impetigo can all be MSSA, which is why penicillin and amoxicillin are the **wrong** treatments.

- MSSA exceeds *Strep pyogenes* (Group A Strep) as the causal organism for cellulitis and impetigo (bullous and non-bullous).
- Group A Strep exceeds MSSA for erysipelas, but MSSA still possible.

Classic outpatient Tx for cellulitis, erysipelas, and severe impetigo: oral dicloxacillin or oral [cephalexin](#). The inpatient equivalents are IV flucloxacillin and IV cephazolin, respectively. First-generation cephalosporins (i.e., cephalexin and cephazolin) are essentially the equivalent of the methicillin-class beta-lactams in that they can cover MSSA.

As stated above, if MSSA → MRSA, the mechanism of resistance is “altered penicillin-binding protein,” **not** “production of beta-lactamase,” because MSSA *already produces* a beta-lactamase.

Beta-lactam efficacy

First and foremost, it should be noted that beta-lactams are considered to be very efficacious drugs. If we *can* use one, we should, because they do the job well. How does that point apply most saliently to real-life medicine?

- Empiric Tx for endocarditis (i.e., before we know the organism) is gentamicin PLUS either vancomycin or ampicillin/sulbactam.
- The gentamicin covers gram-negatives (rods); the vancomycin covers gram-positives (including MRSA).
- However, let's say the culture comes back positive for MSSA (i.e., we *can* use a methicillin-class beta-lactam); the question is:
 - Do we continue the vancomycin in order to minimize the chance of vancomycin resistance (i.e., stopping a drug early can increase this chance), OR do we switch to nafcillin?

- And the answer is: yes, switch to nafcillin (6 weeks).
- This is because beta-lactams are very efficacious. Vancomycin isn't very efficacious; it just happens to be useful against MRSA if that's what the patient has.

If patient has syphilis (normally treated with penicillin) but has history of anaphylaxis, if the patient's condition is latent-secondary, late-secondary, tertiary, or neurosyphilis, the answer is **desensitize and give penicillin**. You might say, "Why can't we just treat with another agent?" It's not that we can't in theory; it's just that penicillin is so efficacious that the benefit of desensitizing and giving it anyway exceeds that of just giving a non-beta-lactam.

Synergism with aminoglycosides

- "Beta-lactams are synergistic with aminoglycosides" is a quote you should remember.
- Beta-lactam will disrupt the cell wall → then aminoglycoside enters cell and disrupts protein synthesis (30s). Just memorize it.

Important allergies

- If Hx of rash to beta-lactam, yes, can still give cephalosporin as alternative (<10% crossover allergy).
- If Hx of anaphylaxis to beta-lactam, **no, cannot** give cephalosporin; use other agent instead (e.g., doxycycline). Exception is advanced syphilis (or pregnant patient with syphilis), in which the patient is desensitized to penicillin.
- Classically cause allergic interstitial nephropathy (aka tubulointerstitial nephritis, or just interstitial nephritis).
 - 55M + treated with nafcillin for MSSA endocarditis + now has rash + urinalysis shows WBCs on dipstick; Dx? → answer = interstitial nephropathy.
 - Although classic, *patient need not present with a rash*. The key is beta-lactam, cephalosporin, or NSAID + now patient has WBCs (eosinophils) in the urine.
 - Methicillin not used clinically because ↑↑ risk of interstitial nephritis.
 - Do not confuse this with acute tubular necrosis (dirty/muddy brown granular casts in someone on gentamicin).

Piperacillin, ticarcillin

- Hard-hitting beta-lactams that cover *Pseudomonas*.
- Tazobactam or clavulanate added as beta-lactamase inhibitors.
- **Piperacillin/tazobactam** (Piptaz) is a HY combo for broad-spectrum nosocomial coverage.
 - E.g., 30M + hospital-acquired pneumonia; how do we treat? → Piptaz + vanc is acceptable combo (covers both *Pseudomonas* as well as MRSA for nosocomial infections).

Carbapenems

- **DIME** → (i.e., **D**oripenem, **I**mipenem, **M**eropenem, **E**rtapenem).
- Imipenem combined with cilastatin (i.e., imipenem/cilastatin), which decreases renal clearance + maintains serum levels.

MOA of carbapenems?

- Similar in structure to beta-lactams except a sulfur atom in position 1 is replaced with a carbon atom + an unsaturation has been introduced.
- Binds to penicillin-binding proteins (i.e., transpeptidase).
- Inhibits cell wall synthesis by blocking transpeptidase-mediated crosslinking of peptidoglycan.

When do we use carbapenems?

- Extremely hard-hitting agents reserved for multi-drug-resistant infections.
- If antibiotics are given for pancreatitis, carbapenems are notably effective.
- Cover *Pseudomonas*; however ertapenem is weak against *Pseudomonas*.

Important side-effects?

- Impipenem causes seizures.

Aztreonam

MOA of aztreonam?

- Mono-bactam antibiotic (similar to beta-lactams) in that disrupts cell wall synthesis.
- Binds to and inhibits penicillin-binding protein 3 (PBP3).

When do we use aztreonam?

- **Can be used safely in persons with beta-lactam allergy.**
- **Only covers gram-negatives.**

Vancomycin

MOA of vancomycin? → answer = disrupts cell wall synthesis by binding to d-Ala-d-Ala portion of cell wall precursors.

Use for vancomycin?

- Classic for MRSA
- Ampicillin-resistant Enterococcus
- Used empirically for meningitis (vanc + ceftriaxone), endocarditis (gentamicin + vanc).

When is vanc notably not effective in terms of MRSA Tx?

- Not good for skin. If patient has MRSA skin infection, use clindamycin, doxycycline, linezolid, or TMP/SMX.

Mechanism of resistance to vanc?

- d-Ala-d-Ala becomes d-Ala-d-Lac in the cell wall.
- Ala = alanine; Lac = lactate.

Toxicity of vanc? → **NOT Red** → **N**ephrotoxicity, **O**totoxicity, **T**hrombophlebitis, **Red man syndrome** (histamine-induced flushing).

- Anti-histamine + slower infusion vanc → decreased risk of Red man syndrome.

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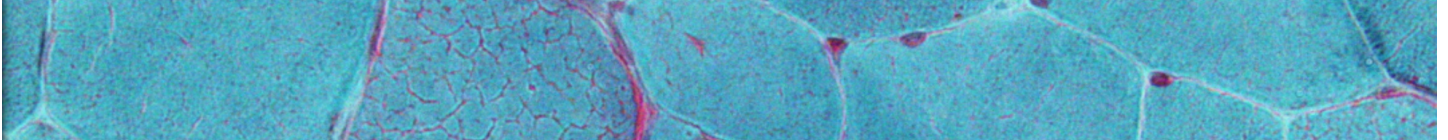


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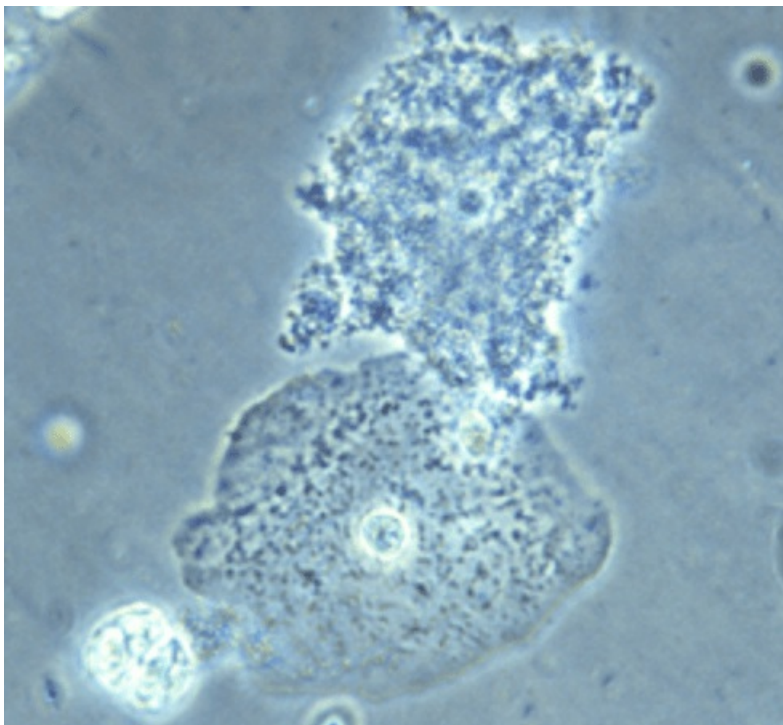
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HY points about each drug followed by a quiz at the end

Polymyxins (i.e., Colistin)

MOA of polymyxins?

- Cationic polypeptide tail that acts as detergent on cell membrane → leakage of cellular components + death.

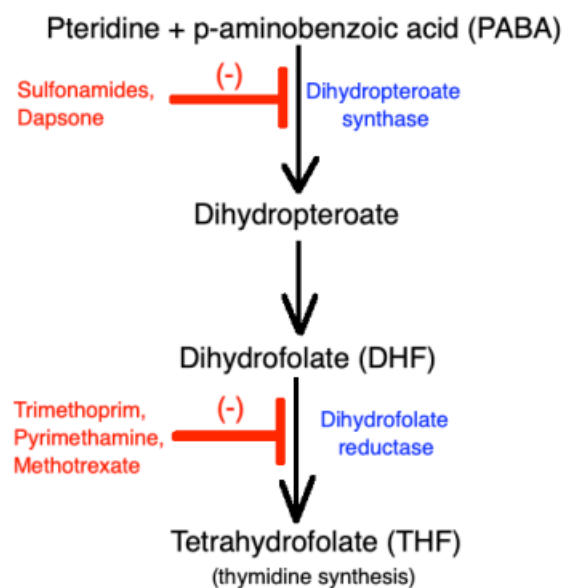
Use for polymyxins?

- Last resort for multi-drug-resistant gram-negatives (i.e., *Pseudomonas*).
- Not effective against gram-negative diplococci (i.e., *Neisseria gonorrhea/ meningitides*).
- Used as part of Thayer-Martin agar for culturing *Neisseria*.
 - Thayer-Martin = Vancomycin + Polymyxin + Nystatin (VPN client).

Toxicity of polymyxins? → answer = neurotoxicity, ototoxicity.

Sulfonamides ("sulfa drugs"; i.e., sulfamethoxazole, sulfadiazine)

MOA of sulfonamides? → answer = inhibit dihydropteroate synthase.



Highest-yield uses for sulfonamides?

- Sulfamethoxazole as part of trimethoprim/sulfamethoxazole (TMP/SMX) → used for many different types of infections, but notably for *Pneumocystis jirovecii* pneumonia (PJP) in immunocompromised patients (both prophylaxis and treatment).
- TMP/SMX classic treatment for simple UTIs.
- TMP/SMX can be used for MRSA skin infections (vanc has poor skin penetration).
- TMP/SMX is prophylaxis for toxoplasmosis (however treatment is sulfadiazine/pyrimethamine).

Important side-effects of sulfa drugs?

- Type I hypersensitivity skin eruptions (IgE crosslinking on mast cells); in increasing order of severity:
 - **Morbilloform rash** (maculopapular; measles-like)
 - **Erythema multiforme** (skin lesions with multiple shapes)
- Type III hypersensitivity (due to immune complex deposition):
 - Serum sickness → rash presenting **3-5 days after** administration of drug (if immediate, the allergy is type I instead).
 - Presents as arthritis or **erythema nodosum**.
 - **Arthus reaction** is a type of serum sickness that is localized to the skin at an injection site (if the

drug was injected).

- Erythema nodosum → **not** a rash; EN is a panniculitis (inflammation of subcutaneous fat) classically of the shins, but can also appear on the arms; USMLE will ask the hypersensitivity type this is (answer = III).
- Type IV hypersensitivity (due to T cell response):
 - **Stevens-Johnson syndrome**
 - Positive Nikolsky sign (sloughing of skin with friction)
 - Diffuse bullae; resemble pemphigus vulgaris.
 - Treat in a burns unit. Use of steroids and IVIG are controversial.
 - **Toxic epidermal necrolysis** (>30% body surface area involvement); often fatal.
 - Treat in a burns unit. Use of steroids and IVIG are controversial.
- Membranous glomerulonephritis (MG):
 - Sulfonamides classically cause MG (nephrotic syndrome; “spike and dome” appearance of subepithelial immunoglobulin and complement deposition).
 - Do not confuse this with aminoglycosides (gentamicin, amikacin) causing acute tubular necrosis (muddy/dirty brown granular casts).

Dapsone

MOA of dapsone? → same as sulfonamides (dihydropteroate synthase inhibitor), but structurally distinct.

When do we use dapsone?

- Clarithromycin + ethambutol +/- dapsone is a combo used for treatment of *Mycobacterium avium intracellulare*.
- Dapsone can be used in Leprosy.

Important side-effect of dapsone?

- **Classically causes hemolysis in G6PD deficiency.**

Trimethoprim, pyrimethamine, methotrexate

MOA of these three agents? → answer = dihydrofolate reductase inhibitors.

- Methotrexate is **not** an antibiotic; it is a DMARD for RA and also an immunosuppressant, but you need to know these three drugs are all dihydrofolate reductase inhibitors.
- Pyrimethamine is used with sulfadiazine to treat Toxoplasmosis (TMP/SMX is prophylaxis).

When do we use trimethoprim?

- Almost always used as the TMP/SMX combo, which has numerous uses. As mentioned above, classic for *Pneumocystis jirovecii*, simple UTIs, and MRSA skin infections.

When do we use pyrimethamine?

- Sulfadiazine + pyrimethamine used to *treat* toxoplasmosis (prophylaxis = trimethoprim/sulfamethoxazole).

Daptomycin

MOA of daptomycin? → answer = forms pores in cell membrane → “disrupts the charge barrier.”

When do we use daptomycin?

- **Notably effective against *Enterococci*.**

- Treatment for *Enterococci* = ampicillin first; if resistant, go to vancomycin; if resistant, use another agent like daptomycin or the streptogramins.

Side-effects of daptomycin?

- Increased creatine kinase (CK) and myopathy.

Fluoroquinolones (e.g., ciprofloxacin, levofloxacin)

MOA of fluoroquinolones?

- Inhibit DNA gyrase (prokaryotic topoisomerase II/IV)
- Topoisomerase is an enzyme that induces a “nicking” in the DNA during replication in order to prevent supercoiling.
- Inhibition of topoisomerase → nicking cannot occur → DNA supercoils + breaks → cell death.
- Should be noted that DNA gyrase specifically refers to the *prokaryotic* enzyme. For instance, etoposide and teniposide are chemotherapeutic topoisomerase II inhibitors, but they do *not* inhibit DNA gyrase.

HY uses for fluoroquinolones?

- **Prostatitis** (ciprofloxacin)
- **Pyelonephritis** (ciprofloxacin)
 - If patient is **septic**, ceftriaxone can be used (also on NBME).
- Pneumonia
 - Empiric Tx for community-acquired pneumonia (CAP): azithromycin (macrolide).
 - Empiric Tx for CAP *if patient has received antibiotics in the past 3 months*: levofloxacin (fluoroquinolone).
 - Empiric Tx for CAP (inpatient; non-ICU): fluoroquinolone, OR azithromycin + a beta-lactam.
 - Empiric Tx for CAP (inpatient; ICU): beta-lactam, PLUS either a fluoroquinolone or azithromycin.
- Diverticulitis
 - Classic Tx is metronidazole PLUS a fluoroquinolone or Augmentin (amoxicillin/clavulanate).

HY side-effect of fluoroquinolones?

- Cartilage damage (tendonopathy).
 - 82M + recently treated for prostatitis + goes metal detecting a lot + sore ankle; what’s the MOA of the drug he was treated with? → answer = inhibition of topoisomerase II.

Metronidazole

MOA of metronidazole? → answer = causes the formation of toxic metabolites that damage DNA.

When do we use metronidazole? → **GET GAP** on the **Metro**

- **G***iardia lamblia* (Giardiasis)
- **E***ntamoeba histolytica* (amoebiasis)
- **T***richomonas vaginalis* (trichomoniasis)
- **G***ardnerella vaginalis* (bacterial vaginosis)
- **A**naerobes below the diaphragm
 - Classic Tx is metronidazole PLUS a fluoroquinolone or Augmentin (amoxicillin/clavulanate).
 - Anaerobes above the diaphragm = **clindamycin**.
- **P***ylori* → *second-line* Tx for *H. pylori* (first-line = CAP = clarithromycin, amoxicillin, PPI; if urease breath test still positive 4 weeks after treatment, assume Abx resistance and replace the clarithromycin and amoxicillin with metronidazole, tetracycline, bismuth). In other words:

- First-line for *H. pylori* = CAP = clarithromycin, amoxicillin, PPI (e.g., omeprazole).
- Second-line for *H. pylori* = metronidazole, tetracycline, bismuth, PPI.
- P used to be able to stand for pseudomembranous colitis (*C. difficile*), but metro is now **no longer used first-line** as of February 2018; oral vancomycin is now used first-line across the board, **not** oral metronidazole.

Any notable side-effect of metronidazole?

- Disulfiram-like reaction if patient drinks alcohol (nausea, vomiting).

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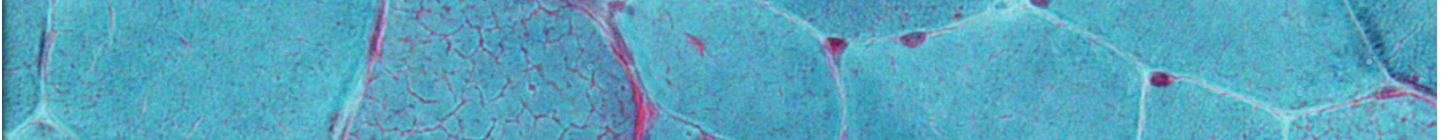


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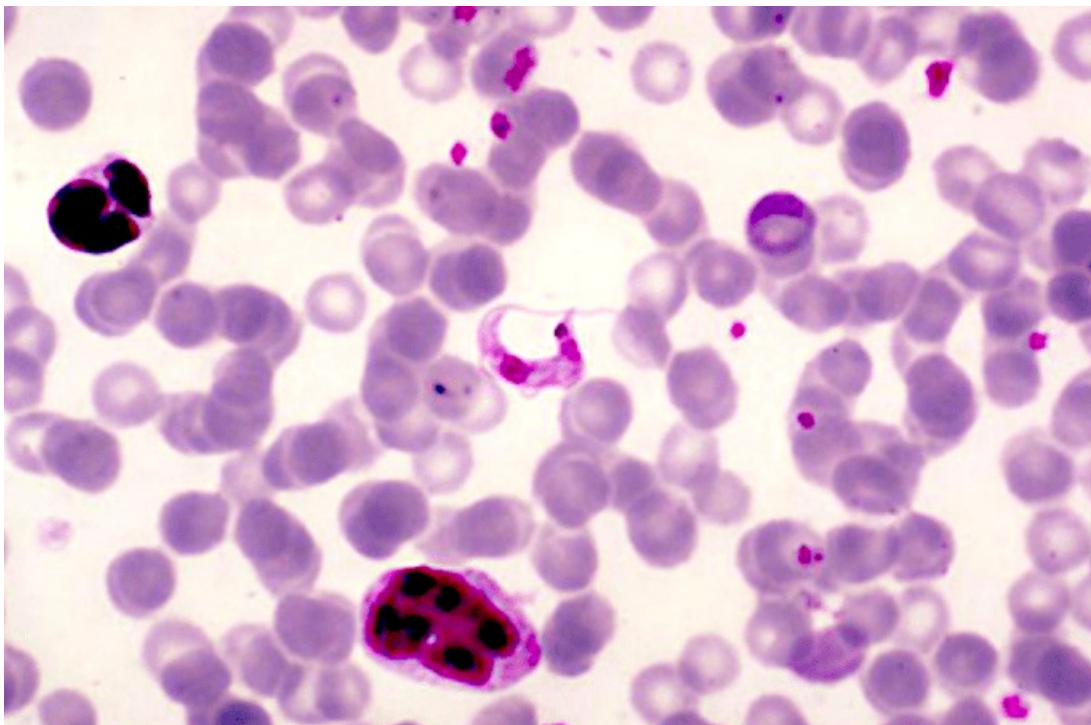
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Microbiology pharm – HY mixed antimicrobials III

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Anti-protozoals

Chloroquine, mefloquine, primaquine

- Tx for malaria (caused by the protozoan *Plasmodium spp.* transmitted by the *Anopheles* mosquito).
- Malaria presents as hemolytic anemia, hepatosplenomegaly, waxing/waning fever patterns, and cerebral malaria (encephalopathy; *P. falciparum*).

MOA of chloroquine?

- Inhibits *Plasmodium* heme polymerase.

Important point about resistance:

- Prophylaxis and treatment of malaria.
- Chloroquine resistance notably high. Mefloquine resistance is emerging.
- 16F + went to Africa + was taking chloroquine prophylactically + still got malaria anyway; why? → answer = “resistance to chloroquine”; the wrong answer is “noncompliance with medication.”

What's special about primaquine as compared with chloroquine and mefloquine?

- Primaquine is used for *P. vivax* and *ovale* to kill hypnozoites (hepatic latent form of organism).
- 20F + went to Africa + got malaria + took chloroquine for Tx and it was effective + weeks later now has resurgence; why? → answer = “presence of extra-erythrocytic form of organism,” which refers to hypnozoites.
- 24F + goes to Africa + is taking primaquine; why? → answer = “primaquine kills hypnozoites.”

Atovaquone

- Atovaquone + proguanil = a known Tx regimen for malaria.
- Atovaquone + azithromycin = newer Tx regimen for Babesia (malaria-like hemolytic disease **but limited to the USA**).
- 23F + went to Africa + came back with hemolytic disease; Dx? → answer = malaria.
- 23F + lives in Pennsylvania + tick bite + hemolytic disease; Dx? → answer = Babesia.

Permethrin

MOA of permethrin?

- Inhibits sodium channel inactivation → depolarization + cell death.

When is permethrin the answer?

- Topical permethrin used for scabies and lice.
- 44M + lived in homeless shelter for 4 months + has itchy hands + photo shows you red dots on hands + Q says topical antifungals were not effective; Tx? → answer = permethrin.
 - “Linear burrows” are classic descriptor for scabies infection, but exam images you get won't really resemble this.
- 9F + head lice; Tx? → answer = permethrin.

Sodium stibogluconate

- Tx for leishmaniasis (caused by the protozoan *Leishmania donovani* transmitted by *Phlebotomus* sandfly).
- Antimony compound.
- Cutaneous leishmaniasis: most common; presents as ulcer on the arm.

- Visceral leishmaniasis (kala azar): starts as cutaneous, then progresses to hepatosplenomegaly + pancytopenia.
- Liposomal **amphotericin B** (antifungal) is another Tx for leishmaniasis.

Nifurtimox

- Tx for Chagas disease (caused by the protozoan *Trypanosoma cruzi* transmitted by reduviid bug ["kissing bug"]).
- Acute Chagas disease: swollen nodule at site of bite (called Romana's sign if on the eyelid; chagoma if elsewhere).
- Chronic Chagas disease (untreated): **dilated cardiomyopathy, secondary achalasia**, megaureter, toxic megacolon.

Suramin + melarsoprol

- Tx for African trypanosomiasis (aka African sleeping sickness, caused by protozoan *Trypanosoma brucei* transmitted by the Tsetse fly).
- Insomnia at night + somnolence during the day; fevers.

Pyrimethamine

MOA of **pyrimethamine**? → inhibits dihydrofolate reductase (same as trimethoprim and methotrexate, albeit these drugs all have very different uses).

- Tx for toxoplasmosis = sulfadiazine + pyrimethamine.
- In contrast, *prophylaxis* for toxo (AIDS with CD4 count <100) = trimethoprim/sulfamethoxazole (TMP/SMX). Patients are usually incidentally covered for toxo prophylaxis because TMP/SMX is commenced for *Pneumocystis jirovecii* prophylaxis at CD4 count <200.
- If cat / cat feces not listed as an answer, choose **consumed pork**.
 - 35F + pregnant + has toxo + Q asks how she got it; cat not listed as an answer; student says "Wtf? I thought it was cats." → answer = pork.

Helminthicides (anti-helminth agents)

Mebendazole, albendazole

MOA of the -bendazoles? **(HY for Step 1)**

- Inhibit microtubules.
 - USMLE also wants you to know the drugs that relate to microtubules:
 - **Prevent assembly/formation of microtubules:**
 - Vinblastine
 - Vincristine
 - Colchicine
 - -Bendazoles (i.e., mebendazole, albendazole), which are anti-helminthic agents; **do not** confuse with -azoles.
 - Griseofulvin
 - **Hyper-stabilize microtubules / prevent microtubule disassembly:**
 - Taxanes (i.e., paclitaxel, docetaxel) → odd one out, because the others all inhibit *assembly/formation* of microtubules, whereas taxanes *hyper-stabilize / prevent disassembly*.

When is a -bendazole the answer?

- Tx for many helminth infections, notably the nematodes (roundworms), such as *Ascaris lumbricoides*, *Enterobius vermicularis*.
- Albendazole is used for neurocysticercosis (caused by the cestode [tapeworm] *Taenia solium*, acquired via consumed pork).
- Students tend to freak out about which anti-helminth agent they're supposed to use when; it's to my observation across all NBME material that the USMLE will not give you multiple anti-helminths as answers; the correct answer will be the only anti-helminth listed. Qbanks may be a bit more nitpicky, but these are not the NBME.

Pyrantel pamoate

When is pyrantel pamoate the answer?

- Tx for many nematode infections.
- Think of this as akin to mebendazole → "Oh ok. Weird drug. But all I need to know is: if I see it, it's basically just the same as mebendazole."

Praziquantel

When is praziquantel the answer?

- Tx for various cestode and trematode (flake) infections.
 - HY cestode Tx: cysticercosis (in contrast, neurocysticercosis is albendazole).
 - HY trematode Tx: schistosomiasis.

Diethylcarbamazine

When is diethylcarbamazine the answer?

- Tx for Filarial infections (nematode infections transmitted by insects).
- Notable use: *Wuchereria bancrofti* (elephantiasis).

Ivermectin

When is ivermectin the answer?

- Tx for onchocerciasis (nematode infection caused by *Onchocerca volvulus* [river blindness]), transmitted by the black fly).
- Can be given orally to treat scabies (normally Tx for scabies is topical permethrin).

—

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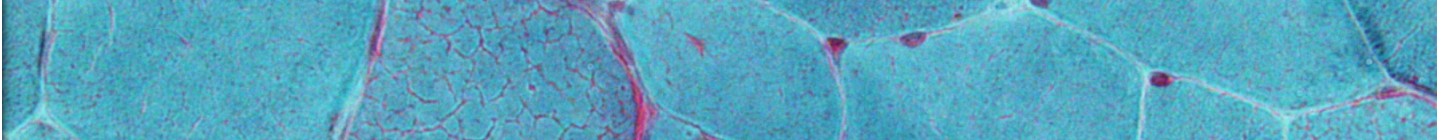


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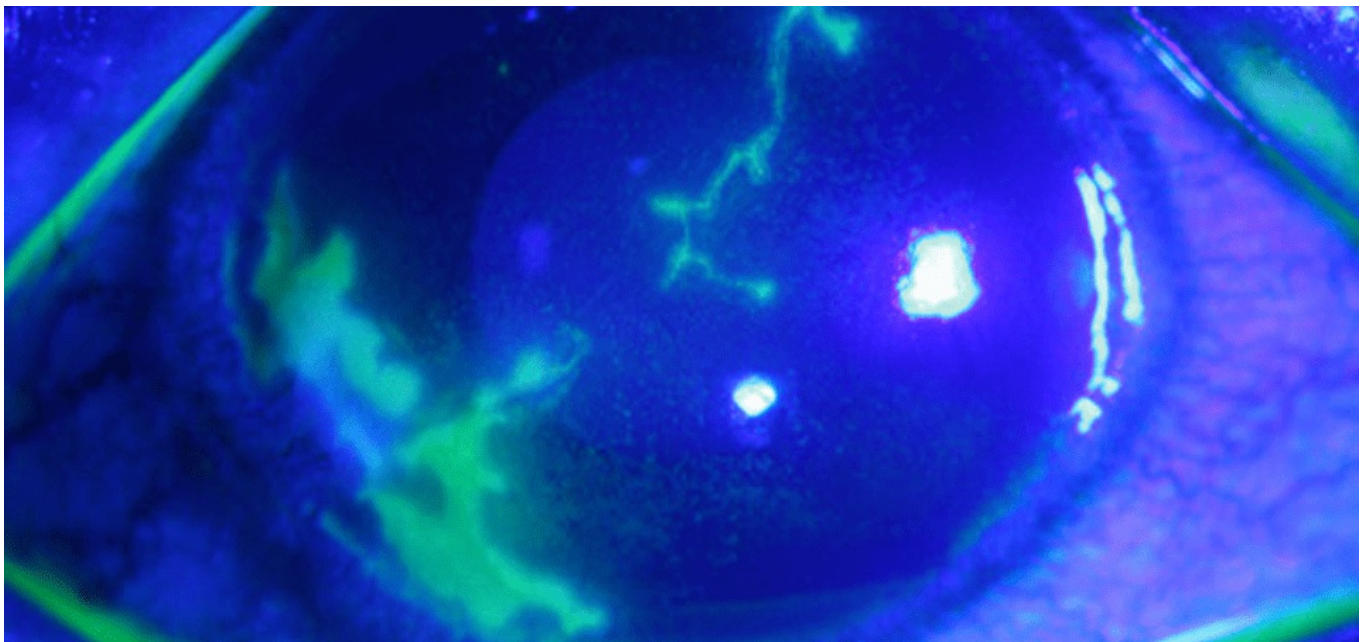
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Microbiology pharm – HY mixed antivirals

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Oseltamivir, Peramivir, Zanamivir

MOA of oseltamivir, peramivir, zanamivir?

- Sialic acid analogues that competitively inhibit influenza A+B neuraminidase.
- **Prevent viral release from the cell.**

When are oseltamivir, peramivir, zanamivir the answer?

- USMLE Q will ask you which agent prevents **viral release**; answer = oseltamivir.
- USMLE Q will ask which agent best decreases **viral spread within a community**; answer = oseltamivir.
- 65F + influenza infection + given a drug for Tx + electron microscopy of upper respiratory tract epithelium shows cells **“packed with virions”**; which drug was she given? → answer = oseltamivir → decreased viral release = cells “packed with virions.”

Ribavirin

MOA of ribavirin?

- Many. But USMLE wants two:
- 1) RNA polymerase inhibitor.
- 2) IMP dehydrogenase inhibitor (purine synthesis).
- Basically say to yourself, “Ribavirin is a weird drug with many MOAs in the literature, but RNA polymerase inhibition is most important. I should also be peripherally aware that it can inhibit IMP dehydrogenase (purine synthesis).”

When is ribavirin the answer?

- It's not. It's almost always the **wrong** answer on USMLE for treatment.
- “Wait. Then why am I learning it?” → USMLE wants you to know the MOA.
- Can be used in select circumstances for RSV bronchiolitis and HepC treatment, but once again, it's almost always the wrong answer.
- Answer for RSV bronchiolitis on USMLE is simply “observation” or “supportive care” almost always.
- If mentioned as part of a hepatitis C, be aware it can cause hemolytic anemia.

Adverse effects of ribavirin?

- USMLE wants you to know it can cause hemolytic anemia.
- 32M + IV drug user + being managed with multiple medications including ribavirin + low Hb + high indirect bilirubin; why? → answer = drug-induced hemolytic anemia” (ribavirin).

Acyclovir, valacyclovir

MOA of acyclovir?

- Competitively inhibits HSV1/2 and VZV DNA polymerase.
- Causes “chain termination.”
- Requires phosphorylation with viral thymidine kinase to a monophosphorylated form.
- The monophosphorylated form is then phosphorylated up to the triphosphate form by host kinases.
- Same MOA as ganciclovir for CMV.

What does “val” mean?

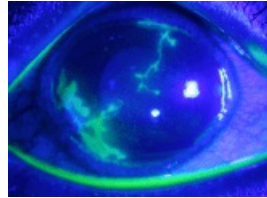
- “Val” means increased oral bioavailability. You'll see this prefix attached to drugs sometimes.

When is acyclovir the answer?

- For any herpes infection (i.e., labialis, genitalis, esophagitis, keratitis, eczema herpeticum etc.). It decreases both

duration and recurrence of symptoms.

- 22F + painful vesicular lesions on labia majora; Tx? → answer = oral acyclovir (herpes genitalis).
- 24M + painful vesicular eruption on the lips; Tx? → answer = oral acyclovir (herpes labialis).
- 34M + HIV + painful swallowing (odynophagia) + endoscopy shows well-demarcated, punched-out ulcers; Tx? → answer = oral acyclovir (herpes esophagitis). Presents as “punched-out” ulcers; in contrast, CMV = “linear” or “confluent” ulcers.
- 12F + blurry vision + Hx of cold sore + fluorescein instillation of the eye shows a dendritic pattern; Tx? → answer = oral acyclovir (herpes keratitis).



- 12F + history of eczema + antecubital fossa on the left has a painful, vesicular eruption superimposed on the eczema; Tx? → answer = oral acyclovir (eczema herpeticum); actual HSV1/2 infection; caused by HSV inoculation into cracked skin of eczema, often from autoinfection from a cold sore. Do not confuse this with dermatitis herpetiformis (not HSV infection; seen in Celiac disease).
- 18M + school wrestling team + painful vesicular eruption down the leg; Tx? → oral acyclovir (also dermatitis herpetiformis).
- 36F + works as a dental hygienist + vesicular eruption on finger; Tx? → answer = oral acyclovir (herpetic whitlow).
- Shingles (herpes zoster). Should be noted that VZV, not HSV, causes shingles. Herpes zoster is just another name for shingles.
 - 48F + painful eruption along a flank, or on the neck; Tx? → oral acyclovir (shingles; herpes zoster; VZV).
 - 50F + diabetic + painful eruption along a flank; Tx? (answers are oral or IV); answer = oral acyclovir.
 - 55M + painful vesicles around the ear + Bells palsy; Tx? → answer = oral acyclovir (herpes zoster oticus; Ramsay-Hunt syndrome type II).
 - 55M + painful vesicles around eye + no prior Hx of herpes infections + fluorescein instillation of the eye shows a dendritic pattern; Tx? → oral acyclovir (herpes zoster ophthalmicus).
 - 45F + Hodgkin lymphoma currently on chemo + painful eruption on the neck; Tx? (answers are oral or IV); answer = IV acyclovir.
 - 2M + painful vesicular eruption on the leg; Tx? → answer = oral acyclovir (pediatric shingles; yes, it's a “thing,” and it's tested; can occur in immunocompromised peds cases if prior Hx of either VZV infection or vaccine).
- Always choose oral (not topical) acyclovir if posed with both as answer choices.
- If you're confronted with both oral and IV, choose oral almost always. IV will be if the patient is immunocompromised (i.e., HIV, lymphoma, on chemotherapy).
- Choose oral, not IV, if the patient merely has diabetes.
- If the patient is septic, give IV acyclovir.
 - Sepsis = SIRS + infection.
 - SIRS = systemic inflammatory response syndrome = two or more of the following: temp <36 or >38; HR >90; RR >20; WBCs <4,000 or >12,000).
- Acyclovir safe to give during pregnancy. Give intrapartum if expectant mother has Sx (even just tingling without visible lesions), then deliver via C-section. If no Sx peripartum, choose vaginal delivery.
 - 38 weeks gestation + typically gets herpes outbreak every 3-4 months + last outbreak was 4 months ago + currently asymptomatic; what should we do? → answer = prophylactic acyclovir.

Adverse-effects of acyclovir?

- Crystal-nephropathy (kidney stones).
- Acyclovir deposits as crystals in the urine.

Mechanism of resistance to acyclovir?

- Altered or absent viral thymidine kinase.
- Same as resistance mechanism against ganciclovir by CMV.

Why can't acyclovir be used for CMV; likewise, why can't ganciclovir be used for HSV1/2 and VZV?

- The viral thymidine kinases have different affinities for the respective drugs.
- Btw, note the difference in spellings of the drugs.

If patient has acyclovir-resistant HSV, which drug can be given?

- In theory, cidofovir. But never an answer on USMLE. I've never once seen this drug come up.
- "Then why the fuck are you mentioning it to me. I don't need info overload." → Because resources sometimes mention it, so rather than wondering if the drugs I've mentioned to you are an incomplete list, I'd rather tell you, "Yes, this drug is mentioned sometimes, but you don't need to know jack shit about it."

Ganciclovir, valganciclovir

MOA of ganciclovir?

- Competitively inhibits CMV DNA polymerase.
- Causes "chain termination."
- Requires phosphorylation with viral thymidine kinase to a monophosphorylated form.
- The monophosphorylated form is then phosphorylated up to the triphosphate form by host kinases.
- Same MOA as acyclovir for HSV1/2 + VZV.

When is ganciclovir the answer?

- 32F + HIV with CD4 count of 80/uL + blurry vision; Tx? → answer = ganciclovir (CMV keratitis).
- 45M + IV drug user + CD4 count of 37/uL + blood in stool + colonoscopy shows linear ulcerations; Tx? → answer = ganciclovir (CMV colitis).
 - USMLE wants you to know CMV colitis comes in with a CD4 count <50/uL.
- 40F + HIV + painful swallowing + endoscopy shows confluent ulcerations; Tx? → answer = ganciclovir (CMV esophagitis).
 - HSV = "punched out" ulcers; CMV = "linear" or "confluent" ulcers.

Adverse-effects of ganciclovir?

- Neutropenia (agranulocytosis).
- Neutropenia presents as **mouth ulcers**, fever, sore throat.
 - Other HY drugs causing neutropenia: methotrexate, clozapine, methimazole, propylthiouracil, ticlopidine.

Mechanism of resistance to ganciclovir?

- Altered or absent viral thymidine kinase.
- Same as resistance mechanism against acyclovir by HSV1/2 or VZV.

Why can't acyclovir be used for CMV; likewise, why can't ganciclovir be used for HSV1/2 and VZV?

- Once again: viral thymidine kinases have different affinities for the respective drugs.
- Reiterating this point because of yieldness on the Step 1.

If patient has ganciclovir-resistant CMV, which drug can be given?

- Foscarnet.
- Probably one of the lowest yield drugs that resources propagate as though it's HY.
- Mentioning it here for the sake of completion, but never actually shows up on NBME material anywhere.
- Pyrophosphate (PPi) analogue; therefore bypasses the need for viral thymidine kinase to perform initial monophosphorylation.
- Can cause hypocalcemia due to chelation.

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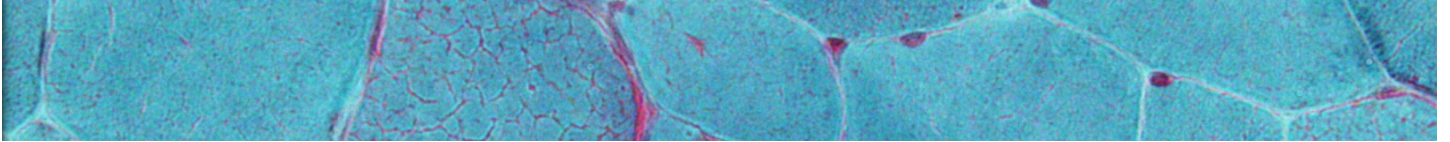


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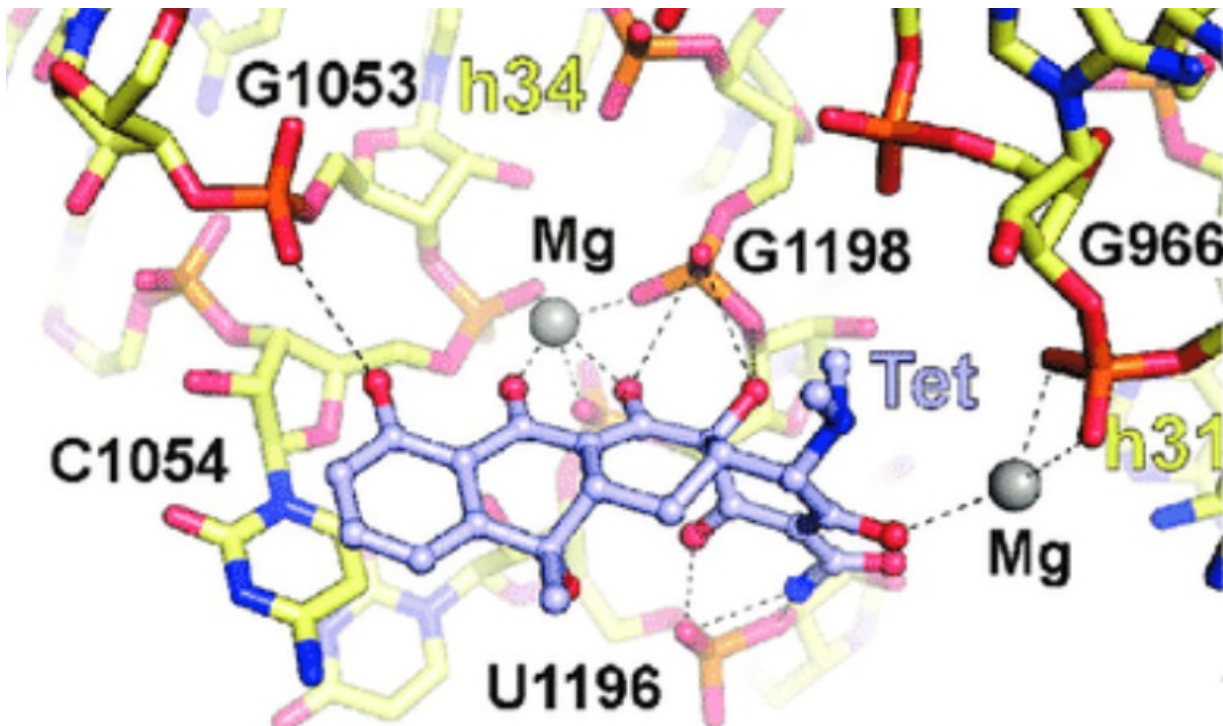
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Microbiology pharm – Protein synthesis (30/50s) inhibitors

by MEHLMANMEDICAL



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HY points about each drug followed by a quiz at the end

Buy **ATT 30**. **CCELS** at **50**. ("Buy at 30. Sell at 50.")

30s ribosomal subunit inhibitors: **A**minoglycosides, **T**etracyclines, **T**igecycline

50s ribosomal subunit inhibitors: **C**hloramphenicol, **C**lindamycin, Macrolides (e.g., **E**rythromycin), **L**inezolid, **S**treptogramins

Aminoglycosides

Highest yield examples: **gentamicin**, tobramycin, amikacin.

MOA of aminoglycosides? → answer = inhibit 30s ribosomal subunit.

- More specifically: prevent formation of initiation complex and causes misreading of mRNA.

What are aminoglycosides effective against? → answer = **gram-negative rods**.

- Aminoglycosides require O₂ for uptake, so occasionally you might hear "aerobic gram-negative rods." But truthfully, this detail about the oxygen requirement is superfluous. I only mention it because it's highly perpetuated. You literally need to take home: **aminoglycosides = gram-negative rods**.

Two important uses of aminoglycosides for USMLE? → answer = 1) empiric endocarditis Tx; 2) *Pseudomonas*.

- Gentamicin, PLUS either vancomycin or ampicillin/sulbactam = empiric therapy for endocarditis.
- Aminoglycosides are notably effective against pseudomonas.

Two important toxicities of aminoglycosides? → answer = nephrotoxicity + ototoxicity.

- The nephrotoxicity associated with aminoglycosides will always be **acute tubular necrosis (ATN)** on the USMLE.
 - E.g., 44M + commenced on gentamicin + vancomycin empirically for endocarditis + now has oliguria + BUN/Cr of 14; what's the most likely explanation for this patient's findings? → answer = acute tubular necrosis.
- Do not confuse aminoglycoside-induced ATN with interstitial nephropathy (tubulointerstitial nephritis) caused by beta-lactams, cephalosporins, and NSAIDs.
 - E.g., 44M + treated with 6 weeks of nafcillin for confirmed MSSA endocarditis +/- rash + urine shows WBCs (eosinophils); Dx? → answer = interstitial nephropathy; wrong answer = acute tubular necrosis.
- Ototoxicity caused by aminoglycosides need not be neurosensory hearing loss or tinnitus; it can also present as vertigo (patient is wobbly).

Tetracyclines

Highest yield examples: doxycycline, minocycline, tetracycline.

MOA of tetracyclines? → answer = inhibit 30s ribosomal subunit.

- More specifically: prevent aminoacyl-tRNA from binding to the A site.

What are tetracyclines effective against? → answer = **VACUUM THE BedRoom** → **V**ibrio spp., **A**cne, **C**hlamydia, **U**reaplasma urealyticum, **M**ycoplasma, **T**ularemia, **H.** pylori, **E**hrlichia, **B**orrelia spp., **R**ickettsia spp.

- *Vibrio cholerae* on USMLE will always be "supportive care" or "rehydration therapy" as the answer. Tetracycline is almost always wrong. It will only be the answer if the question literally forces you into a position where you have to

pick an antibiotic. However, *Vibrio vulnificus* is associated with sepsis and tetracyclines are notably associated with lower mortality.

- **A**cne → tetracyclines are 4th-line Tx for acne (HY for family medicine shelf); standard order for acne Tx for most patients:
 - 1) topical retinoids (i.e., topical tretinoin);
 - 2) benzoyl peroxide;
 - 3) topical clindamycin;
 - 4) **Oral tetracycline**;
 - 5) Oral isotretinoin (not to be confused with topical tretinoin).
- **C**hlamydia → doxycycline is second-line for chlamydial infections. Macrolides (i.e., azithromycin) are used first-line by most practitioners. For the treatment of chlamydial urethritis, azithromycin can be given as a one-off oral 1-gram stat dose, whereas doxycycline must be given BID (twice a day) for a week. Doxycycline can also be given for chlamydial atypical pneumonia (but same as urethritis, azithromycin is first-line by most practitioners).
- **U**reaplasma *urealyticum* → low-yield and you will not get asked; can be a cause of UTIs; more just completes the mnemonic.
- **M**ycoplasma *pneumoniae* → important cause of atypical pneumonia; azithromycin used first-line; doxycycline generally second-line. Should be noted that both *Mycoplasma pneumoniae* and *Chlamydia pneumoniae* cause atypical pneumonia and are treated second-line with doxy.
- **T**ularemia → *Francisella tularensis* → rare cause of atypical pneumonia classically in those with pet rabbits; can also present as painless black skin lesions; can be treated with doxy.
- **H. pylori** → first-line Tx for *H. pylori* is CAP (Clarithromycin, Amoxicillin, Proton pump inhibitor); if CAP combination is not effective (i.e., urease breath test is still positive 4 weeks after Tx), assume resistance, so the clarithromycin and amoxicillin are replaced with the tetracycline, metronidazole, and bismuth. So second-line for *H. pylori* = tetracycline, metronidazole, bismuth, and proton pump inhibitor.
 - 1st-line for *H. pylori* = CAP = clarithromycin, amoxicillin, proton pump inhibitor (e.g., omeprazole).
 - 2nd-line for *H. pylori* = tetracycline, metronidazole, bismuth, and proton pump inhibitor.
- **E**hrlichia *chaffeensis* → Ehrlichiosis → presents similar to Lyme disease with a rash in a patient exposed to a tick, but the difference is Ehrlichia question stems will mention phagocytes with “berry cluster organisms” or “morulae.”
 - Essentially: Ehrlichia = Lyme disease-like presentation PLUS phagocytes with “berry cluster organisms” or “morulae.”
- **B**orrelia *spp.* → doxycycline is first-line for Lyme disease (*Borrelia burgdorferi*) and relapsing fever (*Borrelia recurrentis*).
 - USMLE wants you to know that for children age ≤8 and pregnant women **do not** receive doxy; give amoxicillin instead.
- **R**ickettsia *spp.* → doxy is first-line for *Rickettsia rickettsii* (Rocky Mountain spotted fever; spread by Dermacentor wood tick); *Rickettsia prowazeki* (epidemic typhus; spread by louse); *Rickettsia typhi* (endemic typhus; spread by fleas).

Important side-effects of tetracyclines?

- **Photosensitivity**
 - USMLE might say, e.g., 16M + recently treated for pneumonia + now has blistering of face and lips after soccer tournament; which antibiotic did he receive? → answer = doxycycline (main side-effect is photosensitivity).
 - Topical retinoids in the treatment of acne also cause photosensitivity, but it tends to be more of the rash/desquamation type; in contrast, tetracyclines are more associated with blistering photosensitivity.
- Teeth discoloration (do not give to children age ≤8 or pregnant women [will affect the teeth of fetus]).
- Minocycline is notably associated with drug-induced lupus (anti-histone antibodies).

Tigecycline

MOA of tigecycline?

- Newer, third generation tetracycline derivative. However it's technically considered to be in a newer class called the glycylcyclines.
- Inhibits 30s same as tetracyclines.

Macrolides

Highest yield examples: azithromycin, erythromycin, clarithromycin (do not confuse these with the aminoglycosides, which sound similar).

MOA of macrolides? → answer = inhibit the 50s ribosomal subunit.

- More specifically: binds to the 23S rRNA of peptidyl transferase cavity but does **not** inhibit peptidyl transferase; blocks tunnel that channels nascent peptides away from peptidyl transferase center.

Important uses for macrolides? → empiric Tx for community-acquired pneumonia (CAP); first-line for chlamydia; used as part of the first-line CAP regimen for *H. pylori*.

- Azithromycin is standard first-line for patient with CAP who has not been on antibiotics in the past three months (otherwise we use a fluoroquinolone, such as levofloxacin). This is because it is effective against *Strep pneumoniae* (lobar pneumonia) as well as atypical pneumonia (bilateral interstitial pattern) caused by *Mycoplasma*, *Legionella*, and *Chlamydia pneumoniae*.
- Azithromycin as an one-off, oral 1-gram stat dose is effective against most cases of *Chlamydia trachomatis* urethritis.
- Clarithromycin is first-line as part of CAP (clarithromycin, amoxicillin, proton pump inhibitor) for Tx of *H. pylori*.

Highest yield side-effects of macrolides?

- GI disturbance (diarrhea or constipation).
- Prolongation of QT-interval on ECG (other HY drugs that do this: antipsychotics, metoclopramide).
- P-450 inhibition (**not** azithromycin).

Chloramphenicol

MOA of chloramphenicol? → answer = inhibits 50s ribosomal subunit.

- More specifically: binds to the 23S rRNA of peptidyl transferase cavity + blocks peptidyl transferase; prevents tRNA from binding to A site.

Any notable use for chloramphenicol? → answer = meningitis in the third-world.

- Used for meningitis in some third-world countries; it's a cheap drug that has good blood-brain barrier penetration; not used in the first-world because of its side-effects.

HY side-effects of chloramphenicol? → answer = aplastic anemia; grey baby syndrome.

- Aplastic anemia means RBCs, WBCs, and platelets are all down.
- If given to pregnant women, can cause grey baby syndrome in neonate → neonates have physiologic paucity of bilirubin-uptake enzyme at the liver (UDP glucuronosyl transferase) → this enzyme is needed for elimination of chloramphenicol → therefore drug builds up in blood stream and can deposit in skin, causing blue/grey discoloration.

Clindamycin

MOA of clindamycin? → answer = inhibits 50s ribosomal subunit.

- More specifically: binds to the 23S rRNA of peptidyl transferase cavity + blocks peptidyl transferase; prevents tRNA from binding to A and P sites.

Main use for clindamycin? → answer = anaerobes above the diaphragm (i.e., **pulmonary abscess**).

- 44M + alcoholic + fever + cough + CXR shows air-fluid level; which drug would have utility in treating this patient? –> answer = clindamycin.
- Aspiration pneumonia and pulmonary abscess are classically treated with clindamycin as part of the regimen.
- Clindamycin can also be used for MRSA skin infections (vancomycin has poor skin penetration).

Important side-effect of clindamycin? → pseudomembranous colitis.

- Will only cause pseudomembranous colitis if patient has ingested *Clostridium difficile* spores, but clindamycin, cephalosporins, and ampicillin are HY drugs known to increase the risk of pseudomembranous colitis.

Linezolid

MOA of linezolid? → answer = inhibits 50s ribosomal subunit.

- More specifically: binds to the 23S rRNA of peptidyl transferase cavity + blocks peptidyl transferase; prevents formation of initiation complex.

Use for linezolid? → sometimes used for vancomycin-resistant Enterococcus (VRE); effective against MRSA, but vancomycin preferred in part to avert emergence of linezolid-resistant VRE.

- Vancomycin has poor skin penetration, so patients with MRSA skin infections can be treated with linezolid, sulfonamides, TMP/SMX, or clindamycin.

Side-effects of linezolid? → answer = **HOT** → High risk of serotonin syndrome, **O**ptic neuritis, **T**hrombocytopenia.

Streptogramins (quinupristin/dalfopristin)

MOA of the streptogramins? → answer = inhibit 50s ribosomal subunit.

Use for streptogramins? → answer = vancomycin-resistant *Enterococcus* (VRE).

—

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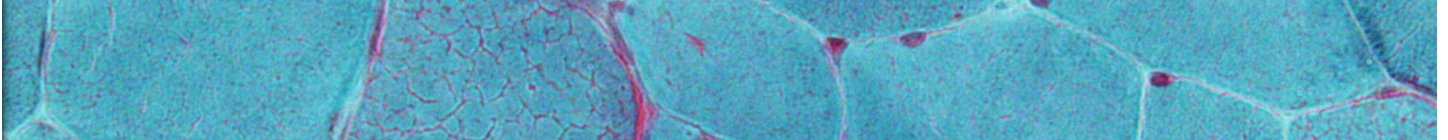


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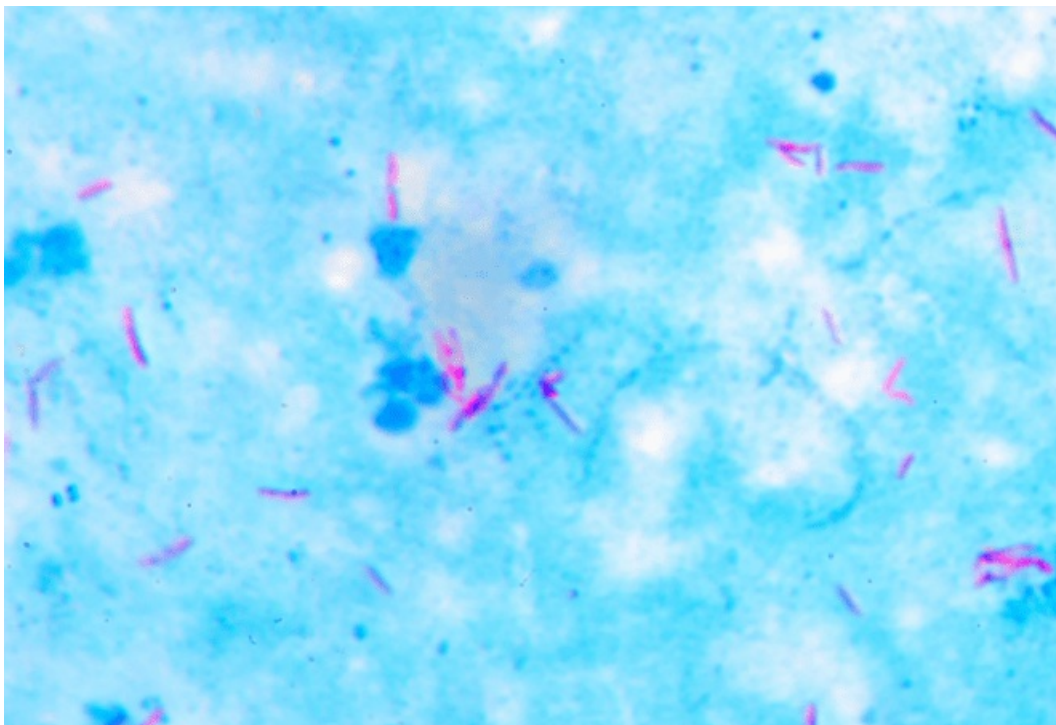
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MICROBIOLOGY / PHARMACOLOGY

Microbiology pharm – TB drugs

by MEHLMANMEDICAL



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[HY points about each drug followed by a quiz at the end](#)

This post is for the TB drugs only. If you want the HY points on the TB treatment algorithm/protocol, see [this post](#).

Simple mnemonic for the TB drugs: **RIPE** → **R**ifampin, **I**soniazid, **P**yrazinamide, **E**thambutol.

Rifampin

MOA of rifampin?

- DNA-dependent RNA polymerase.
- USMLE will simultaneously list the **wrong** answer choice RNA-dependent DNA polymerase. Way to remember the MOA is:
- **RDR** → **R**ifampin is a **D**NNA-dependent **R**NA polymerase.

HY side-effects of rifampin:

- **Orange** bodily secretions.
 - Tears, sweat, and urine can all turn an orange or rusty tinge.
 - This is a normal finding. USMLE can turn this into a pharm-behavioral science type Q, where the patient is concerned that, e.g., his contact lenses turn an orange color, and the correct physician response is: “this is a benign and normal finding of your medication.”
- Upregulation of P-450.
 - If patient has TB and is simultaneously on HAART therapy for HIV, rifampin is avoided because it increases metabolism of the anti-retroviral meds. Rifampin may be replaced with **rifapentine** (does not upregulate P-450).
 - If patient is on anti-epileptic meds (i.e., phenytoin, carbamazepine, valproic acid, etc.), giving rifampin might precipitate seizures → i.e., increased metabolism of the anti-epileptics.
- Hepatotoxicity.
 - Increased LFTs; patient’s LFTs should be checked before commencement.
 - Mild transaminitis is a **normal** finding in patients commencing hepatotoxic agents. For instance: patient commences rifampin (or a statin, or lithium, etc.) + LFTs increase slightly → answer = “maintain dose of the drug”; discontinuation of the drug or decreasing the dose is the wrong answer; once again, mild elevation in LFTs is **normal**.
 - **RIP** → mnemonic for TB drugs that are hepatotoxic → **R**ifampin, **I**soniazid, **P**yrazinamide.

Other HY points about rifampin:

- Used for **prophylaxis in close contacts** of those with *N. meningitidis* meningitis or *H. influenzae type B* epiglottitis.
 - Patients with *N. meningitidis* meningitis or *H. influenzae type B* epiglottitis receive **ceftriaxone** as the treatment; but close contacts get rifampin.
 - *Empiric treatment* for bacterial meningitis (before culture results are known but CSF suggests bacterial): ceftriaxone + vancomycin.
 - *Treatment* for confirmed *Neisseria meningitidis* meningitis = ceftriaxone.
 - *Prophylaxis* given to **close-contacts** of someone with confirmed *Neisseria meningitidis* meningitis (i.e., roommates + those living on the same floor in the dormitory) = rifampin.
- Added to endocarditis treatment when the patient has prosthetic material in the heart.
 - For instance, empiric Tx for endocarditis (i.e., when we don’t yet know the organism) is gentamicin PLUS either vancomycin or ampicillin/sulbactam. If the patient has a prosthetic valve, we add rifampin, so empiric becomes: gent + vanc + rifampin, OR gent + ampicillin/sulbactam + rifampin.

Isoniazid (INH)

MOA of Isoniazid?

- Mycolic acid synthesis inhibitor.

HY side-effects of INH:

- Vitamin B6 (pyridoxine) deficiency → presents as miscellaneous neuropathy and/or seizures → i.e., paresthesias, numbness, seizures.
 - Do not confuse pyridoxine, which is the medical term for vitamin B 6, with the TB drug pyrazinamide.
- Inhibits P-450.
 - Can increase INR in patients on warfarin and/or cause bleeding diathesis.
- Hepatotoxicity.
 - **RIP** → mnemonic for TB drugs that are hepatotoxic → **R**ifampin, **I**soniazid, **P**yrazinamide.
- High anion-gap metabolic acidosis.
 - Part of MUDPILES → Methanol, Uremia, DKA, Phenformin, Iron supplements / **INH**, Lactic acidosis, Ethylene glycol, Salicylates.
- Drug-induced lupus erythematosus (DILE).
 - **Anti-histone antibodies** + arthritis + many other findings such as erythema nodosum, pleuritis, mediastinitis, pericarditis, anti-hematologic cell line antibodies causing thrombocytopenia, erythropenia, and/or leukopenia.
 - Malar rash and renal phenomena are rare in DILE. These are common in systemic lupus erythematosus (SLE).

Notable point about usage of INH:

- USMLE answer for monophylaxis of TB is 9 months INH + vitamin B6.
- That is, when the patient has a (+) PPD test but negative CXR, the answer is “treat for latent TB” or “TB prophylaxis” (same thing, and both are answers). But the Q can also just ask for “9 months INH + pyridoxine” (once again, don’t confuse with pyrazinamide).
- Preferred over rifampin as monophylaxis because it’s not as hepatotoxic.
- When the patient has a (+) PPD *and then also* has a (+) CXR, the answer is “treat for active TB” → RIPE for 2 months, *then* just RI for 4 more months (6 months total of Tx for most patients).
- Once again, for a detailed post on the TB treatment algorithm/protocol, see [here](#).

Pyrazinamide

MOA of pyrazinamide?

- Fatty acid synthetase I inhibitor.
- MOA is low-yield for USMLE.
- Newer research has shown this MOA is dubious at best, but it remains the best answer for USMLE.

“So what do I need to know about pyrazinamide then?”

- Functions best within the acidic pH of phagolysosomes.
- In other words, pyrazinamide is the answer if they ask which TB drug is most effective against *Mycobacteria* that have already been phagocytosed.

Side-effects of pyrazinamide?

- Hepatotoxicity.
 - **RIP** → mnemonic for TB drugs that are hepatotoxic → **R**ifampin, **I**soniazid, **P**razinamide.
- Hyperuricemia → causes gout.

Ethambutol

MOA of ethambutol?

- Inhibits arabinosyl transferase (carbohydrate needed for *Mycobacterium* cell wall) → increased permeability of cell wall → cell death.

Side-effects of ethambutol?

- Ocular toxic → central scotoma; red-green color-blindness; blurry vision.
- Hyperuricemia → causes gout (same as pyrazinamide).

—

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HY USMLE Q #962 – Micro / Pharm

January 23, 2024

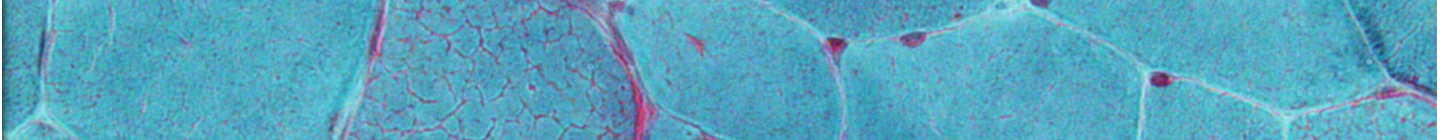


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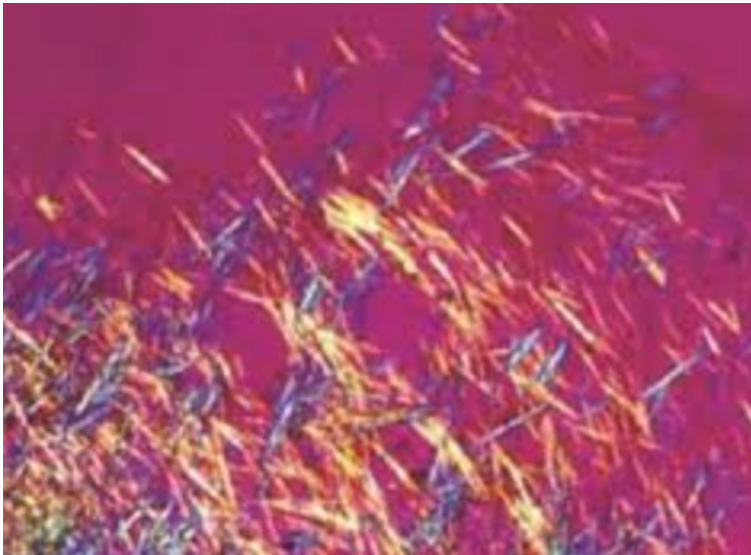
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MSK pharm – Gout drugs

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HY points followed by a quiz at the end

A 54-year-old man with a history of chronic renal disease secondary to granulomatosis with polyangiitis presents with a large, painful, ulcerated tophus located on the first metatarsophalangeal joint. He has a 36-pack-year history of smoking. Which of the following is the most appropriate initial pharmacologic therapy for this patient?

- A) Allopurinol
- B) Colchicine

- C) Corticosteroids
 - D) Febuxostat
 - E) Indomethacin
 - F) Probenecid
 - G) Rasburicase
 - H) Sulfipyrazone
-



This is a classic USMLE “trick” question.

Indomethacin, colchicine, and corticosteroids are all acceptable and comparably effective first-line treatments for acute gout.^{1,2}

In the United States, **indomethacin**, an NSAID, is the standard choice for **acute** gout attacks.¹

On the USMLE, most questions on acute gout will have indomethacin as the answer.

However this patient has a history of renal insufficiency, making NSAIDs and colchicine inferior selections compared to steroids.

Stages of renal failure³:

- **Stage 1** with normal or high GFR (GFR > 90 mL/min)
- **Stage 2** Mild CKD (GFR = 60-89 mL/min)
- **Stage 3A** Moderate CKD (GFR = 45-59 mL/min)
- **Stage 3B** Moderate CKD (GFR = 30-44 mL/min)
- **Stage 4** Severe CKD (GFR = 15-29 mL/min)
- **Stage 5** End Stage CKD (GFR <15 mL/min)

Regarding NSAIDs (e.g., indomethacin) for acute gout:⁴

- **CrCl 30 to 59 mL/min:** avoid or use with caution depending on the kidney disease
- **CrCl <30 mL/min:** relatively contraindicated

Regarding colchicine for acute gout:⁴

- **CrCl ≥30 mL/min:** dosage adjustment not required
- **CrCl <30 mL/min:** consider dosage reduction

Regarding steroids for acute gout:⁴

- **Dosage adjustment for CKD not required**

What the USMLE wants

- If the patient has no apparent contraindications to any medications and indomethacin, steroids, and colchicine are all listed, choose indomethacin.
- If indomethacin and steroids are not listed, colchicine will be the answer they want for acute attacks.
- If the patient has renal insufficiency or history of renal transplant, choose corticosteroids.

Colchicine is a microtubule inhibitor.⁵ This is a high-yield MOA for USMLE.

Allopurinol and febuxostat are xanthine oxidase inhibitors used in the treatment of **chronic gout**, **not** acute flares.⁶ These are not to be used in patients taking 6-mercaptopurine (6-MP), since the latter requires xanthine oxidase for breakdown (i.e., allopurinol or febuxostat can cause ↑ serum 6-MP levels → 6-MP toxicity).⁷

Probenecid and sulfinpyrazone are a uricosurics (i.e., increase urinary excretion of uric acid) used in the treatment of **chronic gout**, **not** acute flares.⁶ They inhibit organic anion transporter (OAT) in the proximal tubules of the kidney, which normally functions to reabsorb uric acid (i.e., ↓ OAT activity → ↓ uric acid reabsorption).

Another high-yield point for USMLE is that OAT functions to secrete beta-lactam antibiotics. In addition to being used as a gout drug, probenecid can sometimes be added to beta-lactam regimens to help maintain serum levels.⁸ USMLE likes this point.

Probenecid and sulfinpyrazone should not be used in patients with history of uric acid stones or in those who are uric acid over-producers (i.e., tumor lysis syndrome; Lesch-Nyhan syndrome). Since these drugs are uricosurics, they increase uric acid in the urine and therefore risk of stones. 90% of patients with gout are under-excretors; 10% are over-producers. These drug are used in under-excretors who do not respond to, or who cannot take, xanthine oxidase inhibitors.⁹

Rasburicase and pegloticase are urate oxidase analogues that directly cleave uric acid. They are used in the prevention and treatment of tumor lysis syndrome.¹⁰

Bottom line: Indomethacin, corticosteroids, and colchicine are all acceptable and comparably effective first-line treatments for acute gout. On the USMLE, indomethacin is usually the answer, however if the patient has renal insufficiency, choose corticosteroids. Colchicine will be the answer if indomethacin and corticosteroids are not listed.

1) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3539261/>

2) <https://annals.org/aim/fullarticle/2578528/>

3) <https://www.davita.com/education/kidney-disease/stages>

4) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5572666/>

5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4656054/>

6) <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858>

7) <https://www.sciencedirect.com/book/9780721694917/peripheral-neuropathy>

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9) <https://www.sciencedirect.com/book/9780323316965/kelley-and-firesteins-textbook-of-rheumatology>

10) <https://www.sciencedirect.com/science/article/abs/pii/S1543291213600090?via%3Dihub>

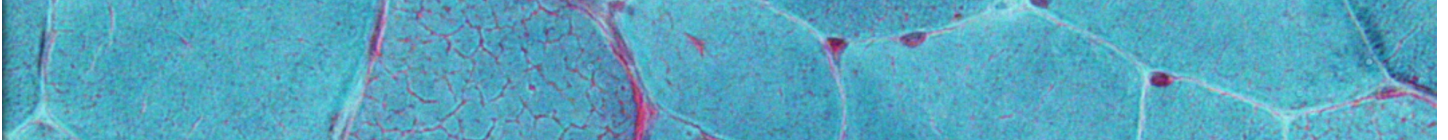


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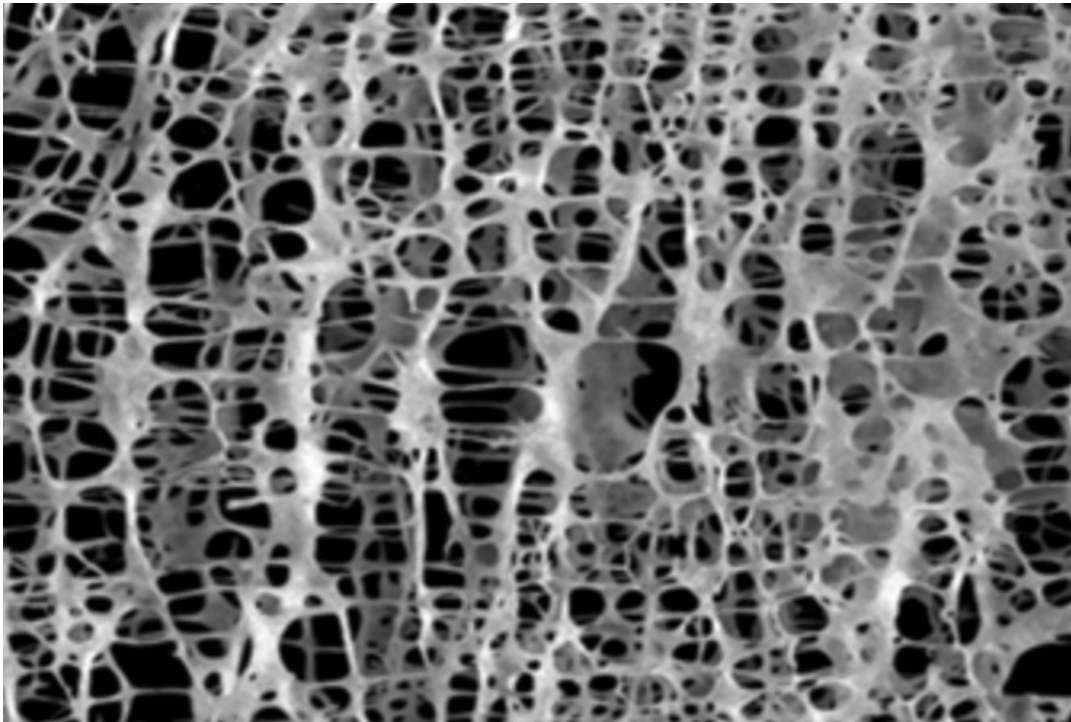
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MSK pharm – Osteoporosis agents

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HY points followed by a quiz at the end

Bisphosphonates (Alendronate, Zoledronic acid)

MOA of bisphosphonates?

- Inhibit osteoclast activity, thereby inhibiting bone resorption.
- Should be noted that the endogenous hormone **calcitonin also inhibits osteoclast activity (HY)**.

HY side-effects of bisphosphonates for USMLE?

- By far and away, **pill-induced esophagitis** is the highest yield side-effect of bisphosphonates. That is, the patient can get a GERD-like presentation with burning in the throat/chest while taking, e.g., alendronate and you need to know this is merely a side-effect of the drug. Should be noted that potassium supplements can also cause pill-induced esophagitis.
- Osteonecrosis of the jaw is a zebra side-effect that most students tend to fixate on, and by all means you should be aware of it, but it's to my observation it's quite low-yield for USMLE compared to pill-induced esophagitis.

When are bisphosphonates the answer on the USMLE for osteoporosis?

- In patients who've already had fracture(s).
- USMLE sequence for osteoporosis Tx = weight-bearing exercise → calcium + vitamin D → bisphosphonate (e.g., alendronate) → teriparatide or denosumab.
- It's HY for USMLE to know that weight-bearing exercise is the number-one treatment + preventative measure for osteoporosis. In other words, even if the patient already has low bone density on DEXA scan, if they ask what should be done for the patient, still choose weight-bearing exercise first if listed.
 - 72F + osteoporosis; which of the following will best decrease fracture risk in this patient? → answer = "going for daily long walks"; wrong answer = "swimming daily" (on NBME exam).
- Calcium + vitamin D are the first-line pharmacologic therapy for osteoporosis. *Then* a bisphosphonate is added if the patient has history of fractures.
 - 74F + osteoporosis + taking calcium and vitamin D; the addition of which pharmacologic agent could best decrease this patient's fracture risk? → answer = alendronate (whichever bisphosphonate is listed).

What are other really HY times on the USMLE that bisphosphonates are used?

- Given to patients who are commencing oral steroids indefinitely, or who are receiving numerous courses of steroids. Remember that corticosteroids cause osteoporosis (compression fracture / point tenderness of the spine) and avascular necrosis (hip pain) in younger patients.
 - 32M + SLE + receiving third two-week course of oral prednisone in the past six months for flare control; what's the next best step in pharmacologic management? → answer = "immediate bisphosphonate therapy," or "alendronate now" (HY in particular on FM shelf).
- First-line agent given after IV normal saline for hypercalcemia.
 - 21M + primary hyperparathyroidism + serum calcium of 14.3 mEq/L (NR 8.4-10.2); what's the next best step in treatment? → answer = 0.9% normal saline.
 - 21M + primary hyperparathyroidism + serum calcium of 14.3 mEq/L (NR 8.4-10.2) + receives infusion of saline + what's the next best step in management? ; what's the next best step in management? → answer = "pamidronate" (frequently the bisphosphonate used for hypercalcemia).

Teriparatide

MOA of teriparatide?

- Increases bone density by stimulating osteoblasts.
- N-terminus analogue of PTH.
- Endogenous PTH normally binds to osteoblasts, which in turn express RANK-L, which binds to RANK receptor on osteoclasts → ↑ bone resorption → *decrease* in bone density. Student says, "Wait, then how does that make sense

then that we'd give a drug that's structurally similar to PTH?" → teriparatide stimulates osteoblasts *without* the concomitant activation of osteoclasts → *increase* in bone density. Weird, but it's what the drug does.

When **not** to give teriparatide?

- Don't give to patients with history of bone tumors.
- Don't give to patients with Paget disease of bone (↑ risk of osteosarcoma).

Denosumab

MOA of denosumab?

- Monoclonal antibody against RANK-L → leads to decreased RANK-L/RANK interaction → better preservation of bone density.
- Used for severe osteoporosis with history of fractures while on multiple medications.

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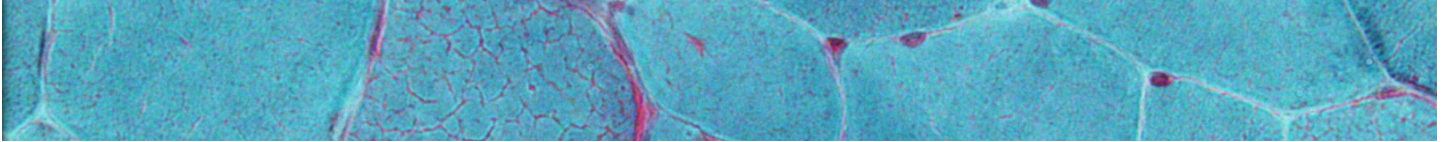
HY USMLE Q #925 – Gastro

December 16, 2023



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MSK pharm – Rheumatoid arthritis drugs

by MEHLMANMEDICAL



A 32-year-old woman has a 7-month history of gradually worsening pain and stiffness in her hands, wrists, and ankles. She began taking 800 mg daily of ibuprofen several months ago which had resulted in temporary amelioration of her symptoms but now is ineffective. Serum studies show positivity for rheumatoid factor and anti-cyclic citrullinated peptide (anti-CCP). She is started on a short course of prednisone that does not relieve her symptoms. The physician discusses his recommendation for commencing a disease-modifying anti-rheumatic drug (DMARD). Which of the following best describes the molecular target of this recommended therapy?

- ☐ CD80/86
- ☐ Dihydroorotate dehydrogenase
- ☐ Dihydrofolate reductase
- ☐ Interleukin-6 (IL-6) receptor
- ☐ Soluble TNF- α



Methotrexate, a competitive and reversible inhibitor of dihydrofolate reductase (DHFR), is the well-established first-line DMARD used to treat rheumatoid arthritis (RA). As soon as a diagnosis of RA is made, it is recommended that DMARD therapy is commenced.¹

NSAIDs, such as ibuprofen, and glucocorticoids may be used for symptomatic relief only, but they do not halt progression of the disease.² Glucocorticoids are generally implemented as a short taper while the initial DMARD is being commenced (or if DMARDs are being switched).¹

Methotrexate causes **pulmonary fibrosis**³, hepatotoxicity ($\uparrow$ ALT, AST)⁴, and mucositis (due to agranulocytosis).⁵

Inhibition of **soluble TNF- α** refers to **infliximab**, **adalimumab**, **etanercept** (a recombinant receptor), golimumab, and certolizumab pegol.⁶

The above three bold TNF- α agents are exceedingly HY for the USMLE. The latter two agents are lower yield but newer. And anything new is fair game.

Inhibition of dihydroorotate dehydrogenase refers to **leflunomide**.⁷

Inhibition of interleukin-6 (IL-6) receptor refers to tocilizumab.⁸

Inhibition of CD80/86 refers to abatacept, a fusion protein that prevents B cell activation by T cells.⁹

Treatment of RA:¹

Early, symptomatic RA

- Methotrexate is first-line DMARD. Commence as soon as diagnosis of RA is made.
- If methotrexate monotherapy is insufficient, add leflunomide or sulfasalazine (i.e., combination DMARD therapy), OR commence TNF- α agent monotherapy.

Established RA

- Methotrexate is first-line DMARD. Commence as soon as diagnosis of RA is made.
- If methotrexate therapy fails, *add* an TNF- α agent.

The USMLE Step 1 and 2CK will **not** make you differentiate between early and established RA. **However they want you to know that methotrexate is used first** and to merely be aware of the other available treatments.

Bottom line: Methotrexate is a competitive, reversible inhibitor of DHFR. It causes pulmonary fibrosis, hepatotoxicity, and mucositis (due to neutropenia). Depending on the severity and progression of RA, anti-TNF- α agents are frequently used as a step-up after methotrexate. NSAIDs and steroids do not slow progression of disease; they are used for symptomatic management only.

- 1) <https://www.rheumatology.org/Portals/0/Files/ACR%202015%20RA%20>
- 2) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3527878/>
- 3) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3814259/>
- 4) <https://www.ncbi.nlm.nih.gov/books/NBK548219/>
- 5) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4851368/>
- 6) <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5877629/>
- 7) <https://www.ncbi.nlm.nih.gov/pubmed/7575649>
- 8) <https://www.ncbi.nlm.nih.gov/pubmed/30427250>
- 9) <https://pubmed.ncbi.nlm.nih.gov/19228144/>

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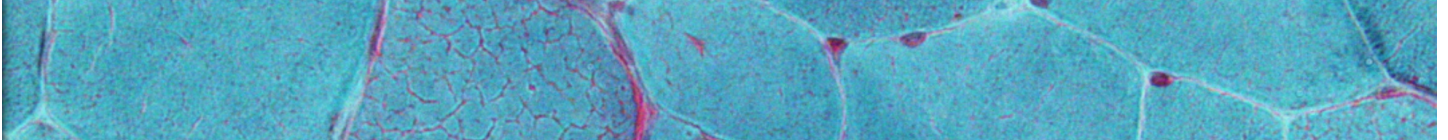


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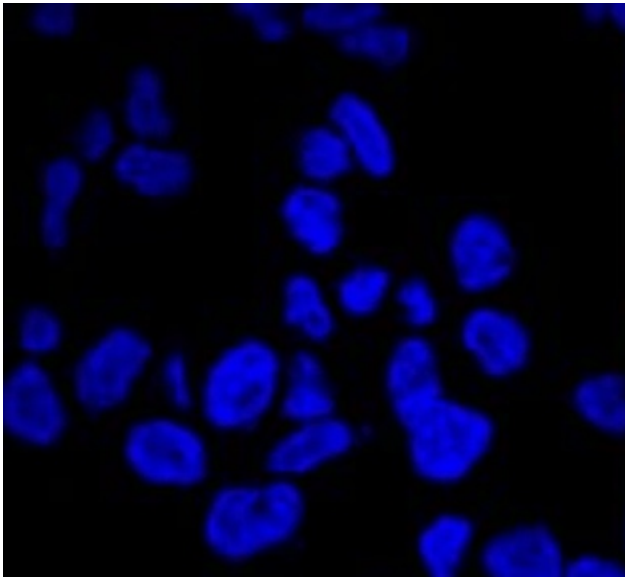
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[NEUROLOGY](#) / [PHARMACOLOGY](#)

Neuro pharm – Epilepsy drugs and hypnotics

by MEHLMANMEDICAL



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HY points followed by a quiz at the end

Hypnotics (sleep-inducing agents)

Benzodiazepines

MOA of benzos?

- Agonize GABAA receptor and cause ↑ *frequency* of chloride channel opening.

Short-acting benzos: **Midazolam**

Intermediate-acting benzos: **Alprazolam, Clonazepam, Lorazepam**

Long-acting benzos: **Chlordiazepoxide, Diazepam**

When are they the answer for USMLE?

- Used for acute anxiety and/or acute insomnia
 - 38F + mother recently diagnosed with breast cancer + recently sent kid off to college + stressed at work + having trouble sleeping; next best step in pharmacologic therapy? → answer = clonazepam (answer on NBME)
 - Never give benzos for chronic use (cause dependence).
- Social phobia **if patient has asthma**
 - 35M + needs to give presentation on stage in front of people; next best step in pharmacologic therapy? → answer = propranolol.
 - 35M + needs to give presentation on stage in front of people + has history of asthma; next best step in pharmacologic therapy? → answer = clonazepam (highly efficacious in social phobia).
 - Psych NBME for 2CK has two nearly identical Qs back-to-back; one has asthma; the other doesn't; propranolol is correct in the non-asthma vignette; clonazepam is correct in patient with asthma Hx.
- Alcohol withdrawal / delirium tremens
 - 44M + in hospital for past 48 hours following surgery + now is tremulous + tachy + high BP; next best step in pharmacologic therapy? → answer = diazepam, lorazepam, or chlordiazepoxide.
 - 44M + cut down to 2 beers a day from 12 beers a day + tremulous and tachy; next best step? → answer = diazepam, lorazepam or chlordiazepoxide → can get delirium tremens even when not cutting down fully (oh wow).
- Panic attack
 - 17M + footballer + tachy + feels like he's going to die + breathing rapidly; next best step in pharmacologic therapy? → benzodiazepine (used for panic *attack*) → should be noted that before benzos, breathing into paper bag is correct if listed (the aim is to calm their respirations).
 - 17M + has third panic + now completely resolves; how to best prevent future attacks pharmacologically? → answer = SSRI, not benzo.
 - SSRIs = Tx for panic *disorder* (prevents recurrent attacks; used between attacks); benzo = Tx for panic *attack* (acute).
- Status epilepticus (seizure lasting >5 minutes)
 - Diazepam is usually used, but can be other benzos.
 - Sequence is: benzo first; if not successful, use phenytoin; if not successful, use barbiturates; if not successful, use general anaesthesia. In other words: diazepam → (fos)phenytoin → phenobarbital → propofol.

Adverse effects of benzos?

- Respiratory depression with a **normal** A-a gradient (arterial oxygen is low *because* the patient is hypoventilating, so alveolar oxygen is also low).
 - 19M + found unconscious with bottle of diazepam; which combination of findings best reflects this patient's current status? → answer = high pCO₂, low pH (respiratory acidosis), normal bicarb (not enough time to change), **normal, not high**, A-a gradient.
- Anterograde amnesia (can't remember new things).
- Dependence (don't give to patients for chronic use).

Treatment for benzo overdose?

- **Flumazenil** (GABAA receptor antagonist).
- Some literature says don't give in the emergency setting because it can precipitate seizures in those who are benzo-dependent, but a 2CK NBME Q has flumazenil as the answer in this setting.

Barbiturates (phenobarbital, thiopental)

MOA of barbiturates?

- Agonize GABAA receptor and cause ↑ *duration* of chloride channel opening.

When are barbiturates the answer for USMLE?

- Truthfully, not often. USMLE wants you to know that they cause **respiratory depression**, similar to benzos and opioids. So the same concepts hold true as far as patients who have high CO₂ and low O₂ with a **normal** A-a gradient.
- Used for status epilepticus after phenytoin, as per above:
 - Sequence is: benzo first; if not successful, use phenytoin; if not successful, use barbiturates; if not successful, use general anaesthesia. In other words: diazepam → (fos)phenytoin → phenobarbital → propofol.
- Treatment for overdose is supportive (ABCs). Flumazenil is for benzos only.
- Have rare uses such as Criger-Najjar syndrome type II (you won't get asked, but students sometimes ask me about this one).
- Contraindicated in porphyrias (i.e., acute intermittent porphyria; porphyria cutanea tarda). Barbiturates upregulate δ-ALA synthase → increased production of heme precursors → patients with porphyrias who lack conversion enzymes will get precursor overload and become symptomatic.

Non-benzo hypnotics (Zolpidem, Zaleplon)

MOA of zolpidem and zaleplon?

- Agonize GABAA receptor at location distinct from benzos.

When are zolpidem and zaleplon the answer on USMLE?

- Can be used for acute insomnia.
- Answer on one of the 2CK Psych forms for insomnia (benzo wasn't listed, but of course other Qs will have benzo used acutely).
- If you then say, "But if USMLE listed both, which one is it then?" My response is: USMLE won't list both like that. If they ever did, it would probably be in the context of a patient with history of substance dependence, where benzo would be a bad choice, even if acutely.

Anti-epileptic agents that block Na⁺ channels

Block sodium channels + ↑ GABA

- **TV** (television) → **Topiramate, Valproic acid**
- Weird mnemonic, but it's better than nothing and helps some students. Watching TV at night can put you to sleep (GABA).
- "TV...okay...well those are the ones that block sodium channels *and* increase GABA."
- Topiramate + valproic acid used for tonic-clonic seizures.

When is topiramate the answer on USMLE?

- Topiramate I've never seen as an answer. Just know it as a common med + its MOA.

When is valproic acid the answer?

- First-line med for **myoclonic seizures** (twitching of limb + no loss of consciousness + no prodrome).
- **Teratogen (causes neural tube defects** → stop drug before getting pregnant; if Hx of taking valproic acid, patients need 10x dose of folic acid prior to + during early pregnancy (400 ug → 4mg/day). Other anti-epileptics (i.e., carbamazepine, phenytoin) can also cause neural tube defects, but the association is notably associated with valproic acid.
- These agents cause folate (B9) deficiency via decreasing intestinal absorption.
 - 39M + taking valproic acid + Q shows you image of hypersegmented neutrophil; what's he deficient in? → answer = B9, not B12. Hypersegmented neutrophils + ↑ MCV are of course seen in both B9 and B12 deficiencies.
- Weird HY detail about valproic acid is that it is the answer for patients with absence seizures *who've also had a history of other seizures*. For example:
 - 5M + stares off blankly into space during class + EEG shows 3Hz spikes; Tx? → answer = ethosuximide (used for absence seizures in isolation; blocks thalamic T-type calcium channels).
 - 5M + stares off blankly into space during class + EEG shows 3Hz spikes + had tonic-clonic seizure several months back; Tx? → answer = valproic acid (kid *also* has Hx of tonic-clonic seizure, so valproic acid is correct over ethosuximide. Oh wow.)

Only block sodium channels

- **Carbamazepine, Lamotrogine, Phenytoin**
- "TV are the ones that block sodium channels *and* increase GABA, so these are the ones that only inhibit sodium channels. Meaning, because they're not TV (putting people to sleep at night), they don't affect GABA."

When is carbamazepine the answer?

- First-line for trigeminal neuralgia.
- For diabetic neuropathic pain, use TCAs (i.e., amitriptyline) first-line. However in elderly (where we want to avoid TCAs), carbamazepine + gabapentin can be combined first-line (on FM form for 2CK).
- Standard first-line Tx for tonic-clonic seizures (as are valproic acid, phenytoin, and others).

When is lamotrogine the answer?

- Is notably associated with **Stevens-Johnson syndrome (SJS)**.
- SJS can occur with any number of agents, but for some reason lamotrogine is known to have a strong association.
- And of course know its MOA (blocks sodium channels).

When is phenytoin the answer?

- Causes **fetal hydrantoin syndrome** in neonates born to mothers taking it while pregnant → **finger nail hypoplasia** + any number of miscellaneous findings, e.g., facial and organ system defects. Carbamazepine and valproic acid can also cause fetal hydrantoin syndrome in theory, but the association is classically phenytoin.
- Can cause neural tube defects similar to valproic acid and carbamazepine (due to B9 deficiency).
- Standard first-line Tx for tonic-clonic seizures (as are valproic acid, carbamazepine, and others).
- Used after benzos in the treatment of status epilepticus (use benzo, then phenytoin, then barbiturate, then propofol).
- Can cause purple glove syndrome (PGS) → swollen purple extremities in patients receiving high-dose phenytoin. Weird, but Google it.

Anti-epileptic agents that block Ca^{2+} channels

Ethosuximide

MOA of ethosuximide?

- Blocks thalamic T-type calcium channels.

When is ethosuximide the answer on USMLE?

- For absence seizures in kids *who don't also have other types of seizures*. Once again:
 - 5M + stares off blankly into space during class + EEG shows 3Hz spikes; Tx? → answer = ethosuximide (used for absence seizures in isolation; blocks thalamic T-type calcium channels).
 - 5M + stares off blankly into space during class + EEG shows 3Hz spikes + had tonic-clonic seizure several months back; Tx? → answer = valproic acid (kid *also* has Hx of tonic-clonic seizure, so valproic acid is correct over ethosuximide. Oh wow.).

Gabapentin

MOA of gabapentin?

- Inhibits high-voltage-activated calcium channels.
- Used for neuropathic pain (e.g., due to injuries, stroke).
- Choose TCA (i.e., amitriptyline) over gabapentin first-line for diabetic neuropathic pain if *not* elderly. In elderly, use gabapentin + carbamazepine, OR just nortriptyline (TCA that can be used in elderly). This is on FM 2CK form.

Levetiracetam

MOA of levetiracetam?

- Blocks pre-synaptic calcium channels. May increase GABA.

Other (lower-yield) anti-epileptic agents

Vigabatrin

MOA of vigabatrin?

- Inhibits GABA transaminase → ↑ GABA.

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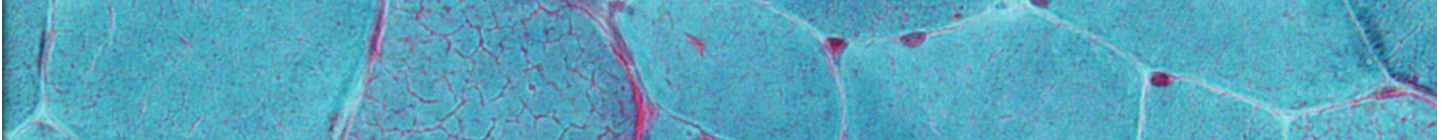


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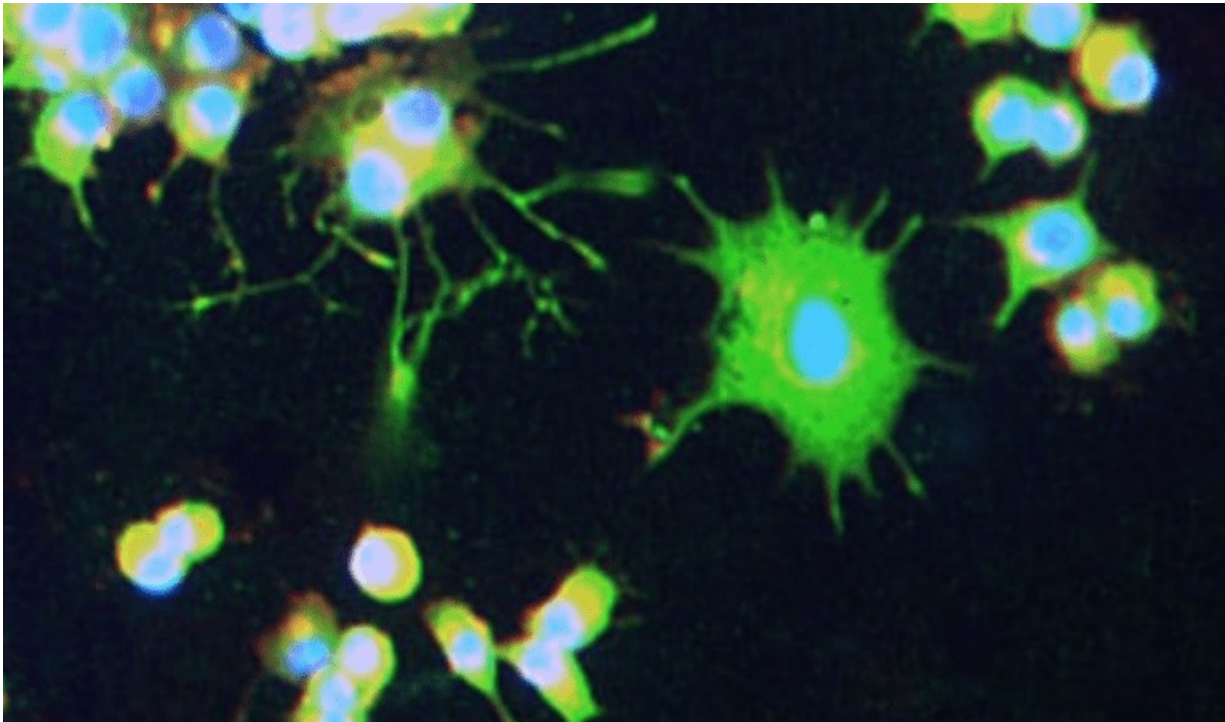
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NEUROLOGY / PHARMACOLOGY

Neuro pharm – Parkinson drugs

by MEHLMANMEDICAL



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Dopamine 2 (D2) receptor agonists

- Parkinson disease is caused by loss of dopamine-secreting neurons in the substantia nigra pars compacta.
- D2 agonists are usually first-line treatment.
- D2 is the main dopamine receptor in the CNS relating to motor function.

Bromocriptine

- Also used for prolactinoma. (Not mandatory that prolactinoma *must be* treated with bromocriptine, but I've seen this agent specifically as the answer on NBME exams).

Ropinirole, Pramipexole

- Also used for restless leg syndrome (have seen these as answers on NBME forms).
- Restless leg syndrome is most commonly caused by iron deficiency anemia. So check serum iron and ferritin first. If abnormal, give iron. If normal, give ropinirole or pramipexole.

Pergolide

- Another D2 agonist used for Parkinson disease. Just know its MOA.

Other Parkinson agents

Amantadine

MOA of amantadine?

- Increases presynaptic release of dopamine.

Notable side-effect of amantadine?

- Can cause livedo reticularis (a weird lacy rash on the legs).

Levodopa/carbidopa

MOA of levodopa/carbidopa?

- Levodopa (L-Dopa) is the precursor of dopamine (i.e., it's converted to dopamine via L-Dopa decarboxylase and vitamin B6).
- L-Dopa crosses the blood-brain barrier (BBB); dopamine does not.
- If we give dopamine to a patient with Parkinson disease, not only will it not cross the BBB, but it has [cardiovascular effects](#) (which we want to avoid).
- So if we give L-Dopa, that will cross the BBB. However, it will be broken down peripherally by an enzyme called catechol-O-methyltransferase (COMT).
- Carbidopa is an analogue of L-Dopa that functions as a competitive inhibitor of COMT (i.e., COMT will act on carbidopa instead of L-Dopa), allowing L-Dopa to cross the BBB.

HY side-effect of levodopa/carbidopa for USMLE?

- Increasing the dose in some patients can cause psychosis. This is HY on the Psych shelf for 2CK. Simply reduce dose as the treatment.
 - 72M + Parkinson disease on L-Dopa/carbidopa + dose was recently increased + now has auditory hallucinations; what do we do? → answer = "decrease the dose of L-Dopa/carbidopa"; the wrong answer is "stop L-Dopa/carbidopa."

Entacapone, Tolcapone

MOA of entacapone and tolcapone?

- Inhibit catechol-O-methyltransferase (COMT).
- Prevent breakdown of L-Dopa peripherally so that it can better cross the BBB.

Selegiline

MOA of selegiline?

- Inhibits monoamine oxidase B (MOAB).
- MOAB has ↑ effect on breaking down dopamine in the synaptic cleft.
- So selegiline prevents breakdown of dopamine in the synaptic cleft.

Benztropine, Trihexyphenidyl

MOA of benztropine and trihexyphenidyl?

- Muscarinic receptor antagonists.
- Don't directly treat Parkinson disease as they don't relate to dopamine.
- Purpose is to relieve symptoms caused by increased muscular rigidity (i.e., cogwheel rigidity).
- Detailed post on anti-muscarinics [here](#).

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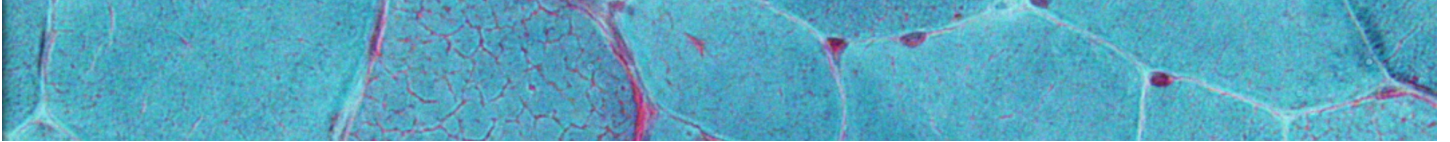


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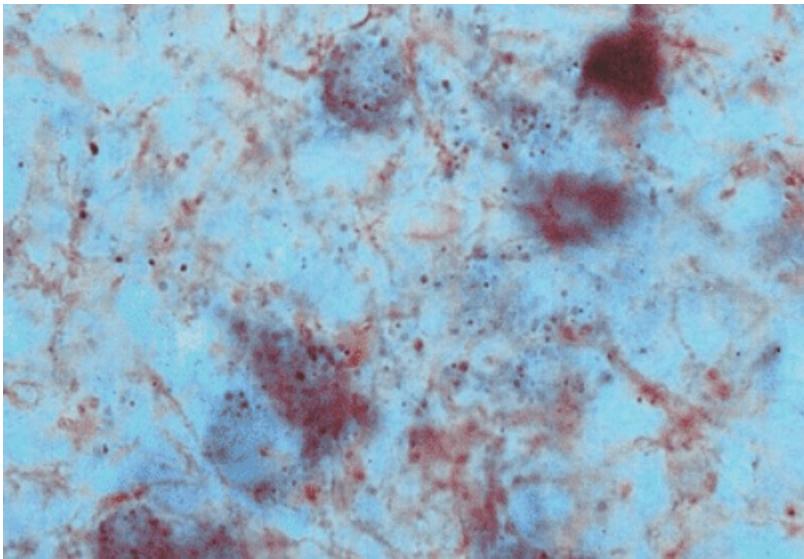
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PHARMACOLOGY / PSYCHIATRY

Psych pharm – Antidepressants + miscellaneous agents

by MEHLMANMEDICAL



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Antidepressants

Selective serotonin reuptake inhibitors (SSRIs)

- Be familiar with a few of these names → **fluoxetine, paroxetine, sertraline, escitalopram, citalopram**

MOA of SSRIs?

- Prevent the presynaptic reuptake of serotonin from the synaptic cleft → increased serotonin in the synaptic cleft → increased serotonin transmission.

What do I need to know for USMLE in terms of use?

- First-line for many psych disorders, including depression, OCD, generalized anxiety disorder (GAD), panic disorder, PTSD.
- Also the first medication given for premature ejaculation (because side-effect is anorgasmia, which is advantageous here).
- Sertraline classically used for post-MI depression.

What are the HY side-effects for USMLE?

- Anorgasmia → major cause of non-compliance, but as mentioned before, this side-effect is advantageous if used for premature ejaculation.
- Sleep/wake disturbance → insomnia/somnolence.
- Serotonin syndrome if combined with St John Wort.
- SIADH → high urinary osmolality + low serum sodium in someone being medicated for (usually) depression.
- Can precipitate mania if erroneously given to patients with bipolar disorder → if a patient with bipolar disorder is in the depressed phase and is not properly screened for history of mania or hypomania, giving an SSRI can swing the patient into the up-phase → patients with depression need to be screened for history of mania/hypomania, and if bipolar is the diagnosis, lithium or valproic acid should be commenced, not an SSRI.

Serotonin-norepinephrine reuptake inhibitors (SNRIs)

- Be familiar with a few of these names → **duloxetine, venlafaxine, desvenlafaxine, milnacipran**

MOA of SNRIs?

- Prevent the presynaptic reuptake of serotonin and norepinephrine from the synaptic cleft.

What do I need to know for USMLE?

- Duloxetine and milnacipran are used first-line for fibromyalgia.
- Can be used for depression after SSRI (e.g., in patients who've failed different SSRIs or who've experienced side-effects).
- SNRIs classically chosen after SSRIs for conditions such as panic disorder, OCD, PTSD, etc.
- USMLE will not make you choose between an SSRI and SNRI. You just need to be aware of the names and that they have broad application in psych.

Tricyclic antidepressants (TCAs)

- Be familiar with a few of these names → **amitriptyline, nortriptyline, clomipramine, imipramine, doxepin**

MOA of TCAs?

- Prevent the presynaptic reuptake of serotonin and norepinephrine from the synaptic cleft (yes, similar to SNRIs).

When are TCAs used?

- First-line for diabetic neuropathic pain.
- Clomipramine can be used for OCD if SSRI not listed as an answer.
- Imipramine can be used in theory as a last-line agent for nocturnal enuresis, but on the USMLE, never choose drugs for this diagnosis.
 - Tx for nocturnal enuresis: spend more time with the kid (behavioral answer) → star chart → enuresis alarm → drugs such as desmopressin or TCAs (once again, drugs are always wrong on the USMLE for nocturnal enuresis); this sequence is HY for 2CK Peds forms.
- 2CK Psych shelf likes doxepin, so be familiar with this name (i.e., don't disregard it as a "weird TCA I probably don't need to know."):
 - 45M + on various meds including doxepin + has delirium; next best step in management? → answer = "discontinue doxepin," or "discontinue anti-cholinergic medications." (explanation about side-effects below)
- Nortriptyline is used in elderly (on FM shelf).
- Only used for depression if other agents have failed first.

Side-effects are hugely HY on the USMLE:

- CCC (the 3Cs) → Coma, Convulsions, Cardiotoxicity (prolonged QT interval → causes ↑ risk of precipitating torsades).
 - TCAs interfere with myocardial sodium channels.
- Causes **the "triad"** → anti-cholinergic (anti-muscarinic), anti-alpha 1, anti-H1.
 - If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms – > **D**iarrrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating. Since antipsychotics are anti-cholinergic, they cause the opposite of **they will cause the opposite of DUMBBELSS** → Constipation, Urinary retention, Mydriasis, Tachycardia, *Not bronchodilation* (β2 agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis.
 - 23M + started on agent for depression + now has suprapubic mass and is hot and dry; which of the following receptor effects is most likely responsible for this patient's condition? → answer = anti-cholinergic (anti-muscarinic); patient has urinary retention (full bladder).
 - Anti-alpha 1 → orthostatic hypotension + fainting (alpha 1 agonism normally constricts peripheral arterioles, maintaining BP and cerebral perfusion).
 - Anti-H1 → sedation.
 - The "triad" is seen with TCAs, antipsychotics, and 1st gen H1 blockers (e.g., diphenhydramine).

How is TCA toxicity treated?

- Sodium bicarb (HY) → causes decreased binding of TCA from myocardial sodium channels.
 - Do not confuse this mechanism with sodium bicarb used for aspirin (salicylate) toxicity; this mechanism is different → increased excretion through urinary alkalinization ($-\text{COOH} \rightarrow -\text{COO}^-$, where the kidney cannot reabsorb charged species as readily).
 - TCAs are basic ($-\text{NH}_3^+ \rightarrow \text{NH}_2$), not acidic, so giving bicarb will actually *decrease* urinary excretion, but the effect on causing dissociation from myocardial sodium channels is how the toxicity is managed.

Monoamine oxidase inhibitors (MOAs)

- Be familiar with a few of these names → **tranylcypromine, phenelzine, selegiline**

MOA of MOAs?

- Prevent the breakdown of neurotransmitters (mostly serotonin and NE) in the synaptic cleft.
- Selegiline is used for Parkinson disease and is a MOAB inhibitor → primarily prevents breakdown of dopamine.

What do I need to know for USMLE?

- MOAs are classically associated with serotonin syndrome. The risk is particularly elevated when the patient is switching to a MOA from a different drug class (usually SSRI) and not enough time has passed. "Washout periods" upwards of 6 weeks are generally required.
 - 22F + switched to tranylcypromine + discontinued fluoxetine a week ago + now has tachycardia, diaphoresis, flushing + temperature of 105F; treatment? → **ciproheptadine** (blocks serotonin receptors).
- MOAs can precipitate serotonin syndrome if the patient is simultaneously taking **St John wort**, meperidine, dextromethorphan, or linezolid.
- MOAs are considered the *most effective* agents for treating atypical depression (depression where the patient notably has hypersomnolence + hyperphagia + the depression is reversible with positive social affirmation), *but SSRIs are still first-line*.

Mirtazapine

MOA of mirtazapine?

- Alpha 2 antagonist → increases presynaptic release of serotonin and NE.
- Alpha 2 is normally a negative feedback receptor on the presynaptic terminal. When NE binds to it, this results in decreased NE and serotonin release.

When is mirtazapine used?

- Patients who have depression + anorexia → mirtazapine stimulates appetite.

Bupropion

MOA of bupropion?

- Inhibits reuptake of NE and dopamine, with minimal effect on serotonin reuptake.

What do I need to know for USMLE?

- Bupropion can cause seizures (i.e., lowers seizure threshold) → do not give to patients with history of eating disorders (because of risk of electrolyte disturbance → further risk of seizure).
- Does not cause sexual dysfunction → notably can be given to patients who decline SSRIs because of the anorgasmia side-effects, or who have had history of erectile dysfunction.
- Can be used as a smoking cessation agent (as with varenicline, mentioned below).

Miscellaneous HY agents

Lithium

MOA of lithium?

- Called a "mood-stabilizer."
- Regulates the rate of phosphoinositide synthesis in neurons.

When is it used?

- Used first-line in bipolar disorder. If lithium isn't used, **valproic acid** is also considered a first-line agent for bipolar.

Why does USMLE care so much about lithium?

- Because of its nasty side-effects. Know these well for USMLE:
- Tremor
 - Thiazides and loop diuretics are known to cause lithium toxicity → the diuretics lead to increased compensatory reabsorption of sodium by the PCT → lithium and sodium are both Group 1 metals and reabsorbed by the same transporters → increased lithium reabsorption.
 - 40F + bipolar disorder + history of calcium nephrolithiasis + started on HCTZ in order prevent recurrent stones → develops tremor; why? → answer = thiazide causing lithium toxicity.
- Nephrogenic diabetes insipidus → insensitivity of the medullary collecting duct to the effects of ADH (vasopressin).
 - 38M + bipolar disorder + started on medication + now has tremor and is peeing a lot; what are the tonicities of the urine at the PCT, JGA, and medullary collecting duct compared to the serum? → answer = PCT isotonic; JGA hypotonic; medullary collecting duct hypotonic.
 - PCT is always isotonic no matter what; JGA is always hypotonic no matter what (connects early-DCT to afferent arterioles; the early-DCT is always hypotonic no matter what because the thick-ascending limb of Henle reabsorbs copious ions); medullary collecting duct is hypotonic in DI (and hypertonic in SIADH and dehydration).
- Hypothyroidism
 - Do **not** discontinue lithium; just add levothyroxine (sounds wrong, but on USMLE).
- Ebstein anomaly
 - Lithium is a teratogen.
 - If a woman takes lithium while pregnant, the fetal heart may develop “atrialization of the right ventricle,” where the tricuspid valve develops inferiorly, making the RV very small (essentially instead of RV, we get a bigger RA).
- Hepatotoxicity
 - ↑LFTs (mild transaminitis) is normal and expected in patients commencing hepatotoxic agents → do not discontinue drug or reduce dose if mild elevation of LFTs occurs.
- Lithium notably has a narrow therapeutic window → means the difference between the minimum dose required for therapeutic effect vs the dose which causes toxicity is very small → in other words, lithium is notably associated with high risk of toxicity because it's easy to overdose.

Buspirone

MOA of buspirone?

- Serotonin (5HT_{1A}) receptor agonist.

When is it used?

- Second-line for generalized anxiety disorder (GAD); **first-line for GAD is SSRIs**, but USMLE wants you to know the MOA and use of buspirone nevertheless.

Trazodone

MOA of trazodone?

- Blocks reuptake of serotonin.

When is it used?

- Treatment of insomnia

What do I need to know for USMLE?

- Can cause serotonin syndrome as a stand-alone agent.
- Can cause priapism.

Varenicline

MOA of varenicline?

- Nicotinic receptor partial agonist

When is it used?

- Used for smoking cessation.

Anything else I need to know for USMLE?

- Varenicline causes wild/intense dreams (similar to the HIV drug efavirenz).

Opioid-detox and -relapse agents

Methadone

- Oral opioid with long half-life used for heroin detox (i.e., helps wean patients off heroin + prevent relapses).
- Binds to mu-receptors.

Buprenorphine

- Sublingual opioid used for heroin detox (i.e., binds to opioid receptors and helps wean patients off heroin + prevent relapses).
- Weak opioid with effects mostly at kappa-receptors.

Naloxone

- Opioid receptor antagonist used to treat heroin overdose.

Naltrexone

- Oral opioid antagonist to prevent cravings in those who have **alcohol dependence** or history of heroin use.
- When the patient is actively weaning off heroin, methadone is used because it's an opioid receptor agonist (same as heroin), but once the patient has weaned off, antagonistic effects at kappa-receptor have been shown to decrease cravings.
- Psych shelf for 2CK likes naltrexone as used for decreasing alcohol cravings.

Disulfiram

- Drug used in alcoholics to decrease drinking.
- Acetaldehyde dehydrogenase inhibitor → produces hangover symptoms → incentivizes non-drinking.
- Normal ethanol metabolism is EtOH → acetaldehyde → acetic acid. If patient drinks EtOH, disulfiram causes a buildup of acetaldehyde, which produces hangover symptoms.

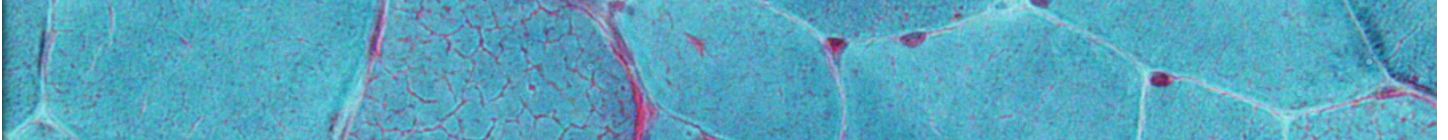


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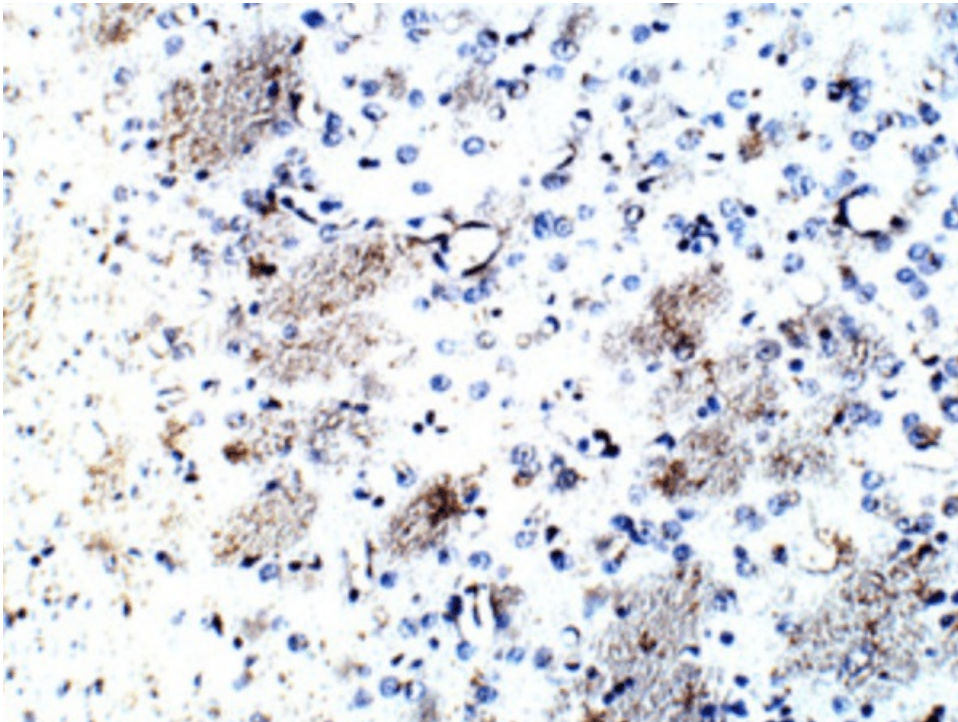
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PHARMACOLOGY / PSYCHIATRY

Psych pharm – Antipsychotics

by MEHLMANMEDICAL



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Antipsychotics are either typical (1st gen) or atypical (2nd gen).

All of the antipsychotics, regardless of class, antagonize dopamine 2 (D2) receptors.

The 1st gen are considered to generate their antipsychotic effect exclusively via antagonizing the D2 receptors; the second generations are considered to have a slightly greater fractional effect at other types of receptors as well, including D4 and serotonin. But regardless, both 1st and 2nd gen primarily antagonize D2.

Antagonism of D2 → treatment for psychosis (and can inadvertently induce Parkinsonism).

Agonism of D2 → treatment for Parkinsonism (and can inadvertently induce psychosis).

Clear yourself of the myth that typicals help more with positive Sx of schizophrenia, whereas atypicals more with negative Sx → literature doesn't support this conclusively anymore, nor does USMLE assess things this way.

You're expected to know the names of essentially all of the antipsychotic drugs for the Step, so the listing of drugs here beyond just one or two examples for each class is not superfluous.

1st gen (typicals)

Haloperidol

- Classic; highest yield.
- **Haloperidol decanoate** is HY on 2CK Psych shelf for patients who have poor compliance with oral antipsychotics; given as a once/month depo injection.
- Low-dose haloperidol is effective for Tourette (also on NBME where they ask Tx for Tourette; clonidine [alpha-2 agonist can be used for Tourette as well]).
- Used in theory for patients who are delirious and at risk of harm to themselves and others, however it is almost always a wrong answer on the USMLE in this context – e.g., if the Q says some guy is on drugs and combative in the emergency room, the answer is “observe in the emergency department,” not “administer haloperidol and observe.”

Trifluoperazine, Fluphenazine

- Nothing special about these. You just need to be familiar with their names.

Prochlorperazine

- Can be used as an anti-emetic / -nausea agent.
- You might see this in a patient's chart and say, “Wait wtf, this patient has schizophrenia?” No, it's just for nausea.

Chlorpromazine

- Can cause **C**orneal deposits.

Thioridazine

- Can cause re**T**inal deposits.

2nd gen (Atypicals)

“Old **C**losets **Q**uietly **R**isper (Whisper) from **A** to **Z**” → **O**lanzapine, **C**lozapine, **Q**uetiapine, **R**isperidone, **A**ripiprazole, **Z**iprasidone

Used first-line in the Tx of schizophrenia (choose one of these over the typicals if both are listed). As mentioned earlier, haloperidol decanoate is correct, however, if the patient has poor compliance.

Olanzapine

- Can cause weight gain / obesity.
- If patient is overweight and/or diabetic, use aripiprazole or ziprasidone instead.
- Can be added to valproic acid or lithium as adjuvant therapy in the Tx of bipolar and schizoaffective disorders.

Clozapine

- Can cause weight gain / obesity.
- **Can cause agranulocytosis (neutropenia) → one of the highest yield adverse drug effects on USMLE.**
 - Can present as **mouth ulcers, fever, and/or sore throat**. The Q can also just simply tell you the patient was started on clozapine and now has a WBC count of 500/uL (NR 4-11,000), where neutrophils are 0 → answer = drug-induced neutropenia (agranulocytosis).
 - If patient has neutropenia while on clozapine, the Tx is stop the drug immediately and give IV antibiotics for neutropenic fever (febrile neutropenia). Then give GM-CSF.

Quetipine

- Can be added to valproic acid or lithium as adjuvant therapy in the Tx of bipolar and schizoaffective disorders.

Risperidone

- Classically used as a first-line agent for schizophrenia.

Aripiprazole, Ziprasidone

- Used in patients who are overweight or have diabetes.

Important side-effects of the antipsychotics:

- Can cause QT prolongation (↑ **risk of torsades** arrhythmia).
- **Neuroleptic malignant syndrome** → muscle rigidity + fever in someone who just commenced an antipsychotic – > fever is usually 103.5+.
 - Mechanism → ryanodine channel becomes constitutively open → calcium is released non-stop from sarcoplasmic reticulum into the cytosol → cell uses lots of ATP in order to get calcium back into the sarcoplasmic reticulum → generates heat.
 - Treat with **dantrolene** (inhibits ryanodine channel); rarely bromocriptine (D2 agonist) can treat (choose if dantrolene not listed).
- Hyperprolactinemia (impotence in men; galactorrhea and amenorrhea in women).
 - D2 agonism normally inhibits prolactin secretion. Therefore D2 antagonism causes increased prolactin.
- **“Rule of 4s”** → After a patient is started on an antipsychotic, a general trend is seen in terms of the onset of particular Sx. The time frame is not strict/rigid; use it as a general trend – i.e., acute dystonia wouldn't just start at 4 months; tardive dyskinesia wouldn't occur as early as 2 weeks.
 - **Acute dystonia at 4 hours** → torticollis, oculogyric crisis, muscle rigidity **without** fever.
 - Torticollis = stiff / crooked neck.
 - Oculogyric crisis = weird eye movements (don't confuse with tongue movements of tardive dyskinesia).
 - Muscle rigidity **without** fever = acute dystonia. Muscle rigidity **with** fever = neuroleptic malignant syndrome.

- Treat acute dystonia with **benztropine** (muscarinic receptor antagonist → decreases muscle tone) or a 1st generation H1 blocker (**diphenhydramine** or **chlorpheniramine**). The latter have nasty anticholinergic (anti-muscarinic) side-effects that are *actually what we want* when we're treating acute dystonia. Maybe 2/3 of acute dystonia Tx Qs will have benztropine as the answer; ~1/3 will have one of the 1st gen H1 blockers as correct.
 - 38F + started on trifluoperazine + now has diffuse muscle rigidity; temperature is 98.6F; treatment? → answer = benztropine, diphenhydramine, or chlorpheniramine; Dx is acute dystonia, not NMS.
 - 38F + started on trifluoperazine + now has diffuse muscle rigidity; temperature is 103.8F; treatment? → dantrolene (or rarely bromocriptine).
- **Akathisia at 4 days** → restlessness.
 - Treat with propranolol (beta-blockade).
- **Parkinsonism at 4 weeks** → akinesia / bradykinesia.
 - Treat with amantadine.
- **Tardive dyskinesia at 4 months** → abnormal facial movements (notably tongue).
 - Risk is greater with typicals compared to atypicals, but TD can be seen in the latter on NBME.
 - Treatment is stop the typical and give an atypical.
 - Couple of weird Psych NBME Qs on this:
 - Patient on *atypical* + gets TD; Tx? → stop the atypical and give another atypical. (Just stop the drug and switch to yet another atypical.)
 - Patient on typical for 15 years and has no problems whatsoever (i.e., does not have TD); patient asks the psychiatrist what can be done to decrease his risk of developing TD; the correct answer is "stop the typical and give an atypical"; wrong answer is "maintain current drug regimen." Apparently even if the patient has been on a typical long term without an issue, switching to an atypical *still* confers a reduction of risk of TD.
- **The "triad"** → anti-cholinergic (anti-muscarinic), anti-alpha 1, anti-H1.
 - If an agent is cholinergic, it will cause **DUMBBELSS** signs/symptoms –
 - > **D**iarrrhea, **U**rination, **M**iosis, **B**radycardia, **B**ronchoconstriction, neuromuscular **E**xcitation, **L**acrimation, **S**alivation, **S**weating. Since antipsychotics are anti-cholinergic, they cause the opposite of **they will cause the opposite of DUMBBELSS** → Constipation, Urinary retention, Mydriasis, Tachycardia, *Not bronchodilation* (β_2 agonism, but not M antagonism, will bronchodilate), Skeletal muscle relaxation, Dry eye, Dry mouth, Anhydrosis.
 - 23M + started on aripiprazole for schizophrenia + now is hot, red, and dry; which of the following receptor effects is most likely responsible for this patient's condition? → answer = anti-cholinergic (anti-muscarinic).
 - Anti-alpha 1 → orthostatic hypotension + fainting (alpha 1 agonism normally constricts peripheral arterioles, maintaining BP and cerebral perfusion).
 - Anti-H1 → sedation.
 - The "triad" is seen with antipsychotics, TCA anti-depressants (e.g., amitriptyline), and 1st gen H1 blockers (e.g., diphenhydramine).

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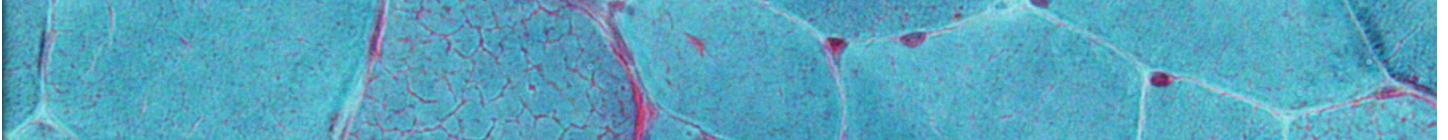


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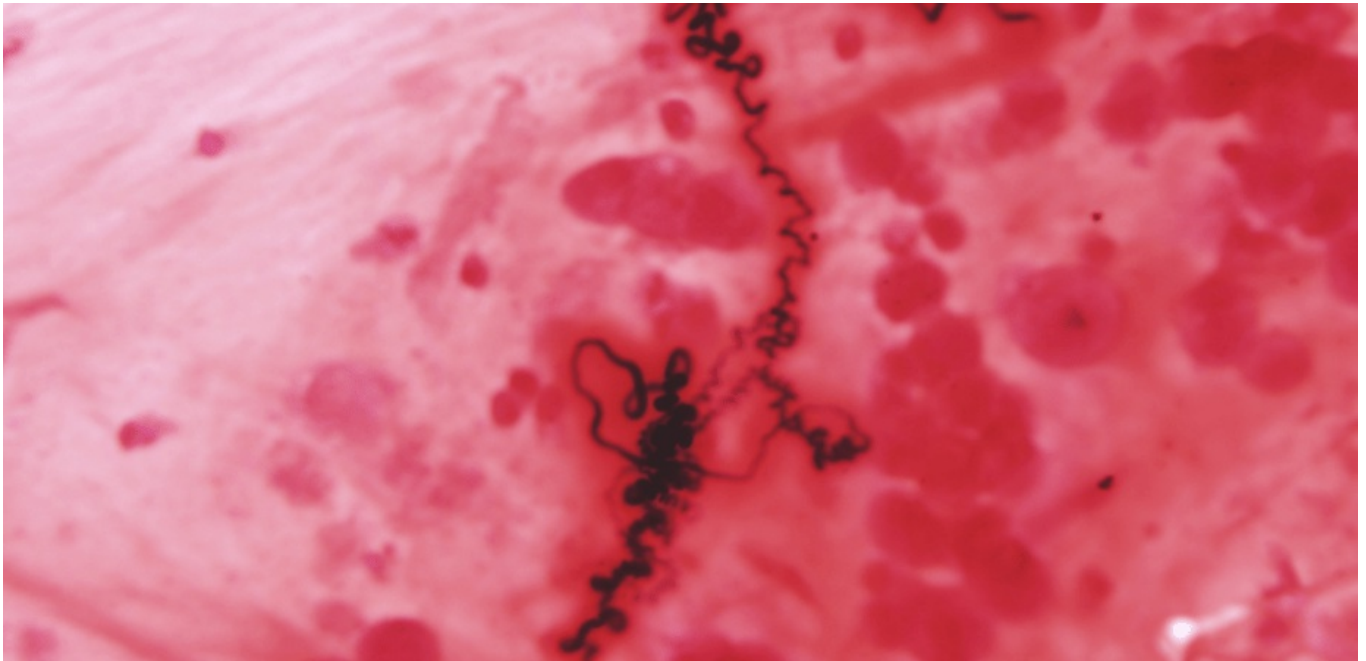
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PHARMACOLOGY / PULMONARY

Pulmonary pharm – Asthma drugs

by MEHLMANMEDICAL



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Asthma drugs

Albuterol (salbutamol)

MOA of albuterol?

- Short-acting beta-2 agonist.

How is it used?

- First-line treatment for acute attacks.
- Patient uses inhaler when feeling asthmatic symptoms.
- Used as a nebulizer (face mask) in the emergency setting.

Fluticasone

MOA of fluticasone?

- Inhaled corticosteroid (ICS).

How is it used?

- Used to prevent recurrences of asthma.
- Added second-line to albuterol for outpatient treatment – i.e., if patient has weekly occurrences of asthma, rather than just receiving an albuterol inhaler to abort acute episodes, a preventative ICS can be added to the regimen.
- **Has no utility in the acute setting.**
- In the acute setting, IV steroids (methylprednisolone) can be given, but ICS have no role.

Salmeterol

MOA of salmeterol?

- Long-acting beta-2 agonist.

How is it used?

- Can be added third-line for outpatient asthma management for patients already using an albuterol and fluticasone inhaler but who still have occurrences.
- In other words, the first three drugs for outpatient management are: albuterol → fluticasone → salmeterol.

Zileuton

MOA of zileuton?

- Lipoxygenase inhibitor (enzyme that normally synthesized leukotrienes, which cause bronchoconstriction).

When is it used?

- Patients already on albuterol, fluticasone, and salmeterol inhalers but who still have intractable asthma.
- Good for patients with history of **NSAID/aspirin-induced asthma** (i.e., Samter triad → NSAID/aspirin hypersensitivity, NSAID/aspirin-induced asthma, nasal polyps):
 - Arachidonic acid is a precursor molecule of both leukotrienes and prostaglandins.
 - Lipoxygenase catalyzes synthesis of leukotrienes.
 - Cyclooxygenase catalyzes synthesis of prostaglandins.
 - NSAIDs (including aspirin) inhibit cyclooxygenase (COX), thereby preventing prostaglandin synthesis. Arachidonic acid is then shunted down the lipoxygenase pathway toward leukotrienes → bronchoconstriction.

- 16M + takes aspirin for sore muscles after snowboarding + gets wheezing. Q asks: what is most likely to be seen on physical exam? → answer = nasal polyps.
- 16M + takes aspirin for sore muscles after snowboarding + gets wheezing. Q asks: which drug should most likely be avoided in this patient? → answer = ibuprofen; wrong answer = acetaminophen (not an NSAID).

Montelukast, zafirlukast

MOA of montelukast and zafirlukast?

- Leukotriene receptor antagonists.
- Leukotrienes cause bronchoconstriction, so antagonizing these receptors decreases bronchoconstriction.

When are they used?

- Same as zileuton.

Nedocromil, cromolyn sodium

MOA of nedocromil and cromolyn sodium?

- Mast cell stabilizers.
- Mast cells normally degranulate to release histamine → stimulate H1 receptors on bronchioles + stimulation of afferent vagal fibers in the airways → bronchoconstriction.

When are they used?

- Patients who have persistent attacks despite already using albuterol, fluticasone, and salmeterol.

Theophylline

MOA of theophylline?

- Phosphodiesterase inhibitor → prevents breakdown of cAMP.

What are its side-effects?

- Not used due to neuro- and cardiotoxicity.
- Just know MOA + toxicities for USMLE.

Omalizumab

MOA of omalizumab?

- Monoclonal antibody against the Fc region of IgE.

When is it used?

- Intractable asthma refractory to multi-drug regimens, including patients already on oral prednisone.
- Patients who qualify for use have high serum IgE levels.

-Lizumab agents → **Benlizumab, Reslizumab, Mepolizumab** → “**Ben** Really likes **Maple**.” → **Ben**lizumab, **Res**lizumab, **Mepo**lizumab

MOA of -lizumabs?

- Monoclonal antibodies against IL-5.
- IL-5 normally promotes eosinophil maturation.

When are they used?

- Intractable asthma in patients with eosinophilia, similar to omalizumab.

Methacholine

MOA of methacholine?

- Muscarinic receptor agonist

When is it used?

- To diagnose asthma. **Not used in the treatment of asthma.**
- Causes bronchoconstriction → reproduces asthma symptoms and obstructive spirometry profile in patients with asthma.
- Patients who do not have asthma will not get symptom production with methacholine.
- Never used during acute attacks; used between attacks to merely reproduce symptoms.

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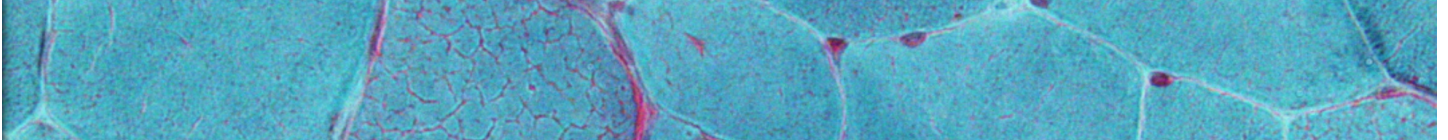


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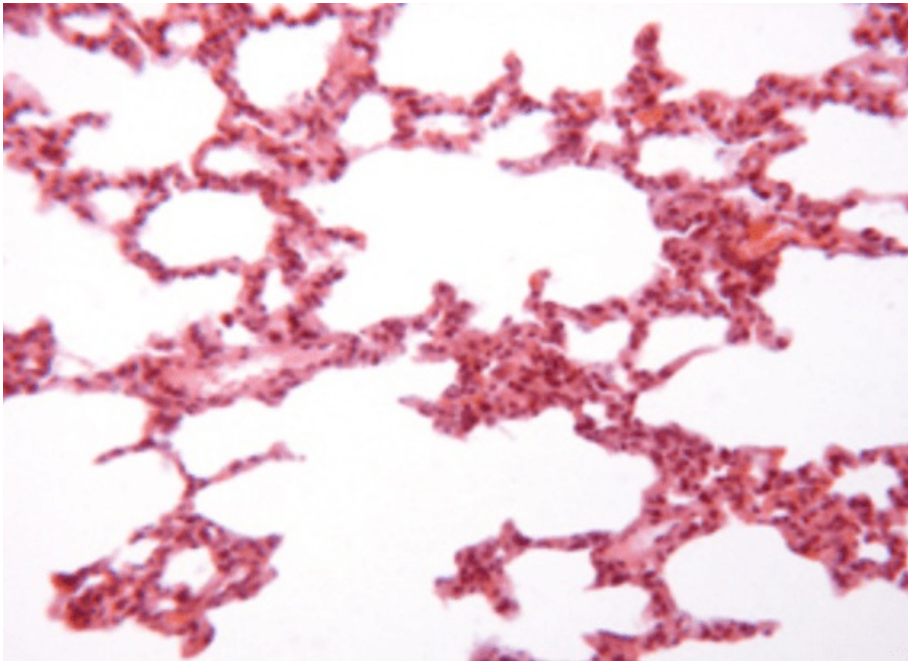
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PHARMACOLOGY / PULMONARY

Pulmonary pharm – HY mixed drugs

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Bosentan

MOA of bosentan?

- Endothelin 1 receptor antagonist.
- Endothelin is a vasoconstrictor of pulmonary vessels.
- Bosentan is used to treat pulmonary hypertension → decreased vasoconstriction of pulmonary vasculature.
- One of the highest yield drugs on the Step 1 NBME forms.

When is endothelin increased?

- In any situation where there is increased pulmonary capillary pressure. This includes:
 - Any cause of pulmonary hypertension (i.e., primary pulmonary hypertension due to BMPR2 mutation; COPD; CF; systemic sclerosis).
 - Cor pulmonale (right heart failure due to a pulmonary cause).
 - Drugs causing pulmonary fibrosis, including methotrexate, amiodarone, bleomycin, busulfan.
 - Left heart failure (due to increased pulmonary capillary wedge pressure, which means increased pulmonary capillary hydrostatic pressure).

How will the USMLE ask about it?

- 68M + lateralized apex beat + S3 heart sound + systemic hypertension; which of the following would be expected in this patient? → answer = up arrow for endothelin; up arrow for beta-myosin gene transcription (due to left ventricular hypertrophy); up arrow for c-jun (a transcription factor, because of left ventricular hypertrophy).
 - Patient has dilated cardiomyopathy secondary to HTN → increased left-sided heart pressure → increased left atrial pressure → increased pulmonary capillary wedge pressure (since LAP = PCWP) → increased pulmonary vascular pressure means constriction of pulmonary vessels (mediated by endothelin).
- 32F + non-smoker + 6-month Hx of worsening dyspnea + loud pulmonary component of S2 (loud P2); which of the following is most likely associated with her condition? → answer = “increased endothelin activity.”
 - Patient has primary pulmonary hypertension (genetic; usually due to BMPR2 mutations); will present as young adult with pulmonary hypertension who’s a non-smoker.
 - If you know that bosentan treats pulmonary hypertension by increasing the diameter of the vasculature, then you can infer that in pulmonary hypertension, where the vasculature is more constricted, endothelin must be higher.
 - Loud P2 means pulmonary hypertension → pulmonic valve is slamming shut due to increased pressure distal to the valve.

Epoprostenol, iloprost

MOA of epoprostenol and iloprost?

- PGI₂ analogues (prostacyclin).

When are they used?

- Pulmonary hypertension.
- Should be noted that dihydropyridine calcium channel blockers, such as nifedipine, are actually first-line for most cases of pulmonary hypertension. Patients non-responsive to calcium channel blockers are trialed on other agents such as bosentan, PGI₂ analogues, and sildenafil (yes, Viagra; it’s a phosphodiesterase 5 inhibitor → increases cGMP → vasodilation).

Pseudoephedrine, ephedrine

MOA of pseudoephedrine, ephedrine?

- Various; but main one is alpha-adrenergic receptor agonism.

When are they used?

- Nasal congestion
- Prevention of barotrauma in scuba divers (opens Eustachian tubes) → weird, but a detail that's associated with these drugs.

Guaifenesin

- Expectorant → thins bronchial secretions → facilitates coughing up mucous.

Dextromethorphan

- Mu-opioid receptor agonist
- Used for cough suppression
- Low addictive potential
- Can cause constipation

Dornase alfa (correct, not alpha)

- Recombinant human deoxyribonuclease → cleaves nucleic acids in bronchial mucous → thins mucous in patients with cystic fibrosis.

N-acetylcysteine

- Used as the antidote for acetaminophen toxicity.
- Also can be used to thin bronchial secretions by disrupting disulfide bonds.
- Cysteine has a thiol (-SH) side-group that can bind to -S-S- and break open the bonds.

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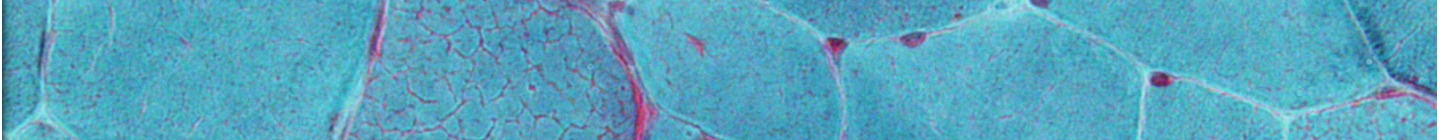


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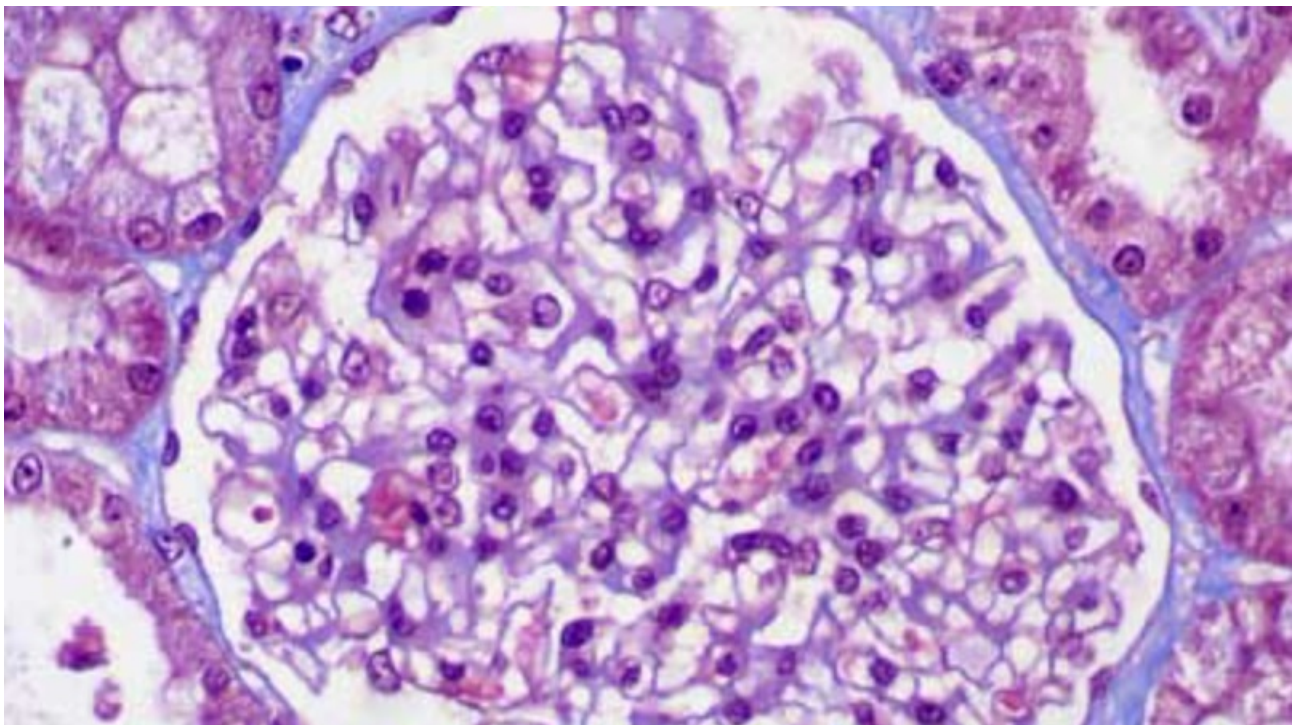
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Renal pharm – Diuretics HY points

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Mannitol

MOA of mannitol?

- Osmotic diuretic; acts in the proximal convoluted tubule (PCT), proximal straight tubule (PST), and descending loop of Henle.
- Osmotic diuretic means it functions to retain water in the lumen of the nephron by exerting an oncotic effect (i.e., the mere presence of more solute in the lumen means water is less likely to be reabsorbed → more is micturated).

When is it used?

- Can be used to treat increased intracranial pressure.
- The sequence of treatment is: elevate head of the bed → intubation + hyperventilation (decreased CO₂ causes decreased cerebral perfusion) → mannitol. In other words, mannitol is never the first answer for increased intracranial pressure; just know that it can be used for this purpose.

When is mannitol notably contraindicated?

- Don't give to patients with heart failure → increased oncotic pressure occurs in the peripheral vasculature as well → transient increase in plasma volume → increased venous return → increased oxygen demand on the myocardium → increased risk of cardiac decompensation and ischemia.

Acetazolamide

MOA of acetazolamide?

- Carbonic anhydrase inhibitor in the PCT of the kidney.
- Carbonic anhydrase functions to facilitate CO₂ and H₂O combining to form H₂CO₃. Once H₂CO₃ is formed, it can equilibrate into bicarb and a proton. The bicarb is then reabsorbed by the PCT tubular cell.
- In other words, if we inhibit carbonic anhydrase with acetazolamide, less bicarb is reabsorbed, thereby creating a metabolic acidosis for the patient.

When is it used?

- Altitude sickness → person goes to high altitude where there's fractionally less O₂ in the air → breathes faster to compensate → also results in blowing off more CO₂ → patient becomes transiently alkalotic because it takes time for the kidney to naturally excrete more bicarb to compensate for the reduced serum pCO₂. The role of acetazolamide is to create a transient metabolic acidosis by forcing bicarb excretion; this compensates for the respiratory alkalosis faster than the kidney can naturally excrete bicarb.
- Topical carbonic anhydrase inhibitors can be used for glaucoma by decreasing aqueous humor production.

Adverse effects?

- Sulfa drug (contraindicated in sulfa allergy).
- Because it increased the pH of the urine, it can theoretically increase the risk of calcium phosphate stones (which form at higher pH). Calcium oxalate stones, in contrast, form at lower pH, so risk of these would not be increased.

Loop diuretics

- Be familiar with a few names → **furosemide, bumetanide, torsemide, ethacrynic acid**

MOA of loops?

- Inhibit the Na/K/2Cl (one sodium, one potassium, and two chloride) symporter on the apical membrane of the thick ascending limb of the loop of Henle.

- When ions are reabsorbed, water follows, so inhibition of this four-ion symporter causes massive diuresis (increased urination).
- This effect also decreases paracellular (between the cells, without a transporter) reabsorption of calcium and magnesium. That is, loops also cause hypercalciuria (and can cause hypocalcemia). The USMLE likes you to know the effect of loops on calcium.

When are they used?

- Most effective agents for fluid unloading in patients with peripheral and pulmonary edema due to heart, liver, or renal failure.
- Must be pointed out that they do not decrease mortality in heart failure; they are merely used for symptomatic patients (i.e., dyspnea from pulmonary edema, or peripheral edema) who need fluid unloading.
- Ethacrynic acid is a non-sulfa loop – i.e., it can be used in patients with sulfa allergy. It carries the same adverse effects of other loops, including ototoxicity.

Adverse effects?

- Ototoxicity → need not be neurosensory hearing loss → can be vertigo (room is spinning). This ototoxic effect is synergistic if the patient is simultaneously on aminoglycosides (e.g., gentamicin), as the latter are also ototoxic.
- Hypokalemia → due to a combination of the loop directly preventing potassium reabsorption + the effects of the renin-angiotensin-aldosterone system (RAAS) being upregulated to compensate for the drop in plasma volume → increased distal secretion of potassium under the action of aldosterone.
 - Aldosterone causes sodium + water reabsorption, and potassium + proton secretion (excretion) in the distal kidney.
- Dehydration with orthostatic hypotension (i.e., if you're causing a high level of diuresis, you're prone to fainting).
- Contraction alkalosis → since loops cause significant diuresis, the RAAS will upregulate and the late-DCT and collecting duct will attempt to reabsorb more fluid to compensate → however the action of aldosterone leads to secretion of protons distally, resulting in increased serum pH and bicarb.
- Hypocalcemia (due to hypocalciuria).
- Loops are said to “abolish the effects of vasopressin (ADH).” → ADH functions to insert aquaporins in the medullary collecting duct to promote free-water reabsorption → the high ion concentrations in the renal interstitium *due to* the ion reabsorption in the thick ascending limb are what enable the free-water reabsorption to occur passively. If loops prevent ion reabsorption and in turn abolish the ion concentrations in the interstitium, less free-water reabsorption will occur due to ADH. Once again, just memorize: “loops abolish the effects of vasopressin.”
- Sulfa allergy.

Thiazides

- Be familiar with a few names → **hydrochlorothiazide (HCTZ), chlorthalidone, metolazone, indapamide**

MOA of thiazides?

- Block the Na/Cl symporter on the apical membrane of the early-DCT.
- This results in increased calcium reabsorption in the DCT. That is, thiazides cause hypocalciuria and hypercalcemia. Note that this is the opposite effect of loop diuretics. The USMLE wants you to know this contrast of effect urinary and serum on calcium.

When are they used?

- First-line treatment for essential hypertension in patients without pre-diabetes, diabetes, renal, or cardiovascular disease. In other words, thiazides or dihydropyridine calcium channel blockers (e.g., nifedipine) are used for

otherwise healthy patients who have essential hypertension. This shows up on 2CK FM material (i.e., patient has HTN and hydrochlorothiazide is the answer).

- If the patient has pre-diabetes, diabetes, renal, or cardiovascular disease, use an ACE inhibitor or angiotensin II receptor blocker (ARB). As an example, if the patient has elevated fasting glucose or evidence of protein in the urine, use an ACEi or ARB, not a thiazide or dCCB.
- Used to prevent recurrent calcium urolithiasis (because they cause hypocalciuria).

Adverse effects?

- Hyper**GLUC**
 - Hyper**G**lycemia → can cause diabetes, or push pre-diabetics into diabetes; so never use in patients with elevated fasting glucose. Use with caution in overweight patients.
 - Hyper**L**ipidemia → have never seen this on NBME material, but part of the mnemonic.
 - Hyper**U**ricemia → can cause gout → have also never seen this on NBME material, but part of the mnemonic.
 - Hyper**C**alcemia → due to increased reabsorption of calcium in the DCT; likewise, thiazides cause hypocalciuria; this effect on calcium is HY for Step.

Amiloride, triamterine

MOA of amiloride and triamterine?

- ENaC inhibitors (inhibit sodium reabsorption in the late-DCT and cortical collecting duct).
- Water follows sodium, so inhibition of ENaC promotes diuresis.
- Potassium-sparing (i.e., do not cause a decrease in serum potassium).
 - Aldosterone normally promotes potassium secretion distally in the kidney. It accomplishes this by upregulating the basolateral Na/K-ATPase pump on the basolateral surface of the cortical collecting duct. This causes sodium to be reabsorbed from the tubular cell into the blood, and potassium to be secreted from the blood into the tubular cell. The reduction in intra-tubular cell sodium causes a favorable (high-to-low) gradient from the urine into the cell for sodium. ENaC, on the apical membrane, then opens to enable sodium to flow from the urine into the cell. The movement of sodium through ENaC is considered secondary active transport because sodium is able to move down its apical gradient only because energy was first utilized by the basolateral ATPase pump.
 - Inhibition of ENaC on the apical membrane means the basolateral ATPase pump can't continue to function, since there's less intracellular sodium to move across the basolateral membrane. In turn, this means less potassium can be secreted, so serum potassium is maintained (i.e., ENaC inhibitors are potassium-sparing).

When are they used?

- Triamterine (and spironolactone, mentioned below) is also used as a second agent for fluid unloading when the patient is already on a loop diuretic (usually furosemide). Since furosemide can cause hypokalemia, if the patient needs more diuretic therapy, a potassium-sparing diuretic is chosen next, and triamterine or spironolactone are common answers on USMLE for this purpose.

Adverse effects?

- Can cause hyperkalemia (since it's potassium-sparing).

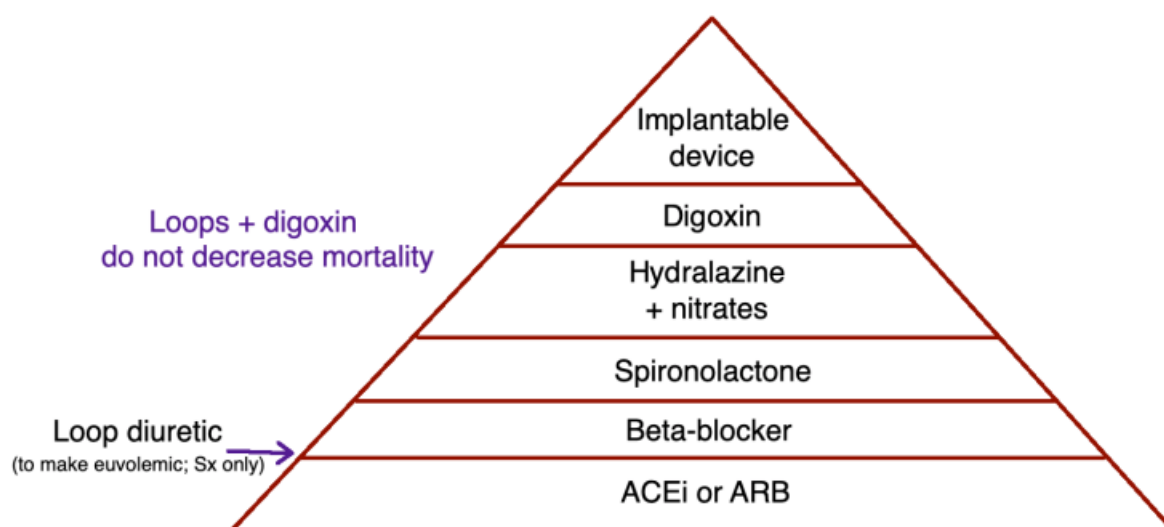
Spironolactone, eplerenone

MOA of spironolactone and eplerenone?

- Aldosterone receptor antagonists.
- Potassium-sparing.
- By blocking aldosterone receptors, there's decreased activity of the Na/K-ATPase pumps on the basolateral membrane of the late-DCT and cortical collecting duct → less Na reabsorption → since water follows sodium, there's less water reabsorption → decreased plasma volume. Likewise, potassium and protons are not secreted – > serum potassium does not go down (potassium-sparing).

When are they used?

- Spironolactone is classically used in heart failure after the patient is already on an ACEi/ARB and beta-blocker.
- Hierarchy for drugs in heart failure (simplified but HY for USMLE):
 - Start with ACE inhibitor or an angiotensin II receptor blocker (ARB) to improve ejection fraction (normal is 55-70%).
 - If the patient is fluid overloaded (i.e., peripheral edema or pulmonary edema), attempt to achieve euvolemia by *adding* furosemide (loop diuretic).
 - If insufficient ejection fraction with the ACEi or ARB, *add* a beta-blocker (metoprolol XR, bisoprolol, carvedilol, or nebivolol).
 - If insufficient ejection fraction with ACEi/ARB + beta-blocker, *add* spironolactone (aldosterone receptor antagonist).
 - If insufficient ejection fraction, *add* the **combination** of hydralazine + nitrates. Only the combo decreases mortality in HF. It is especially efficacious in African Americans (tend to have stiffer vessels).
 - If insufficient ejection fraction when already on ACEi/ARB + beta-blocker + spironolactone + hydralazine + nitrates, *add* digoxin.
 - If pharmacologic therapy insufficient, use implantable device.



- Spironolactone is also used as a second agent for fluid unloading when the patient is already on a loop diuretic (usually furosemide). Since furosemide can cause hypokalemia, if the patient needs more diuretic therapy, a potassium-sparing diuretic is chosen next, and spironolactone or triamterine are common answers on USMLE for this purpose.
- Can be used for adrenal cortical hyperplasia where aldosterone is high, or for aldosteronoma prior to surgery.

Adverse effects?

- **Spironolactone causes gynecomastia (HY).** It is anti-androgenic and can block androgen receptors. Eplerenone apparently doesn't carry the same risk.
- Both can cause hyperkalemia.

Angiotensin-converting enzyme inhibitors (ACEi)

- Be familiar with a few names → **enalapril, lisinopril, captopril**

MOA of ACEi?

- Inhibit ACE, which normally converts angiotensin I to angiotensin II.
- ACE acts **in the lungs**. Therefore ACEi act **in the lungs**.
- By decreasing angiotensin II levels, ACEi intercept the RAAS and therefore decrease aldosterone production and effects (angiotensin II normally upregulates aldosterone synthase in the zona glomerulosa of the adrenal cortex).
- Angiotensin II on its own is a potent vasoconstrictor. It acts on systemic arterioles and at the efferent arterioles of the kidney. So giving an ACEi causes a decrease in afterload (decreased constriction of systemic arterioles) and improves cardiac output and ejection fraction.
- Potassium-sparing because they intercept RAAS (and therefore aldosterone-induced distal renal secretion of potassium is hindered).

When are they used?

- First-line for essential hypertension in patients with pre-diabetes, diabetes, renal, or cardiovascular disease.
- Also used in patients who don't have hypertension if they have pre-diabetes or diabetes and evidence of proteinuria or increased creatinine.
- First-line in heart failure.
- Oral captopril can be used for hypertensive urgency (>180/110, either number; without evidence of end-organ damage; if retinopathy, nephropathy, etc., then it's hypertensive emergency, and nitroprusside is used, among others).

Adverse effects?

- **Dry cough (HY)** → ACE is also known as bradykininase → therefore ACEi can cause increased bradykinin in the lungs → can cause cough.
- Hyperkalemia (because they're potassium-sparing).
- Angioedema (avoid in patients with hereditary angioedema).
- Can cause increased serum creatinine and renin in patients with renal artery stenosis or fibromuscular dysplasia.
 - 32F + HTN + given enalapril + now creatinine rises to 2.1 from 1.0; diagnosis? → fibromuscular dysplasia.
 - In other words, if you observe in a vignette that creatinine or renin jumps in response to an ACEi or ARB, you know the answer is renal artery stenosis (patients 50s+ with cardiovascular disease) or fibromuscular dysplasia (women 20s-40s without cardiovascular disease).

Angiotensin II receptor blockers (ARBs)

- Be familiar with a few names → **losartan, valsartan, candesartan**

MOA of ARBs?

- Block angiotensin II receptors.
- Effects are the same as ACEi for all intents and purposes.

When are they used?

- Same as ACEi.
- If patient was given an ACEi and gets dry cough, switch to ARB.

Adverse effects?

- Hyperkalemia (because they're potassium-sparing).
- Can cause increased serum creatinine and renin in patients with renal artery stenosis or fibromuscular dysplasia.
 - 32F + HTN + given enalapril + now creatinine rises to 2.1 from 1.0; diagnosis? → fibromuscular dysplasia.
 - In other words, if you observe in a vignette that creatinine or renin jumps in response to an ACEi or ARB, you know the answer is renal artery stenosis or fibromuscular dysplasia.

Vasopressin (ADH) receptor antagonists (tolvaptan, conivaptan)

MOA of tolvaptan, conivaptan?

- Block vasopressin receptors in the kidney.

When are they used?

- SIADH (e.g., from small cell carcinoma of the lung, or post-head trauma).
 - Should be noted that demeclocycline (a tetracycline antibiotic) can cause nephrogenic diabetes insipidus and is also an answer on NBME for the treatment of SIADH.

Aliskiren

MOA of aliskiren?

- Direct renin inhibitor.
- Renin normally cleaves angiotensinogen into angiotensin I in the plasma. So aliskiren decreases angiotensin I production.
- Can be used for hypertension.
- Just know this drug's MOA.

—

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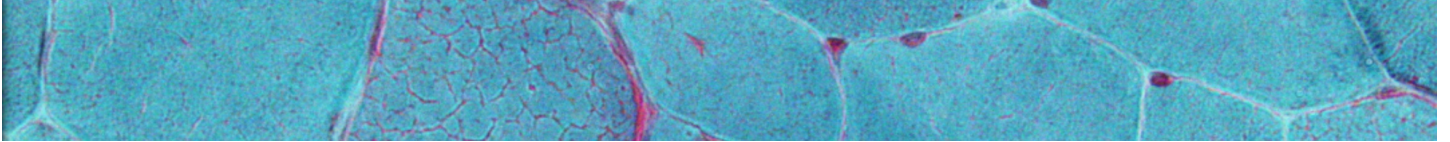


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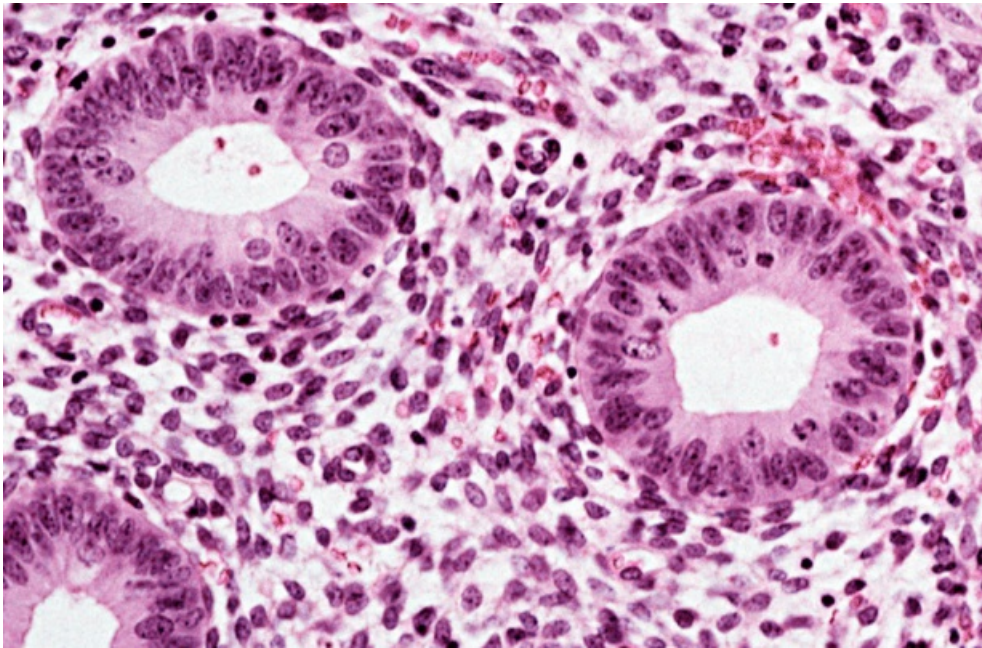
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HY points followed by a quiz at the end

GnRH receptor agonists

- Be familiar with a few of the names → **goserelin, nafarelin, leuprolide**

MOA of GnRH receptor agonists?

- By agonizing the GnRH receptors continuously, despite leading to a transient increase in LH and FSH production, the net result is a desensitization of the receptors and suppression of LH and FSH. This means these drugs *act as antagonists* despite technically being agonists at the GnRH receptor → ultimately leads to decreased LH + FSH – > decreased androgen production by the gonads.

When are they used?

- Classically for prostate cancer (after flutamide).
 - Prostate cancer is normally stimulated by androgens, so by giving a continuous GnRH receptor agonist (usually leuprolide), the net result is decreased androgen production. However, during the first couple days of therapy, since it takes time for the desensitization of the GnRH receptors to occur, there will be a transient *rise* in LH + FSH → *increased* androgens. Therefore, we co-administer leuprolide with **flutamide**, an androgen receptor blocker, which prevents the transient rise in androgens from having an agonistic effect.
- Can be used for uterine fibroids, adenomyosis, and endometriosis. However it should be noted they are *not* first-line therapies. You should just be peripherally aware that they can be used in theory.

Flutamide

MOA of flutamide?

- Androgen receptor blocker

When is it used?

- Given simultaneous to (or before) GnRH receptor agonists (e.g., leuprolide) in the treatment of prostate cancer.
- Since leuprolide will cause a transient increase in androgens, flutamide will block this effect.

Degarelix

MOA of degarelix?

- GnRH receptor antagonist

When is it used?

- Prostate cancer
- Unlike GnRH receptor agonists, it does not cause a transient rise in androgens. Therefore flutamide need not be co-administered.

Clomiphene

MOA of clomiphene?

- Technically a selective estrogen receptor modulators (SERM), which means it has different effects at different tissues, but for USMLE purposes, memorize it as an estrogen receptor partial agonist at the hypothalamus.
- The partial agonist effect is akin to **being an antagonist** because it does not induce as much efficacy at the receptor compared to estrogen (i.e., the hypothalamus interprets the binding of clomiphene as insufficient estrogen binding).
- Result is increased GnRH secretion → increased LH + FSH → induction of ovulation.

When is it used?

- Induces ovulation in those with abnormal cycles (i.e., enables women to time copulation for fertilization) and subfertility.
- Can be used in women with subfertility.

Adverse effects?

- Hot flashes
- Multiple gestation pregnancies (e.g., quadruplets due to release of multiple ova; although some argue this isn't actually a bad thing).

Tamoxifen vs raloxifene

MOA of tamoxifen and raloxifene?

- Selective estrogen receptor modulators (SERMs)
- Bind to estrogen receptors with varying degrees of antagonism and agonism depending on the tissue the receptors are located in.
- Both are antagonists in breast tissue and agonists at bone.
- **Tamoxifen is a partial agonist at endometrium; raloxifene is an antagonist at endometrium.**

When are they used?

- Estrogen receptor positive (ER+) breast cancer.

If they are agonists at bone, does that mean they're used to preserve bone density?

- **No.** Whilst in theory they could be used to preserve bone density, they are not used for this purpose because they increase risk of thromboembolic events (i.e., DVT, stroke, MI). This is because estrogen upregulates fibrinogen and factors V and VIII.

What's the highest yield point for USMLE regarding these drugs?

- Because tamoxifen is a partial agonist at endometrium, there is **increased risk of endometrial cancer** with its use. The only time tamoxifen should ever be considered in a patient who has breast cancer is if the patient has history of hysterectomy.
 - 54F with ER(+) breast cancer + long paragraph of a vignette including surgical history + commenced on tamoxifen; Q asks what aspect of the patient's history makes her a good candidate for tamoxifen → answer = "history of hysterectomy," or "surgical history."

Aromatase inhibitors

- Be familiar with a few names → **anastrozole, exemestane**

MOA of anastrozole and exemestane?

- Aromatase converts androgens to estrogens. Therefore aromatase inhibitors decrease estrogen synthesis.

When are they used?

- Estrogen receptor + (ER+) breast cancer.
- Usually used in postmenopausal women (but USMLE won't assess you on this detail; you just need to know the

drug names + MOA).

Hormone replacement therapy (HRT)

- Estrogen and progestin (any number of progesterone compounds) given cyclically.
- If the woman does not have a uterus, just give continuous estrogen (only reason progesterone is normally given is to prevent unopposed estrogen increasing the risk of endometrial cancer, but if the woman has history of hysterectomy, give estrogen alone).

When is it used?

- **Only approved indication is severe vasomotor symptoms** (i.e., hot flashes, urge incontinence).
- Is **not** used to preserve bone density. This is because estrogen increases risk of **thromboembolic events** (i.e., DVT, stroke, MI), since estrogen upregulates fibrinogen and factors V and VIII. It also **increases risk of breast cancer**. The risk of breast cancer is related to the estrogen exposure; it has nothing to do with whether progesterone is added or not; progesterone is only added for women without Hx of hysterectomy in order to prevent endometrial cancer.
- If a peri- or postmenopausal woman has vaginal dryness alone (atrophic vaginitis) resulting in dyspareunia (painful sex), topical non-hormonal lubricants are used first, followed by **topical estrogen**. That is, oral HRT is not needed for mere atrophic vaginitis. But it's HY for USMLE 2CK obgyn component to know topical estrogen is for atrophic vaginitis.

Mifepristone

MOA of mifepristone?

- Progesterone receptor antagonist used as an abortifacient (i.e., to induce abortion in early first trimester).
- Do not confuse with **misoprostol**, which is a prostaglandin E1 analogue.
- The combination of mifepristone, followed by misoprostol, is used to induce abortion.

Ulipristal

MOA of ulipristal?

- Selective progesterone receptor modulator (SPRM) that acts **as an antagonist at progesterone receptor**.
- Used for emergency contraception.
- Most effective pharmacologic agent for emergency contraception. Most effective emergency contraception overall is copper IUD.

Danazol

MOA of danazol?

- Androgen receptor partial agonist

When is it used?

- Treatment for hereditary angioedema (deficiency of C1 esterase inhibitor). Weird, but it causes the liver to produce more C1 esterase inhibitor.
- Can be used for endometriosis (won't be an answer on the USMLE, but just be peripherally aware of this).

Side-effects?

- Danazol is known to cause pseudotumor cerebri (increased intracranial pressure; ICP).
- When are you asked what can cause increased ICP, things that should come to mind are: obesity, combined oral contraceptive pills, danazol, high-dose vitamin A (bear liver, or isotretinoin).
- Can cause hirsutism. Despite being a partial agonist (i.e., does not achieve full efficacy at the androgen receptor), it still causes hirsutism and this will show up in vignettes.

Finasteride

MOA of finasteride?

- 5 α -reductase inhibitor.
- 5 α -reductase normally converts testosterone into dihydrotestosterone (DHT), so finasteride decreases production of DHT.

When is it used?

- Treats benign prostatic hyperplasia (BPH).
- Prostatic tissue growth is stimulated by DHT, so decreased DHT production $\rightarrow$ treatment for BPH.
- Can be used as monotherapy, or in combo with tamsulosin or terazosin.

Tamsulosin, terazosin

MOA of tamsulosin and terazosin?

- α 1 receptor antagonists, with greater binding to the receptors on the internal sphincter of the bladder.
- α 1 agonism normally causes constriction of the sphincter, so antagonism causes decreased tone of the sphincter $\rightarrow$ better ability to void the bladder. This does not cause incontinence since the external sphincter is under somatic (voluntary) control; in contrast, the internal sphincter is under sympathetic (autonomic; involuntary) control.

When are they used?

- BPH, same as finasteride. Can be used as monotherapy, or in combo with finasteride.
- Can lower blood pressure slightly, so if the Q gives HTN in a guy with BPH, and finasteride and the α 1 blocker are both listed, choose the latter.

Sildenafil

MOA of sildenafil?

- Phosphodiesterase-5 inhibitor $\rightarrow$ prevents breakdown of cGMP.
- cGMP normally activates protein kinase G $\rightarrow$ leads to relaxation of smooth muscle in the corpus cavernosum $\rightarrow$ increased blood flow $\rightarrow$ tumescence (erection).

When is it used?

- To achieve penile tumescence (erection).

Minoxidil

- Opens potassium channels on vascular smooth muscle $\rightarrow$ hyperpolarization of the cell (potassium flows out) $\rightarrow$ relaxation of smooth muscle $\rightarrow$ decreases blood pressure (can be used for hypertension).
- Can be used topically for androgenic alopecia (male pattern balding).